Keraunos-E100 Architecture

Version 0.9.14, 04/14/2025: stable

Table of Contents

Chapter 1. Revision History

Revision	Date	Author	Description
0.7	07/31/2024	Suresh Hariharan	POR feature signoff
0.7.1	08/30/2024	Suresh Hariharan	Added clock and voltage specifications
0.7.2	09/20/2024	Suresh Hariharan	Added power and PTP requirements
0.7.3	09/23/2024	Suresh Hariharan	Updates based on internal review feedback: Harvesting, address map
0.7.4	09/30/2024	Suresh Hariharan	Updates based on internal review feedback: PTP, address map, Ethernet TX/RX controller
0.7.5	10/01/2024	Suresh Hariharan	Updates based on internal review feedback: Address translation, warm reset, telemetry
0.7.6	10/02/2024	Suresh Hariharan	Updates based on cross-functional review feedback: Increase SRAM size from 4MB to 8MB, add two more cores to DM-DISC complex for link management
0.7.7	10/03/2024	Suresh Hariharan	Updates based on cross-functional review feedback: Add JTAG details
0.7.8	10/04/2024	Suresh Hariharan	Updates based on cross-functional review feedback: Change controller architecture to two parallel 400G datapaths to ensure performance with BlackHole Ethernet TX/RX controller logic. Clarify that only Quasar-to-Quasar use cases are POR.
0.7.9	10/07/2024	Suresh Hariharan	Add composition diagrams. Update profiling section.
0.7.10	11/01/2024	Suresh Hariharan	Update JTAG information.
0.7.11	12/03/2024	Suresh Hariharan	Add ethernet controller details, add register descriptions
0.7.12	12/11/2024	Neeraj Kulkarni	Add microarchitecture details for HSIO tile
0.7.13	12/12/2024	Suresh Hariharan	Add register descriptions
0.7.14	12/13/2024	Neeraj Kulkarni	Add microarchitecture details for HSIO tile
0.7.15	12/13/2024	Neeraj Kulkarni	Updates based on review feedback
0.7.16	1/23/2025	Neeraj Kulkarni	Updated address map, added Keraunos NOC section, updated HSIO uarch, and HSIO fabric connectivity
0.9	1/24/2025	Suresh Hariharan, Neeraj Kulkarni	Add PCIe subsystem and SEP, update compositions, update clocks, additional fabric details.
0.9.1	1/29/2025	Neeraj Kulkarni	Updated PCIe tile interfaces, add HSIO fabric paths, updated HSIO figures
0.9.2	2/20/2025	Suresh Hariharan	Added MSI mailbox info from Yasuo. Removed harvesting muxes. Added KPI change for D2D sharing between PCIe and one HSIO subsystem.
0.9.3	2/21/2025	Suresh Hariharan	Update voltages to match new fixed VDD_SOC decision.
0.9.4	3/7/2024	Suresh Hariharan	Update clock table and add note on placement and PLL selection.
0.9.5	3/10/2025	Neeraj Kulkarni	Add DFD, Profiler, Error bank, SERDES control address spaces and respective ports in HSIO fabric. Route control traffic between HSIO tiles via SMN.
0.9.6	3/15/2025	Neeraj Kulkarni	Clean up clocks and add clock domain figures for keraunos soc and hsio tile, add address remap entries for all keraunos remaps
0.9.7	3/21/2025	Neeraj Kulkarni	Removed AWM PLLs, updated PLL placement, fixed DFS clocks
0.9.8	3/24/2025	Neeraj Kulkarni	Removed NOC2AXI from HSIO tile. Use NOC2AXI in D2D to route between HSIO tiles. Updated D2D, HSIO, Keraunos Addressing, and Keraunos NOC sections.
0.9.9	3/27/2025	Neeraj Kulkarni	Updated Interrupt Handling. Added wired interrupts to SMC PLIC.
0.9.10	3/27/2025	Neeraj Kulkarni/Jeff Mitchem	Added HSIO Clock and Reset block (HSIO and Addressing sections). Fixed Ethernet SerDes reference clock.
0.9.11	4/1/2025	Alan Carr	Added PTP timesync to GPIO

Revision	Date	Author	Description
0.9.12	4/4/2025	Neeraj Kulkarni	Added Reset section (Reset signals, Reset Sequences, and subsystem resets).
0.9.13	4/5/2025	Neeraj Kulkarni	HSIO fabric updates: CCE MMIO AXI ID, Profiler to SRAM connectivity
0.9.14	4/7/2025	Zheng Liu	Added Keraunos Perf tests
0.9.15	4/14/2025	Suresh Hariharan	Added clock gaters to HSIO tile
0.9.16	8/1/2025	Neeraj Kulkarni	Section 7.1: Add address regions for HSIO/D2D temperature sensors, HSIO/SMN ATB2AXI, SRAM config, and removed Profiler 1 address space. Subsumed DFD Trace Sink address in DFD address space. Section 7.2: Fixed PCIE remap table. Fixed SMC/SEP address map in remap tables.
0.9.17	8/1/2025	Neeraj Kulkarni	Section 8.1: Add AxID for SMN ports. Section 8.3: Add PacketProbetoATB INIU and SRAM config TNIU. Added connections from CCE0/CCE1 MMIO to CCE1/CCE0 NOC port, SRAM config, and Profiler config. Updated outstanding entries for each initiator and target. Section 6.5: Added ATB2AXI bridge and SRAM config connectivity in Figure 6.
0.9.18	8/1/2025	Neeraj Kulkarni	Section 4.7: Removed any references to warm reset and hold signal functionality.
0.9.19	8/2/2025	Neeraj Kulkarni	Section 4.6.2: Replaced Athena mailbox functionality for interrupts from Host to SMC with existing SMC mailboxes.
0.9.20	8/3/2025	Neeraj Kulkarni	Section 13.1: Updated Power supply specifications to match Grendel specification.
0.9.21	8/4/2025	Neeraj Kulkarni	Section 8.1: Updated SMN to partitioned design with HSIO SMN tile and SMN Bridge. Added ports, address targets and connectivity for all destination ports on both tiles.
0.9.22	8/5/2025	Neeraj Kulkarni	Section 5.1 Updated D2D tile micro-architecture figure, with error bank, west QNP ports, and legend. Section 7.1 Added address space for D2D Error bank registers.
0.9.23	8/6/2025	Neeraj Kulkarni	Section 10. Added multiple subsections on features, usage, record formats, profiler architecture and profiler events.
0.9.24	8/6/2025	Neeraj Kulkarni	Section 17.3 Added Error banks and signaling in HSIO and Keruanos SOC.
0.9.25	8/8/2025	Neeraj Kulkarni	Section 14.2 Added clock gaters and reference clock muxes. Updated HSIO clock source to CGM3/4 in SMC. Section 6.8 Added reference clock muxes and CCE DFS. Section 6.5.4 Updated key components of HSIO Clock Reset Unit.
0.9.25	8/8/2025	Neeraj Kulkarni	Section 4.7.1 Updated Keraunos CCE section; Added reset domains and reset sequence
0.9.26	8/9/2025	Neeraj Kulkarni	Section 18 Added more details to security, including SEP, ISS, fabric support for SEP/ISS and HSIO virtualization. Section 8.1 and 8.2: Added Traffic Filtering/Firewalls and AxUSER field definition.
0.9.26	8/10/2025	Neeraj Kulkarni	Section 8.1 and 8.2: Added SMN and HSIO fabric timeout, packet tracing and performance monitoring sections. Section 4.6.1 Updated interrupts for HSIO fabric and HSIO SMN tile. Added interrupts for SMN bridge and temperatue sensors.
0.9.27	8/11/2025	Neeraj Kulkarni	Section 14.2: Updated HSIO and QNP clock sources. Use existing 2 CGMs + clock mux in SMC. Section 14.2 and Section 6.8: Updated HSIO clock domain figure.
0.9.28	8/11/2025	Kai Qin/Neeraj Kulkarni	Section 4.1: Updated SMU figure.
0.9.29	8/11/2025	Neeraj Kulkarni	Section 4.3: Updated SMC with CPU configuration and mailboxes. Section 4.4: Updated temperature sensor placement. Updated performance counters description. Section 4.8: Added PCIE/Ethernet SERDES and CCE fimrware loading plan.
0.9.30	8/11/2025	Alan Carr/Neeraj Kulkarni	Section 6.1/6.2: Updated text to reflect 1x800G, 2x400G, 2x200G, 2x100G and 2x10G as POR Ethernet configs. Section 6.5.7: Added MAC/PCS high-level arch figure and Ethernet configurations table.

Revision	Date	Author	Description
0.9.31	8/12/2025	Cameron McNairy/Neeraj Kulkarni	Section 21: Fixed Keruanos RAS section
0.9.32	8/12/2025	Neeraj Kulkarni	Section 3: Updated Keraunos top diagram to include harvesting muxes. Section 11: Added SERDES Harvesting muxes.
0.9.33	8/12/2025	Neeraj Kulkarni	Section 8.3: Added Group ID tagging in HSIO fabric. Section 6.5.4: Added Group ID register in HSIO Clock Reset Unit.
0.9.34	8/14/2025	Shibo Chen	Section 7.2.4 PCIe Remapper programming sequence. Section 8.3.5: Added timeout programming sequence. Section 8.3.6/8.3.7: Added NoC probe programming sequence.
0.9.35	8/14/2025	Jonghyup Lee/Neeraj Kulkarni	Section 15.1 Clean up of PTP timestamp to GPIO section. Added HSIO PTP Mux microarchitecture.
0.9.36	8/14/2025	Neeraj Kulkarni	Section 20.1 Added CCE in Keraunos JTAG network; Section 22.1 Used Tx/Rx instead of Read/Write for Ethernet transfer KPIs, Added packet/s for 1KB frames
0.9.37	8/15/2025	Richard Roze	Section 9: Major refresh of this whole chapter, moving entire Keraunos Ethernet Controller spec into this doc.
0.9.38	8/15/2025	Neeraj Kulkarni	Section 9: Added description for cmd_snoop support in TT Ethernet Controller. Section 6.5.5 Removed duplicated TT Ethernet Controller content.
1.0.0	8/15/2025	Neeraj Kulkarni	Promoted to spec v1.0 (no changes from previous revision 0.9.38)
1.0.1	9/8/2025	Neeraj Kulkarni	Section 6.8 and Section 14.2: Fixed Ethernet SERDES lane clocks
1.0.2	9/23/2025	Neeraj Kulkarni	Section 14.2: Updated QNP clock SS target (GREN-839)
1.0.3	9/23/2025	Neeraj Kulkarni	Section 4.7: Updated CCE debug resets (GREN-509) and PTP reset (GREN-556)
1.0.4	9/23/2025	Neeraj Kulkarni	Section 6.5.1: Added CCE clock DFS (GREN-653)
1.0.5	10/24/2025	Neeraj Kulkarni	Section 7.1: Fix HSIO[i] Error bank to HSIO[i] MAC/PCS Error bank, Section 8.3.1: Remove erroneous connectivity from i_noc2axi2hsio_rd/wr to t_hsio2smn, Section 8.1: Clarified the usage of HSIO express lane in SMN
1.0.6	10/24/2025	Alan Carr/Neeraj Kulkarni	Section 11: Added allowed lane swapping configurations
1.0.7	10/24/2025	Neeraj Kulkarni	Section 6.8, 14.2: Move CCE to single clock and update CCE clock to 1.65GHz (GREN-1089)
1.0.8	10/24/2025	Neeraj Kulkarni	Section 8.3: Connect Eth Ctrl0 Data to CCE 1 Eth port for 800G (GREN-1075)

Chapter 2. Overview

Keraunos is a family of I/O interface chiplets in the Grendel chiplet ecosystem. This document describes Keraunos-E100, the first chiplet in the Keraunos chiplet family. Keraunos-E100 provides a glueless scale-out interface for Quasar, where Quasar-to-Quasar interconnect across Grendel packages is carried over 400G or 800G Ethernet.

Keraunos-E100 will be implemented by Tenstorrent and manufactured by Samsung in SF4.

This revision of the spec describes the plan of record for the Keraunos-E100 chiplet in a manner that unblocks implementation and allows microarchitecture, including IP selection and configuration, floorplanning, and resource allocation. Where there are opens in the spec, these are explicitly called out in each section.

This document is intended for internal use within Tenstorrent only.

Chapter 3. Block Diagram

A high-level block diagram of Keraunos-E100 is shown below.

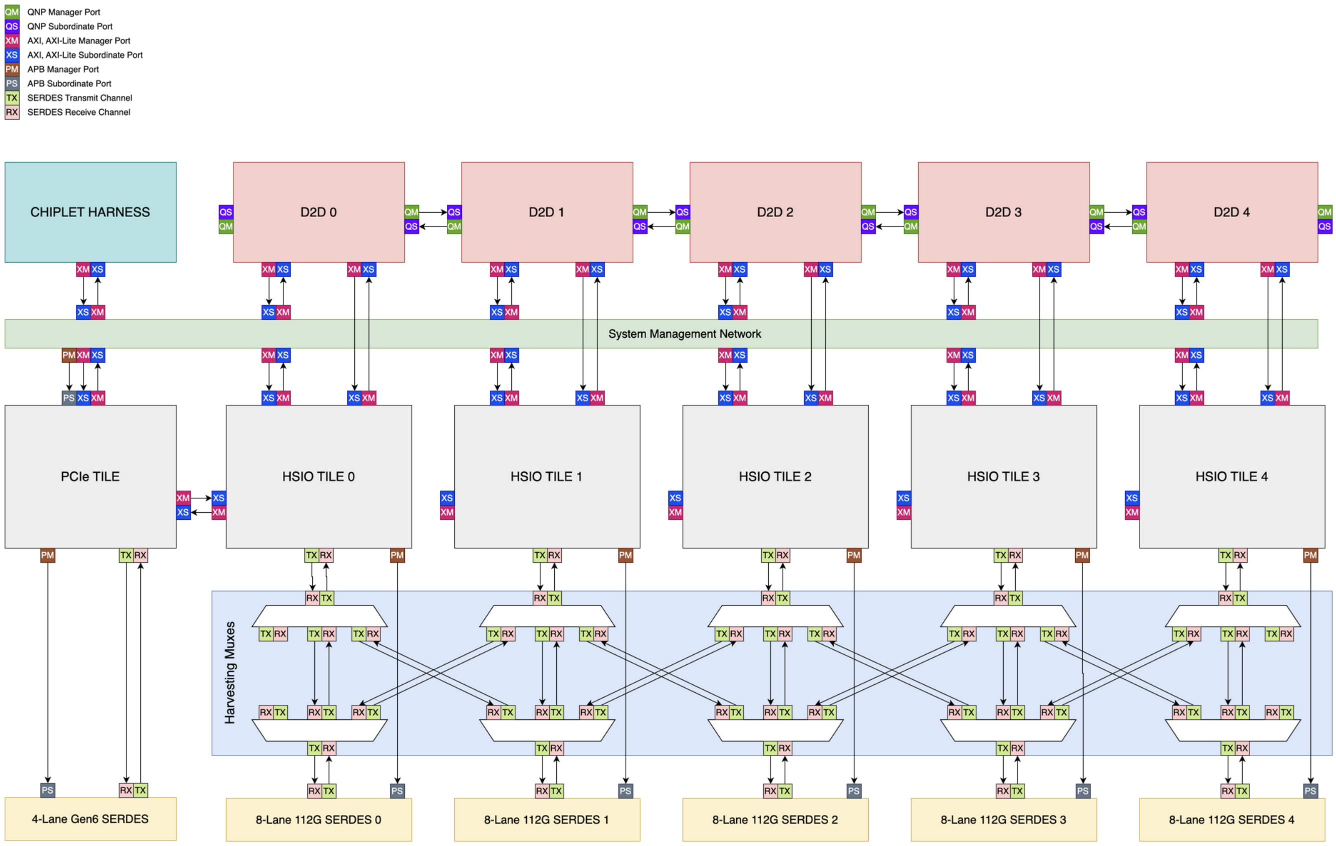


Figure 1. Simplified Keraunos-E100 Block Diagram

Chapter 4. Chiplet Harness

Keraunos-E100 uses the chiplet harness that is common to all peripheral chiplets in the Grendel chiplet ecosystem.

4.1. Block Diagram

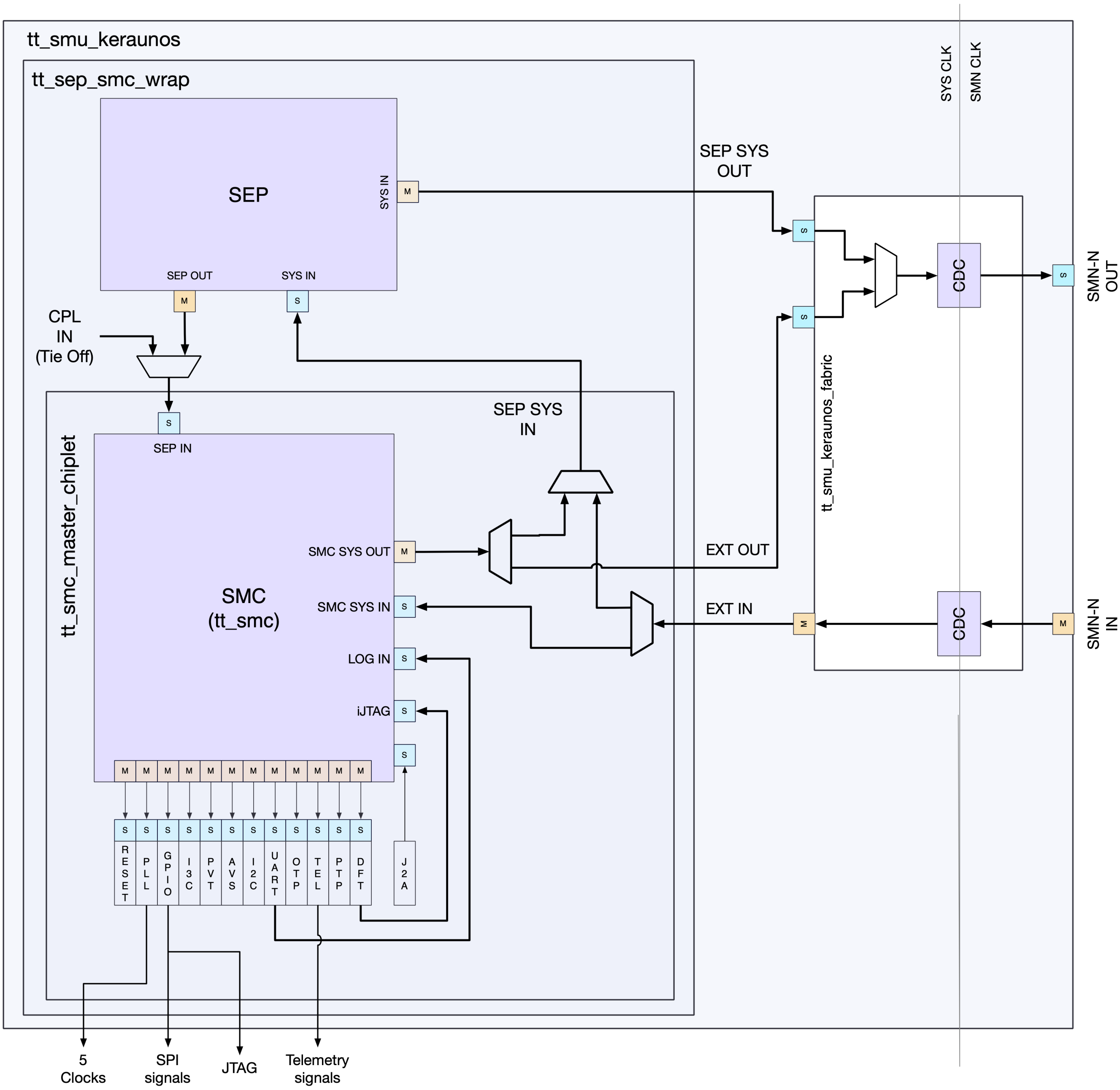


Figure 2. Chiplet Harness Block Diagram

4.2. SEP

Keraunos-E100 contains an instance of SEP and can thus be used as the boot loader for initializing all the chiplet in a package. Its main responsibilities include secure boot, access filtering, product lifecycle management, etc.

4.3. SMC

The chiplet harness contains an instance of the Tenstorrent SMC.

This SMC is responsible for management of all of the hardware components contained onboard. This management includes, but is not limited to:

- Fuse load and distribution
- Hardware reset sequencing
- Hardware configuration
- Power management

Keraunos-E100 could be primary or secondary chiplet in a package. If it is the primary chiplet, it is responsible for initializing itself and triggering initialization of other chiplets in the package.

4.3.1. CPU and Memory

SMC integrates a Rocket RISC-V CPU and a local SRAM. [tbl:smc_cpu_config] shows the configuration for Mimir SMC, which is identical to Athena, Quasar and Keraunos SMCs.

Feature	Value
Cores	4
L1I Cache	4KB
L1D Cache	4KB
SRAM	1MB
Frequency	800MHz

Table 1. SMC Rocket Core Configuration

4.3.2. Mailboxes

SMC uses mailboxes to communicate with the rest of the system, e.g. CCE, PCIE host. Each mailbox has a pair of inbox and outbox. The inbox sends wire based interrupt to SMC CPU through local interrupt controller (PLIC). In Keraunos SMC, only inboxes on both sides are used. Keraunos SMC has 32 mailboxes with 1 entry in each mailbox.

	Value
Number of mailboxes	32
Number of entries per mailbox	1

Table 2. SMC mailboxes

4.4. Power Management

Keraunos-E100 supports the full set of Grendel hierarchical power management features. SMC is capable of monitoring and limiting power consumption of each of the subsystems. The IO Chiplet allows clock gating of each of the subsystems.

For more details, please refer to the Grendel power management architecture.

4.4.1. Power Management Knobs

HSIO fabric/QNP Mesh/CCE Frequency

Keraunos allows the DFS capability on QNP clock, HSIO clock, CCE clock.

Ethernet PHY power modes

Keraunos-E100 support use of all power states and operating conditions supported by the Alphawave ZeusCore SERDES.

The following power states are supported by the SERDES:

Power State	Description
PD	Full power down on all blocks.
P2	Lowest power / longest recovery time state. Everything powered down except beaconing and receiver detection.

Power State	Description
P1	Low power / medium recovery time state. All P2 functionality with synthesizer powered up but transmit driver and receive paths powered down.
P0s	Some power saving / fast recovery state. All P1 functionality with receive path powered up and transmit driver powered down.
P0	Normal full operation mode. Everything powered up.

Power gating

Keraunos-E100 does not support any on-die power gating.

Ethernet lane scaling

Keraunos-E100 supports using SERDES in the following Ethernet configurations: 1x800G, 2x400G, 2x200G, 2x100G and 2x10G Ethernet configurations (More details in [xref:sec-hsio-mac-pcs]). Changing configurations cannot be done on the fly. It requires reconfiguring the controllers at both ends of the link and bringing the links back up with the new configuration.

Data bus inversion

Keraunos-E100 implements DBI on all long and wide datapath links. See DBI for Quasar for details.

4.4.2. Telemetry

Sensors

Keraunos-E100 implements the following sensors:

- Eleven temperature sensors:
  - 1 in SMU, which acts as the master
  - 1 in each D2D tile (total of 5)
  - 1 in each HSIO tile inside MAC/PCS partition (total of 5)
- Voltage droop detectors

Performance counters

Keraunos-E100 supports bandwidth measurement at each initiator and target of the SMN and the HSIO fabrics, which includes the on- and off-ramps on QNP mesh. Bandwidth measurement is enabled by the performance counters in the INIUs and TNIUs of the SMN and HSIO fabrics (detailed further in Section 8.1.6.1 and Section 8.3.7.1). SMC can periodically read the performance counters to extract the bandwidth usage of each initiator and target. Alternatively, SMC can also configure the SMN and HSIO fabric to periodically dump performance counter values as packets into its SRAM, using the ATB2AXI bridge described in Section 8.1.6.1 and Section 8.3.7.1.

4.5. Fuses

The chiplet harness contains one Samsung fuse block with 4k fuses. We do not anticipate using all of the fuses in this block, but choose a standard size of 4k fuses for all peripheral chiplets for consistency. The size difference between a 4k-fuse block and a smaller fuse block is negligible. The 4k fuse block is similar functionally to the 64k fuse block.

4.6. Interrupt Handling

4.6.1. Wired Interrupts

Wired interrupts from various blocks in Keraunos are connected to PLIC in SMC as described in Table 3.

Block	Interrupt mapping	# signals	SMC PLIC indices	Signal name	Description
D2D0	Secure irq from NOC2AXI	1	[0]	o_noc_sec_fence_irq	irq from NOC2AXI
	ECC error from NOC2AXI	1	[1]	o_ecc_error	ECC error from NOC2AXI
	Timeout from NOC2AXI	1	[2]	o_niu_timeout_intp	Timeout report from NOC2AXI
D2D1	Secure irq from NOC2AXI	1	[3]	o_noc_sec_fence_irq	irq from NOC2AXI
	ECC error from NOC2AXI	1	[4]	o_ecc_error	ECC error from NOC2AXI
	Timeout from NOC2AXI	1	[5]	o_niu_timeout_intp	Timeout report from NOC2AXI
D2D2	Secure irq from NOC2AXI	1	[6]	o_noc_sec_fence_irq	irq from NOC2AXI
	ECC error from NOC2AXI	1	[7]	o_ecc_error	ECC error from NOC2AXI
	Timeout from NOC2AXI	1	[8]	o_niu_timeout_intp	Timeout report from NOC2AXI
D2D3	Secure irq from NOC2AXI	1	[9]	o_noc_sec_fence_irq	irq from NOC2AXI
	ECC error from NOC2AXI	1	[10]	o_ecc_error	ECC error from NOC2AXI
	Timeout from NOC2AXI	1	[11]	o_niu_timeout_intp	Timeout report from NOC2AXI
D2D4	Secure irq from NOC2AXI	1	[12]	o_noc_sec_fence_irq	irq from NOC2AXI
	ECC error from NOC2AXI	1	[13]	o_ecc_error	ECC error from NOC2AXI
	Timeout from NOC2AXI	1	[14]	o_niu_timeout_intp	Timeout report from NOC2AXI
HSIO0 CCE0/1	Watchdog Timeout	1	[15]	o_smc_events[0]	Watchdog timeout from DMRISC complex
	HW error uncorrectable	1	[16]	o_smc_events[1]	Uncorrectable error in I/D or L2 cache
	HW error correctable	1	[17]	o_smc_events[2]	Correctable error in I/D cache
	LLK error	1	[18]	o_smc_events[3]	Error in LLK counters
	iDMA error	1	[19]	o_smc_events[4]	Error in iDMA backends
	AXI error	1	[20]	o_smc_events[5]	SLVERR or DECERR received on AXI port(s)
HSIO0 TT Eth controller		1	[21]		Reduction of 7 different errors
HSIO0 MAC/PCS	SW interrupt	1	[22]		
	SRAM ECC error	1	[23]		Reduction of ECC errors in all SRAMs
HSIO0 SerDes	Correctable error	1	[24]		Reduction of correctable ECC errors in SRAMs
	Uncorrectable error	1	[25]		Reduction of uncorrectable ECC errors in SRAMs
HSIO0 SMN tile	NOC Timeout	1	[26]	o_hsio_smn_timeout_irq	NoC Timeout from SMN tile in HSIO0
	NOC Stat Alarm	1	[27]	o_hsio_smn_statalarm_irq	NoC Stat Alarm from SMN tile in HSIO0
HSIO0 fabric	NOC Timeout	1	[28]	o_hsio_timeout_irq	NoC Timeout from HSIO0 fabric
	NOC Stat Alarm	1	[29]	o_hsio_statalarm_irq	NoC Stat Alarm from HSIO0 fabric
Reserved	N/A	5	[30:34]		Reserved

Block	Interrupt mapping	# signals	SMC PLIC indices	Signal name	Description
HSIO1 CCE0/1	Watchdog Timeout	1	[35]	o_smc_events[0]	Watchdog timeout from DMRISC complex
	HW error uncorrectable	1	[36]	o_smc_events[1]	Uncorrectable error in I/D or L2 cache
	HW error correctable	1	[37]	o_smc_events[2]	Correctable error in I/D cache
	LLK error	1	[38]	o_smc_events[3]	Error in LLK counters
	iDMA error	1	[39]	o_smc_events[4]	Error in iDMA backends
	AXI error	1	[40]	o_smc_events[5]	SLVERR or DECERR received on AXI port(s)
HSIO1 TT Eth controller		1	[41]		Reduction of 7 different errors
HSIO1 MAC/PCS	SW interrupt	1	[42]		
	SRAM ECC error	1	[43]		Reduction of ECC errors in all SRAMs
HSIO1 SerDes	Correctable error	1	[44]		Reduction of correctable ECC errors in SRAMs
	Uncorrectable error	1	[45]		Reduction of uncorrectable ECC errors in SRAMs
HSIO1 SMN tile	NOC Timeout	1	[46]	o_hsio_smn_timeout_irq	NoC Timeout from SMN tile in HSIO1
	NOC Stat Alarm	1	[47]	o_hsio_smn_statalarm_irq	NoC Stat Alarm from SMN tile in HSIO1
HSIO1 fabric	NOC Timeout	1	[48]	o_hsio_timeout_irq	NoC Timeout from HSIO1 fabric
	NOC Stat Alarm	1	[49]	o_hsio_statalarm_irq	NoC Stat Alarm from HSIO1 fabric
Reserved	N/A	5	[50:54]		Reserved
HSIO2 CCE0/1	Watchdog Timeout	1	[55]	o_smc_events[0]	Watchdog timeout from DMRISC complex
	HW error uncorrectable	1	[56]	o_smc_events[1]	Uncorrectable error in I/D or L2 cache
	HW error correctable	1	[57]	o_smc_events[2]	Correctable error in I/D cache
	LLK error	1	[58]	o_smc_events[3]	Error in LLK counters
	iDMA error	1	[59]	o_smc_events[4]	Error in iDMA backends
	AXI error	1	[60]	o_smc_events[5]	SLVERR or DECERR received on AXI port(s)
HSIO2 TT Eth controller		1	[61]		Reduction of 7 different errors
HSIO2 MAC/PCS	SW interrupt	1	[62]		
	SRAM ECC error	1	[63]		Reduction of ECC errors in all SRAMs
HSIO2 SerDes	Correctable error	1	[64]		Reduction of correctable ECC errors in SRAMs
	Uncorrectable error	1	[65]		Reduction of uncorrectable ECC errors in SRAMs
HSIO2 SMN tile	NOC Timeout	1	[66]	o_hsio_smn_timeout_irq	NoC Timeout from SMN tile in HSIO2
	NOC Stat Alarm	1	[67]	o_hsio_smn_statalarm_irq	NoC Stat Alarm from SMN tile in HSIO2
HSIO2 fabric	NOC Timeout	1	[68]	o_hsio_timeout_irq	NoC Timeout from HSIO2 fabric
	NOC Stat Alarm	1	[69]	o_hsio_statalarm_irq	NoC Stat Alarm from HSIO2 fabric

Block	Interrupt mapping	# signals	SMC PLIC indices	Signal name	Description
Reserved	N/A	5	[70:74]		Reserved
HSIO3 CCE0/1	Watchdog Timeout	1	[75]	o_smc_events[0]	Watchdog timeout from DMRISC complex
	HW error uncorrectable	1	[76]	o_smc_events[1]	Uncorrectable error in I/D or L2 cache
	HW error correctable	1	[77]	o_smc_events[2]	Correctable error in I/D cache
	LLK error	1	[78]	o_smc_events[3]	Error in LLK counters
	iDMA error	1	[79]	o_smc_events[4]	Error in iDMA backends
	AXI error	1	[80]	o_smc_events[5]	SLVERR or DECERR received on AXI port(s)
HSIO3 TT Eth controller		1	[81]		Reduction of 7 different errors
HSIO3 MAC/PCS	SW interrupt	1	[82]		
	SRAM ECC error	1	[83]		Reduction of ECC errors in all SRAMs
HSIO3 SerDes	Correctable error	1	[84]		Reduction of correctable ECC errors in SRAMs
	Uncorrectable error	1	[85]		Reduction of uncorrectable ECC errors in SRAMs
HSIO3 SMN tile	NOC Timeout	1	[86]	o_hsio_smn_timeout_irq	NoC Timeout from SMN tile in HSIO3
	NOC Stat Alarm	1	[87]	o_hsio_smn_statalarm_irq	NoC Stat Alarm from SMN tile in HSIO3
HSIO3 fabric	NOC Timeout	1	[88]	o_hsio_timeout_irq	NoC Timeout from HSIO3 fabric
	NOC Stat Alarm	1	[89]	o_hsio_statalarm_irq	NoC Stat Alarm from HSIO3 fabric
Reserved	N/A	5	[90:94]		Reserved
HSIO4 CCE0/1	Watchdog Timeout	1	[95]	o_smc_events[0]	Watchdog timeout from DMRISC complex
	HW error uncorrectable	1	[96]	o_smc_events[1]	Uncorrectable error in I/D or L2 cache
	HW error correctable	1	[97]	o_smc_events[2]	Correctable error in I/D cache
	LLK error	1	[98]	o_smc_events[3]	Error in LLK counters
	iDMA error	1	[99]	o_smc_events[4]	Error in iDMA backends
	AXI error	1	[100]	o_smc_events[5]	SLVERR or DECERR received on AXI port(s)
HSIO4 TT Eth controller		1	[101]		Reduction of 7 different errors
HSIO4 MAC/PCS	SW interrupt	1	[102]		
	SRAM ECC error	1	[103]		Reduction of ECC errors in all SRAMs
HSIO4 SerDes	Correctable error	1	[104]		Reduction of correctable ECC errors in SRAMs
	Uncorrectable error	1	[105]		Reduction of uncorrectable ECC errors in SRAMs
HSIO4 SMN tile	NOC Timeout	1	[106]	o_hsio_smn_timeout_irq	NoC Timeout from SMN tile in HSIO4
	NOC Stat Alarm	1	[107]	o_hsio_smn_statalarm_irq	NoC Stat Alarm from SMN tile in HSIO4
HSIO4 fabric	NOC Timeout	1	[108]	o_hsio_timeout_irq	NoC Timeout from HSIO4 fabric

Block	Interrupt mapping	# signals	SMC PLIC indices	Signal name	Description
	NOC Stat Alarm	1	[109]	o_hsio_statalarm_irq	NoC Stat Alarm from HSIO4 fabric
Reserved	N/A	3	[110:112]		Reserved
SMN bridge	NOC Timeout	1	[113]	o_smn_bridge_timeout_irq	NoC Timeout from SMN Bridge
	NOC Stat Alarm	1	[114]	o_smn_bridge_statalarm_irq	NoC Stat Alarm from SMN Bridg
PCIe Tile	Function Level Reset	1	[115]		Function Level Reset (FLR) request
	Hot Reset	1	[116]		Hot Reset request
	Configuration Update	1	[117]		Configuration update is captured
	RAS error	1	[118]		RAS error
	DMA Completion	1	[119]		DMA transfer completion
	Controller Misc	1	[120]		Misc interrupt from Controller IP
	NOC timeout	3	[121:123]		NOC/SMN timeout
Temperature sensors	Alert from Temp Sensors	3	[124:126]		Alert signals for 3 threshold levels (temp_interrupts_ored[2:0]). Each level is the 'OR' of alertes from all sensors

Table 3. Wired interrupts to SMC

4.6.2. Interrupts from Host

PCIe-connected x86 Host can send interrupts to Keraunos SMC using SMC mailboxes.

4.6.3. Interrupts to Host

Keraunos allows any software running on Keraunos SMC, CCE, or other controllers to send MSI messages to a PCIe-connected x86 Host using the MSI Relay Unit in Keraunos PCIE tile. MSI Relay Unit enables controllers to set interrupt bits in it's memory-mapped MSI-X PBA register via APB port. Once an interrupt is registered in the MSI-X PBA, it generates a memory write request (MSI) based on the MSI-X table contents. More details can be found in Section 2.10 of Keraunos PCIE tile Specification.

4.7. Reset

4.7.1. Reset signals

Keraunos has two reset pins (reset_n and cool_reset_n_i), isolation request pin (isolate_req), and a reset output (cool_reset_n_o), as decribed in Section 3.1.2 in Grendel Power-On and Reset Specification. cool_reset_n_i is an external cool reset pin and is used only when Keraunos is the secondary chiplet, and is tied to 1 when Keraunos is the primary chiplet. The isolate_req pin is intented to be used by BMC to send PCIe isolation request. It is only connected to package's isolate_req when Keraunos is primary chiplet, and tied to 0 if Keraunos is the secondary chiplet.

SMC holds FLR block which drives external output pin cool_reset_n_o and an internal cool_reset_n through its flr_reset_n_reg register. When Keraunos is the primary chiplet, cool_reset_n_o is connected to cool_reset_n_i of the secondary chiplets. The cool_reset_n is used to reset the internal SMC. SMC's primary_reset_n is logical OR of the extended reset_n, extended cool_reset_n_i, and extended cool_reset_n as described in Section 4.9 in Grendel System Management Architecture Specification.

SMC holds a reset unit which drives reset signals to various subsystems in Keraunos. SMC is responsible for writing to a memory-mapped register (ss_cold_reset_n) in the reset unit to deassert the cold reset signals to SMN, D2D, PCIE and HSIO tiles. Note, warm reset signals and hold signals implemented in SMC are not used in Keraunos.

Each HSIO tile implements reset register(s) in the Clock Reset Unit, that drive reset signals to the subsystems within HSIO tile. The HSIO cold reset signals from SMC only reset the Clock Reset Unit in the HSIO tiles.

All the reset signals in Keraunos are summarized in Table 4.

Signal Name	Source	Targets	Description
reset_n	External pin	SEP, SMC, DAP/TAP	External cold reset pin

Signal Name	Source	Targets	Description
cool_reset_n_i	External pin	SMC, SEP	External cool reset pin. Only used when Keraunos is secondary chiplet. Otherwise, tied to 1.
isolate_req	External pin	SMC, SEP	External PCIe isolation request to Grendel package. Only connected to package's isolate_req, when Keraunos is primary chiplet. Otherwise, tied to 0.
cool_reset_n_o	SMC register	Secondary chiplets	Only driven when Keraunos is the primary chiplet
ss_cold_reset_n[31:0]	SMC register	SMN,D2D,PCIE,HSIO tiles	Used to cold reset a subsystem. Mapping in Table 5
ss_warm_reset_n[31:0]	SMC register		Not used in Keraunos
ss_critical_signal_hold[31:0]	SMC register		Not used in Keraunos
hsio_cold_reset_n[31:0]	HSIO tile[i], i=0,..,4	Subsystems within HSIO tile	Used to cold reset a subsystem within HSIO tile. Mapping in Table 6

Table 4. Keraunos reset signals

SMC cold reset signals mapping are described in Table 5.

Subsystem	SMC reset mapping	Comments
Reserved	ss_cold_reset_n[0]	Reserved for SMU reset
SMN	ss_cold_reset_n[1]	
PCIE Tile Management Reset	ss_cold_reset_n[2]	
PCIE Tile Main Reset	ss_cold_reset_n[3]	
D2D0 tile	ss_cold_reset_n[4]	SMC also needs to deassert reset in NOC2AXI
D2D1 tile	ss_cold_reset_n[5]	SMC also needs to deassert reset in NOC2AXI
D2D2 tile	ss_cold_reset_n[6]	SMC also needs to deassert reset in NOC2AXI
D2D3 tile	ss_cold_reset_n[7]	SMC also needs to deassert reset in NOC2AXI
D2D4 tile	ss_cold_reset_n[8]	SMC also needs to deassert reset in NOC2AXI
HSIO0 tile	ss_cold_reset_n[9]	SMC also needs to deassert resets in HSIO Clock Reset Unit
HSIO1 tile	ss_cold_reset_n[10]	SMC also needs to deassert resets in HSIO Clock Reset Unit
HSIO2 tile	ss_cold_reset_n[11]	SMC also needs to deassert resets in HSIO Clock Reset Unit
HSIO3 tile	ss_cold_reset_n[12]	SMC also needs to deassert resets in HSIO Clock Reset Unit
HSIO4 tile	ss_cold_reset_n[13]	SMC also needs to deassert resets in HSIO Clock Reset Unit
Global PTP reset	ss_cold_reset_n[14]	Used to simulateneously reset TT Ethernet Controller PTP instances in all HSIO tiles

Table 5. SMC reset signals mapping

HSIO Clock Reset Unit holds hsio_cold_reset_n that drive cold reset signals to various subsystems within HSIO tile. HSIO cold reset signals mapping are described in Table 5. Each subsystem's HSIO cold reset signal listed below are ANDed with HSIO tile cold reset signal.

Subsystem	HSIO register mapping	Comments
CCE0	hsio_cold_reset_n[0]	SMC also needs to assert warm resets in CCE
CCE1	hsio_cold_reset_n[1]	SMC also needs to assert warm resets in CCE
SRAM	hsio_cold_reset_n[2]	
DFD0	hsio_cold_reset_n[3]	
DFD1	hsio_cold_reset_n[4]	
Error bank	hsio_cold_reset_n[5]	
Fabric	hsio_cold_reset_n[6]	Used for HSIO fabric and NOC2AXI remap
TT Eth Ctrl0	hsio_cold_reset_n[7]	

Subsystem	HSIO register mapping	Comments
TT Eth Ctrl0 PTP	hsio_cold_reset_n[8]	Default set to 1. So this reset is not used during powerup. Global PTP reset should be used to simultaneously reset PTP instances.
TT Eth Ctrl1	hsio_cold_reset_n[9]	
TT Eth Ctrl1 PTP	hsio_cold_reset_n[10]	Default set to 1. So this reset is not used during powerup. Global PTP reset should be used to simultaneously reset PTP instances.
Profiler0	hsio_cold_reset_n[11]	
Profiler1	hsio_cold_reset_n[12]	
MAC/PCS	hsio_cold_reset_n[13]	
SERDES	hsio_cold_reset_n[14]	
CCE0 Debug	hsio_cold_reset_n[15]	Reset to CCE0 debug module
CCE1 Debug	hsio_cold_reset_n[16]	Reset to CCE1 debug module

Table 6. HSIO reset signals mapping

As specified in Grendel Power-On and Reset Specification, since all TT subsystems/IPs uses synchronous resets and we must guarantee timing (e.g. deassertion of reset reaches all flip flops within a subsystem in the same clock cycle), we need to force clock of the subsystem being reset to reference clock (ref_clk, 100MHz). This implies all reset signals in Keraunos could be set as multi-cycle paths. The only exception to this rule in Keraunos are the CCE core resets, which can be deasserted/asserted at CCE clock's mission-mode frequency (described further in [sec-ker-subsystem-reset]).

Note, each of the CGMs in Keraunos has the ability to force the clock to ref_clk, described further in [sec-ker-clock-refclk-mux].

4.7.2. Reset Sequences

Normal Cold Reset

This sequence is described in Section 3.2 of Grendel Power-On and Reset Specification and follows the following broad steps:

1. External reset_n
2. SMC and SEP Initialization
   - SMC's primary_reset_n
3. Power-On Primary SoC Supplies
4. Fabric Reset, Initialization (SMN and HSIO fabric in Keraunos)
   - SMN: SMC deasserts ss_cold_reset_n[1], which drives the reset signal to SMN.
   - HSIO fabric: SMC deasserts ss_cold_reset_n[9:13], which drives the reset signal to all HSIO tiles' Clock Reset Unit. SMC then deasserts hsio_cold_reset_n[6] in each HSIO tile, which drives the reset signal to its HSIO fabric.
5. D2D Power-On and Reset
6. Other subsystems Reset, Initialization
   - PCIE: SMC deasserts ss_cold_reset_n[2:3], which drives the reset signals to PCIE tile.
   - HSIO subsystems: SMC deasserts hsio_cold_reset_n[14:7] and hsio_cold_reset_n[5:0] in each HSIO tile, which drives the reset signals to other subsystems in each HSIO tile.

BMC-based Cool Reset

Grendel's BMC-based Cool Reset is described in Section 3.3.1 of Grendel Power-On and Reset Specification. In Keraunos, the isolate_req signal from BMC is routed to PCIE tile and is used to isolate the PCIe tile from the rest of the chiplet. The isolation block within the PCIE tile starts to complete all pending transactions and rejects future transactions with DECERR.

PCIE Function Level Reset (FLR)/SMC-based Cool Reset

Section 3.3.2 in Grendel Power-On and Reset Specification describes the cool reset flow when host requests a PCIE FLR. In Keraunos, the PCIe controller (EP) asserts cfg_flr_pf_active in SMC FLR, which drives an interrupt Keraunos SMC. The details of the cool reset flow orchestrated by SMC FLR are provided in Section 4.9 in Grendel System Management Architecture Specification. On completion of the cool reset, SMC SW updates app_flr_pf_done_reg in PCIE tile to indicate completion of FLR.

SMC-based Warm Reset

Keraunos does not support SMC-based warm reset or hold signal functionality.

4.7.3. Keraunos subsystem resets

Many subsystems in Keraunos have their own reset control block with warm reset signals to components within the susbsystem. This section describes the resets for each of the subsystem that has such capability. Note, software has ability to reset subsystems independently, but there is no systematic solution to isolate each subsystem that ensures that the reset of a subsystem doesn't cause unexpected impact to the rest of the chiplet. So software should reset subset of subsystems with caution.

D2D

D2D resets are listed in Grendel chiplets D2D specification

Keraunos NOC

Keraunos NOC consists of SMN and HSIO fabric, that are built using Arteris FlexNoC. Each of SMN and HSIO fabric exposes only a single cold reset input that resets the respective fabric entirely.

Keraunos CCE

Reset Domains

Keraunos CCE supports multiple reset domains, as detailed in Table 7. The table shows which sub-components are in each reset domain and which resets need the clocks to be forced to refclk before assertion/deassertion. The CCE cold reset and the CCE uncore warm reset are multi-cycled and require the HSIO clock to be lowered before asserting/de-asserting the reset. In contrast, the CCE core warm resets are not multi-cycled and can be asserted/deasserted in CCE clock's mission-mode frequency. To put one or more DMRISC cores in reset, the cores must first be brought to an idle condition (for example, by executing a `wfi` instruction) before the corresponding CCE core warm reset can be asserted.

CCE sub-components	Reset Domain	Driven by	Force to refclk
DMRISC core	8 cce core warm resets	Reset register in CCE	No
DMRISC uncore	cce uncore warm reset	Reset register in CCE	Yes
AXI-DMA	cce uncore warm reset	Reset register in CCE	Yes
AXI snoop block	cce cold reset	ss_cold_reset_n in SMC	Yes
Register xbar	cce cold reset	ss_cold_reset_n in SMC	Yes
Keraunos CCE network	cce cold reset	ss_cold_reset_n in SMC	Yes

Table 7. CCE Clock and Reset Domains

Reset Sequence

The reset sequence for Keraunos CCE is as follows:

1. De-assert cce cold reset: This allows access to all CCE registers and Banked SRAM via CCE's AXI Snoop block.
2. De-assert cce uncore warm reset: This brings CCE IP's DMRISC uncore, AXI-DMA, address remaps, etc. out of reset.
3. De-assert cce core warm resets: This brings DMRISC cores out of reset and allows them to execute instructions.

More details (i.e., reset sequence for CCE) can be found in Chiplet Communication Engine (CCE) Specification.

Banked SRAM

TL1 IP, that is used to build Banked SRAM, only supports cold reset, which resets the entire block. Internally, out of reset the flex client ports wait for about 30 cycles to deassert req_full to assure that the L1 phase offset signals, which synchronize the Superbank time division multiplexing, has propagated.

It is recommended to reset the Banked SRAM prior to CCE core warm resets as they start fetching instructions from the SRAM immeditately after coming out of reset.

4.8. Firmware loading

This section describes the firmware loading plan for various controllers in Keraunos.

4.8.1. PCIE SERDES Firmware

PCIE is required to be initialized at boot. Thus, PCIE firmware loading will be done by SMC during power-on boot. PCIE SERDES firmware is loaded into the SMC SRAM by SEP from the boot flash, and SMC firmware loads the image into the PCIE SERDES SRAM.

4.8.2. Ethernet SERDES Firmware

Since Ethernet links are not required to be up at boot, Ethernet SERDES firmware is loaded after boot. Ethernet SERDES firmware should be loaded during bring-up and mission mode in the following ways:

1. Bring-up scenario: The Ethernet SERDES firmware is loaded into the SERDES SRAM through JTAG2AXI in SMC, bypassing the SMC cores. The data path followed in this case is JTAG→SMC JTAG2AXI→SMN→HSIO fabric→SERDES SRAM. Alternatively, Ethernet SERDES firmware can also be loaded through JTAG2AXI in HSIO tile directly into SERDES SRAM, bypassing SMN and HSIO fabric.
2. Mission mode: The Ethernet SERDES firmware is loaded into the SERDES SRAM through SMC FW. Firmware image is loaded into SMC SRAM through PCIE and then, SMC firmware loads the image into the SERDES SRAM. Alternatively, the firmware can also be loaded directly into SERDES SRAM via PCIE by the PCIE driver.

4.8.3. CCE Firmware

CCE primarily runs data-plan software (also known as tt-fabric). CCE firmware is loaded into the HSIO SRAM by User mode driver (UMD) or runtime software (fast dispatch). In either case, CCE firmware is loaded physically via the PCIE interface.

Chapter 5. D2D

Keraunos-E100 uses a Tenstorrent-proprietary version of the BoW D2D interface for its connection to Quasar.

Keraunos-E100 uses the QNP variant of the BCA BoW link layer. There is no AXI functional traffic transmitted over D2D. The only use of AXI in the link layer is for management traffic.

The D2D subsystem in Keraunos-E100 is identical to the N/S version used in Quasar. Figure 3 shows the key components in the Keraunos D2D tile. The NOC2AXI in D2D subsystem is used route to traffic a) from BC LL+PHY block to the corresponding HSIO tile as well as b) between the HSIO tiles. Note there is no NOC2AXI in HSIO subsystem, so all traffic between HSIO tiles is routed via D2D.

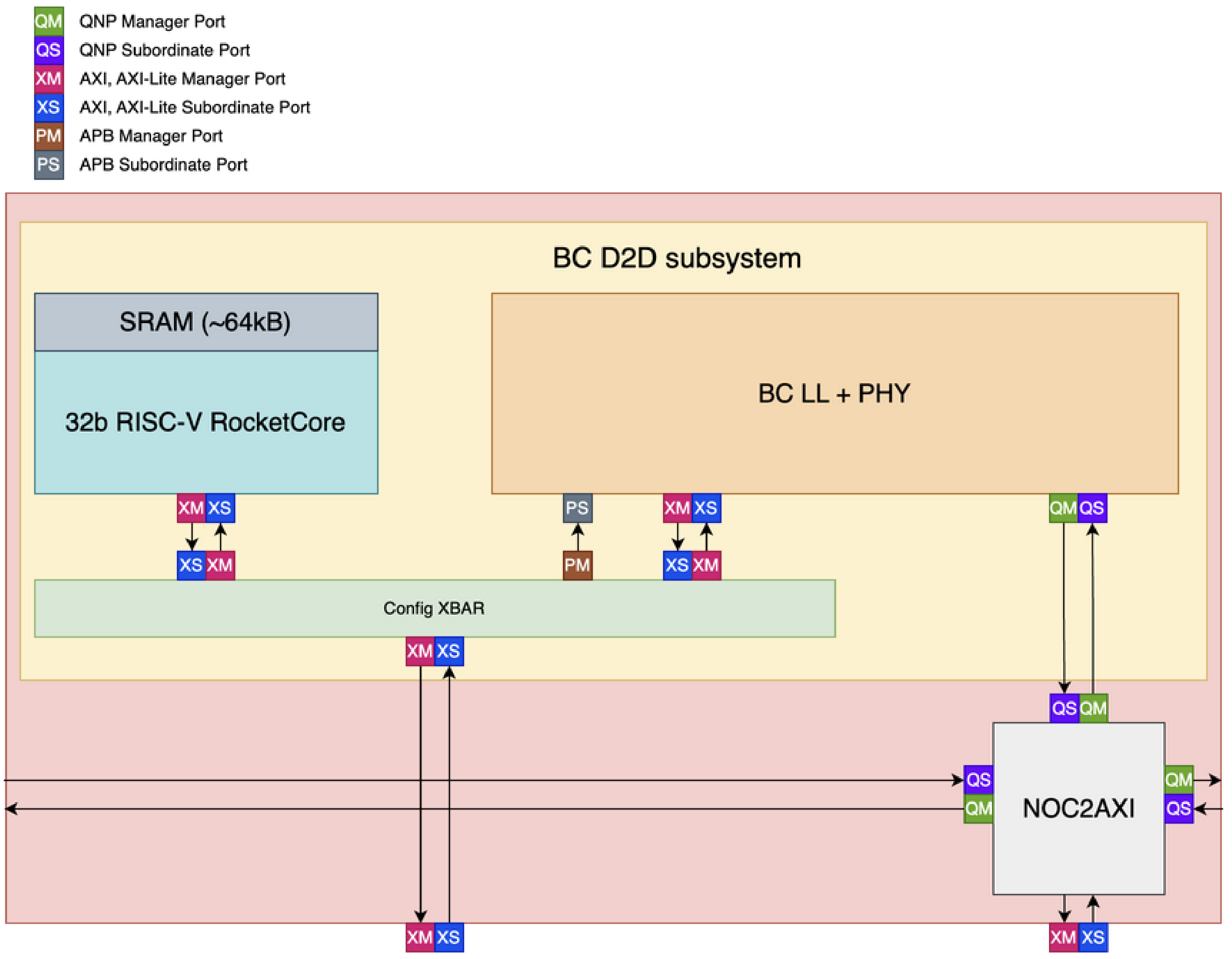


Figure 3. Keraunos D2D Tile Block Diagram

Keraunos-E100 will only be manufactured with 130um pitch for the D2D subsystem and will only be mounted on the substrate during packaging. There is no advanced packaging use case for Keraunos-E100.

In the 130um pitch standard packaging variant of the D2D subsystem, the raw throughput of the D2D subsystem is 144GB/s RX + 144GB/s TX. We estimate the usable throughput of the D2D subsystem after accounting for FLIT encoding overhead to be 112GB/s read + 112GB/s write. Note that this bandwidth is shared between I/O functional traffic and chiplet management traffic.

5.1. Microarchitecture

Figure 4 illustrates the microarchitecture for the Keraunos D2D tile, adding more details on the internal components in the Keraunos D2D tile.

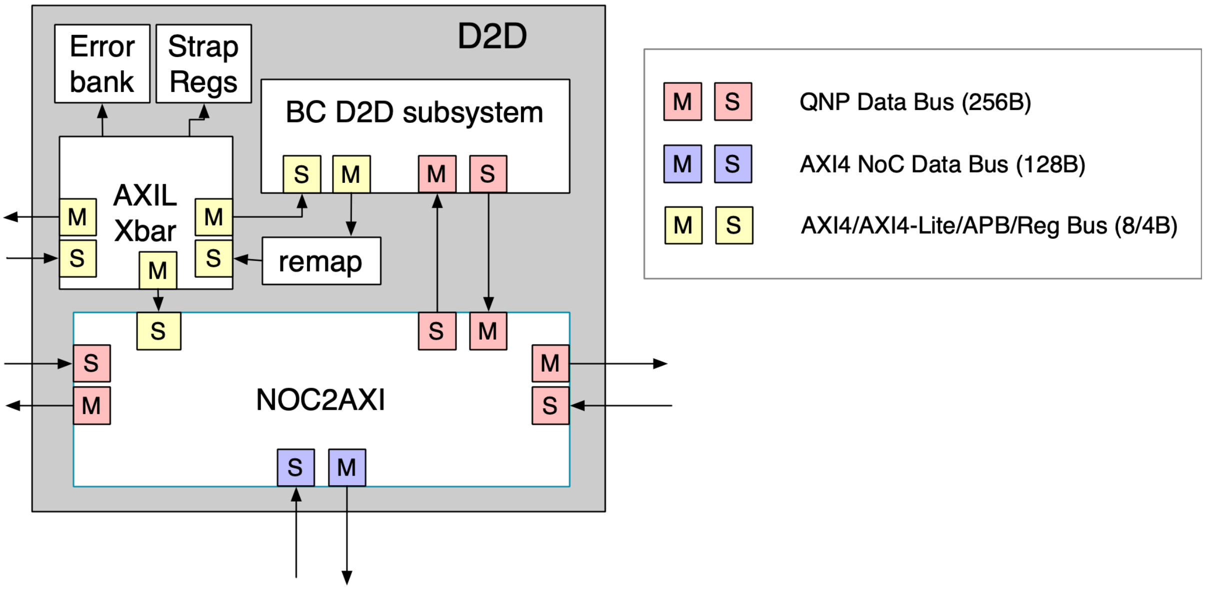


Figure 4. Keraunos D2D Tile Microarchitecture

Chapter 6. HSIO Tile

6.1. Overview

The HSIO Tile is the primary data path tile in Keraunos, facilitating data movement between D2D and Ethernet links, as well as between Ethernet links. The HSIO tile architecture, depicted in Figure 5, includes components such as CCE, SRAM, TT Ethernet Controllers, and AlphaWave MAC/PCS. It connects to the D2D tile, and to Ethernet SERDES through the MAC/PCS block. HSIO tile is primarily designed to operate in 1x800G or 2x400G Ethernet configuration, and optionally can operate in 2x200G, 2x100G and 2x10G Ethernet configurations.

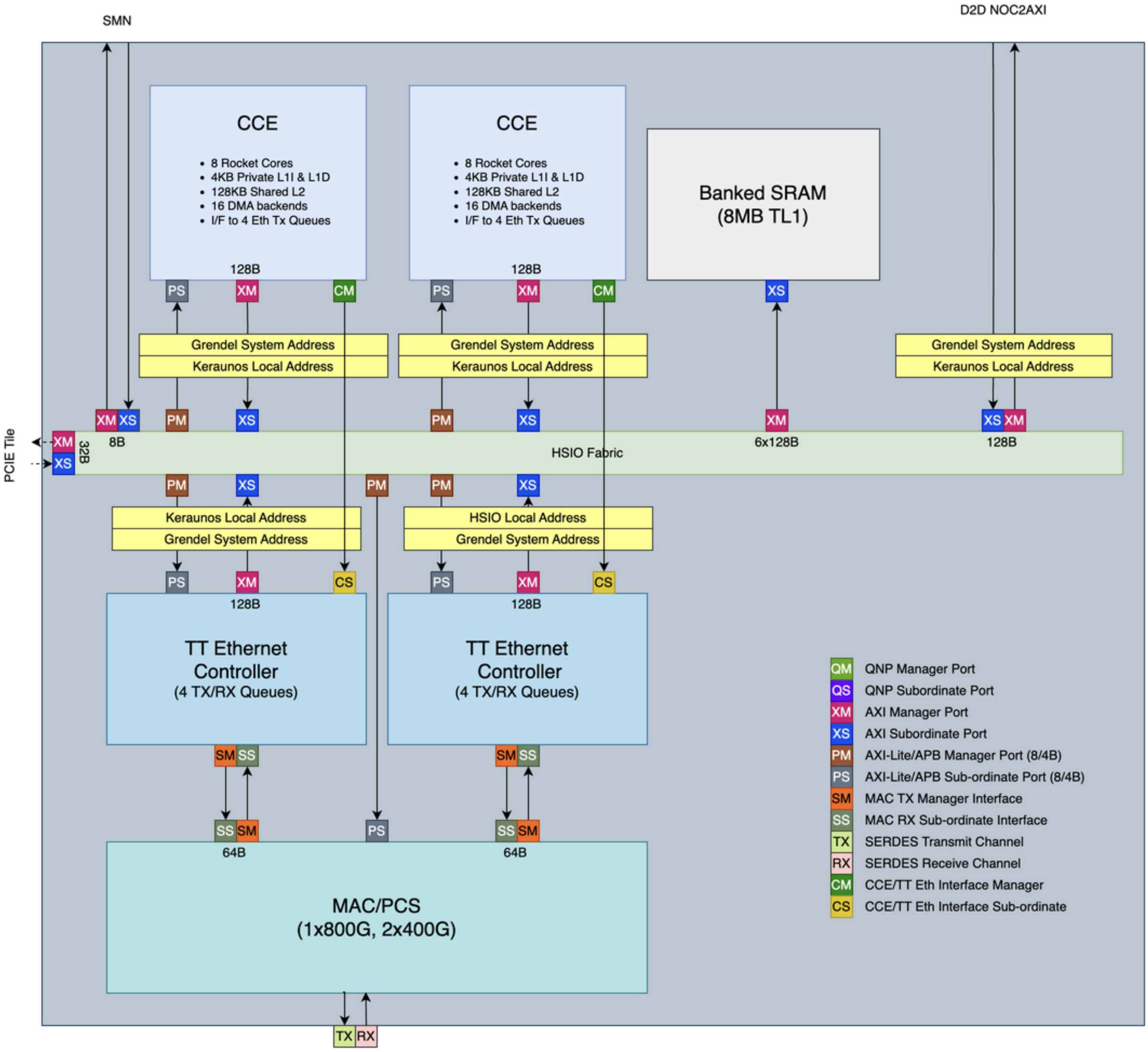


Figure 5. HSIO Tile Block Diagram

6.2. Features

This section outlines the main features of the Keraunos HSIO tile:

1. Support 1x800G, 2x400G, 2x200G, 2x100G and 2x10G Ethernet configurations. (Other configurations are also possible and detailed in [sec-hsio-mac-pcs]).
    - MAC, PCS and SERDES IPs from AlphaWave
2. Substitute ERISC cores and HW overlay complex with CCE IP
3. Uniform programming interface for sending data over Ethernet, Quasar NOC and also AXI-based NOCs in Mimir/Athena.

    - Achieved by considering TT Ethernet Controller as CCE's DMA backend and mapping HW overlay interface of the TT Ethernet controller to the DMA backend interface.
    - Ethernet controller register interface retained to maintain compatibility with BH TT-Metalium stack
4. Enhance communication between Ethernet endpoints by supporting RISC cores initiated packet transmission in TT-Link mode (described in TT-Fabric specification).
    - This is also achieved by mapping the HW overlay interface of the TT Ethernet controller to the DMA backend interface of the CCE mentioned above.
5. Single distributed SRAM in Keraunos
    - SRAMs in all HSIO tiles to be viewed as a single distributed SRAM
    - Data buffers can reside in any SRAM within Keraunos and do not need to be copied to the local SRAM of the Ethernet Controller.
    - Eliminates redundant buffering and reduces latency when transferring data between different HSIO tiles
6. Support inline register write commands from DMRISC cores
7. Increase the TT Ethernet Tx/Rx controllers, which are conceptually equivalent to QNP VCs, to 4 identical instances
    - BH has 2 for bulk data and 1 for register writes.
8. Capability to write to any register or memory on the remote Ethernet endpoint
9. Capability to update read/write pointers cached in the DMRISC complex
    - Enables consistent programming semantics across Ethernet links and within the Grendel SoP.
    - This was not applicable in BH Ethernet subsystem.
10. Addition of doorbell registers to interrupt RISC cores in CCE
    - Useful for data plane software
11. Addition of general-purpose mailboxes in CCE
    - Useful for control plane software

6.3. Components

This section highlights the components in HSIO tile and their usage. The architecture of each component is described in detail in Section 6.5.

Keraunos CCE

The Keraunos CCE is the primary data movement coordinator in Keraunos and is responsible for moving data across the following paths:

1. Receive data from TensixNEOs in Quasar via D2D and transmit it to Ethernet Tx.
2. Receive data from Ethernet Rx and transmit it to TensixNEOs in Quasar via D2D.
3. Receive data from Ethernet Rx and transmit it to Ethernet Tx in same or another HSIO tile.

Keraunos CCE provides the same programming interface to the DMRISCs for sending read/write requests to the D2D/neighboring HSIO tiles through HSIO fabric as the one used for sending read/write requests over Quasar NOC. This is achieved by using the same command buffer accelerators in the DMRISC complex as in Quasar Overlay, and generating 1-D data movement commands for the AXI-DMA equivalent to the QNP commands used in Quasar NOC/NIU.

Moreover, Keraunos CCE also provides the same programming interface to the DMRISCs for sending data over Ethernet as the one used for sending data over Quasar NOC. As noted above this is also the same as the one used for sending data over AXI-based NOCs in Keraunos HSIO/Mimir/Athena. This is described in more detail in Section 6.5.5.

Each HSIO tile includes two instances of Keraunos CCE, one for transmitting and receiving data to/from each 400G Ethernet link.

Banked SRAM

Each HSIO tile has a banked SRAM, 8MB in size, that is used to store Keraunos CCE firmware as well as data buffers for outstanding requests. The SRAM size is calculated as follows:

- The Ethernet throughput is 100GB/s in each direction
- At full bidirectional utilization, we will fill 200kB (100kB per direction) in 1us.

- To match Blackhole latency assumptions (10us) across ethernet endpoints, we will need 10us * 200kB/us, or 2MB
- To support double buffering, we need 2x this amount, or 4MB
- To provide margin and to have enough buffering for future use cases, double this minimum 4MB size to 8MB .

The banked SRAMs across all HSIO tiles function as a unified distributed SRAM, accessible by Keraunos CCE and TT Ethernet Controllers in any HSIO tile. This design prevents redundant buffering, particularly when transferring data between different HSIO tiles.

HSIO Fabric

The HSIO Fabric is an 128B AXI interconnect fabric that connects components within HSIO tile, specifically

- Request paths from Keraunos CCEs to the banked SRAM and D2D tile.
- Request paths from TT Ethernet Controllers to Keraunos CCE MMRs, banked SRAM and D2D tile.
- Request paths from D2D tile to banked SRAM and Keraunos CCE MMRs.

More details on connectivity are provided in Section 8.3.

TT Ethernet Controller

The primary purpose of the TT Ethernet controller block is to perform hardware assist for header insertion on Ethernet Tx path and packet classification on Ethernet Rx path to decrease the software packet processing overhead. Specifically, it does the following:

- Transmit packets to Ethernet Tx: Keraunos CCE programs the TT Ethernet Controller, which then fetches data from SRAM, adds packet headers, and transmits to Ethernet Tx.
- Receive packets from Ethernet Rx: TT Ethernet Controller receives packets from Ethernet Rx, classifies them, and stores them in SRAM for Keraunos CCE to process.

The only defined use case for Keraunos-E100 is Quasar-to-Quasar communication, which needs CCE-to-CCE point-to-point data transfers. So Keraunos-E100 only supports a store-and-forward data transfer model. Multi-hop transfers are still possible with CCE doing software routing of the packet to next hop. However, cut through and multi-hop hardware data transfer support is deferred to a future generation of the scaleout chiplet hardware.

Keraunos-E100 is able to sustain continuous Ethernet line rate for frame sizes of 4kB or higher. For an 800G link, 100GB/s of data transfer at 4kB frame size is 24.4M packet/s. To ensure that we can sustain line rate with the BH TX/RX controller, which was designed for 400G link use, we instantiate two copies of the TX/RX controller per 800G link. In other words, each HSIO tile in Keraunos-E100 behaves similarly to two BH Ethernet tiles. We will attempt to enhance the BH Ethernet TX/RX controller logic to support 800G line rate but this is only a P1 requirement for Keraunos-E100. The KPI for line rate performance is that the HSIO tile can support line rate when used as two 400G links.

MAC/PCS

We use the Alphawave OmegaCore MAC/PCS block for Ethernet connectivity, which is responsible for encoding/decoding Ethernet frames. On the client side, the MAC/PCS block interfaces with the TT Ethernet Controller, while on the other side it interfaces with the Ethernet SERDES.

Keraunos Profiler

The HSIO tile contains an instance of the Keraunos Profiler to generate records on interesting transaction events. Such transaction profiling is useful for software tuning and visibility. The architecture of the Keraunos Profiler is described in [sec-keraunos-profiling-arch].

6.4. Addressing

The HSIO tile follows two addressing schemes: Grendel System Address and Keraunos Local Address. The addressing boundaries are shown in Figure 5. As seen, all initiators in HSIO tile follow the Grendel System Address map, while HSIO fabric uses Keraunos Local Address map (described in Section 7.1) to route requests to local/remote HSIO tiles. Address remap blocks are added on the request path to HSIO fabric that remap addresses to Keraunos local address. Section 7.2 details the address remap entries needed in each of the address remap blocks.

6.5. Micro-architecture

This section describes the micro-architecture of the HSIO tile, as shown in Figure 6, and provides detailed information about each component.

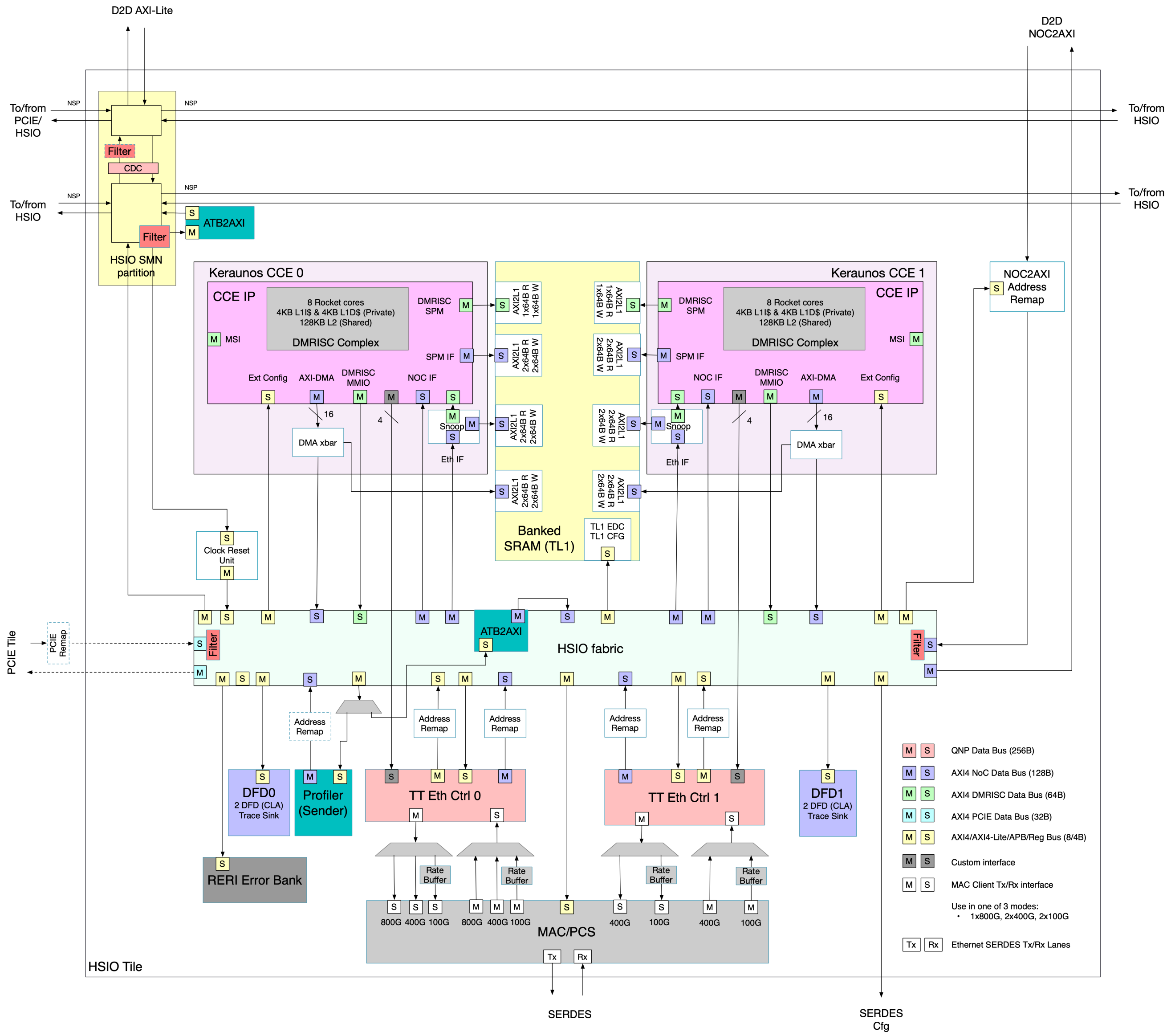


Figure 6. HSIO Tile Micro-architecture

6.5.1. Keraunos CCE

The Keraunos CCE primarily consists of the CCE IP (detailed in CCE IP Specification) along with additional components required for connectivity and functionality within the Keraunos HSIO tile. Keraunos CCE has the following components:

CCE IP

CCE has the following key components:

- DMRISC complex: 8 in-order RISCV Rocket cores, 4 KB L1 I-cache, 4KB L1 D-cache, 128KB shared L2, 3 command buffer accelerators per Rocket core
- AXI-DMA: DMA engine with 16 DMA backends that gets 1-D data movement commands from DMRISC complex and moves data between different memory locations
- AXI Snoop Coherence Master: Used to update cached data in DMRISC complex, when an external agent modifies the data (specifically read/write pointers) in the SRAM.
- Supports only incoming requests from the CCE's NOC interface, which are routed from D2D tile.
- Additional DMA backend interfaces to interface with TT Ethernet Controllers.

- 32 descriptor ring interfaces with doorbells.
- 8 general-purpose mailboxes for communicating with CCE firmware

CCE usage, including data movement dataflow, transaction sizing, number of outstanding transactions and buffering requirements for various workloads will be documented in a future version of this spec.

AXI Snoop Coherence Master

An additional AXI Snoop Coherence Master is added in Keraunos CCE for requests coming from TT Ethernet Controllers. As described above, this component is used to update cached data in the DMRISC complex when an remote Ethernet endpoint modifies the read/write pointers in the SRAM.

AXI xbar for DMA backends

CCE IP supports 16 DMA backends which 16 separate AXI interfaces. These DMA backends are conceptually equivalent to QNP VCs. An AXI crossbar is added in Keraunos CCE to multiplex these 16 AXI interfaces and route to either HSIO fabric or CCE SRAM.

Interfaces

The Keraunos CCE has the following interfaces. The exact widths of the AXI interfaces are specified in the (CCE IP Specification).

Interface	Port	AxData	Description
DMRISC Mem	AXI4 Manager	64B	DMRISCs access firmware in SRAM
DMRISC MMIO	AXI4 Manager	64B	DMRISCs access any Grendel memory
DMA	AXI4 Manager	128B	AXI-DMA read/write data
Ext Config	AX4-Lite Sub-ordinate	8B	Access CCE MMRs
NOC IF	AXI4 Sub-ordinate	128B	Incoming requests from D2D tile
SPM IF (NOC)	AXI4 Manager	128B	Forward requests from D2D tile
Eth IF	AXI4 Sub-ordinate	128B	Incoming requests from TT Ethernet controller
SPM IF (Eth)	AXI4 Manager	128B	Forward requests from TT Ethernet controller
DMA CMD	Custom	N/A	Command interface to send commands to TT Ethernet Tx controller

Table 8. Keraunos CCE Interfaces

CCE clock DFS

HSIO tile supports slow-speed DFS for CCE core clock by using two CGMs multiplexed with a glitch-free mux (described in [sec-hsio-clock-reset-unit]). The CCE core clock CGMs are placed in each of the HSIO tile, and is shared by all the DMRISC cores in both CCE0 and CCE1 in an HSIO tile.

In an HSIO tile, the two CGMs can be programmed to run at a different frequency, and the select signal to the mux can be used to switch the clock source/frequency. Typically, one CGM can be programmed to run at peak frequency and the other at a much lower frequency, representing idle condition. Note, each HSIO tile has a separate CCE core clock PLL, which allows the CCE core clock to be switched independently of the CCE core clock in other HSIO tiles.

The CCE core clock can be switched in two ways:

1. SMC can program the cce_mux_select register in HSIO clock reset to select the CGM to be used as the clock source.
2. CCE can switch the clock source/frequency in the following ways:
    1. Program the cce_mux_select register in HSIO clock reset.
    2. Program internal registers and use the CCE clock DFS control block shown in Figure 7.

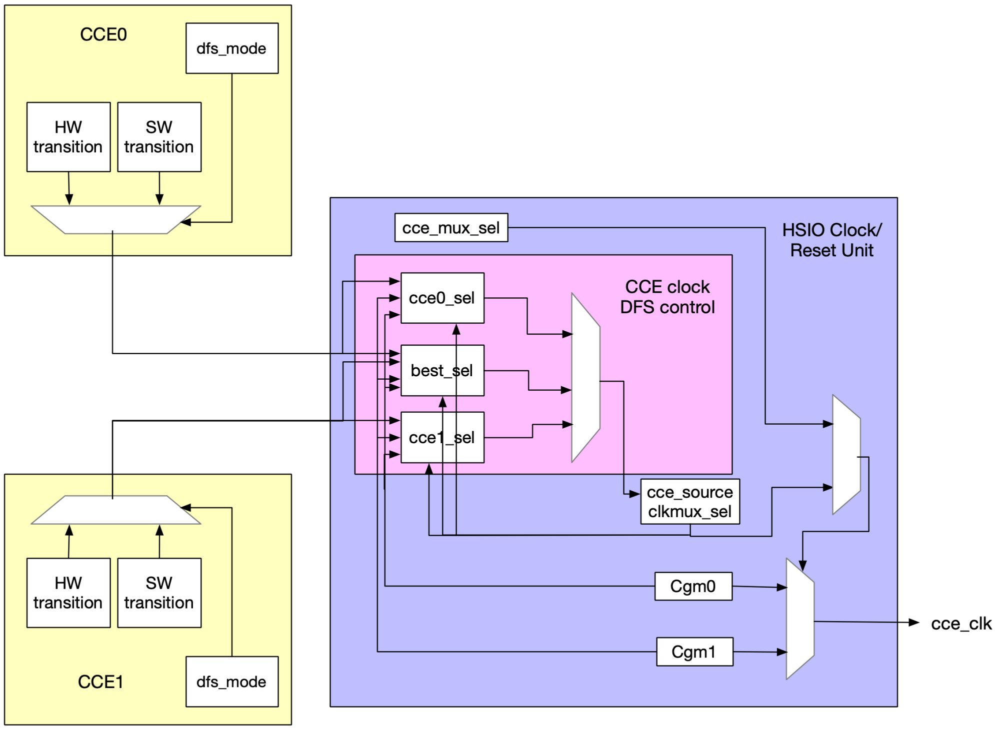


Figure 7. CCE clock DFS control block

CCE clock DFS control block takes in frequency transition signals from each CCE and determines which clock source/frequency to use for CCE core clock. The CCE clock DFS control block implements the following logic:

```
//cce_transition - 0: no transition, 1: transition to lower frequency, 2: transition
to higher frequency
//cce_source_sel - 0: cce0_transition, 1: cce1_transition, 2: consider both cce0 and
cce1 transitions
//cce_source_clkmux_sel - 0: cgm0, 1: cgm1



If cce_source_sel is 0, use cce0_transition.
     If cce0_transition is 0, set next_cce_source_clkmux_sel to cce_source_clkmux_sel
     If cce0_transition is 1 and cgm1 > cgm0, set next_cce_source_clkmux_sel to 0,
else set to 1.
     If cce0_transition is 2 and cgm1 > cgm0, set next_cce_source_clkmux_sel to 1,
else set to 0.
 If cce_source_sel is 1, use cce1_transition.
     If cce1_transition is 0, set next_cce_source_clkmux_sel to cce_source_clkmux_sel
     If cce1_transition is 1, if cgm1 > cgm0, set next_cce_source_clkmux_sel to 0,
else set to 1.
     If cce1_transition is 2, if cgm1 > cgm0, set next_cce_source_clkmux_sel to 1,
else set to 0.
 If cce_source_sel is 2, use both cce0_transition and cce1_transition.
     If either of cce0_transition or cce1_transition is 2, if cgm1 > cgm0, set
next_cce_source_clkmux_sel to 1, else set to 0.
     If both of cce0_transition or cce1_transition is 1, if cgm1 > cgm0, set
next_cce_source_clkmux_sel to 0, else set to 1.
     In all other cases, set next_cce_source_clkmux_sel to cce_source_clkmux_sel.
```

The following programming steps should be followed to switch the CCE core clock using the CCE clock DFS control block:

1. Program the clock_dfs_ctrl register in CCE and set the clock_dfs_mode to either HW/SW. In the current generation, we only

support SW mode and HW mode is tied mode.

2. Program the cce_mux_en register to HSIO clock reset unit to allow CCE DFS block to select the CGM to be used for CCE core clock.
3. Program the cce_source_select register in HSIO clock reset unit to select which cce transitions to consider in the CCE clock DFS control block.
4. Program the cce_cgm_higher_freq register in HSIO clock reset unit to let CCE clock DFS control block know which CGM is running at a higher frequency.
5. Program the clock_dfs_sw_ctrl in CCE to notify whether to increase or decrease the frequency in SW DFS mode.

6.5.2. Kerauno SRAM

Quasar L1 (TL1) IP is re-used as the banked SRAM in Keraunos HSIO tile. This is the main memory used to store Keraunos CCE firmware, and data buffers. The TL1 IP is configured to be 8MB in size, with 512 sub-banks (16B width each) and 28 64B (14 read and 14 write) flex client ports.

With only 28 64B ports (TL1 supports 61 64B ports with 256 sub-banks), we could potentially consider doubling the SRAM depth and halving the number of banks which would be more area efficient. However, the current SRAM in Quasar TL1 IP is already a custom one from Samsung and fairly large (1024x137b). To use a different # banks, we would want to see a) the timeline for other custom SRAMs (eg. 2048x137b), b) area reduction %, and c) timing impact. The recommendation is to use the same SRAM depth and double the sub-banks compared to Quasar TL1 instance (4MB in size), to increase the size of the SRAM.

Each 64B flex client port consists of 4 16B independent ports. With respect to connectivity to the 16B ports, recommendations from Tom (TL1 design owner) are as follows:

1. Spread the 112 (=28x4) 16B ports in a 3:4 (or 2:4) ratio on the 64B ports i.e. 112 16B ports spread amongst 38 64B flex client ports.
2. Not to mix read only and write only ports in the same 64B flex client port.
3. Good design point: 56 16B read ports as 3-3-3-…-3-3-3-2 (19 64B flex client ports) and 56 16B write ports as 3-3-3-…-3-3-3-2 (19 64B flex client ports). This distribution connects 2 or 3 16B ports on each 64B flex client port

This will improve throughput and latency at the cost of more client-side buffering, wiring, and flops.

An example connection from one AXI 128B port to TL1 is shown in Figure 8. Six 128B AXI ports and two 64B AXI port can be connected together to 112 16B ports as described above.

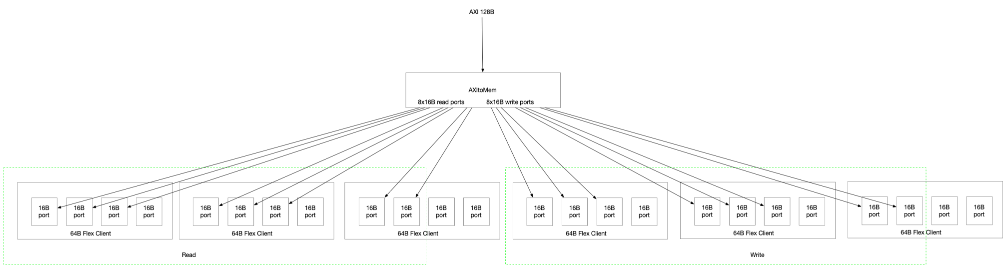


Figure 8. Keraunos TL1 Connection

6.5.3. HSIO Fabric

Keraunos HSIO fabric is used to route traffic within each HSIO tile. It is described in more detail in Section 8.3.

6.5.4. HSIO Clock Reset Unit

This block, shown in Figure 9, holds the following key components:

1. Two CGMs with glitch-free clock mux for cce_clk, which provide slow-DFS functionality for CCE clock.
2. One CGM for mac_clk.
3. Clock muxes to enable force reference clock functionality for both cce_clk and mac_clk.
4. Clock gaters to halt both cce_clk and mac_clk using a single halt clock port, which will be connected to the debug clock stop logic in SMU.
5. Supporting blocks for CGMs like clock counters, control/status registers etc.
6. HSIO reset registers to generate resets for various blocks in HSIO tile, described in Table 6.

7. Group ID register which is used by all requests exiting the HSIO tile.

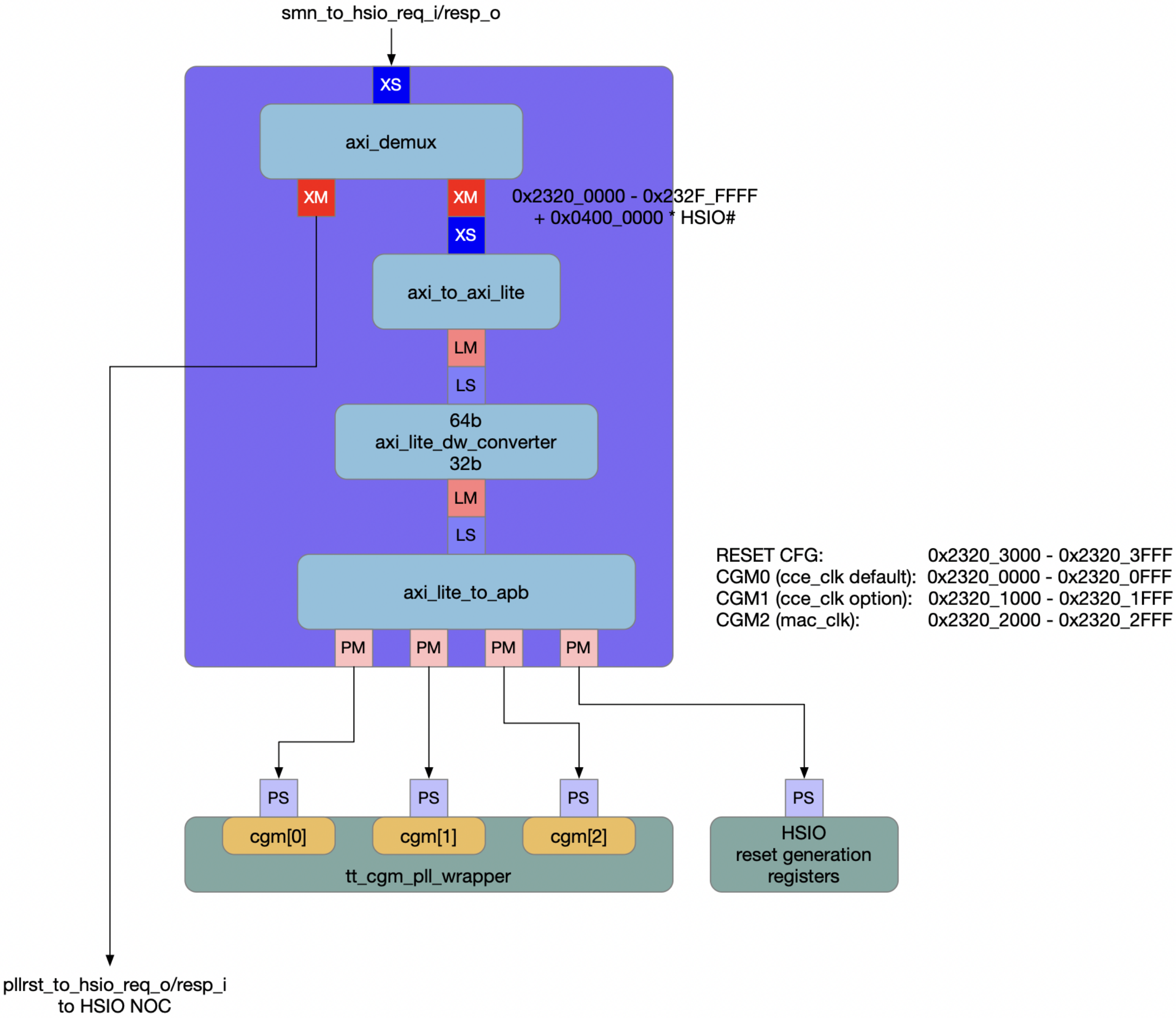


Figure 9. HSIO Clock Reset Unit

6.5.5. TT Ethernet Controller

Keraunos uses TT Ethernet Controller described in Chapter 9. TT Ethernet Controller supports in Keraunos the same hardware capabilities as Blackhole with a few enhancements.

Chapter 9 covers all the details of the TT Ethernet Controller microarchitecture. This section only covers the interface between TT Ethernet Controller and Keraunos CCE.

Keraunos CCE and TT Ethernet Controller interface

Keraunos provides the same programming interface to the DMRISCs for sending data over Ethernet as the one used for sending data over Quasar NOC and also AXI-based NOCs in Keraunos HSIO/Mimir/Athena. This is achieved by considering Ethernet Controller as CCE's DMA backend and mapping the Tx Data Forward and Remote update interface Ethernet controller to the CCE's DMA backend interface. The DMA BE and TT Ethernet controller interface can be connected as shown in Table 9.

TT Ethernet Controller interface inputs	CCE DMA interface output	Notes
i_eth_data_fwd_req_data.src_addr	o_req_head_flit.src_addr	
i_eth_data_fwd_req_data.dst_addr	o_req_head_flit.dest_addr	
i_eth_data_fwd_req_data.num_words	o_req_head_flit.length >> 4	
i_eth_data_fwd_req_data.last_msg	o_req_head_flit.cmd_user	Last message bit in the TT-Link header is repurposed as the command snoop bit, which is used to hint CCE to update its caches.

i_eth_data_fwd_req	o_req_head_flit_valid && o_req_head_flit.src_protocol=AXI	
i_eth_remote_update_req_data.addr	o_req_head_flit.payload.dst_address	
i_eth_remote_update_req_data.data	o_req_head_flit.payload.src_address	Since source address is not needed in the remote update request, it is used to carry the payload data.
i_eth_remote_update_req_data.last_msg	o_req_head_flit.cmd_user	This is the command snoop bit in the TT-Link heade
i_eth_remote_update_req	o_req_head_flit_valid && o_req_head_flit.src_protocol = INIT	
CMD DMA interface input	**BH HW overlay interface output**	
i_req_head_flit_ready	o_eth_data_fwd_ack || o_eth_remote_update_ack	
i_resp_flit_valid	o_eth_data_fwd_sent || o_eth_remote_update_sent	

Table 9. TT Ethernet Controller and CCE DMA interface mapping

This CCE/TT Ethernet controller also provides following advantages to the Ethernet cores:

1. Expose the HW retransmission to Ethernet cores: This is basically acheived by connecting the o_eth_data_fwd_ack, and o_eth_remote_update_ack signals, which are set by the TT Ethernet Controller when the sequence ack pointer is updated by the HW retransmission logic.
2. Inline register writes to remote Ethernet endpoints: DMRISCs already send inline write commands to DMA BE by tweaking the fields in the same o_req_head_flit. i_eth_remote_update_req_data and i_eth_remote_update_req are appropriately connected, enabling inline register writes to remote Ethernet endpoint.

6.5.6. NOC2AXI Address Remap

Incoming request from D2D NOC2AXI to HSIO tile use Grendel System address while HSIO fabric uses Keraunos local address. NOC2AXI Address Remap in HSIO tile remaps the Grendel System address to Keraunos local address.

6.5.7. MAC/PCS

Keraunos-E100 uses the AXIS interface version of the Alphawave OmegaCore MAC/PCS.

Figure 10 shows the high-level architecture of the MAC/PCS block and its connectivity with the TT Ethernet Controller and Ethernet SERDES. MAC/PCS supports 3 separate interfaces: 800G, 400G (2 ports) and 100G (8 ports). In Keraunos HSIO tile, only up to two ports/links in each of the interfaces of the MAC/PCS block are connected to the TT Ethernet Controller as shown in Figure 10 and Figure 6.

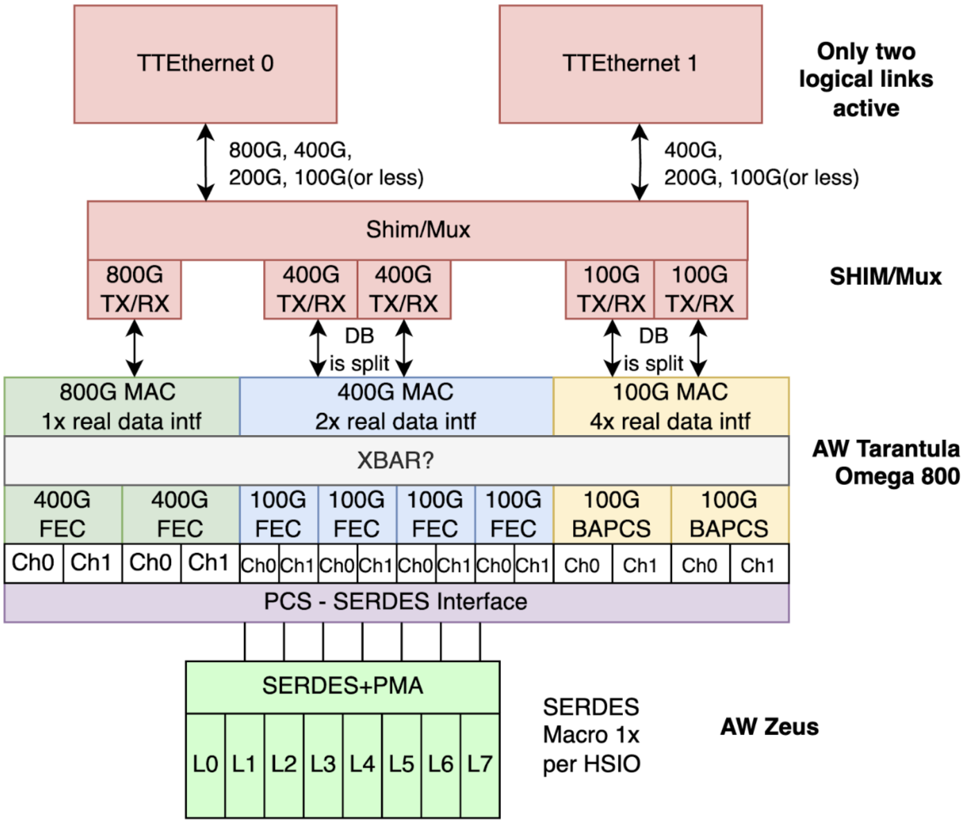


Figure 10. MAC/PCS High-Level Architecture

Alphawave OmegaCore MAC/PCS IP can be configured in various different Ethernet configurations detailed in Figure 11. Primarily, Keraunos supports 1x800G, 2x400G, 2x200G, 2x100G and 2x10G Ethernet configurations as POR. The rest of the configurations are possible but need not be prioritized by DV.

			Per Logical Link								
Config Name	PCS Naming	Standard	Lane Speed	Lanes	Throughput (Gbps)	SERDES Rate Gbps	Signaling	Lanes	PCS - FEC	MACs	TT Eth/MAC interface
POR CONFIGS											
1x800G	800G-BASE-K(R)8		100G	8	800	106.25	PAM4	0-7	400G+400G	800G	1xTT Eth to 800G MAC
2x400G	400G-BASE-K(R)4	802.3df-2024	100G	4	400	106.25	PAM4	0-3, 4-7	400G, 400G	400G, 400G	2xTT Eth to 400G MAC
2x200G	200G-BASE-K(R)4	802.3cd - 2018	50G	4	200	53.125	NRZ	0-3, 4-7	400G, 400G	400G, 400G	2xTT Eth to 200G MAC
2x100G	100G-BASEK(R)4	802.3bj	25G	4	100	25.78125	NRZ	0-3, 4-7	100G, 100G	100G, 100G	2xTT Eth to 100G MAC
2x10G	10G-BASEK(R)		10G	1	10	10.3125	NRZ	0, 4	100G, 100G	100G, 100G	2xTT Eth to 10G MAC
OTHER CONFIGS											
	200G-BASE-K(R)2		100G	2	200	106.25	PAM4				
	100G-BASE-K(R)1	802.3df-2024	100G	1	100	106.25	PAM4				
	100G-BASE-K(R)2		50G	2	100	53.125	NRZ				
	50G-BASE-K(R)	802.3cd - 2018	50G	1	50	53.125	NRZ				
	400G-BASE-K(R)8		50G	8	400	53.125	NRZ				
	200G-BASE-K(R)4	802.3bs - 2017	50G	4	200	53.125	NRZ				
	25G-BASE-K(R)	802.3by - 2014	25G	1	25	25.78125	NRZ				
MIXED MODES											
	400G-BASE-KR4 200G-BASE-KR4	Mixed	100G, 50G	4, 4	400, 200	106.25, 53.125	PAM4, NRZ	0-3, 4-7	400G, 400G	400G, 400G	
	200G-BASE-KR4 100G-BASE-KR4	Mixed	50G, 25G	4, 4	200, 100	53.125, 25.78125	NRZ, NRZ	0-3, 4-7	400G, 100G	400G, 100G	
	100G-BASE-KR4 10G-BASE-KR	Mixed	25G, 10G	4, 1	100, 10	25.78125, 10.3125	NRZ, NRZ	0-3, 4	100G, 100G	100G, 100G	

Figure 11. Ethernet configurations

6.6. Bandwidth Analysis

In this section, a bandwidth analysis for the Keraunos HSIO tile is provided. Table 10 lists the possible 4 end-to-end data movement paths from HSIO0 tile's perspective. Table 10 considers three data movement models for each end-to-end data path: A) pull model, B) push model, and C) pull model with distributed SRAM where TT Ethernet Controller can pull from remote SRAM. Each path needs data to be moved across multiple pairs of source and destination, with two options of initiators for each source/destination pair, depending on pull or push model used. Table 10 lists the intiator and source/destination pair for each end-to-end data path and the data movement model considered. It is assumed that Ethernet Rx always puts data in the local SRAM since the software router running on local CCE needs to process it further to determine the next hop.

Sr No.	Path	Pull Model (A)	Push Model (B)	Pull Model with Distributed SRAM ©
1	HSIO1 Eth to HSIO0 Eth	1. HSIO1 Rx puts data in HSIO1 SRAM 2. HSIO0 CCE gets data from HSIO1 SRAM to HSIO0 SRAM 3. HSIO0 Tx gets data from HSIO0 SRAM	1. HSIO1 Rx puts data in HSIO1 SRAM 2. HSIO1 CCE puts data from HSIO1 SRAM to HSIO0 SRAM 3. HSIO0 Tx gets data from HSIO0 SRAM	1. HSIO1 Rx puts data in HSIO1 SRAM 2. HSIO0 Tx gets data from HSIO1 SRAM
2	HSIO0 Eth to HSIO1 Eth	1. HSIO0 Rx puts data in HSIO0 SRAM 2. HSIO1 CCE gets data from HSIO0 SRAM to HSIO1 SRAM 3. HSIO1 Tx gets data from HSIO1 SRAM	1. HSIO0 Rx puts data in HSIO0 SRAM 2. HSIO0 CCE puts data from HSIO0 SRAM to HSIO1 SRAM 3. HSIO1 Tx gets data from HSIO1 SRAM	1. HSIO0 Rx puts data in HSIO0 SRAM 2. HSIO1 Tx gets data from HSIO0 SRAM
3	D2D0 to HSIO0 Eth	1. HSIO0 CCE gets data from D2D0 and puts in HSIO0 SRAM 2. HSIO0 Eth Tx gets data from HSIO0 SRAM	1. D2D0 puts data in HSIO0 SRAM 2. HSIO0 Eth Tx gets data from HSIO0 SRAM	N/A
4	HSIO0 Eth to D2D0	1. HSIO0 Eth Rx puts data in HSIO0 SRAM 2. D2D0 gets data from HSIO0 SRAM	1. HSIO0 Eth Rx puts data in HSIO0 SRAM 2. HSIO0 CCE puts data from HSIO0 SRAM to D2D0	N/A

Table 10. Data movement paths from HSIO0 tile's perspective

Figure 12 considers uniform simulataneous flows on the 4 end-to-end paths listed in Table 10 using pull model and allocates bandwidth utilization on each of the links in HSIO0 tile. A value of 1 refers to 128B/cycle while a value of 0.5 refers to 64B/cycle. There are two paths trying to send data over Ethernet links (D2D0 to Eth Tx and HSIO1 to Eth Tx), which can only sustain an aggregate bandwidth of 128B/cycle (into to MAC and 800G on the wire). Thus, each of the two paths are allocated a bandwidth utilization of 0.5. The same logic is applied to the paths where data is being received from Ethernet Rx. From Figure 12, it can be concluded that the bandwidth utilization on the links is less than or equal to the provisioned bandwidth and that the bottleneck is Ethernet Tx and Rx links, which is desired. In fact, as can be seen from Figure 12, there is no traffic on write channels on HSIO fabric to D2D NOC2AXI and D2D NOC2AXI to HSIO fabric links. So we can add additional push model flows on these links to utilize the bandwidth as shown in Figure 12.

Simultaneous flows if using pull model (from HSIO0 point of view)																	
				Traffic on each link (Legend: 0.5 = 64B/cycle, 1 = 128B/cycle)													
				Ethernet Link		Eth Ctrl to HSIO		HSIO to D2D NOC2AXI		D2D NOC2AXI to HSIO		CCE to HSIO		CCE to SRAM		HSIO to SRAM	
Path	Initiator	Src	Dest	Tx	Rx	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write
1A	HSIO0 Tx	HSIO0 SRAM	MAC	0.5		0.5										0.5	
1A	HSIO0 CCE	HSIO1 SRAM	HSIO0 SRAM					0.5				0.5			0.5		
2A	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
2A	HSIO1 CCE	HSIO0 SRAM	HSIO1 SRAM							0.5						0.5	
3A	HSIO0 Tx	HSIO0 SRAM	MAC	0.5		0.5										0.5	
3A	HSIO0 CCE	D2D0	HSIO0 SRAM					0.5				0.5			0.5		
4A	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
4A	D2D0	HSIO0 SRAM	D2D0							0.5						0.5	
			Total Traffic	**1**	**1**	**1**	**1**	**1**		**1**		**1**			**1**	**2**	**1**
Additional push model flows																	
1B/3B	HSIO1 CCE/D2D0	HSIO1 SRAM/D2D0	HSIO0 SRAM								1						1
2B/4B	HSIO0 CCE	HSIO0 SRAM	HSIO1 SRAM/D2D0						1				1	1			
			Total Traffic	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**2**	**2**

Figure 12. Bandwidth utilization in HSIO0 tile links for pull model

Similar analysis is done with simulataneous flows on the 4 end-to-end paths listed in Table 10 using push model and pull model with distributed SRAMs in Figure 13 and Figure 14 respectively.

Simultaneous flows if using push model (from HSIO0 point of view)																	
				Traffic on each link (Legend: 0.5 = 64GB/s, 1 = 128GB/s)													
				Ethernet Link		Eth Ctrl to HSIO		HSIO to D2D NOC2AXI		D2D NOC2AXI to HSIO		CCE to HSIO		CCE to SRAM		HSIO to SRAM	
Path	Initiator	Src	Dest	Tx	Rx	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write
1B	HSIO0 Tx	HSIO0 SRAM	MAC	0.5		0.5										0.5	
1B	HSIO1 CCE	HSIO1 SRAM	HSIO0 SRAM								0.5						0.5
2B	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
2B	HSIO0 CCE	HSIO0 SRAM	HSIO1 SRAM						0.5				0.5	0.5			
3B	HSIO0 Tx	HSIO0 SRAM	MAC	0.5		0.5										0.5	
3B	D2D0	D2D0	HSIO0 SRAM								0.5						0.5
4B	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
4B	HSIO0 CCE	HSIO0 SRAM	D2D0						0.5				0.5	0.5			
			Total Traffic	**1**	**1**	**1**	**1**		**1**		**1**		**1**	**1**		**1**	**2**
Additional pull model flows																	
1A/3A	HSIO0 CCE	HSIO1 SRAM/D2D0	HSIO0 SRAM					1				1			1		
2A/4A	HSIO1 CCE/D2D0	HSIO0 SRAM	HSIO1 SRAM/D2D0							1						1	
			Total Traffic	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**2**	**2**

Figure 13. Bandwidth utilization in HSIO0 tile links for push model

Simultaneous flows if using pull model with distributed SRAM (from HSIO0 point of view)																	
				Traffic on each link (Legend: 0.5 = 64GB/s, 1 = 128GB/s)													
				Ethernet Link		Eth Ctrl to HSIO		HSIO to D2D NOC2AXI		D2D NOC2AXI to HSIO		CCE to HSIO		CCE to SRAM		HSIO to SRAM	
Path	Initiator	Src	Dest	Tx	Rx	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write	Read	Write
1C	HSIO0 Tx	HSIO1 SRAM	MAC	0.5		0.5		0.5									
2C	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
2C	HSIO1 Tx	HSIO0 SRAM	MAC (remote)							0.5						0.5	
3A	HSIO0 Tx	HSIO0 SRAM	MAC	0.5		0.5										0.5	
3A	HSIO0 CCE	D2D0	HSIO0 SRAM					0.5				0.5			0.5		
4A	HSIO0 Rx	MAC	HSIO0 SRAM		0.5		0.5										0.5
4A	D2D0	HSIO0 SRAM	D2D0							0.5						0.5	
			Total Traffic	**1**	**1**	**1**	**1**	**1**		**1**		**0.5**			**0.5**	**1.5**	**1**
Additional push model flows																	
1B/3B	HSIO1 CCE/D2D0	HSIO1 SRAM/D2D0	HSIO0 SRAM								1						1
2B/4B	HSIO0 CCE	HSIO0 SRAM	HSIO1 SRAM/D2D0						1				1	1			
			Total Traffic	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0.5**	**1.5**	**2**

Figure 14. Bandwidth utilization in HSIO0 tile links for pull model with distributed SRAM

The analysis is based on the following assumptions:

6.7. Reset configuration

At reset, the interfaces are configured by the SMC according to the desired usage of the chiplet in the particular configuration of Grendel.

The configuration of Keraunos-E100 for the PCIe card variants of Grendel is:

Interface	Mode	Speed	Width
HSIO Tile 0 / SERDES 0	PAM4	100Gbps	x8
HSIO Tile 1 / SERDES 1	PAM4	100Gbps	x8
HSIO Tile 2 / SERDES 2	PAM4	100Gbps	x8
HSIO Tile 3 / SERDES 3	PAM4	100Gbps	x8
HSIO Tile 4 / SERDES 4	Powered Down	OFF	OFF

The configuration of Keraunos-E100 for the Galaxy variant of Grendel is:

Interface	Mode	Speed	Width
HSIO Tile 0 / SERDES 0	PAM4	100Gbps	x8
HSIO Tile 1 / SERDES 1	PAM4	100Gbps	x8
HSIO Tile 2 / SERDES 2	PAM4	100Gbps	x8
HSIO Tile 3 / SERDES 3	PAM4	100Gbps	x8
HSIO Tile 4 / SERDES 4	PAM4	100Gbps	x8

6.7.1. Warm Reset

Some registers in the HSIO Tile are preserved through a warm reset to allow Ethernet links to stay up through a partial reset of the compute portions of a Grendel node. This is a requirement for resilience of a network of Grendel devices.

The list of registers that are preserved through a warm reset is pending.

6.8. Clock

6.8.1. Clock Domains

Table 11 summarizes the clocks used in the HSIO tile. More details on the clock sources, DFS ability, and voltage is provided in Table 56.

Name	Signal Name	Frequency	Usage
CCE clock	cce_clk	1650 MHz	Clock for CCE
HSIO clock	hsio_clk	1650 MHz	Clock for CCE snoop blocks, HSIO fabric, TT Eth Ctrl, SRAM, DFD, Profiler, Error bank

Name	Signal Name	Frequency	Usage
MAC/PCS clock	mac_clk	1600 MHz	Clock for Ethernet MAC/PCS
SMN clock	smn_clk	400 MHz	Clock for MAC/PCS and SERDES configuration
Ethernet SerDes lane clocks	serdes_lane_clk	645-831 MHz	Clock for MAC/PCS <-> Ethernet SerDes interface

Table 11. HSIO Tile Clocks

Figure 15 shows the clock domains along with the placement of CDCs in Keraunos HSIO tile.

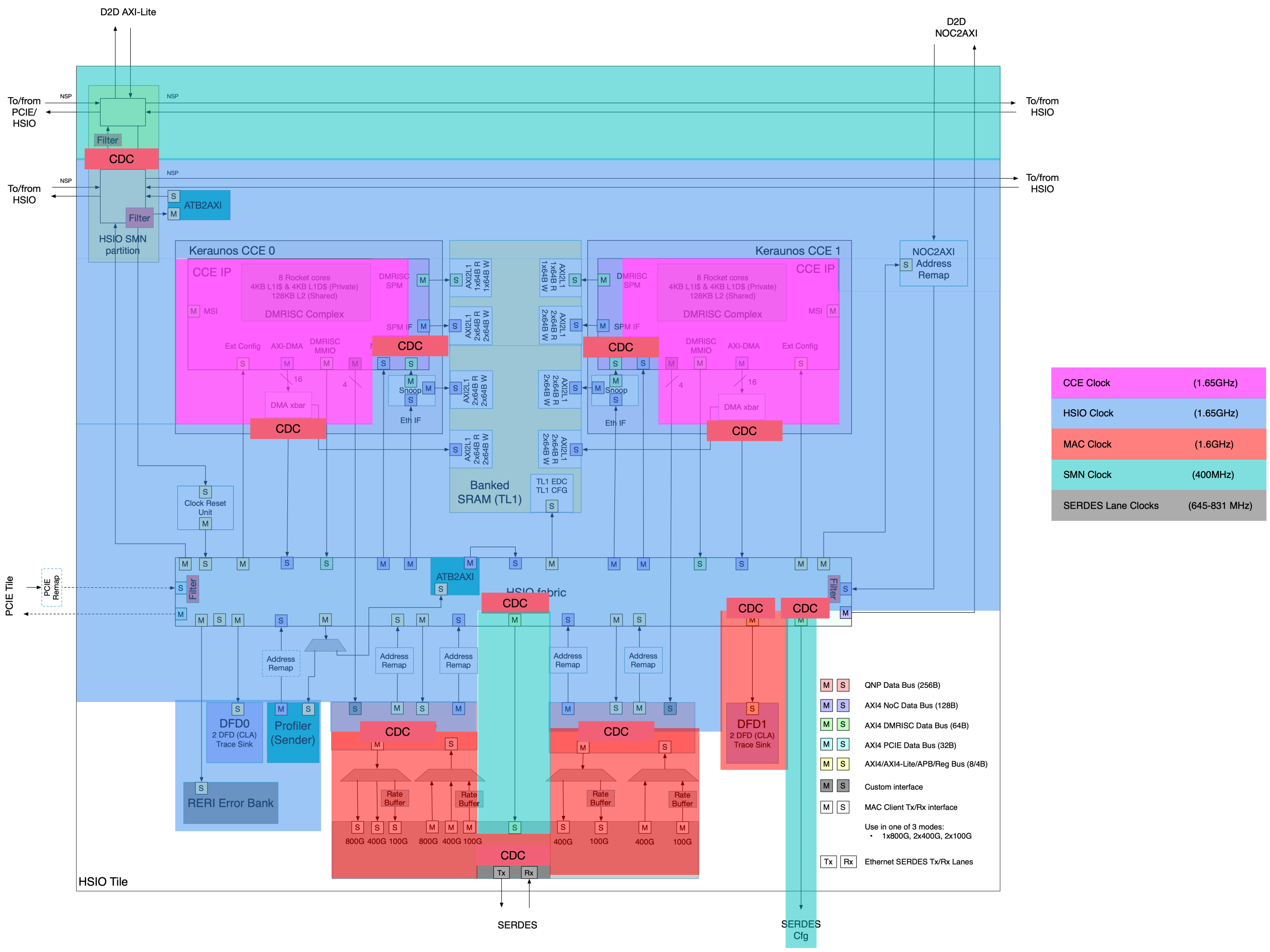


Figure 15. HSIO Tile Clock Domains

6.8.2. Clock gaters

Each clock in the HSIO tile supports a functional clock gater. A single halt port, that is connected to the debug clock stop logic in SMU, drives the clock gating logic on both cce_clk and mac_clk.

6.8.3. Reference clock muxes

Each of clocks generated in HSIO tile, cce_clk and mac_clk, have a glitch-free clock mux to select between the reference clock and the generated clock from the CGM. These clock muxes enable the force reference clock functionality needed for reset assertion/deassertion.

6.8.4. CCE Clock DFS

Keraunos supports slow-speed DFS for the CCE core clock, which enables the CCE DMRISC cores to operate in a lower frequency when they are not doing useful work (e.g. spinning in a busy-wait loop). This is achieved by using two CGMs multiplexed with a glitch-free mux placed in HSIO clock reset unit.

Chapter 7. Keraunos Addressing

7.1. Keraunos Local Address Map

Keraunos SMN and HSIO fabrics use the following local address map to route traffic within Keraunos chiplet.

Note, all initiators in HSIO tile (Keraunos CCE, TT Ethernet Controller, NOC2AXI, and Profiler), follow the Grendel System Address map i.e. address space above 64GB (Chiplet config, Quasar L1/Memory, etc.) to target resources both internal and external to Keraunos chiplet. HSIO fabric internally uses Keraunos local address map to route requests to local/remote HSIO tiles. The address remap blocks on the request path to HSIO fabric are used to remap addresses to Keraunos local address.

Address Region	Start	Size	Description
SEP	0x0_0000_0000_0000	32MB	Config region for Keraunos-E100 SEP
SMC	0x0_0000_0800_0000	32MB	Config region for Keraunos-E100 SMC
HSIO0 SMN MMR	0x0_0000_1000_0000	128KB	Config space for HSIO0 SMN partition
HSIO1 SMN MMR	0x0_0000_1002_0000	128KB	Config space for HSIO1 SMN partition
HSIO2 SMN MMR	0x0_0000_1004_0000	128KB	Config space for HSIO2 SMN partition
HSIO3 SMN MMR	0x0_0000_1006_0000	128KB	Config space for HSIO3 SMN partition
HSIO4 SMN MMR	0x0_0000_1008_0000	128KB	Config space for HSIO4 SMN partition
SMN bridge MMR	0x0_0000_100a_0000	128KB	Config space for SMN bridge between SMU and PCIE tile
HSIO0 SMN ATB2AXI MMR	0x0_0000_100c_0000	1KB	Config space for HSIO0 SMN ATB2AXI registers
HSIO1 SMN ATB2AXI MMR	0x0_0000_100c_0400	1KB	Config space for HSIO1 SMN ATB2AXI registers
HSIO2 SMN ATB2AXI MMR	0x0_0000_100c_0800	1KB	Config space for HSIO2 SMN ATB2AXI registers
HSIO3 SMN ATB2AXI MMR	0x0_0000_100c_0c00	1KB	Config space for HSIO3 SMN ATB2AXI registers
HSIO4 SMN ATB2AXI MMR	0x0_0000_100c_1000	1KB	Config space for HSIO4 SMN ATB2AXI registers
SMN bridge ATB2AXI MMR	0x0_0000_100c_1400	1KB	Config space for SMN bridge ATB2AXI bridge registers
PCIE remap MMR	0x0_0000_100c_1800	1KB	Config space for PCIE remap registers
SMU Temp Sensor	0x0_0000_1010_0000	1KB	Config space for SMU temperature sensor
Reserved	0x0_0000_1018_0000	512KB	Reserved
D2D0 MMR	0x0_0000_1020_0000	512KB	Config space for D2D in D2D0 tile
D2D0 NOC2AXI MMR	0x0_0000_1028_0000	512KB	Config space for NOC2AXI in D2D0 tile
D2D0 NOC2AXI strap MMR	0x0_0000_1030_0000	512KB	Config space for NOC2AXI strap registers in D2D0 tile
D2D0 Error bank MMR	0x0_0000_1038_0000	4KB	Config space for Error bank registers in D2D0 tile
D2D1 MMR	0x0_0000_1040_0000	512KB	Config space for D2D in D2D1 tile
D2D1 NOC2AXI MMR	0x0_0000_1048_0000	512KB	Config space for NOC2AXI in D2D1 tile
D2D1 NOC2AXI strap MMR	0x0_0000_1050_0000	512KB	Config space for NOC2AXI strap registers in D2D1 tile
D2D1 Error bank MMR	0x0_0000_1058_0000	4KB	Config space for Error bank registers in D2D1 tile
D2D2 MMR	0x0_0000_1060_0000	512KB	Config space for D2D in D2D2 tile
D2D2 NOC2AXI MMR	0x0_0000_1068_0000	512KB	Config space for NOC2AXI in D2D2 tile
D2D2 NOC2AXI strap MMR	0x0_0000_1070_0000	512KB	Config space for NOC2AXI strap registers in D2D2 tile

Address Region	Start	Size	Description
D2D2 Error bank MMR	0x0_0000_1078_0000	4KB	Config space for Error bank registers in D2D2 tile
D2D3 MMR	0x0_0000_1080_0000	512KB	Config space for D2D in D2D3 tile
D2D3 NOC2AXI MMR	0x0_0000_1088_0000	512KB	Config space for NOC2AXI in D2D3 tile
D2D3 NOC2AXI strap MMR	0x0_0000_1090_0000	512KB	Config space for NOC2AXI strap registers in D2D3 tile
D2D3 Error bank MMR	0x0_0000_1098_0000	4KB	Config space for Error bank registers in D2D3 tile
D2D4 MMR	0x0_0000_10a0_0000	512KB	Config space for D2D in D2D4 tile
D2D4 NOC2AXI MMR	0x0_0000_10a8_0000	512KB	Config space for NOC2AXI in D2D4 tile
D2D4 NOC2AXI strap MMR	0x0_0000_10b0_0000	512KB	Config space for NOC2AXI strap registers in D2D4 tile
D2D4 Error bank MMR	0x0_0000_10b8_0000	4KB	Config space for Error bank registers in D2D4 tile
D2D0 Temp Sensor MMR	0x0_0000_10c0_0000	1KB	Config space for D2D0 temperature sensor
D2D1 Temp Sensor MMR	0x0_0000_10c0_0400	1KB	Config space for D2D1 temperature sensor
D2D2 Temp Sensor MMR	0x0_0000_10c0_0800	1KB	Config space for D2D2 temperature sensor
D2D3 Temp Sensor MMR	0x0_0000_10c0_0c00	1KB	Config space for D2D3 temperature sensor
D2D4 Temp Sensor MMR	0x0_0000_10c0_1000	1KB	Config space for D2D4 temperature sensor
Miscellaneous	0x0_0000_10d0_0000	3MB	Misc functions like TAP, DAP
PCIe Mgmt MMR	0x0_0000_1800_0000	8MB	Config space for PCIe Management
PCIe App MMR	0x0_0000_1880_0000	8MB	Config space for PCIe Application Logic
PCIe PLL Config	0x0_0000_1900_0000	1MB	Config space for PCIe PLL
HSIO0 Keraunos CCE0 MMR	0x0_0000_2000_0000	16MB	Config space for HSIO0 CCE0
HSIO0 Keraunos CCE1 MMR	0x0_0000_2100_0000	16MB	Config space for HSIO0 CCE1
HSIO0 SERDES MMR	0x0_0000_2200_0000	8MB	Config space for HSIO0 SERDES
HSIO0 SRAM	0x0_0000_2280_0000	8MB	HSIO0 SRAM (TL1)
HSIO0 fabric	0x0_0000_2300_0000	1MB	Config space for HSIO0 fabric internal MMR
HSIO0 NOC2AXI remap MMR	0x0_0000_2310_0000	1MB	Config space for HSIO0 NOC2AXI Address Remap
HSIO0 Clock/Reset MMR	0x0_0000_2320_0000	1MB	Config space for HSIO0 CGM and reset MMRs
HSIO0 Eth Ctrl0 MMR	0x0_0000_2330_0000	64KB	Config space for HSIO0 Eth Ctrl0
HSIO0 Eth Ctrl1 MMR	0x0_0000_2331_0000	64KB	Config space for HSIO0 Eth Ctrl1
HSIO0 PCS MMR	0x0_0000_2332_0000	32KB	Config space for HSIO0 PCS
HSIO0 MAC MMR	0x0_0000_2332_8000	32KB	Config space for HSIO0 MAC
HSIO0 MAC/PCS ctrl MMR	0x0_0000_2333_0000	1KB	Config space for HSIO0 MAC/PCS control registers
HSIO0 SERDES ctrl MMR	0x0_0000_2333_0400	1KB	Config space for HSIO0 SERDES control registers
HSIO0 MAC/PCS Error bank MMR	0x0_0000_2333_0800	1KB	Config space for HSIO0 MAC/PCS error bank registers
HSIO0 Profiler MMR	0x0_0000_2333_1000	4KB	Config space for HSIO0 Profiler Sender registers
HSIO0 ATB2AXI MMR	0x0_0000_2333_2000	4KB	Config space for HSIO0 ATB2AXI bridge registers
HSIO0 DFD0 MMR	0x0_0000_2334_0000	128KB	Config space for HSIO0 DFD0 config registers (includes 2 DFDs internally) and Trace sink (SRAM)
HSIO0 DFD1 MMR	0x0_0000_2336_0000	128KB	Config space for HSIO0 DFD1 config registers (includes 2 DFDs internally) and Trace sink (SRAM)

Address Region	Start	Size	Description
HSIO0 SRAM Config	0x0_0000_2337_8000	4KB	Config space for HSIO0 TL1 CSRs
HSIO0 Temp Sensor	0x0_0000_2337_9000	1KB	Config space for HSIO0 Temp sensor
HSIO0 Reserved	0x0_0000_2340_0000	12MB	Reserved
HSIO1 Keraunos CCE0 MMR	0x0_0000_2400_0000	16MB	Config space for HSIO1 CCE0
HSIO1 Keraunos CCE1 MMR	0x0_0000_2500_0000	16MB	Config space for HSIO1 CCE1
HSIO1 SERDES MMR	0x0_0000_2600_0000	8MB	Config space for HSIO1 SERDES
HSIO1 SRAM	0x0_0000_2680_0000	8MB	HSIO1 SRAM (TL1)
HSIO1 fabric	0x0_0000_2700_0000	1MB	Config space for HSIO1 fabric internal MMR
HSIO1 NOC2AXI remap MMR	0x0_0000_2710_0000	1MB	Config space for HSIO1 NOC2AXI Address Remap
HSIO1 Clock/Reset MMR	0x0_0000_2720_0000	1MB	Config space for HSIO1 CGM and reset MMRs
HSIO1 Eth Ctrl0 MMR	0x0_0000_2730_0000	64KB	Config space for HSIO1 Eth Ctrl0
HSIO1 Eth Ctrl1 MMR	0x0_0000_2731_0000	64KB	Config space for HSIO1 Eth Ctrl1
HSIO1 PCS MMR	0x0_0000_2732_0000	32KB	Config space for HSIO1 PCS
HSIO1 MAC MMR	0x0_0000_2732_8000	32KB	Config space for HSIO1 MAC
HSIO1 MAC/PCS ctrl MMR	0x0_0000_2733_0000	1KB	Config space for HSIO1 MAC/PCS control registers
HSIO1 SERDES ctrl MMR	0x0_0000_2733_0400	1KB	Config space for HSIO1 SERDES control registers
HSIO1 MAC/PCS Error bank MMR	0x0_0000_2733_0800	1KB	Config space for HSIO1 MAC/PCS error bank registers
HSIO1 Profiler MMR	0x0_0000_2733_1000	4KB	Config space for HSIO1 Profiler Sender registers
HSIO1 ATB2AXI MMR	0x0_0000_2733_2000	4KB	Config space for HSIO1 ATB2AXI bridge registers
HSIO1 DFD0 MMR	0x0_0000_2734_0000	64KB	Config space for HSIO1 DFD0 config registers (includes 2 DFD internally)
HSIO1 DFD0 Trace Sink	0x0_0000_2735_0000	32KB	Trace Sink (SRAM) for HSIO1 DFD0
HSIO1 DFD1 MMR	0x0_0000_2736_0000	64KB	Config space for HSIO1 DFD1 config registers (includes 2 DFDs internally)
HSIO1 DFD1 Trace Sink	0x0_0000_2737_0000	32KB	Trace Sink (SRAM) for HSIO1 DFD1
HSIO1 SRAM Config	0x0_0000_2737_8000	4KB	Config space for HSIO1 TL1 CSRs
HSIO1 Temp Sensor	0x0_0000_2737_9000	1KB	Config space for HSIO1 Temp sensor
HSIO1 Reserved	0x0_0000_2740_0000	12MB	Reserved
HSIO2 Keraunos CCE0 MMR	0x0_0000_2800_0000	16MB	Config space for HSIO2 CCE0
HSIO2 Keraunos CCE1 MMR	0x0_0000_2900_0000	16MB	Config space for HSIO2 CCE1
HSIO2 SERDES MMR	0x0_0000_2a00_0000	8MB	Config space for HSIO2 SERDES
HSIO2 SRAM	0x0_0000_2a80_0000	8MB	HSIO2 SRAM (TL1)
HSIO2 fabric	0x0_0000_2b00_0000	1MB	Config space for HSIO2 fabric internal MMR
HSIO2 NOC2AXI remap MMR	0x0_0000_2b10_0000	1MB	Config space for HSIO2 NOC2AXI Address Remap
HSIO2 Clock/Reset MMR	0x0_0000_2b20_0000	1MB	Config space for HSIO2 CGM and reset MMRs
HSIO2 Eth Ctrl0 MMR	0x0_0000_2b30_0000	64KB	Config space for HSIO2 Eth Ctrl0
HSIO2 Eth Ctrl1 MMR	0x0_0000_2b31_0000	64KB	Config space for HSIO2 Eth Ctrl1
HSIO2 PCS MMR	0x0_0000_2b32_0000	32KB	Config space for HSIO2 PCS
HSIO2 MAC MMR	0x0_0000_2b32_8000	32KB	Config space for HSIO2 MAC
HSIO2 MAC/PCS ctrl MMR	0x0_0000_2b33_0000	1KB	Config space for HSIO2 MAC/PCS control registers

Address Region	Start	Size	Description
HSIO2 SERDES ctrl MMR	0x0_0000_2b33_0400	1KB	Config space for HSIO2 SERDES control registers
HSIO2 MAC/PCS Error bank MMR	0x0_0000_2b33_0800	1KB	Config space for HSIO2 MAC/PCS error bank registers
HSIO2 Profiler MMR	0x0_0000_2b33_1000	4KB	Config space for HSIO2 Profiler Sender registers
HSIO2 ATB2AXI MMR	0x0_0000_2b33_2000	4KB	Config space for HSIO2 ATB2AXI bridge registers
HSIO2 DFD0 MMR	0x0_0000_2b34_0000	64KB	Config space for HSIO2 DFD0 config registers (includes 2 DFDs internally)
HSIO2 DFD0 Trace Sink	0x0_0000_2b35_0000	32KB	Trace Sink (SRAM) for HSIO3 DFD0
HSIO2 DFD1 MMR	0x0_0000_2b36_0000	64KB	Config space for HSIO2 DFD1 config registers (includes 2 DFDs internally)
HSIO2 DFD1 Trace Sink	0x0_0000_2b37_0000	32KB	Trace Sink (SRAM) for HSIO2 DFD1
HSIO2 SRAM Config	0x0_0000_2b37_8000	4KB	Config space for HSIO2 TL1 CSRs
HSIO2 Temp Sensor	0x0_0000_2b37_9000	1KB	Config space for HSIO2 Temp sensor
HSIO2 Reserved	0x0_0000_2b40_0000	12MB	Reserved
HSIO3 Keraunos CCE0 MMR	0x0_0000_2c00_0000	16MB	Config space for HSIO3 CCE0
HSIO3 Keraunos CCE1 MMR	0x0_0000_2d00_0000	16MB	Config space for HSIO3 CCE1
HSIO3 SERDES MMR	0x0_0000_2e00_0000	8MB	Config space for HSIO3 SERDES
HSIO3 SRAM	0x0_0000_2e80_0000	8MB	HSIO3 SRAM (TL1)
HSIO3 fabric	0x0_0000_2f00_0000	1MB	Config space for HSIO3 fabric internal MMR
HSIO3 NOC2AXI remap MMR	0x0_0000_2f10_0000	1MB	Config space for HSIO3 NOC2AXI Address Remap
HSIO3 Clock/Reset MMR	0x0_0000_2f20_0000	1MB	Config space for HSIO3 CGM and reset MMRs
HSIO3 Eth Ctrl0 MMR	0x0_0000_2f30_0000	64KB	Config space for HSIO3 Eth Ctrl0
HSIO3 Eth Ctrl1 MMR	0x0_0000_2f31_0000	64KB	Config space for HSIO3 Eth Ctrl1
HSIO3 PCS MMR	0x0_0000_2f32_0000	32KB	Config space for HSIO3 PCS
HSIO3 MAC MMR	0x0_0000_2f32_8000	32KB	Config space for HSIO3 MAC
HSIO3 MAC/PCS ctrl MMR	0x0_0000_2f33_0000	1KB	Config space for HSIO3 MAC/PCS control registers
HSIO3 SERDES ctrl MMR	0x0_0000_2f33_0400	1KB	Config space for HSIO3 SERDES control registers
HSIO3 MAC/PCS Error bank MMR	0x0_0000_2f33_0800	1KB	Config space for HSIO3 MAC/PCS error bank registers
HSIO3 Profiler MMR	0x0_0000_2f33_1000	4KB	Config space for HSIO3 Profiler Sender registers
HSIO3 ATB2AXI MMR	0x0_0000_2f33_2000	4KB	Config space for HSIO3 ATB2AXI bridge registers
HSIO3 DFD0 MMR	0x0_0000_2f34_0000	64KB	Config space for HSIO3 DFD0 config registers (includes 2 DFDs internally)
HSIO3 DFD0 Trace Sink	0x0_0000_2f35_0000	32KB	Trace Sink (SRAM) for HSIO3 DFD0
HSIO3 DFD1 MMR	0x0_0000_2f36_0000	64KB	Config space for HSIO3 DFD1 config registers (includes 2 DFDs internally)
HSIO3 DFD1 Trace Sink	0x0_0000_2f37_0000	32KB	Trace Sink (SRAM) for HSIO3 DFD1
HSIO3 SRAM Config	0x0_0000_2f37_8000	4KB	Config space for HSIO3 TL1 CSRs
HSIO3 Temp Sensor	0x0_0000_2f37_9000	1KB	Config space for HSIO3 Temp sensor
HSIO3 Reserved	0x0_0000_2f40_0000	12MB	Reserved
HSIO4 Keraunos CCE0 MMR	0x0_0000_3000_0000	16MB	Config space for HSIO4 CCE0
HSIO4 Keraunos CCE1 MMR	0x0_0000_3100_0000	16MB	Config space for HSIO4 CCE1
HSIO4 SERDES MMR	0x0_0000_3200_0000	8MB	Config space for HSIO4 SERDES

Address Region	Start	Size	Description
HSIO4 SRAM	0x0_0000_3280_0000	8MB	HSIO4 SRAM (TL1)
HSIO4 fabric	0x0_0000_3300_0000	1MB	Config space for HSIO4 fabric internal MMR
HSIO4 NOC2AXI remap MMR	0x0_0000_3310_0000	1MB	Config space for HSIO4 NOC2AXI Address Remap
HSIO4 Clock/Reset MMR	0x0_0000_3320_0000	1MB	Config space for HSIO4 CGM and reset MMRs
HSIO4 Eth Ctrl0 MMR	0x0_0000_3330_0000	64KB	Config space for HSIO4 Eth Ctrl0
HSIO4 Eth Ctrl1 MMR	0x0_0000_3331_0000	64KB	Config space for HSIO4 Eth Ctrl1
HSIO4 PCS MMR	0x0_0000_3332_0000	32KB	Config space for HSIO4 PCS
HSIO4 MAC MMR	0x0_0000_3332_8000	32KB	Config space for HSIO4 MAC
HSIO4 MAC/PCS ctrl MMR	0x0_0000_3333_0000	1KB	Config space for HSIO4 MAC/PCS control registers
HSIO4 SERDES ctrl MMR	0x0_0000_3333_0400	1KB	Config space for HSIO4 SERDES control registers
HSIO4 MAC/PCS Error bank MMR	0x0_0000_3333_0800	1KB	Config space for HSIO4 MAC/PCS error bank registers
HSIO4 Profiler MMR	0x0_0000_3333_1000	4KB	Config space for HSIO4 Profiler Sender registers
HSIO4 ATB2AXI MMR	0x0_0000_3333_2000	4KB	Config space for HSIO4 ATB2AXI bridge registers
HSIO4 DFD0 MMR	0x0_0000_3334_0000	64KB	Config space for HSIO4 DFD0 config registers (includes 2 DFDs internally)
HSIO4 DFD0 Trace Sink	0x0_0000_3335_0000	32KB	Trace Sink (SRAM) for HSIO4 DFD0
HSIO4 DFD1 MMR	0x0_0000_3336_0000	64KB	Config space for HSIO4 DFD1 config registers (includes 2 DFDs internally)
HSIO4 DFD1 Trace Sink	0x0_0000_3337_0000	32KB	Trace Sink (SRAM) for HSIO4 DFD1
HSIO4 SRAM Config	0x0_0000_3337_8000	4KB	Config space for HSIO4 TL1 CSRs
HSIO4 Temp Sensor	0x0_0000_3337_9000	1KB	Config space for HSIO4 Temp sensor
HSIO4 Reserved	0x0_0000_3340_0000	12MB	Reserved
Chiplet config region	0x0_0010_0000_0000	64GB	Global config space for all chiplets
Quasar L1	0x0_0100_0000_0000	64GB	Quasar L1 SRAM
Quasar Memory	0x1_0000_0000_0000	256TB	Quasar/Mimir DRAM
PCIe IO memory space	0x2_0000_0000_0000	256TB	PCIe IO memory space
Remote Athena IOVA	0x8_0000_0000_0000	2048TB	Athena resources

Table 12. Keraunos Local Address Map

7.2. Keraunos Address Remaps

The section describes the address remap entries needed in each of the address remap blocks added in Keraunos. For each address remap block, the incoming addresses are in Grendel System Address space and the remapped addresses are in Keraunos Local Address space. The Keraunos Local Address regions are fixed at design time, but the Grendel System Addresses are expected to be programmable at boot time.

7.2.1. HSIO NOC2AXI Remap

Table 13 lists the remap entries needed in the NOC2AXI address remap block in HSIO tile. This is applicable to NOC2AXI remap in all HSIO tiles. Note HSIO0-HSIO4 entries can be combined together since they belong to a contiguous space.

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
0	SEP	0x0_0012_0000_0000	32MB	0x0_0000_0000_0000
1	SMC	0x0_0012_0200_0000	32MB	0x0_0000_0800_0000
2	SMN,D2D	0x0_0012_1C00_0000	16MB	0x0_0000_1000_0000

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
3	PCIE	0x0_0012_1800_0000	32MB	0x0_0000_1800_0000
4	HSIO0	0x0_0012_0400_0000	64MB	0x0_0000_2000_0000
5	HSIO1	0x0_0012_0800_0000	64MB	0x0_0000_2400_0000
6	HSIO2	0x0_0012_0C00_0000	64MB	0x0_0000_2800_0000
7	HSIO3	0x0_0012_1000_0000	64MB	0x0_0000_2c00_0000
8	HSIO4	0x0_0012_1400_0000	64MB	0x0_0000_3000_0000

Table 13. HSIO NOC2AXI Address Remap Entries

7.2.2. Keraunos CCE MMIO Remap

Table 14 lists the remap entries needed in the CCE IP's DMRISC address remap block for the requests going out on the DMRISC MMIO port. This is applicable to Keraunos CCE in all HSIO tiles. Note HSIO0-HSIO4 entries can be combined together since they belong to a contiguous space.

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
0	SEP	0x0_0012_0000_0000	32MB	0x0_0000_0000_0000
1	SMC	0x0_0012_0200_0000	32MB	0x0_0000_0800_0000
2	SMN,D2D	0x0_0012_1C00_0000	16MB	0x0_0000_1000_0000
3	PCIE	0x0_0012_1800_0000	32MB	0x0_0000_1800_0000
4	HSIO0	0x0_0012_0400_0000	64MB	0x0_0000_2000_0000
5	HSIO1	0x0_0012_0800_0000	64MB	0x0_0000_2400_0000
6	HSIO2	0x0_0012_0C00_0000	64MB	0x0_0000_2800_0000
7	HSIO3	0x0_0012_1000_0000	64MB	0x0_0000_2c00_0000
8	HSIO4	0x0_0012_1400_0000	64MB	0x0_0000_3000_0000

Table 14. HSIO CCE MMIO Address Remap Entries

7.2.3. TT Ethernet Controller Remap

TT Ethernet Controller Remap has two manager ports, a) Data, and b) Control, corresponding to data plane and control plane traffic. Table 15 and tab-eth-ctrl-remap-data[] list the remap entries needed for the TT Ethernet Controller Control port and Data port respectively.

Table 15 is applicable to TT Ethernet Controller Control port in all HSIO tiles. Note HSIO0-HSIO4 entries can be combined together since they belong to a contiguous space.

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
0	SEP	0x0_0012_0000_0000	32MB	0x0_0000_0000_0000
1	SMC	0x0_0012_0200_0000	32MB	0x0_0000_0800_0000
2	SMN,D2D	0x0_0012_1C00_0000	16MB	0x0_0000_1000_0000
3	PCIE	0x0_0012_1800_0000	32MB	0x0_0000_1800_0000
4	HSIO0	0x0_0012_0400_0000	64MB	0x0_0000_2000_0000
5	HSIO1	0x0_0012_0800_0000	64MB	0x0_0000_2400_0000
6	HSIO2	0x0_0012_0C00_0000	64MB	0x0_0000_2800_0000
7	HSIO3	0x0_0012_1000_0000	64MB	0x0_0000_2c00_0000
8	HSIO4	0x0_0012_1400_0000	64MB	0x0_0000_3000_0000

Table 15. TT Ethernet Controller Address Remap Entries for Control port

Table 16 lists the address remap entries specifically for the TT Ethernet Controller Data port in HSIO Tile[i]. AXI requests on TT Ethernet Controller Data port only target SRAMs in Keraunos. Only the request to the local SRAM is remapped to the Keraunos Local Address space, while the requests to neighboring SRAMs continue to use the Grendel System Address space and are routed to

D2D NOC2AXI.

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
0	HSIO[i] SRAM	0x0_0012_0680_0000 + i*400_0000	8MB	0x0_0000_2280_0000 + i*400_0000

Table 16. TT Ethernet Controller Address Remap Entries for Data port (HSIO Tile[i])

7.2.4. PCIE Remap

Read/write requests from the PCIE port to HSIO[0] are expected to be remapped as listed in Table 17.

1. All the requests to HSIO[0] MMRs and SRAM are remapped to Keraunos Local Address.
2. All the requests to HSIO[j] SRAMS (j = 1,2,3,4) need to be routed to D2D NOC2AXI without any remap, while requests HSIO[j] MMRs need to be remapped to the Keraunos Local Address space and routed via SMN.

Entry #	Region	Start Address (System Address)	Size	Remap Start Address (Keraunos Local Address)
0	SEP	0x0_0012_0000_0000	32MB	0x0_0000_0000_0000
1	SMC	0x0_0012_0200_0000	32MB	0x0_0000_0800_0000
2	SMN,D2D	0x0_0012_1C00_0000	16MB	0x0_0000_1000_0000
3	HSIO0	0x0_0012_0400_0000	64MB	0x0_0000_2000_0000
4	HSIO1 (CCE0/1, SERDES)	0x0_0012_0800_0000	40MB	0x0_0000_2400_0000
5	HSIO1 (Other MMRs)	0x0_0012_0B00_0000	16MB	0x0_0000_2700_0000
6	HSIO2 (CCE0/1, SERDES)	0x0_0012_0C00_0000	40MB	0x0_0000_2800_0000
7	HSIO2 (Other MMRs)	0x0_0012_0F00_0000	16MB	0x0_0000_2B00_0000
8	HSIO3 (CCE0/1, SERDES)	0x0_0012_1000_0000	40MB	0x0_0000_2C00_0000
9	HSIO3 (Other MMRs)	0x0_0012_1300_0000	16MB	0x0_0000_2F00_0000
10	HSIO4 (CCE0/1, SERDES)	0x0_0012_1400_0000	40MB	0x0_0000_3000_0000
11	HSIO4 (Other MMRs)	0x0_0012_1700_0000	16MB	0x0_0000_3300_0000

Table 17. PCIE Address Remap Entries

7.2.5. Enabling and Programming Address Remappers

To program and enable the remappers

1. Write the addr[51:23] of the original region's start address to REMAP_START
2. Write the addr[51:23] of the original region's end address to REMAP_END
3. Write the {1, 2'b0, target_addr[51:23]} to REPLACE_ADDR

Chapter 8. Keraunos NOC

Keraunos consists of the following 3 NOCs, 1) Keraunos SMN, 2) Keraunos HSIO fabric, and 3) Keraunos QNP mesh.

8.1. Keraunos SMN

The Keraunos System Management Network (SMN) is a dual-purpose interconnect fabric built using Arteris FlexNoC for routing a) system management traffic and b) CCE sync traffic between HSIO tiles. Keraunos SMN is partitioned into 6 tiles, one SMN Bridge and 5 HSIO SMN tiles as shown in Figure 16.

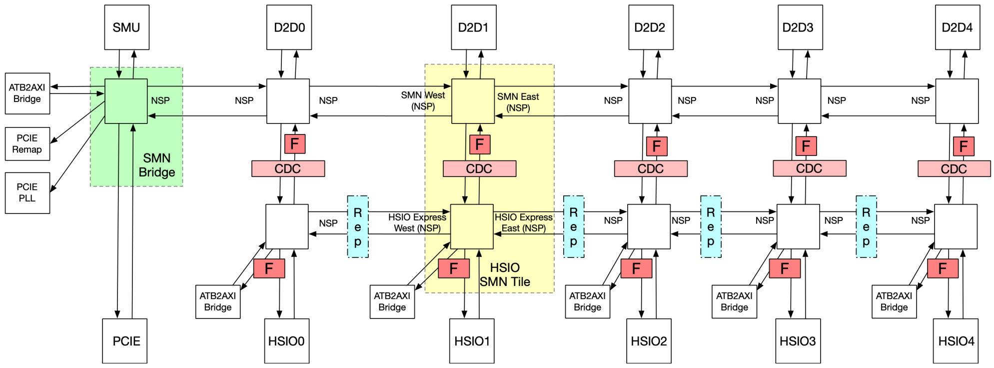


Figure 16. Keraunos SMN

As shown in Figure 16, there two lanes:

1. SMN lane: This is used for system management traffic initiated by SMC on Keraunos or other chiplets in the package.
2. HSIO Express lane: This is intended to be used for small sync messages between CCE in different HSIO tiles. This is called an express path as it allows moving messages between HSIO tiles at a much lower latency compared to the QNP mesh.

8.1.1. Ports and Connectivity

Logical SMN Connectivity

Figure 17 shows the logical connectivity between source and destination ports in the Keraunos SMN.

	smn_fabric_regs	t_smn2d2d_cfg_0	t_smn2d2d_cfg_1	t_smn2d2d_cfg_2	t_smn2d2d_cfg_3	t_smn2d2d_cfg_4	t_smn2hsio_0	t_smn2hsio_1	t_smn2hsio_2	t_smn2hsio_3	t_smn2hsio_4	t_smn2axiprob_smnbridge	t_smn2axiprob_hsio_0	t_smn2axiprob_hsio_1	t_smn2axiprob_hsio_2	t_smn2axiprob_hsio_3	t_smn2axiprob_hsio_4	t_smn2pcie_mgmt	t_smn2pcie_pll	t_smn2pcie_remap_mirr	t_smn2smc	Outstanding requests
i_d2d2smn_cfg_0	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	8
i_d2d2smn_cfg_1	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	8
i_d2d2smn_cfg_2	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	8
i_d2d2smn_cfg_3	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	8
i_d2d2smn_cfg_4	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	8
i_hsio2smn_0	Y	Y	Y	Y	Y	Y	Y*	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	32
i_hsio2smn_1	Y	Y	Y	Y	Y	Y	Y	Y*	Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	32
i_hsio2smn_2	Y	Y	Y	Y	Y	Y	Y	Y	Y*	Y	Y	Y	Y	Y	Y	Y	Y				Y	32
i_hsio2smn_3	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y*	Y	Y	Y	Y	Y	Y	Y				Y	32
i_hsio2smn_4	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y*	Y	Y	Y	Y	Y	Y				Y	32
i_axiprob2smn_smnbridge																		Y			Y	32
i_axiprob2smn_hsio_0							Y														Y	32
i_axiprob2smn_hsio_1							Y	Y													Y	32
i_axiprob2smn_hsio_2							Y		Y												Y	32
i_axiprob2smn_hsio_3							Y			Y											Y	32
i_axiprob2smn_hsio_4							Y				Y										Y	32
i_pcie2smn	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	32
i_smc2smn	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y							Y	Y	Y		32
Outstanding requests		32	32	32	32	32	32	32	32	32	32	1	1	1	1	1	1	8	8	1	8	

Figure 17. Logical SMN Connectivity (Y: connected)

HSIO SMN tile

Table 18 specifies the manager and sub-ordinate ports on the HSIO SMN tile and Table 19 lists the target addresses for each manager port in the HSIO SMN tile.

Manager	Sub-ordinate	AxID	AxADDR	AxDATA	Protocol
D2D Cfg/Mgmt	HSIO SMN	N/A	52	8B	AXI4-Lite

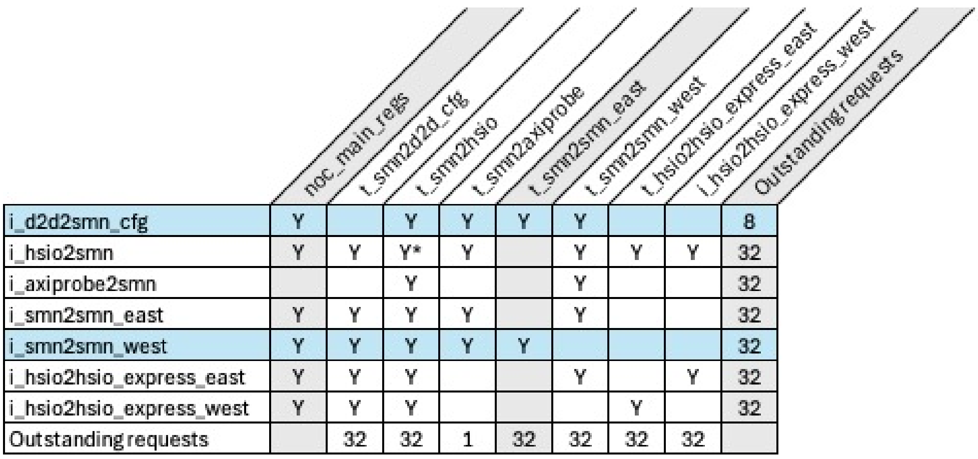
Manager	Sub-ordinate	AxID	AxADDR	AxDATA	Protocol
HSIO Fabric	HSIO SMN	10	52	8B	AXI4
AXI Probe	HSIO SMN	10	52	8B	AXI4
SMN East	HSIO SMN	10	52	8B	AXI4
SMN West	HSIO SMN	10	52	8B	AXI4
HSIO Express East	HSIO SMN	10	52	8B	AXI4
HSIO Express West	HSIO SMN	10	52	8B	AXI4
HSIO SMN	D2D Cfg/Mgmt	N/A	52	8B	AXI4-Lite
HSIO SMN	HSIO Fabric	10	52	8B	AXI4
HSIO SMN	AXI Probe	N/A	52	8B	APB
HSIO SMN	SMN East	10	52	8B	AXI4
HSIO SMN	SMN West	10	52	8B	AXI4
HSIO SMN	HSIO Express East	10	52	8B	AXI4
HSIO SMN	HSIO Express West	10	52	8B	AXI4

Table 18. HSIO SMN Tile Ports

Manager	Sub-ordinate	Target Address
HSIO SMN	D2D Cfg/Mgmt	D2D[i] MMRs (includes all MMR in D2D[i]), Chiplet Config Region
HSIO SMN	HSIO Fabric	HSIO[i] MMR(includes all MMRs in HSIO[i] tile)
HSIO SMN	AXI Probe	HSIO[i] SMN ATB2AXI MMR
HSIO SMN	SMN East	D2D[i] MMRs (includes all MMR in D2D[i]), where j > i
HSIO SMN	SMN West	D2D[i] MMRs (includes all MMR in D2D[i]), where j < i, Keraunos SMC/SEP, PCIE Mgmt MMR, PCIE remap MMR, SMN bridge ATB2AXI MMR
HSIO SMN	HSIO Express East	HSIO[j] MMR (includes all MMRs in HSIO[j] tile), HSIO[j] SMN ATB2AXI MMR, where j > i
HSIO SMN	HSIO Express West	HSIO[j] MMR (includes all MMRs in HSIO[j] tile), HSIO[j] SMN ATB2AXI MMR, where j < i

Table 19. HSIO SMN Tile Target Addresses (for HSIO[i] SMN tile, where i = 0..4)

Connectivity between source and destination ports in HSIO SMN tile is detailed in Figure 18. Note, i_hsio2smn to t_smn2hsio connectivity, indicated as Y^{*}, only allows i_hsio2smn to access the HSIO Clock Reset Unit registers from the t_smn2hsio port. This is enabled to allow CCE cores to access the HSIO Clock Reset Unit registers.



	noc_main_regs	t_smn2d2d_cfg	t_smn2hsio	t_smn2axiprobe	t_smn2smn_east	t_smn2smn_west	t_hsio2hsio_express_east	t_hsio2hsio_express_west	Outstanding requests
i_d2d2smn_cfg	Y		Y	Y	Y	Y			8
i_hsio2smn	Y	Y	Y*	Y		Y	Y	Y	32
i_axiprobe2smn			Y			Y			32
i_smn2smn_east	Y	Y	Y	Y		Y			32
i_smn2smn_west	Y	Y	Y	Y	Y				32
i_hsio2hsio_express_east	Y	Y	Y			Y		Y	32
i_hsio2hsio_express_west	Y	Y	Y				Y		32
Outstanding requests		32	32	1	32	32	32	32	

Figure 18. HSIO SMN Connectivity (Y: connected)

SMN Bridge

Table 20 specifies the manager and sub-ordinate ports on the SMN Bridge and Table 21 lists the target addresses for each manager port in the SMN Bridge.

Manager	Sub-ordinate	AxID	AxADDR	AxDATA	Protocol
Keraunos SMC	SMN Bridge	10	52	8B	AXI4
PCIE	SMN Bridge	7	52	8B	AXI4
AXI Probe	SMN Bridge	10	52	8B	AXI4
SMN East	SMN Bridge	10	52	8B	AXI4
SMN Bridge	Keraunos SMC	10	52	8B	AXI4
SMN Bridge	PCIE	7	52	8B	AXI4
SMN Bridge	PCIE Remap	N/A	52	8B	APB
SMN Bridge	PCIE PLL	N/A	52	8B	APB
SMN Bridge	AXI Probe	N/A	52	8B	APB
SMN Bridge	SMN East	10	52	8B	AXI4

Table 20. SMN Bridge Ports

Manager	Sub-ordinate	Target Address
SMN Bridge	Keraunos SMC	SEP/SMC registers
SMN Bridge	PCIE	PCIe Mgmt MMR
SMN Bridge	PCIE Remap	PCIE remap MMR
SMN Bridge	PCIE PLL	PCIE PLL config
SMN Bridge	AXI Probe	SMN bridge ATB2AXI MMR
SMN Bridge	SMN East	HSIO[i] MMR (includes all MMRs in HSIO[i] tile), D2D[i] MMR (includes all MMRs in D2D[i] tile), where i = 0..4

Table 21. SMN Bridge Target Addresses

Connectivity between source and destination ports in SMN Bridge is detailed in Figure 19.

	noc_main_regs	t_smn2smc	t_smn2pcie	t_smn2pcie_remapper_mmr	t_smn2pcie_pllctrl	t_smn2axiprobe	t_smn2smn_east	Outstanding requests
i_smc2smn	Y		Y	Y	Y	Y	Y	32
i_pcie2smn	Y	Y		Y		Y	Y	32
i_axiprobe2smn		Y	Y					32
i_smn2smn_east	Y	Y	Y	Y	Y	Y		32
Outstanding requests	16	8	8	1	1	1	32	

Figure 19. SMN Bridge Connectivity (Y: connected)

8.1.2. Traffic Filtering/Firewalls

Keraunos SMN supports traffic filtering by Source ID (used for SEP/ISS security) and Group ID (used for HSIO virtualization) using the NOC Traffic Filter described in Grendel Security Specification (Section 4.1.4).

Keraunos SMN includes the following NOC Traffic Filters in HSIO SMN tiles (denoted by F in Figure 16):

1. On HSIO to SMN path: Traffic entering the SMN from initiators in the HSIO tiles (CCEs, Ethernet Rx, and Profilers) and from the PCIe tile.
2. On SMN to HSIO path: Traffic entering the HSIO tiles from other HSIO tiles, and D2D tiles. Note, the path to the ATB2AXI bridge registers also goes through this filter.

Each filter has 16 entries and each entry has 3 64-bit configuration registers: *filter_configX*, *filter_start_addrX*, and *filter_end_addrX* (X ranges from 0 to 15). *filter_configX* register is defined in Grendel Security Specification (Section 4.1.4). NOC Traffic Filters in HSIO SMN tiles only use the source ID and group ID fields in *filter_configX* for filtering. Source ID allocation for Keraunos is defined in Table 57, while group IDs are allocated at runtime depending on how HSIO tiles are allocated to different VMs.

8.1.3. AxUSER field definition

Security features supported by Keraunos SMN, source ID and group ID, are carried in AxUSER field of AW and AR channels in Keraunos SMN.Keraunos SMN also carries CMD snoop hint in AxUSER field to indicate Keraunos CCEs to update/invalidate their caches. The AxUSER field definition is shown in Table 22.

Field	Name	Description
[3:0]	Source ID	Used for SEP/ISS secured accesses to private assets not in SEP/SMC instances
[7:4]	Group ID	Used for HSIO virtualization
[8]	CMD Snoop	Hint information to Keraunos CCE to update/invalidate its caches.
[11:9]	Reserved	Reserved for future use

Table 22. AxUSER field definition

Note that W-channel, B-channel, R-channel do not carry any user defined information.

8.1.4. Timeout

Keraunos SMN fabric supports timeout capability on all TNIUs of SMN bridge and HSIO SMN tile. The timeout duration is configurable with timeout ranging from 1us to 10s. When a target endpoint does not respond within the timeout duration, the following actions are taken:

1. Timeout signal is asserted at the TNIU. Sideband manager in the SMN bridge/HSIO SMN tile aggregates timeouts from all TNIUs using logical OR operation, and asserts the noc timeout interrupt to SMC.
2. All outstanding transactions to the target endpoint are aborted, and error responses are sent back to the initiators.
3. TNIU does not accept any new transactions to the target endpoint until the timeout condition is cleared.

Enabling and Programming Timeout Interval in NoC

Three steps are required to enable timeout and program timeout interval in Keraunos NoC (including SMN bridge and HSIO SMN tile):

1. Set faultEN register in the TNIU to 1 to enable timeout.
2. Enable timeout in each TNIU by setting the timeout FLAGINEN0 register. Each TNIU corresponds to 1 bit in this register. Setting 1 to the corresponding bit enables timeout for that TNIU. <tab:smn_tile_timeout_en_bit_correspondence>, <tab:smn_bridge_timeout_en_bit_correspondence> and <tab:hsio_timeout_en_bit_correspondence> show the correspondence between TNIU and the bit in FLAGINEN0 register.
3. Program the timeout interval in each NoC by setting the FLAGOUT register. <tab:keraunos_noc_timeout_csr> shows the address map of the timeout CSR in Keraunos NoC and an example of programming it. The timeout interval is `2^threshold*4096` cycles based on the clocks it connects.

TNIU	Bit	Description	clock
smn2pcie	0	SMN to PCIe	SMN Clock
smn2pcie_pllctrl	1	SMN to PCIe PLL Control	SMN Clock
smn2smc	2	SMN to SMC	SMN Clock
smn2smn_east	3	SMN to SMN east	SMN Clock
smn2axiprobe	4	SMN to ATB2AXI bridge	SMN Clock
smn2pcie_remapper_mmr	5	SMN to PCIe remapper mmr	SMN Clock

Table 23. Keraunos SMN Bridge Timeout Enable Bit Correspondence

TNIU	Bit	Description	clock
smn2d2d	0	SMN to D2D	SMN Clock
smn2hsio_express_east	1	SMN to HSIO Express East	HSIO Clock
smn2hsio_express_west	2	SMN to HSIO Express West	HSIO Clock
smn2smn_east	3	SMN to SMN East	SMN Clock
smn2smn_hsio	4	SMN to HSIO	HSIO Clock

TNIU	Bit	Description	clock
smn2smn_west	5	SMN to SMN West	SMN Clock
smn2axiprobe	6	SMN to ATB2AXI bridge	HSIO Clock

Table 24. Keruanos SMN Timeout Enable Bit Correspondence

TNIU	Bit	Description	clock
hsio2cce_cfg_0	0	HSIO to CCE0 Config	HSIO Clock
hsio2cce_cfg_1	1	HSIO to CCE1 Config	HSIO Clock
hsio2cce_eth_rx_0_rd	2	HSIO to CCE0 Eth Rx0 Read	HSIO Clock
hsio2cce_eth_rx_0_wr	3	HSIO to CCE0 Eth Rx0 Write	HSIO Clock
hsio2cce_eth_rx_1_rd	4	HSIO to CCE1 Eth Rx1 Read	HSIO Clock
hsio2cce_eth_rx_1_wr	5	HSIO to CCE1 Eth Rx1 Write	HSIO Clock
hsio2cce_noc_0_rd	6	HSIO to CCE0 NOC0 Read	HSIO Clock
hsio2cce_noc_0_wr	7	HSIO to CCE0 NOC0 Write	HSIO Clock
hsio2cce_noc_1_rd	8	HSIO to CCE1 NOC1 Read	HSIO Clock
hsio2cce_noc_1_wr	9	HSIO to CCE1 NOC1 Write	HSIO Clock
hsio2cla_0	10	HSIO to CLA0	HSIO Clock
hsio2cla_1	11	HSIO to CLA1	MAC/PCS clock
hsio2eth_cfg_0	12	HSIO to Eth0 Config	HSIO Clock
hsio2eth_cfg_1	13	HSIO to Eth1 Config	HSIO Clock
hsio2noc2axi_rd	14	HSIO to NOC2AXI Read	HSIO Clock
hsio2noc2axi_wr	15	HSIO to NOC2AXI Write	HSIO Clock
hsio2omega	16	HSIO to Omega	SMN Clock
hsio2pcie_rd	17	HSIO to PCIe Read	HSIO Clock
hsio2pcie_wr	18	HSIO to PCIe Write	HSIO Clock
hsio2prof_0	19	HSIO to Profiler0	HSIO Clock
hsio2sram_cfg	20	HSIO to SRAM Config	HSIO Clock
hsio2smn	21	HSIO to SMN	HSIO Clock
hsio2zeus	22	HSIO to Zeus	SMN Clock
hsio2noc2axi_cfg	23	HSIO to NOC2AXI Config	HSIO Clock
hsio2error_bank	24	HSIO to Error Bank	HSIO Clock

Table 25. Keruanos HSIO Timeout Enable Bit Correspondence

Cmd	bit_position	8	7	6	5	4	3	2	1	0	val	write value
	description	tmo_rst	tmo_stop	tmo_start	tmo_en	threshold[4]	threshold[3]	threshold[2]	threshold[1]	threshold[0]		
out of reset		1	1	1	1	1	1	1	1	1	0x1FF	n/a
clear bits to start		1	1	0	0	1	1	1	1	0	0x19E	0x19E to FlagOutClr0 register (setting threshold to 0b00001)
set bits to reset		1	1	1	1	1	1	1	1	1	0x1FF	0x1FF to FlagOutSet0 register
clear bits to start		1	1	0	0	1	1	0	1	1	0x19B	0x19B to FlagOutClr0 register (setting threshold to 0b00100)

Table 26. Keraunos NoC Timeout CSR Programming

8.1.5. Packet tracing

In Keraunos SMN, all INIUs in SMN bridge and HSIO SMN tile have packet tracing capability, which can be used to dump packet headers trace on to their respective ATB ports. This feature refers to the packet tracing capability of packet probes in Arteris

FlexNoC.

In Kerauno SMN, both SMN bridge and HSIO SMN tile connect to an ATB2AXI bridge which routes the packets on the ATB port back into the SMN bridge and HSIO SMN tile respectively. For SMN bridge, software can configure the ATB2AXI bridge to route the packets to any address targeting PCIE or SMC (e.g. SMC SRAM). For the HSIO SMN tile, software can configure the ATB2AXI bridge to route the packets to any address targeting PCIE, HSIO SRAM, or SMC.

The packet tracing capability is primarily intented to be used for debugging. SMN also allows for packets to be filtered based on address range, length etc. to reduce the amount of trace data generated.

Programming sequence to enable packet tracing

1. Set field TraceEn of register MainCtl to 1 to enable forwarding of traced packets to the connected observer. Optionally set field PayloadEn of register MainCtl to 1 if the packet payload should be included in the trace.
2. Set register TracePortSel to the value corresponding to the probe point of interest.
3. Set field GlobalEn of register CfgCtl to 1.
4. Set register FilterLut to -1 (all bits set).

To enable tracing all packets, additional steps are required: . Program one of the filters to accept all packet header field values. Set the filter registers to the values given in the table <tab:keraunos_exclude_packet_header_fields>. . Program register FilterLut to select the output of filter programmed in the previous step.

Register name	Setting
Filters_N_RouteIdMask	0
Filters_N_WindowSize	:stem:`-1^1`
Filters_N_SecurityMask	0
Filters_N_Opcode	0xF
Filters_N_Status	0x3
Filters_N_Length	0xF
Filters_N_Urgency	0

Table 27. Keraunos NoC Excluding packet header fields from filter criteria

8.1.6. Performance monitoring

Performance counters

In Keraunos SMN, all INIUs in SMN bridge and HSIO SMN tile instantiate performance counters, which can be used to extract performance metrics like number of packets, number of bytes, number of idle/busy cycles, etc. Each INIU is provisioned with 32 32-bit performance counters, two of which could be chained together to get a 64-bit performance counter. This feature refers to the statistics collection capability of packet probes in Arteris FlexNoC.

The performance counters are primarily intended to be used for performance tuning and software is expected to directly read the counters to extract performance metrics.

Alternatively, since the packet probes on all INIUs are connected to the ATB port, packet probes can be configured to send the performance counter values aggregated over a period of time as packets on the ATB port. Further, the ATB2AXI bridge can be configured to route the performance counter packets to any address targeting PCIE or SMC for SMN bridge and PCIE, HSIO SRAM, or SMC for HSIO SMN tile.

1. Set field StatEn to 1 in register MainCtl.
2. Set register Counters_0_PortSel to the value corresponding to the probe point of interest.
3. Set register Counters_0_Src to 0x6 (PKT) to count packets.
4. Specify the period during which the packets should be counted by setting register StatPeriod to: :stem:`log2(interval expressed in number of probe clock cycles)`
5. Set field GlobalEn of register CfgCtl to 1 to enable packet counting. Once time :stem:`2^StatPeriod` has elapsed, the number of packets counted is dumped to the observer.

To enable performance counters dumping to ATB, additional steps are required:

1. Specify the required period between STP synchronization sequences in register AsyncPeriod. Ensure that the value of

AsyncPeriod is large enough to allow STPv2 cells, encapsulating Trace / Statistic Packets, to be inserted in between STPv2 synchronization sequences.

2. Ensure that register STPv2En is set to 1 to enable the STPv2 interface.
3. Set register AtbId to a value that is unique not only within the NoC but also within the debug system.
4. Set register AtbEn to 1 to enable the ATB function.

To enable ATB2AXI bridge, additional steps are required: . Program the dump destination addresss in DEST_MEM_REGION_LSB and DEST_MEM_REGION_MSB registers. . Enable the ATB2AXI bridge by setting register CONFGI.enable to 1.

Transaction Profiling

Keraunos SMN fabric supports transaction profiling to generate histograms of transaction latencies or the number of pending transactions. This feature uses the transaction profiling capability of transaction probes in Arteris FlexNoC. The 32 32-bit performance counters provisioned for each INIU are shared for this purpose.

Programming sequence to enable transaction profiling meansuing latency

The following example of a run-time programming sequence shows how a transaction probe can be used to measure transaction latency. The example assumes appropriate probe configuration at design time. The measurement specification is:

1. For init0, monitor delays for read transactions only.
2. For init1, monitor delays for read and write transactions.

To program the transaction profiler unit

1. Ensure that register En is 0.
2. Set register ObservedSel_0 to 0 to monitor transactions from init0 on observed transaction port 0.
3. Set register ObservedSel_1 to 1 to monitor transactions from init1 on observed transaction port 1.
4. Set register Mode to 2’b00 to set the event mode to DELAY for both transaction ports.
5. Set register nTenureLines_0 to 1 to allocate one "tenure line", that is, four transaction profiling counters, to init0. The remaining five tenure lines are automatically allocated to init1.
6. Set the threshold registers for transaction port 0. These select the actual delay thresholds, from the set of thresholds defined in parameter delayThresholds, which is used to categorize the delay in transactions issued by init0.
   a. Thresholds_0_0 = 0 → delay threshold(0) = 2
   b. Thresholds_0_1 = 1 → delay threshold(1) = 5
   c. Thresholds_0_2 = 2 → delay threshold(2) = 10
   d. Thresholds_0_3 = 3 → delay threshold(3) = 100
7. Set the threshold registers for transaction port 1. These select the actual delay thresholds, from the set of thresholds defined in parameter delayThresholds, which is used to categorize the delay in transactions issued by init1.
   a. Thresholds_1_0 = 0 → delay threshold(0) = 2
   b. Thresholds_1_1 = 1 → delay threshold(1) = 5
   c. Thresholds_1_2 = 2 → delay threshold(2) = 10
   d. Thresholds_1_3 = 3 → delay threshold(3) = 100

The statistics counters in the packet probe unit of the transaction probe must be programmed to count the delay events recorded by the transaction profiler unit.

1. Map the five delay event bins, associated with transactions issued by init0, from the transaction profiler unit to statistics counters in the packet probe unit by programming the following registers:
   a. Counters_0_Src = 0x20 → Counts the number of transactions from init0 with a delay between 0 and 1.
   b. Counters_1_Src = 0x21 → Counts the number of transactions from init0 with a delay between 2 and 4.
   c. Counters_2_Src = 0x22 → Counts the number of transactions from init0 with a delay between 5 and 9.
   d. Counters_3_Src = 0x23 → Counts the number of transactions from init0 with a delay between 10 and 99.
   e. Counters_4_Src = 0x24 → Counts the number of transactions from init0 with a delay greater than 100.
2. Map the five delay event bins, associated with transactions issued by init1, from the transaction profiler unit to statistics counters in the packet probe unit by programming the following registers:

a. Counters_5_Src = 0x25 → Counts the number of transactions from init1 with a delay between 0 and 1.
b. Counters_6_Src = 0x26 → Counts the number of transactions from init1 with a delay between 2 and 4.
c. Counters_7_Src = 0x27 → Counts the number of transactions from init1 with a delay between 5 and 9.
d. Counters_8_Src = 0x28 → Counts the number of transactions from init1 with a delay between 10 and 99.
e. Counters_9_Src = 0x29 → Counts the number of transactions from init1 with a delay greater than 100.

3. Set register StatPeriod to log2(period), that is, the period during which delay events should be counted before being sent to the observer.
4. Set field StatEn of register MainCtl to 1 to enable the statistics counters to count delay events.

To enable the transaction probe . Set field GlobalEn of register CfgCtl to 1 to enable the statistics counters. . Set register En to 1 to enable the transaction profiling counters.

To program the init0 NIU transaction filter . Set register Mode to 1 to set the delay mode to LATENCY. . Set field RdEn of register Opcode to 1 to monitor read transactions only.

To program the init1 NIU transaction filter . Set register Mode to 1 to set the delay mode to LATENCY. . Set fields RdEn and WrEn of register Opcode both to 1 to monitor read and write transactions.

8.2. Keraunos QNP mesh

Keraunos QNP mesh, shown in Figure 20, is QNP interconnect used for to route traffic between 1) D2D and HSIO tiles, and 2) between HSIO tiles. Traffic routing in QNP mesh follows the following rules:

1. Traffic between D2D and HSIO tile: Requests from each D2D can only be routed the HSIO tile connected to it. e.g. D2D0 can only route traffic to HSIO0. Similarly, requests from each HSIO tile can only be routed to the D2D connected to it. e.g. HSIO0 can only route traffic to D2D0.
2. Traffic between HSIO tiles: AXI requests from any HSIO tile can be routed to any other HSIO tile, by setting force dimension routing bit in QNP header.

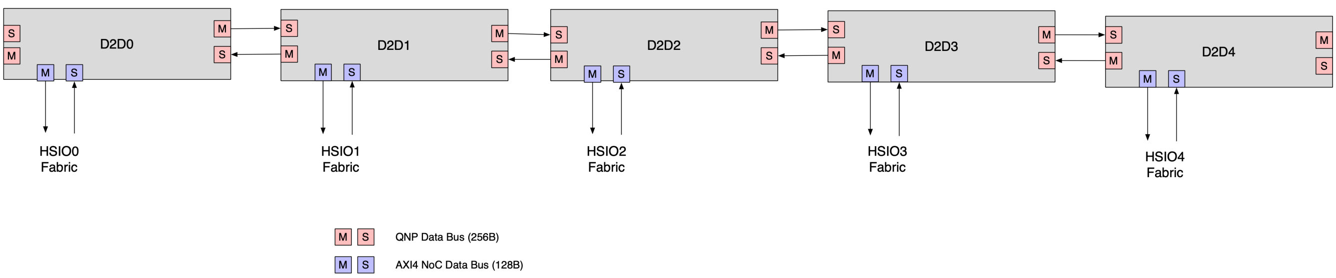


Figure 20. Keraunos QNP mesh

8.3. Keraunos HSIO fabric

The HSIO Fabric is an 128B AXI interconnect fabric built using Arteris FlexNoC. The HSIO fabric is identical across all HSIO tiles in Keraunos, and receives HSIO fabric ID via pinstraps. The HSIO fabric ID is used to route traffic to local components or to neighboring HSIO tiles.

8.3.1. Ports and Connectivity

Table 28 specifies the manager and sub-ordinate ports on the HSIO fabric.

Manager	Sub-ordinate	AxID	AxADDR	AxDATA	Protocol
CCE0 DMA	HSIO Fabric	8	52	128B	AXI4
CCE1 DMA	HSIO Fabric	8	52	128B	AXI4
Eth_Ctrl0 Data	HSIO Fabric	10	52	128B	AXI4
Eth_Ctrl1 Data	HSIO Fabric	10	52	128B	AXI4
D2D NOC2AXI	HSIO Fabric	10	52	128B	AXI4
Profiler Sender	HSIO Fabric	10	52	64B	AXI4
PacketProbeATB	HSIO Fabric	10	52	64B	AXI4

Manager	Sub-ordinate	AxID	AxADDR	AxDATA	Protocol
PCIE	HSIO Fabric	10	52	32B	AXI4
CCE0 MMIO	HSIO Fabric	9	52	64B	AXI4
CCE1 MMIO	HSIO Fabric	9	52	64B	AXI4
Eth_Ctrl0 Cfg	HSIO Fabric	N/A	52	8B	AXI4-Lite
Eth_Ctrl1 Cfg	HSIO Fabric	N/A	52	8B	AXI4-Lite
SMN	HSIO Fabric	10	52	8B	AXI4
HSIO Fabric	CCE0 NOC	10	52	128B	AXI4
HSIO Fabric	CCE1 NOC	10	52	128B	AXI4
HSIO Fabric	CCE0 Eth	10	52	128B	AXI4
HSIO Fabric	CCE1 Eth	10	52	128B	AXI4
HSIO Fabric	D2D NOC2AXI	10	52	128B	AXI4
HSIO Fabric	CCE0 Cfg	N/A	52	8B	AXI4-Lite
HSIO Fabric	CCE1 Cfg	N/A	52	8B	AXI4-Lite
HSIO Fabric	Eth_Ctrl0 Cfg	N/A	32	4B	APB
HSIO Fabric	Eth_Ctrl1 Cfg	N/A	32	4B	APB
HSIO Fabric	MAC/PCS Cfg	N/A	32	4B	APB
HSIO Fabric	SERDES Cfg	N/A	32	4B	APB
HSIO Fabric	DFD0 Cfg	N/A	32	4B	APB
HSIO Fabric	DFD1 Cfg	N/A	32	4B	APB
HSIO Fabric	Profiler Cfg	N/A	32	4B	APB
HSIO Fabric	HSIO SRAM Cfg	N/A	32	4B	APB
HSIO Fabric	Error Bank Cfg	N/A	32	4B	APB
HSIO Fabric	NOC2AXI Remap Cfg	10	32	4B	APB
HSIO Fabric	PCIE	10	52	32B	AXI4
HSIO Fabric	SMN	10	52	8B	AXI4

Table 28. HSIO Fabric Ports

Table 29 lists the target addresses for each manager port in the HSIO fabric.

Manager	Sub-ordinate	Target Address
HSIO Fabric	CCE0 NOC	HSIO SRAM 0-4MB
HSIO Fabric	CCE1 NOC	HSIO SRAM 4-8MB
HSIO Fabric	CCE0 Eth	HSIO SRAM 0-8MB
HSIO Fabric	CCE1 Eth	HSIO SRAM 0-8MB
HSIO Fabric	D2D NOC2AXI	Grendel chiplet cfg + Quasar L1/memory + Remote Athena IOVA
HSIO Fabric	CCE0 Cfg	HSIO[i] CCE0 MMR
HSIO Fabric	CCE1 Cfg	HSIO[i] CCE1 MMR
HSIO Fabric	Eth_Ctrl0 Cfg	HSIO[i] Eth Ctrl0 MMR
HSIO Fabric	Eth_Ctrl1 Cfg	HSIO[i] Eth Ctrl1 MMR
HSIO Fabric	MAC/PCS Cfg	HSIO[i] MAC, PCS, and MAC/PCS ctrl MMR
HSIO Fabric	SERDES Cfg	HSIO[i] SERDES, and SERDES ctrl MMR
HSIO Fabric	DFD0 Cfg	HSIO[i] DFD0 MMR, and Trace Sink
HSIO Fabric	DFD1 Cfg	HSIO[i] DFD1 MMR, and Trace Sink
HSIO Fabric	Profiler Cfg	HSIO[i] Profiler Sender and ATB2AXI bridge MMR
HSIO Fabric	HSIO SRAM Cfg	HSIO[i] SRAM MMR
HSIO Fabric	Error Bank Cfg	HSIO[i] Error Bank MMR

Manager	Sub-ordinate	Target Address
HSIO Fabric	NOC2AXI Remap Cfg	HSIO[i] NOC2AXI Remap MMR
HSIO Fabric	PCIE	PCIe IO memory space + PCIe App MMR if i == 0
HSIO Fabric	SMN	SMC MMRs, and D2D[k] MMRs, k = 0..4
		HSIO[j] CCE0/1, Eth Ctrl0/1, SRAM, MAC/PCS, SERDES, DFD0/1, Profiler, Error bank, SRAM MMRs, where j ≠ i

Table 29. HSIO Fabric Target Addresses (for HSIO[i] tile)

Figure 21 shows the connectivity matrix between the ports in the HSIO fabric, and Table 30 explains the usage for each access path.

	hsio_fabric_regs	t_hsio2cce_noc_0_rd	t_hsio2cce_noc_0_wr	t_hsio2cce_noc_1_rd	t_hsio2cce_noc_1_wr	t_hsio2cce_eth_0_rd	t_hsio2cce_eth_0_wr	t_hsio2cce_eth_1_rd	t_hsio2cce_eth_1_wr	t_hsio2noc2axi_rd	t_hsio2noc2axi_wr	t_hsio2cce_cfg_0	t_hsio2cce_cfg_1	t_hsio2eth_cfg_0	t_hsio2eth_cfg_1	t_hsio2omega	t_hsio2zeus	t_hsio2dfd_0	t_hsio2dfd_1	t_hsio2prof_0	t_hsio2sram_cfg_t	t_hsio2noc2axi_cfg	t_hsio2error_bank	t_hsio2pcie_rd	t_hsio2pcie_wr	t_hsio2smn	Outstanding entries
i_cce2hsio_dma_0_rd										Y																	32
i_cce2hsio_dma_0_wr											Y																32
i_cce2hsio_dma_1_rd										Y																	32
i_cce2hsio_dma_1_wr											Y																32
i_eth2hsio_data_0_rd						Y		Y		Y																	32
i_eth2hsio_data_0_wr							Y		Y		Y																32
i_eth2hsio_data_1_rd								Y		Y																	32
i_eth2hsio_data_1_wr									Y		Y																32
i_noc2axi2hsio_rd	Y	Y		Y								Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			32
i_noc2axi2hsio_wr	Y		Y		Y							Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y		32
i_prof2hsio_0			Y		Y						Y														Y		8
i_axiprobe2hsio_wr			Y		Y						Y														Y	Y	32
i_pcie2hsio_rd		Y		Y						Y		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			Y	32
i_pcie2hsio_wr			Y		Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			Y	32
i_cce2hsio_mmio_0	Y			Y	Y					Y	Y		Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y	Y	8
i_cce2hsio_mmio_1	Y	Y	Y							Y	Y	Y		Y	Y	Y	Y		Y	Y	Y	Y	Y	Y	Y	Y	8
i_eth2hsio_cfg_0							Y				Y	Y	Y													Y	32
i_eth2hsio_cfg_1									Y		Y	Y	Y													Y	32
i_smn2hsio	Y	Y	Y	Y	Y							Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		32
Outstanding entries		32	32	32	32	32	32	32	32	32	32	8	8	1	1	1	1	1	1	1	1	1	1	32	32	32	

Figure 21. HSIO fabric connectivity (Y: connected)

Note, Eth Ctrl0 Data (i_eth2hsio_data_0_rd/wr) connects to both CCE0 Eth (t_hsio2cce_eth_0_rd/wr) and CCE1 Eth (t_hsio2cce_eth_1_rd/wr) ports, while Eth Ctrl1 Data (i_eth2hsio_data_1_rd/wr) connects to CCE1 Eth (t_hsio2cce_eth_1_rd/wr) port. In 2x400G, both TT Ethernet controllers are active and each of them connects to their respective CCE Eth port. In 800G mode, since only TT Ethernet controller 0 is active, we enable connectivity from Eth Ctrl0 Data to both CCE0 and CCE1 Eth ports. This allows software to efficiently utilize DMRISC cores on both CCE0 and CCE1 For example, software can use CCE0 for Tx functionality while CCE1 for Rx functionality.

The traffic from Eth Ctrl0 targeting HSIO SRAM is routed to either CCE0/1 Eth ports based on the address and pre-defined configuration in the HSIO fabric. We support 3 configurations:

1. All traffic to CCE0 Eth: Entire 8MB HSIO SRAM address region mapped to CCE0 Eth
2. All traffic to CCE1 Eth: Entire 8MB HSIO SRAM address region mapped to CCE1 Eth
3. Traffic to both CCE0 and CCE1 Eth: [0-4MB) HSIO SRAM address region mapped to CCE0 Eth and [4-8MB) HSIO SRAM address region mapped to CCE1 Eth

Initiator port	Target port(s)	Usage
Data plane		
CCE0 DMA	D2D NOC2AXI	Data path for CCE0 to read/write data to HSIO[j] SRAM (j ≠ i), or across D2D
CCE1 DMA	D2D NOC2AXI	Data path for CCE1 to read/write data to HSIO[j] SRAM (j ≠ i), or across D2D
Eth Ctrl0 Data (Read)	CCE0/1 Eth	Data path for Tx controller in Eth Ctrl0 to read data from HSIO[i] SRAM
Eth Ctrl0 Data (Write)	CCE0/1 Eth	Data path for Rx controller in Eth Ctrl0 to send data (received from Ethernet Rx) to HSIO[i] SRAM
Eth Ctrl0 Data (Read)	D2D NOC2AXI	Data path for Tx controller in Eth Ctrl0 to read data from HSIO[j] SRAM (j ≠ i)
Eth Ctrl0 Data (Write)	D2D NOC2AXI	Data path for Rx controller in Eth Ctrl0 to send data (received from Ethernet Rx) to HSIO[j] SRAM (j ≠ i)
Eth Ctrl1 Data (Read)	CCE0 Eth	Data path for Tx controller in Eth Ctrl1 to read data from HSIO[i] SRAM

Initiator port	Target port(s)	Usage
Eth Ctrl1 Data (Write)	CCE0 Eth	Data path for Rx controller in Eth Ctrl1 to send data (received from Ethernet Rx) to HSIO[i] SRAM
Eth Ctrl1 Data (Read)	D2D NOC2AXI	Data path for Tx controller in Eth Ctrl1 to read data from HSIO[j] SRAM (j ≠ i)
Eth Ctrl1 Data (Write)	D2D NOC2AXI	Data path for Rx controller in Eth Ctrl1 to send data (received from Ethernet Rx) to HSIO[j] SRAM (j ≠ i)
D2D NOC2AXI	CCE0/1 NOC	Data path for D2D NOC2AXI to access HSIO [i] SRAM via CCE0/1
Profiler	D2D NOC2AXI	Data path for Profiler Sender to write to Quasar memory (DRAM)
PacketProbeATB	D2D NOC2AXI	Data path for PacketProbeATB to write to Quasar memory (DRAM)
PCIE	CCE0/1 NOC	Data path for PCIE tile to read/write HSIO[i] SRAM (i==0) via CCE0/1
PCIE	D2D NOC2AXI	Data path for PCIE tile HSIO[j] SRAM (j ≠ i), or across D2D
Control plane		
D2D NOC2AXI	HSIO fabric MMR	Ctrl path for external initiators to access HSIO fabric MMRs
D2D NOC2AXI	CCE0/1 Cfg	Ctrl path for external initiators to access CCE0/1 MMRs
D2D NOC2AXI	Eth Ctrl 0/1 Cfg	Ctrl path for external initiators to access Eth Ctrl 0/1 MMRs
D2D NOC2AXI	MAC/PCS Cfg	Ctrl path for external initiators to access MAC/PCS MMRs
D2D NOC2AXI	SERDES Cfg	Ctrl path for external initiators to access SERDES MMRs
D2D NOC2AXI	CLA0/1	Ctrl path for external initiators to access CLA0/1 MMRs
D2D NOC2AXI	Profiler Cfg	Ctrl path for external initiators to access Profiler Sender and ATB2AXI bridge MMRs
D2D NOC2AXI	SRAM Cfg	Ctrl path for external initiators to access SRAM configuration MMRs
D2D NOC2AXI	NOC2AXI Remap Cfg	Ctrl path for external initiators to access NOC2AXI Remap MMRs
D2D NOC2AXI	Error Bank	Ctrl path for external initiators to access Error Bank MMRs
D2D NOC2AXI	HSIO PCIE	Ctrl path for external initiators to access PCIe App MMRs (i == 0)
PCIE tile	CCE0/1 Cfg	Ctrl path for PCIE tile to access CCE0/1 MMRs
PCIE tile	Eth Ctrl0/1 Cfg	Ctrl path for PCIE tile to access Eth Ctrl 0/1 MMRs
PCIE tile	MAC/PCS Cfg	Ctrl path for PCIE tile to access MAC, PCS, and MAC/PCS ctrl MMRs
PCIE tile	SERDES Cfg	Ctrl path for PCIE tile to access SERDES + SERDES ctrl MMRs
PCIE tile	DFD0/1 Cfg	Ctrl path for PCIE tile to access DFD0/1 MMRs
PCIE tile	Profiler Cfg	Ctrl path for PCIE tile to access Profiler Sender and ATB2AXI bridge MMRs
PCIE tile	HSIO SRAM Cfg	Ctrl path for PCIE tile to access HSIO SRAM MMRs
PCIE tile	Error bank Cfg	Ctrl path for PCIE tile to access Error bank MMRs
PCIE tile	NOC2AXI Remap Cfg	Ctrl path for PCIE tile to access NOC2AXI Remap MMRs
PCIE tile	SMN	Ctrl path for PCIE tile to access all MMRs in HSIO[j] (j ≠ i)
CCE0/1 MMIO	HSIO fabric MMR	Ctrl path for CCE0/1 to access HSIO fabric MMRs
CCE0/1 MMIO	D2D NOC2AXI	Ctrl path for CCE0/1 to access MMRs/memory on other chiplets
CCE0 MMIO	CCE1 Cfg	Ctrl path for CCE0 to access CCE1 MMRs
CCE1 MMIO	CCE0 Cfg	Ctrl path for CCE1 to access CCE0 MMRs
CCE0/1 MMIO	Eth Ctrl0/1 Cfg	Ctrl path for CCE0/1 access to Eth Ctrl 0 and Eth Ctrl 1 MMRs
CCE0/1 MMIO	MAC/PCS Cfg	Ctrl path for CCE0/1 to access MAC + PCS + MAC/PCS ctrl MMRs
CCE0/1 MMIO	SERDES Cfg	Ctrl path for CCE0/1 to access SERDES + SERDES ctrl MMRs
CCE0 MMIO	DFD0 Cfg	Ctrl path for CCE0 to access DFD0 MMRs + Trace sink
CCE1 MMIO	DFD1 Cfg	Ctrl path for CCE1 to access DFD1 MMRs + Trace sink
CCE0/1 MMIO	Profiler Cfg	Ctrl path for CCE0/1 to access Profiler Sender and ATB2AXI bridge MMRs
CCE0/1 MMIO	HSIO SRAM Cfg	Ctrl path for CCE0/1 to access HSIO SRAM MMRs
CCE0/1 MMIO	NOC2AXI Remap Cfg	Ctrl path for CCE0/1 to access NOC2AXI Remap MMRs

Initiator port	Target port(s)	Usage
CCE0/1 MMIO	Error Bank Cfg	Ctrl path for CCE0/1 to access Error Bank MMRs
CCE0/1 MMIO	SMN	Ctrl path for CCE0/1 to access all MMRs/SRAM in HSIO[j] (j ≠ i)
CCE0/1 MMIO	PCIE	Ctrl path for CCE0/1 to access PCIe App MMRs (i == 0)
Eth Ctrl 0/1 Cfg (Write)	CCE0/1 Eth	Ctrl path for Eth Ctrl 0/1 to write to SRAM via CCE0/CCE1
Eth Ctrl 0/1 Cfg (Write)	D2D NOC2AXI	Ctrl path for Eth Ctrl 0/1 to write MMRs/memory on other chiplets
Eth Ctrl 0/1 Cfg (Write)	CCE0/1 Cfg	Ctrl path for Eth Ctrl 0/1 to write to CCE0/1 MMRs
Eth Ctrl 0/1 Cfg (Write)	SMN	Ctrl path for Eth Ctrl 0/1 to all MMRs/SRAM in HSIO[j] (j ≠ i)
SMN	HSIO fabric MMR	Ctrl path for SMC to access HSIO fabric MMRs
SMN	CCE0/1 NOC	Ctrl path for SMC to access SRAM via CCE0/1
SMN	CCE0/1 Cfg	Ctrl path for SMC to access CCE0/1 MMRs
SMN	Eth Ctrl 0/1 Cfg	Ctrl path for SMC to access Eth Ctrl 0/1 MMRs
SMN	MAC/PCS Cfg	Ctrl path for SMC to access MAC/PCS MMRs
SMN	SERDES Cfg	Ctrl path for SMC to access SERDES MMRs
SMN	DFD0/1 Cfg	Ctrl path for SMC to access CLA0/1 MMRs
SMN	Profiler Cfg	Ctrl path for SMC to access Profiler Sender and ATB2AXI bridge MMRs
SMN	HSIO SRAM Cfg	Ctrl path for SMC to access HSIO SRAM MMRs
SMN	NOC2AXI Remap Cfg	Ctrl path for SMC to access NOC2AXI Remap MMRs
SMN	Error Bank Cfg	Ctrl path for SMC to access Error Bank MMRs
SMN	PCIE	Ctrl path for SMC to access MMRs PCIe App MMRs, i == 0

Table 30. HSIO Fabric Access Paths (for HSIO[i])

8.3.2. Traffic Filtering/Firewalls

Keraunos HSIO fabric supports traffic filtering by Source ID (used for SEP/ISS security) and Group ID (used for HSIO virtualization) using the NOC Traffic Filter described in Grendel Security Specification (Section 4.1.4).

Keraunos HSIO fabric includes the following NOC Traffic Filters (denoted by Filters in Figure 6):

1. On NOC2AXI to HSIO fabric path: Traffic entering the HSIO tiles from other HSIO tiles, and D2D tiles.
2. On PCIE to HSIO fabric path: Traffic entering the HSIO fabric from PCIE tile.

Each filter has 16 entries and each entry has 3 64-bit configuration registers: *filter_configX*, *filter_start_addrX*, and *filter_end_addrX* (X ranges from 0 to 15). *filter_configX* register is defined in Grendel Security Specification (Section 4.1.4). NOC Traffic Filters in HSIO fabric only use the source ID and group ID fields in *filter_configX* for filtering. Source ID allocation for Keraunos is defined in Table 57, while group IDs are allocated at runtime depending on how HSIO tiles are allocated to different VMs.

8.3.3. Group ID tagging

To support HSIO virtualization, all requests exiting HSIO tile towards NOC2AXI or SMN need to carry the group ID allocated to the HSIO tile. The group ID for the HSIO tile is programmed by SMC in the Group ID register in the HSIO clock reset unit. HSIO fabric simply attaches this group ID in the AxUSER field (AxUSER[7:4]) of all outgoing requests to NOC2AXI and SMN.

8.3.4. AxUSER field definition

AxUSER field definition for HSIO fabric is same as that on Keraunos SMN (shown in Table 22).

8.3.5. Timeout

Keraunos HSIO fabric supports timeout capability on all TNIUs. The timeout duration is configurable with timeout ranging from 1us to 10s. When a target endpoint does not respond within the timeout duration, the following actions are taken:

1. Timeout signal is asserted at the TNIU. Sideband manager in the HSIO fabric aggregates timeouts from all TNIUs using logical OR operation, and asserts the noc timeout interrupt to SMC.
2. All outstanding transactions to the target endpoint are aborted, and error responses are sent back to the initiators.
3. TNIU does not accept any new transactions to the target endpoint until the timeout condition is cleared.

8.3.6. Packet tracing

In Keraunos HSIO fabric, all INIUs have packet tracing capability, which can be used to dump packet headers trace on to its ATB ports. This feature refers to the packet tracing capability of packet probes in Arteris FlexNoC.

HSIO fabric has an ATB2AXI bridge which routes the packets on the ATB port back into itself. Software can configure the ATB2AXI bridge to route the packets to any address targeting PCIE, HSIO SRAM, or Grendel DRAM via D2D.

The packet tracing capability is primarily intented to be used for debugging. HSIO fabric also allows for packets to be filtered based on address range, length etc. to reduce the amount of trace data generated.

8.3.7. Performance monitoring

Performance counters

In Keraunos HSIO fabric, all INIUs in HSIO fabric instantiate performance counters, which can be used to extract performance metrics like number of packets, number of bytes, number of idle/busy cycles, etc. Each INIU is provisioned with 32 32-bit performance counters, two of which could be chained together to get a 64-bit performance counter. This feature refers to the statistics collection capability of packet probes in Arteris FlexNoC.

The performance counters are primarily intended to be used for performance tuning and software is expected to directly read the counters to extract performance metrics.

Alternatively, since the packet probes on all INIUs are connected to the ATB port, packet probes can be configured to send the performance counter values aggregated over a period of time as packets on the ATB port. Further, the ATB2AXI bridge can be configured to route the performance counter packets to any address targeting PCIE or SMC for SMN bridge and PCIE, HSIO SRAM, or SMC for HSIO SMN tile.

Transaction Profiling

The Keraunos HSIO fabric supports transaction profiling to generate histograms of transaction latencies or the number of pending transactions. This feature uses the transaction profiling capability of transaction probes in Arteris FlexNoC. The 32 32-bit performance counters provisioned for each INIU are shared for this purpose.

8.4. Traffic routing

Figure 22 shows the connectivity for routing control traffic, which includes register reads/writes, within the Keraunos chiplet. Table 31 shows the various paths possible within the Keraunos chiplet as well as going out of the chiplet via D2D. Note, Table 31 shows only the key paths that have connectivity using SMN. Some additional paths are also possible from one HSIO fabric to another HSIO fabric via QNP mesh but are not efficient.

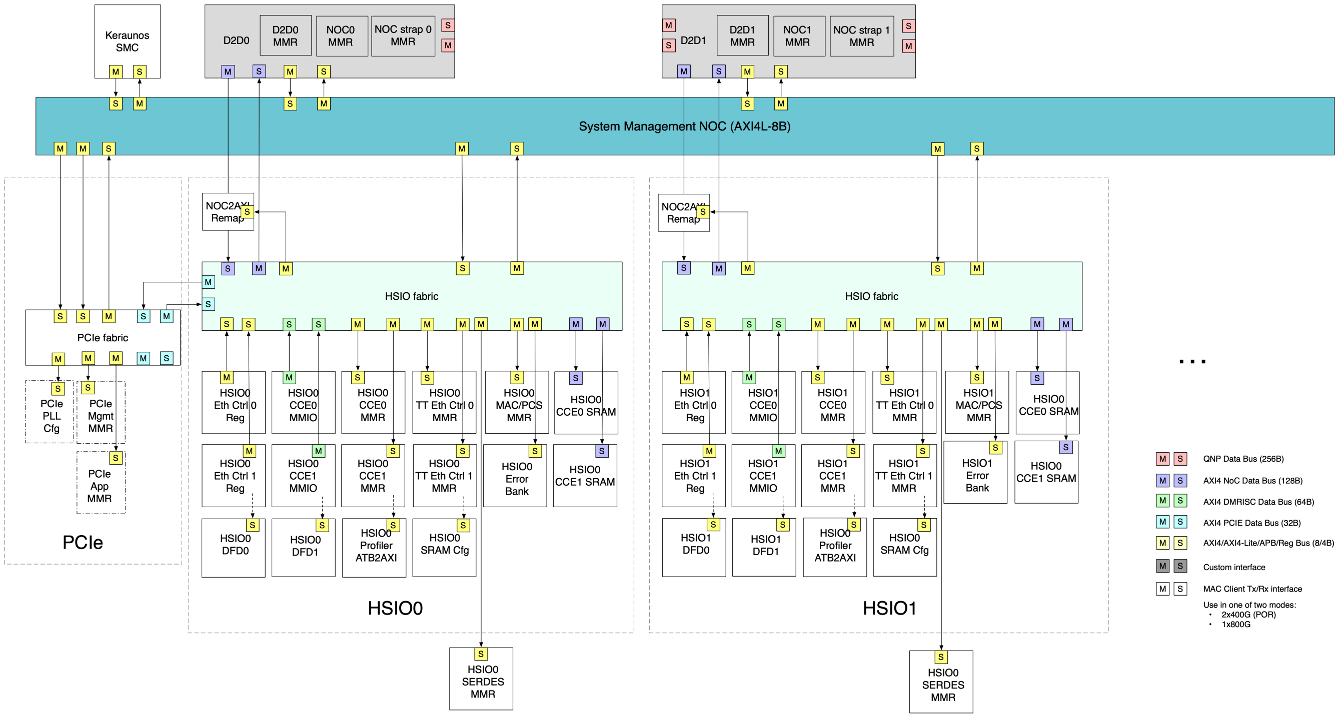


Figure 22. Keraunos control traffic connectivity

Source	Destination	Path
Keraunos SMC	D2D[i] Cfg/Mgmt	Keraunos SMN
Keraunos SMC	HSIO[i] CCE0/1 MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] Eth Ctrl 0/1 MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] MAC/PCS MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] SERDES MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] CLA0/1 MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] Profiler Sender/ATB2AXI MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] SRAM MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] NOC2AXI Remap MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] Error Bank MMR	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	HSIO[i] SRAM	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	PCIe App	Keraunos SMN → HSIO[i] fabric
Keraunos SMC	PCIe Mgmt	Keraunos SMN
Keraunos SMC	PCIe PLL	Keraunos SMN
HSIO[i] CCE0/1 MMIO	HSIO[i] CCE1/0 MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] Eth Ctrl 0/1 MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] MAC/PCS MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] SERDES MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] CLA0/1 MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] Profiler Sender/ATB2AXI MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] SRAM MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] NOC2AXI Remap MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] Error Bank MMR	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	PCIe App, i==0	HSIO[i] fabric
HSIO[i] CCE0/1 MMIO	PCIe App, i ≠ 0	HSIO[i] fabric → SMN → HSIO[0] fabric
HSIO[i] CCE0/1 MMIO	HSIO[j] CCE0/1 (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[j] Eth Ctrl 0/1 (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric

Source	Destination	Path
HSIO[i] CCE0/1 MMIO	HSIO[j] MAC/PCS (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[j] SERDES (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] CLA0/1 (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] Profiler Sender/ATB2AXI (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] SRAM (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] NOC2AXI Remap (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] Error Bank (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	HSIO[i] SRAM	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	D2D[i] MMR	HSIO[i] fabric → SMN
HSIO[i] CCE0/1 MMIO	D2D[i] Data	HSIO[i] fabric → D2D NOC2AXI
HSIO[i] Eth Ctrl 0/1	HSIO[i] CCE0/1	HSIO[i] fabric
HSIO[i] Eth Ctrl 0/1	HSIO[i] SRAM	HSIO[i] fabric
HSIO[i] Eth Ctrl 0/1	HSIO[j] CCE0/1 (j ≠ i) MMR	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] Eth Ctrl 0/1	HSIO[j] SRAM (j ≠ i)	HSIO[i] fabric → SMN → HSIO[j] fabric
HSIO[i] CCE0/1 MMIO	D2D[i] Data	HSIO[i] fabric → D2D NOC2AXI
D2D[i] Cfg/Mgmt	Keraunos SMC	Keraunos SMN
D2D[i] Cfg/Mgmt	HSIO[j] CCE1/0 (j = i or j ≠ i)	Keraunos SMN → HSIO fabric
D2D[i] Cfg/Mgmt	HSIO[j] Eth Ctrl 0/1 (j = i or j ≠ i)	Keraunos SMN → HSIO fabric
D2D[i] Cfg/Mgmt	HSIO[j] MAC/PCS (j = i or j ≠ i)	Keraunos SMN → HSIO fabric
D2D[i] Cfg/Mgmt	HSIO[j] SERDES MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	HSIO[j] CLA0/1 MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	HSIO[j] Profiler Sender/ATB2AXI MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	HSIO[j] SRAM MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	HSIO[j] NOC2AXI Remap MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	HSIO[j] Error Bank MMR (j = i or j ≠ i)	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	PCIe App	Keraunos SMN → HSIO[i] fabric
D2D[i] Cfg/Mgmt	PCIe Mgmt	Keraunos SMN

Table 31. Keraunos control paths

Figure 23 shows the fabric connectivity for routing main data traffic in the Keraunos chiplet. Table 32 shows the various paths possible in the Keraunos chiplet.

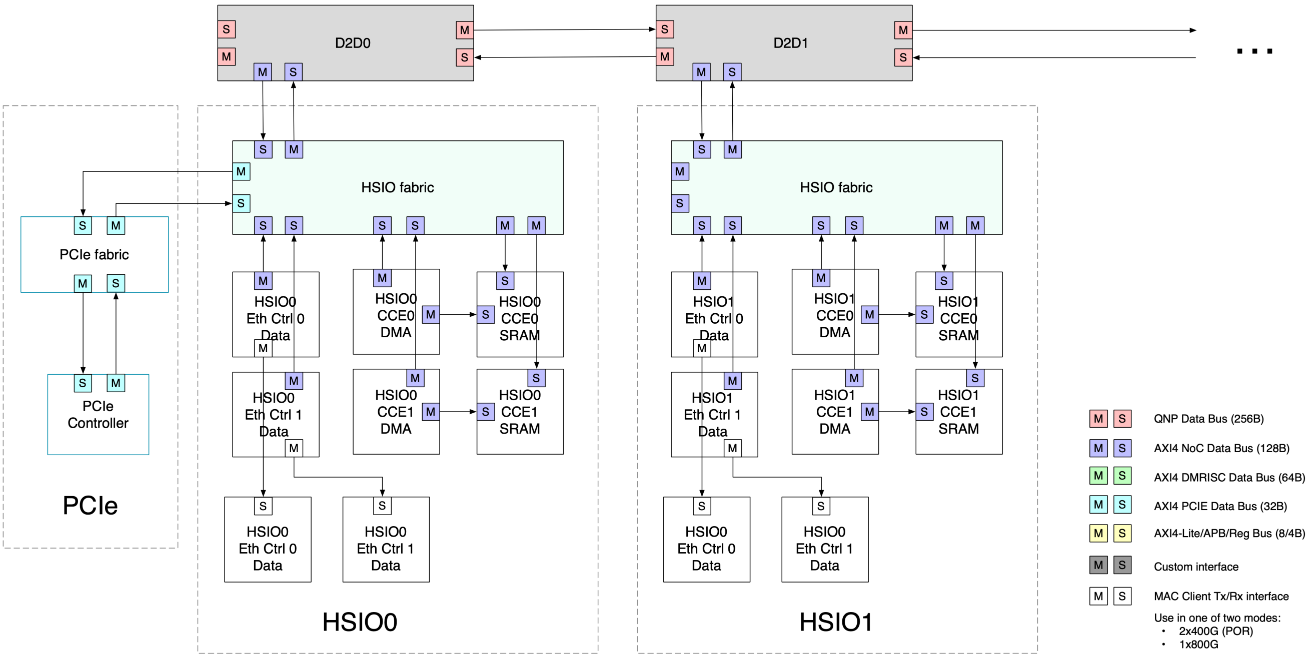


Figure 23. Keraunos data traffic connectivity

Source	Destination	Path
HSIO[i] CCE0/1	HSIO[i] SRAM	Direct link
HSIO[i] CCE0/1	HSIO[j] SRAM (j ≠ i)	HSIO fabric → D2D NOC2AXI → D2D NOC2AXI → HSIO fabric
HSIO[i] Eth Ctrl 0/1	HSIO[i] SRAM	HSIO fabric
HSIO[i] Eth Ctrl 0/1	HSIO[j] SRAM (j ≠ i)	HSIO fabric → D2D NOC2AXI → D2D NOC2AXI → HSIO fabric
HSIO[i] CCE0/1	D2D[i]	HSIO fabric → D2D NOC2AXI
D2D[i]	HSIO[i] CCE0/1	D2D NOC2AXI → HSIO fabric

Table 32. Keraunos data paths

Chapter 9. TT Ethernet Controller

9.1. Introduction

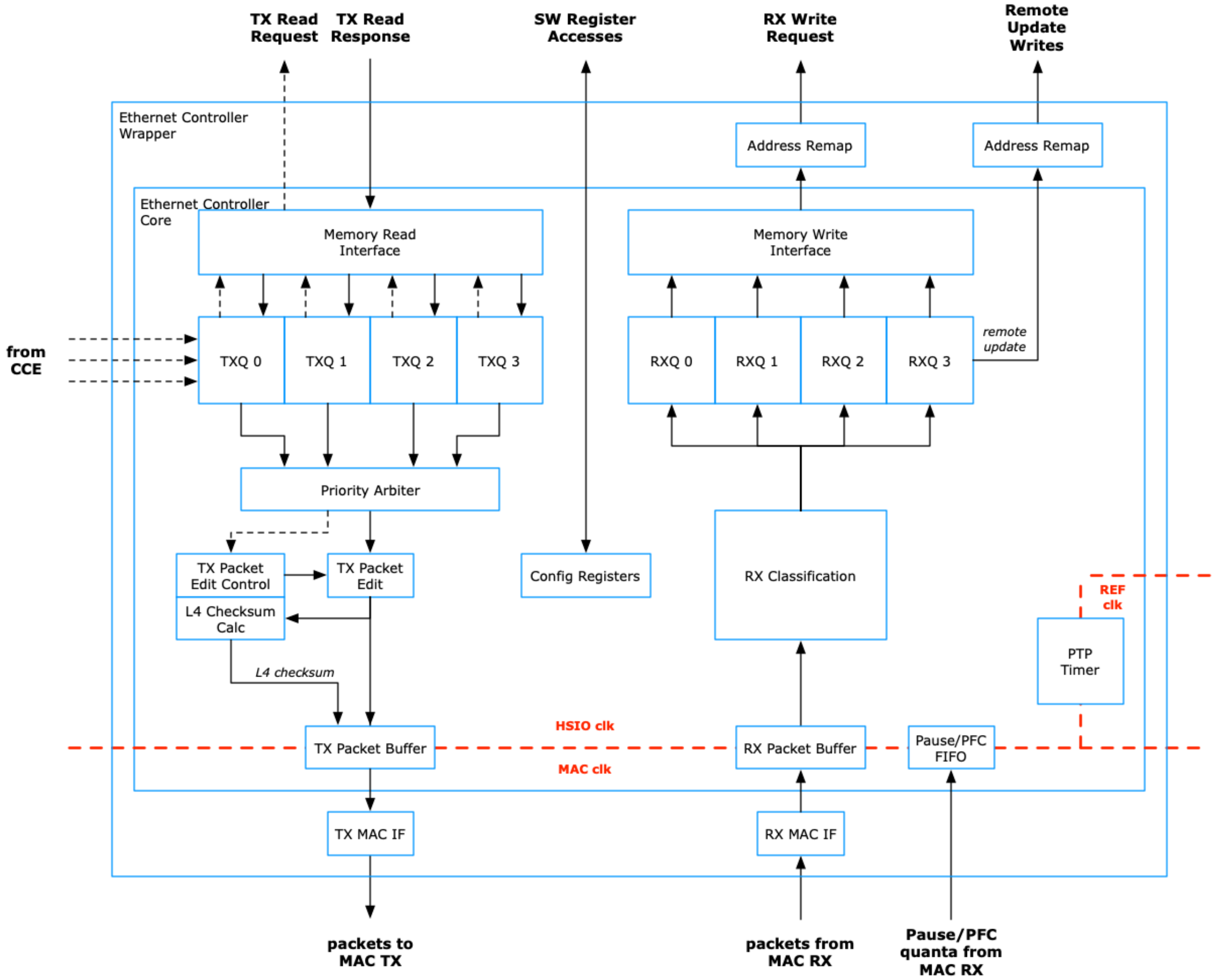


Figure 24. Ethernet Controller

The Ethernet Controller manages the transfer of data between memory and the TX and RX interfaces of the Ethernet MAC. Figure 24 shows an overview of the structure and data paths in the Ethernet Controller.

The Ethernet Controller comprises the following for each of the TX and RX directions: four queues that manage the data formatting and transfer of packets, an arbiter to manage access to the single interface to/from memory, and an arbiter to manage access to the MAC interface.

The presence of four separate queues allows for different types of traffic to be in progress at the same time. For example, it might be useful to have a separate software-controlled stream of traffic operating at the same time as the hardware-controlled traffic using the CCE.

The four queues are identical in terms of their usability. However, traffic should only be transferred between queues with the same index: TX0 → RX0, TX1 → RX1, TX2 → RX2, and TX3 → RX3.

9.1.1. Changes From Previous Version

This document describes a version of the Ethernet Controller that is being modified from its implementation in Black Hole for use in Keraunos. This section lists the main set of changes in this version, not including any changes that are not visible externally.

800G Support

The Black Hole Ethernet Controller supports a maximum of 400G Ethernet. The Keraunos Ethernet Controller supports a maximum of 800G Ethernet.

TT-Link Header Support

The Black Hole Ethernet Controller uses a particuar format for the 16-byte TT-Link header. The Keraunos Ethernet Controller changes the format of the TT-Link header in a few ways, including:

- overlay tile support is removed
- sequence number fields are now 16 bits wide instead of 8 bits
- destination word address is now 48 bits wide instead of 32 bits
- support for command snoop bit, which is used by CCE as hint to update its caches

The Keraunos Ethernet Controller has a backwards-compatible mode that allows it to generate and receive packets with the Black Hole TT-Link header format, but this mode is not enabled by default.

L1 / Memory Interface

The Black Hole Ethernet Controller interfaces directly to an L1 memory via a dedicated memory interface. The interface comprised a number of independent ports, where each port was 16 bytes wide.

The Keraunos interface to all external SOC clients (including memory) is via an AXI interface.

Number of Queues

The Black Hole Ethernet Controller has three queues in both TX and RX directions, where only queues 0 and 1 could interact with Overlay. The Keraunos Ethernet Controller now supports four queues in each direction, where all queues can interact with the CCE.

Packet Preemption

The Black Hole Ethernet Controller supports IET packet preemption on both TX and RX. The Keraunos Ethernet Controller does not support packet preemption.

Ethernet IP Provider

In Black Hole there are two IPs that implement the corresponding set of Ethernet functionality: an IP from Rianta implements the MAC and PCS functionality, and a separate Serdes IP from AlphaWave implements the PMA and PMD functionality. The Keraunos Ethernet Controller interfaces with the AlphaWave (OmegaCORE) Ethernet IP for MAC/PCS functionality and an updated version of the AlphaWave serdes.

RX Classification Engine

The Black Hole version of the RX Classification Engine does not have support for checking the L3/L4 checksums of received packets. The Keraunos RX Classification Engine now supports checking the L3 and L4 checksums of received packets, and can drop packets with bad checksums.

RoCE and RoCEv2 Support

The Black Hole Ethernet Controller contains support for optionally inserting (TX) and removing (RX) the RoCE/RoCEv2 invariant CRC (ICRC) field, in order to create packets compliant with the RoCE standard.

The Keraunos Ethernet Controller does not support RoCE.

9.2. Functional Operation

9.2.1. Interfaces

The following table shows a simplified list of the interfaces of the Ethernet Controller:

Interface	Main Data Flow Direction	Notes
register slave	input from HSIO fabric	32-bit AXI Lite read & write register transactions to Ethernet Controller
register master	output to HSIO fabric	64-bit AXI Lite remote update register transactions from RX
register values	input/output from/to MAC/PCS	various status and control register values
CCE data forward	input from / output to CCE	interface for CCE to request bulk TX data transfers
CCE remote update	input from / output to CCE	interface for CCE to request individual 64-bit remote register writes
MAC TX data path	output to MAC TX	1024-bit data and MAC frame control
MAC RX data path	input from MAC RX	1024-bit data and MAC frame control
memory TX data path	output to / input from memory	1024-bit AXI read interface to memory, to fetch TX data
memory RX data path	output to memory	1024-bit AXI write interface to memory, to write RX data

Table 33. Interfaces of Ethernet Controller

CCE TX Data Forward Interface

The CCE makes requests of the Ethernet Controller to transmit data across the Ethernet link via the CCE Data Forward Interface. Each TXQ has a separate set of interface signals, as shown in the following table:

Signal	Direction	Width	Notes
eth_data_fwd_req_valid	CCE to Eth	1	asserted to request transfer
eth_data_fwd_req_data	CCE to Eth	114	requested transfer details
eth_data_fwd_ack	Eth to CCE	1	acknowledgement that TX request has been accepted
eth_data_fwd_sent	Eth to CCE	1	acknowledgement that entire transfer has been successfully received by remote side

Table 34. CCE Data Forward Interface: Single Data Stream

The assertion of a request signal indicates that a CCE stream wants to transmit data from memory across the Ethernet link, and the request is accompanied by a 114-bit data signal that contains details about the requested transaction. The following table shows the content of the data portion of this interface for each interface.

Signal	Width	Notes
num_words	17	number of 16-byte words to be transmitted
dest_addr	48	(16-byte) word address in destination device memory for payload
src_addr	48	(16-byte) word address in source memory for payload
cmd_snoop	1	flag indicating the remote CCE to update its caches for this write

Table 35. CCE Data Forward Interface: Request Data

There are two acknowledgements that are sent from the Ethernet Controller back to the CCE for data forward requests. When the requested transaction is accepted by the Ethernet Controller, a first acknowledgement is sent back to the CCE to indicate that the transfer request has been accepted. A second ("sent") acknowledgement is sent to the CCE when the entire transaction has been completed (i.e., the data has been transmitted) and acknowledged by the remote node (i.e., the final packet has been acknowledged via the RX sequence numbers).

CCE Remote Update Interface

The CCE makes requests of the Ethernet Controller to perform register writes in the remote device via the CCE Data Forward Interface. Each TXQ has a separate set of interface signals, as shown in the following table:

Signal	Direction	Width	Notes
eth_rem_upd_req_valid	CCE to Eth	1	asserted to request transfer
eth_rem_upd_req_data	CCE to Eth	116	requested transfer details

Signal	Direction	Width	Notes
eth_rem_upd_ack	Eth to CCE	1	acknowledgement that TX request has been accepted
eth_rem_upd_sent	Eth to CCE	1	acknowledgement that transfer has been successfully received by remote side

Table 36. CCE Remote Update Interface: Single Data Stream

The assertion of a request signal indicates that a CCE stream wants to perform a 64-bit register write in the remote device's address space. The request is accompanied by a 116-bit data signal that contains details about the requested transaction. The following table shows the content of the data portion of this interface for each interface.

Signal	Width	Notes
cmd_snoop	1	flag indicating the remote CCE to update its caches for this write
addr	52	52-bit byte address of memory/register location in remote device
data	64	64-bit data value to be written

Table 37. CCE Remote Update Interface: Request Data

There are two acknowledgements that are sent from the Ethernet Controller back to the CCE for remote update requests. When the requested transaction is accepted by the Ethernet Controller, a first acknowledgement is sent back to the CCE to indicate that the transfer request has been accepted. A second ("sent") acknowledgement is sent to the CCE when the transaction has been completed (i.e., the data has been transmitted) and acknowledged by the remote node (i.e., the packet has been acknowledged via the RX sequence numbers).

9.2.2. Traffic Types

There are two broad categories of traffic passing through the Ethernet Controller:

- Raw data traffic initiated by software writes to registers:
  - TX payload read directly from memory, no address/alignment restrictions
  - RX payload written directly to memory at destination
  - Auto-incrementing destination pointer, buffer can also be set to wrap
  - Flow control and retransmission must be handled by software
- Packets with sequence numbers and automatic retransmission control:
  - Can be initiated either by software writes or CCE streams
  - Tenstorrent-proprietary protocol on top of L2 Ethernet with 16-byte TT-Link header
  - 16B restriction for address alignment and data length
  - separate packet type used for 64-bit immediate register writes
  - 1024-packet retransmission window

Table 38 summarizes the types of traffic sent by the Ethernet Controller.

Initiator	Type	TX Initiation
Software	Raw	Register write (eth_txq.CMD.0)
	Normal Packet	Register write (eth_txq.CMD.1)
	Remote Register Write	Register write (eth_txq.CMD.2)
CCE	Data Forward Packet Request	Request from CCE
	Remote Update Request	Request from CCE
TX Queue Controller	Sequence Number Update	Sequence number timeout detected by TX queue controller

Table 38. Traffic Types

9.2.3. Raw Packets

Figure 25 shows the format of raw packets that are sent to the MAC. In the transmit direction, the first three fields of the packet (the 14-byte Ethernet header comprising MAC destination address, MAC source address, and type/length) are taken from TX configuration registers; the rest of the packet is read out from memory. The first word of data from the memory is realigned and placed immediately following the Ethertype field. During reception, the payload of a raw packet is written into memory; the Ethernet header is not written to memory.

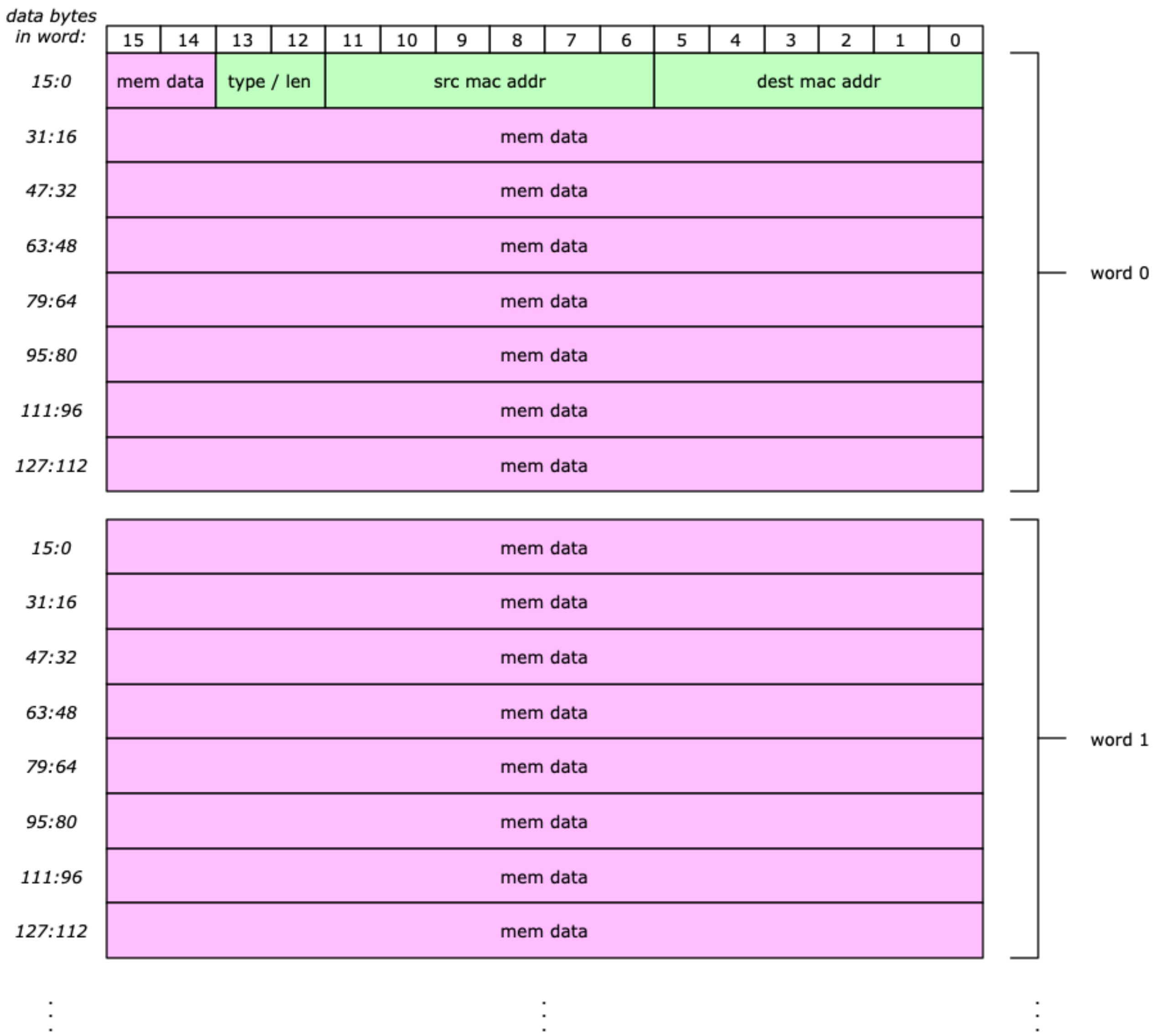


Figure 25. Raw Packet Format

The payload data for raw packets can be located at any byte address / alignment in memory, with arbitrary packet sizes up to certain limits.

9.2.4. Non-Raw Packets

Figure 26 shows the format of non-raw packets that are sent to the MAC. The main difference compared to raw packets is the presence of a sixteen byte "TT-Link header". This header is inserted into the stream by the hardware before the start of the normal payload data during transmission and is removed from the data stream by the hardware during reception.

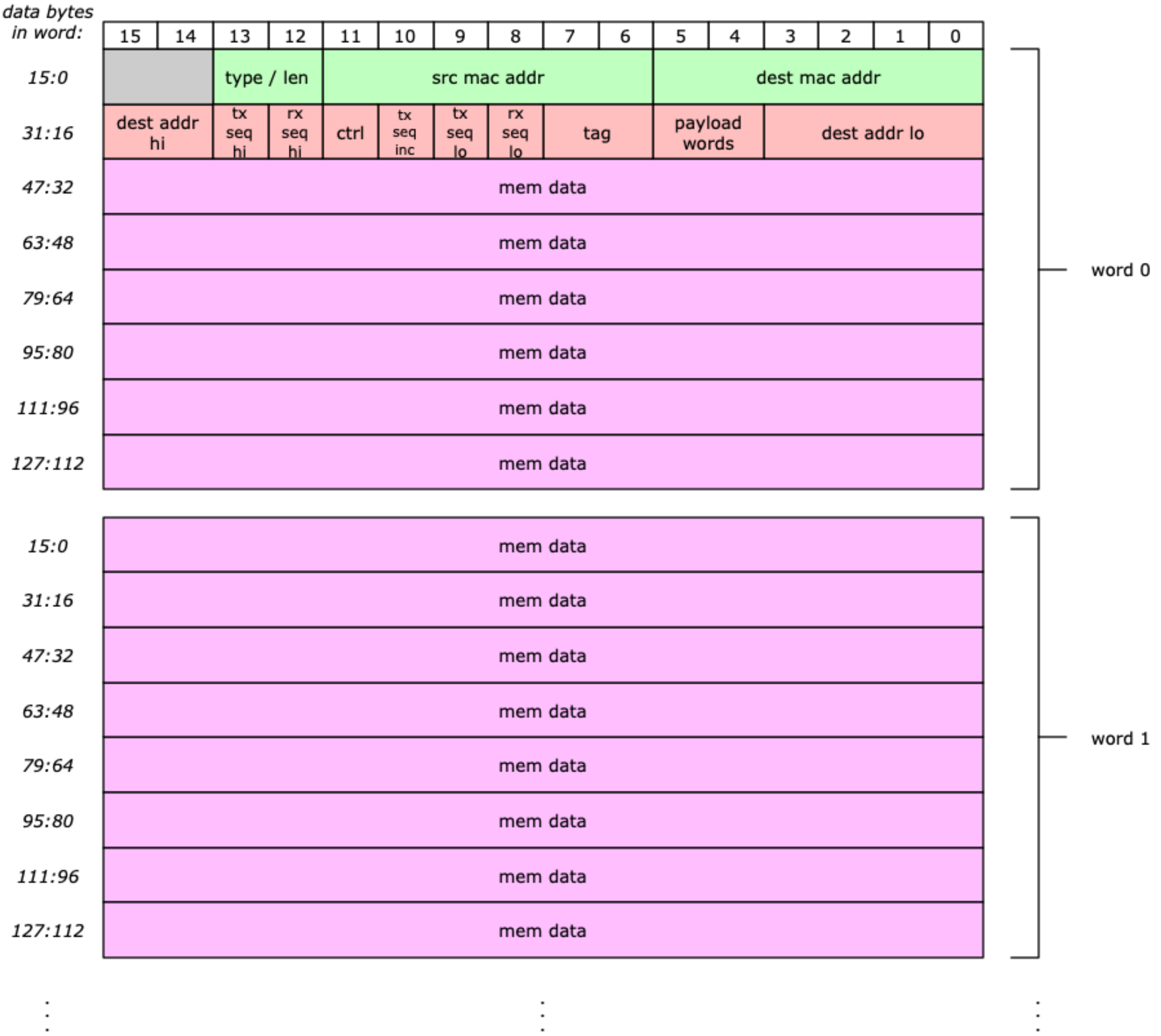


Figure 26. Normal Packet Format

The payload data for non-raw packets must be located starting at a 16-byte boundary in memory, and the size must also be a multiple of 16 bytes.

A variation of non-raw packets are the remote remote update packets, whose format is shown in Figure 27. Note that memory is not accessed in the TX device for register write packets.

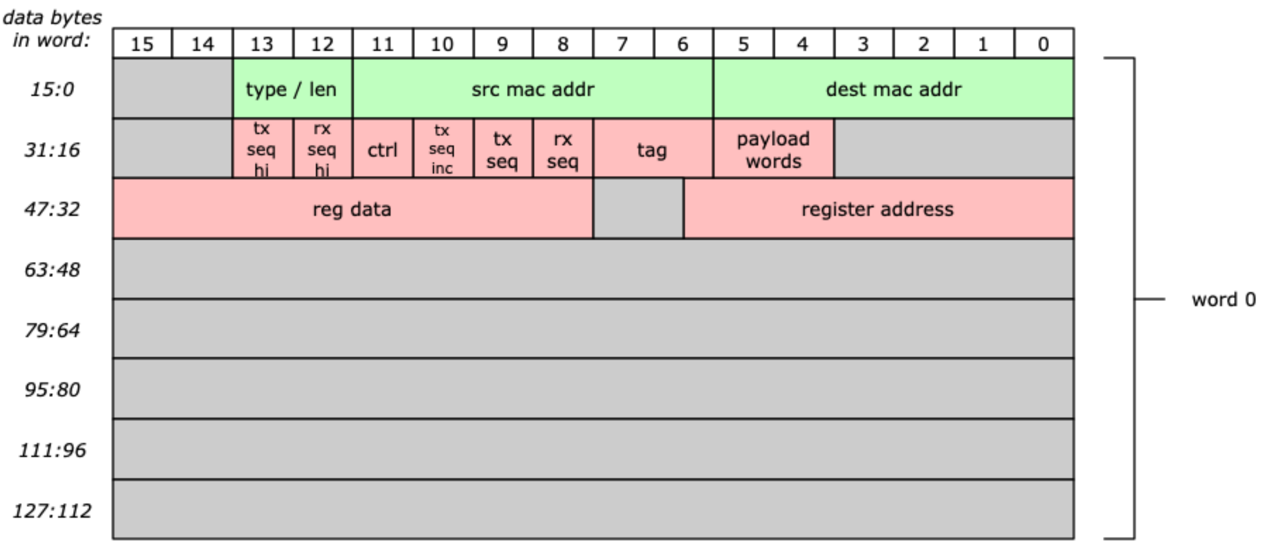


Figure 27. Remote Update Packets

Table 39 shows the fields of the TT-Link header that is added to non-raw packets.

Bytes	Name	Description
15:14, 3:0	header destination addr	48-bit destination word (16B) memory address
5:4	payload words	number of 16-byte payload data words in this packet (following TT-Link header)
7:6	tag	tag for CCE-initated packets: bit[8]: flag for cmd_snoop bit
12, 8	RX sequence number	Sequence number acknowledge for opposite-direction traffic
13, 9	TX sequence number	Sequence number identifying this packet
10	TX sequence number incr	1 for all packet types, 0 for sequence number update packets
11	control	bit[0]: set to 1 for reg write / remote update bit[1]: set to 1 for sequence number update packet bit[2]: local RX indicated packet drop

Table 39. TT-Link Header Fields

9.2.5. Ethernet Layer 3/4 Packets

Headers for advanced networking capabilities can optionally be inserted into packets, to support transfer of packets through various types of real-world networking equipment. Figure 28 shows an example of a non-raw packet into which two VLAN tags, an IPv6 header, and a TCP header have all been inserted.

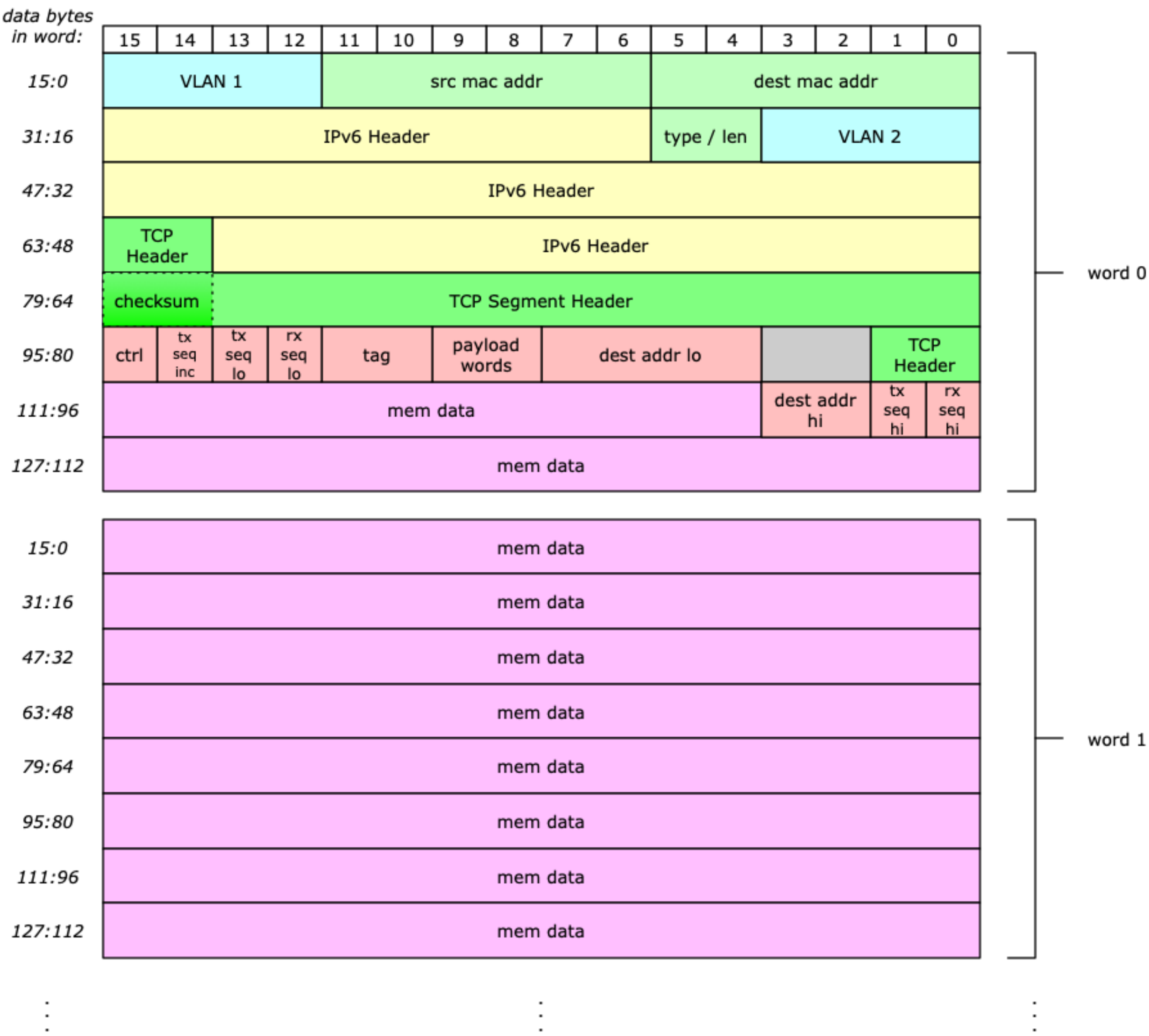


Figure 28. Advanced Networking Packet Example

These advanced headers can be inserted for both raw packets and non-raw packets.

9.2.6. Sequence Number Tracking

The Ethernet Controller implements a protocol that supports retransmission, using sequence numbers to report and detect which packets have been sucessfully transferred from one node to another. This protocol is not used for raw transfers.

Basic Operation

For every packet sent, the TX queue controller puts two sequence numbers into the TT-Link: a TX sequence number and an RX sequence number. The TX sequence number identifies the packet being transmitted, and is incremented by one each time a packet is sent. The RX sequence acts as an acknowledgement of packets that were received from the remote node; the RX sequence number is set to a value that is one larger than the TX sequence number that was set by the remote transmitter in the most recently-received packet.

The following figure depicts an example of how two nodes set the values of the TX and RX sequence numbers when transfering packets.

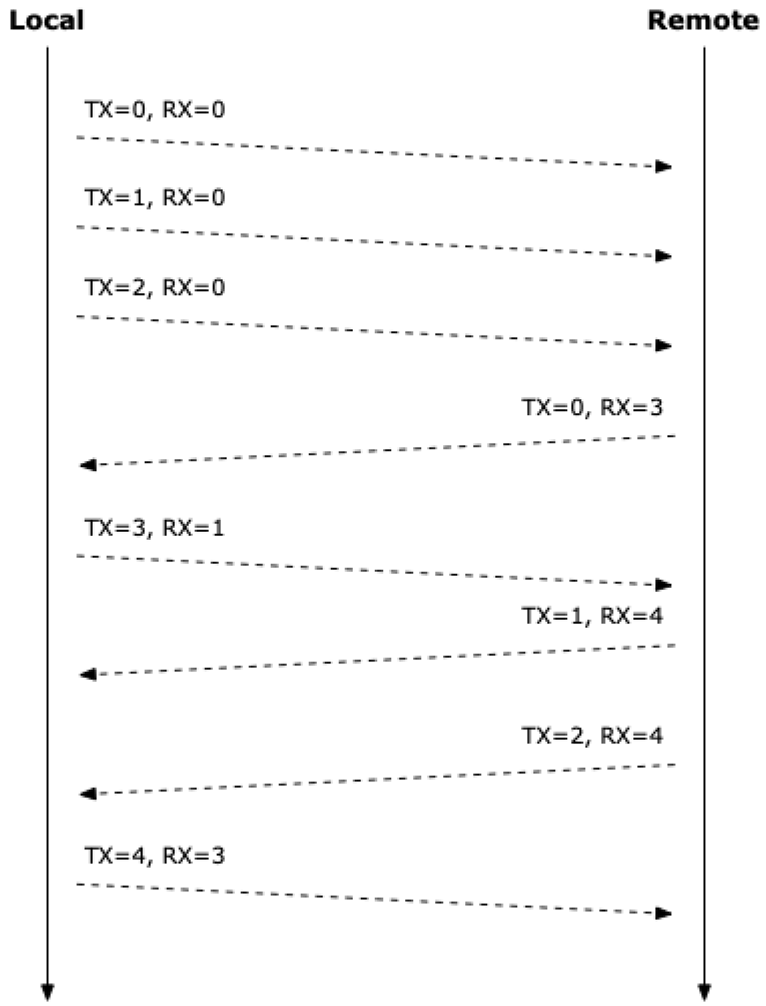


Figure 29. Packet Sequencing Example

If the packet traffic between two nodes is primarily in one direction, it is possible that one node may not receive acknowledgements for the packets it has transmitted. For this reason, the TX queue controller supports sending special packets whose sole purpose is to give acknowledgements to the other node if no normal packets have been transmitted within a certain time period. These sequence number update packets contain no payload and do not contain a valid TX sequence number. Figure 30 depicts an example of how a remote node that is not transmitting packets will occasionally send sequence number updates to the local node.

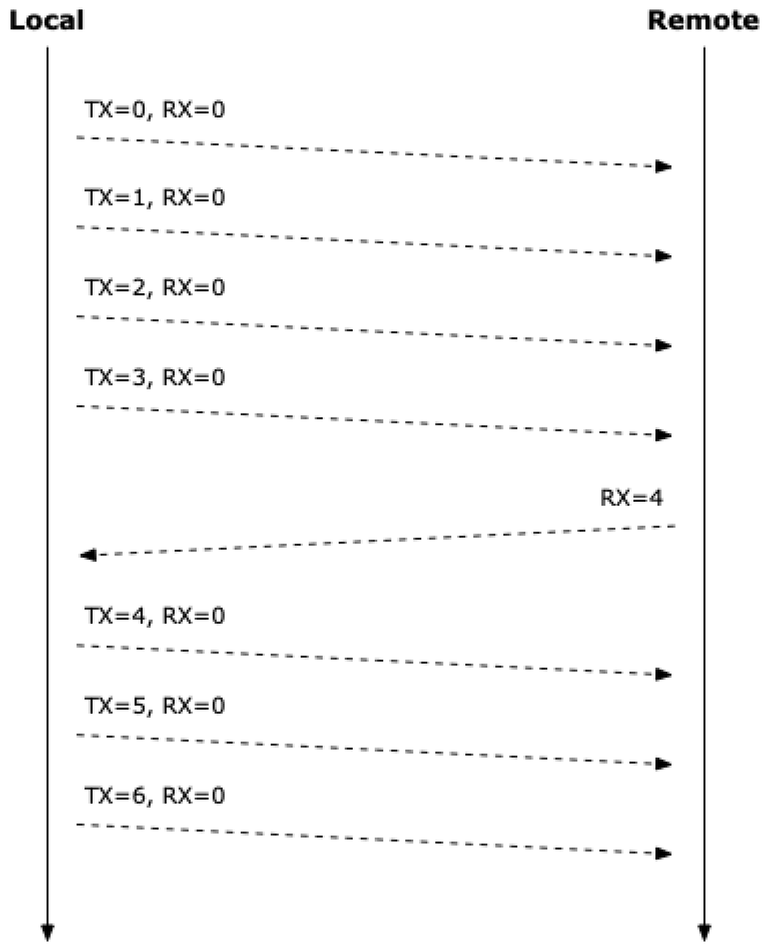


Figure 30. Sequence Number Update Example

Packet Transmission Errors

The transmitted sequence numbers are useful in case of any errors in transmission. If a packet is not successfully transferred between nodes, the queue controllers can arrange for retransmission of any packets that failed to be delivered.

A transmission error is detected at a receiver when the TX sequence number of the received packet does not match the receiver's expectation. In case of mismatch the entire RX packet is discarded ("dropped") by the receiver, thereby ensuring no negative side-effects caused by the reception of an unexpected packet. In the next packet that is transmitted by the node that detected the error, a "drop" flag is set in the TT-Link header. Receiving a packet with the "drop" flag set causes the original transmitter to resend all packets since the last successfully acknowledged packet.

Figure 31 depicts an example of retransmission that is caused by a sequence order error.

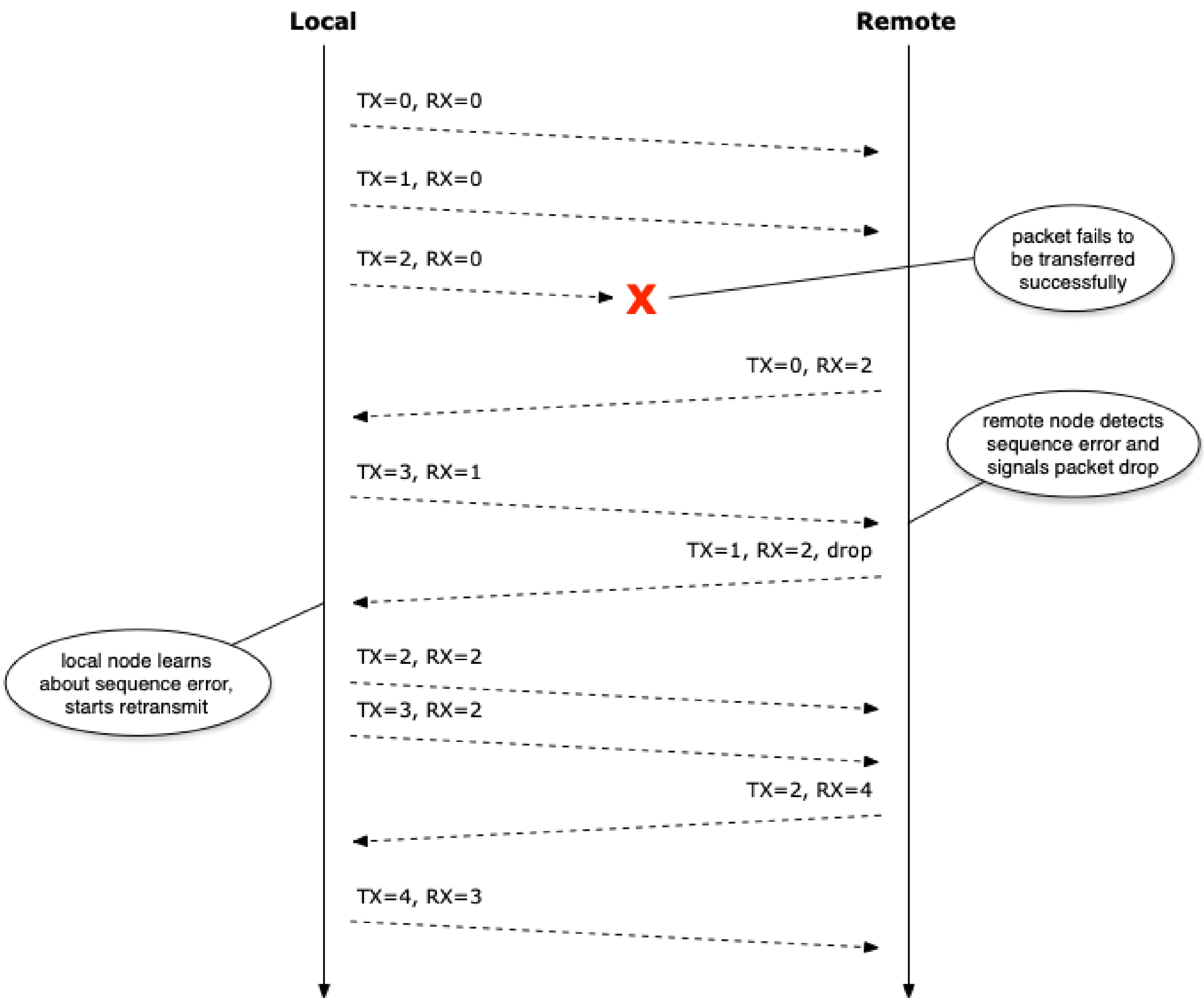


Figure 31. Packet Drop Example

In order to handle the scenario where a remote node stops sending packets entirely, the TX queue controller also supports starting retransmission if no packets have been received (i.e., with an updated RX sequence number) within a certain amount of time.

9.2.7. TX Data Path

TX Packet Config

Certain parts of the transmitted packet are assembled dynamically by the TX queue controller hardware. A TX packet config table contains the following content that affects the generated packet:

Protocol	Element (Column)	Description
Ethernet	MAC Source Address	48-bit MAC source address
	MAC Destination Address	48-bit MAC destination address
	Use Ethertype	1-bit flag specifying whether to calculate length or use Ethertype Value
	Ethertype	16-bit Ethertype value
VLAN	Insert VLAN 1	1-bit flag specifying whether a (first) VLAN tag should be inserted
	VLAN 1	16-bit content of first VLAN tag
	Insert VLAN 2	1-bit flag specifying whether a second VLAN tag should be inserted
	VLAN 2	16-bit content of second VLAN tag

Protocol	Element (Column)	Description
Network Layer 3	Insert L3 Header	1-bit flag specifying whether a Layer 3 header should be inserted
	L3 Header Type	2-bit field indicating the type of Layer 3 header to be inserted (IPv4 or IPv6)
	L3 Header	40-byte field containing static content of Layer 3 header
Network Layer 4	Insert L4 Header	1-bit flag specifying whether a Layer 4 header should be inserted
	L4 Header Type	2-bit field indicating the type of Layer 4 header to be inserted (UDP or TCP)
	L4 Header	20-byte field containing static content of Layer 4 header
	L4 Header Checksum	1-bit flag specifying whether L4 checksum should be inserted into header

Table 40. TX Packet Config Table

The TX packet config table has ten rows to allow different types/sources of traffic to use different configurations. A single table is shared by all queues, but the selection of which row to use for a particular type of traffic can be different for each queue. Each queue has separate programmable table indices for each of the following types of traffic:

- raw traffic
- software-initiated remote update transfers
- software-initiated data packet transfers
- CCE-initiated traffic

Of the settings contained in the TX packet config table, only the Ethernet protocol contents are set within the TX queue controllers; the remaining contents are added in the TX packet editing block (see Section 9.2.7.5).

Memory Read Interface

The Ethernet memory read interface converts the memory-style data read requests from the TX queue controllers into AXI read transactions. This block also includes an arbiter for managing access to the AXI interface.

TX Queue Controller

The TX queue controller is the main block in the Ethernet Controller that manages the generation and sending of Ethernet frames to the MAC TX data path. Figure 32 shows an overview of the TX queue controller structure and data paths.

Access to the single memory path is managed by the Ethernet memory read interface, and access to the single MAC TX data path is controlled by an arbitration function downstream of the TX queue controller.

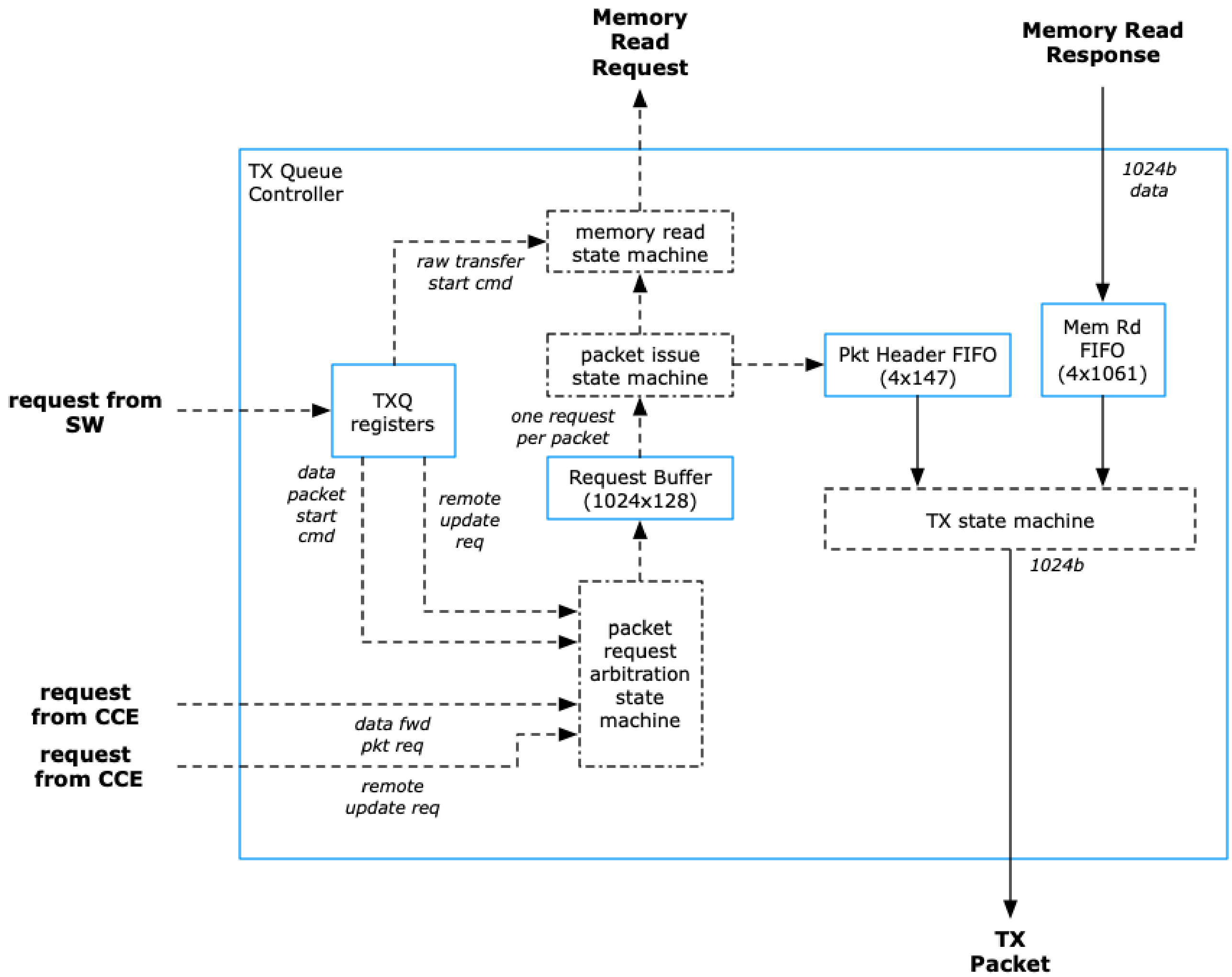


Figure 32. TX Queue Controller

Read Request Generation

There are two broad classes of traffic that can be generated by the TX queue controller: "raw" transfers where the entire contents of the packet are fetched directly from memory; and TT-Link packets whose construction is more closely managed by the hardware. Four types of packets can be generated: data packets that are started by writes to a configuration register; remote update packets that are initiated by writes to a configuration register; data forwarding packets that are initiated by a request from the CCE; and remote update packets that are initiated by a request from the CCE. The four sources of packet data are controlled by an arbiter whose state machine is shown in Figure 33.

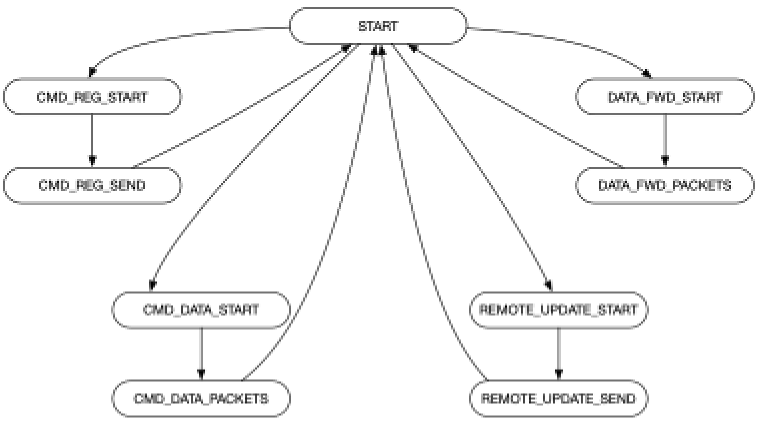


Figure 33. TX Request Arbitration State Machine

Requests for the four types of packet transfers are placed by the TX request arbitration state machine into a read request buffer, which is implemented as a FIFO that has 1024 entries. One request is added to the FIFO for each packet that is to be transmitted. Depending on the type of packet, the following information about the packet may be included in each packet request: source address, destination address, size of payload, remote update write data, etc.

Each TX queue controller can be configured to issue packets with specific minimum and maximum payload sizes. If a transfer request would result in a packet being issued with a payload size that is less than the specified minimum, the TX queue controller will pad the payload with extra data. If a transfer request would result in a packet being issued with a payload size that is larger than the specified maximum, the TX queue controller will split the transfer into multiple packets. Separately, there is an additional constraint on the amount of data that is requested from memory: in order to avoid potential problems with deadlock in the request-to-response loop, the TX queue controller will by default break up memory read requests at 4KB address boundaries to match AXI transaction boundaries.

For all types of TX traffic that require payload data from memory, a memory read state machine manages the generation of memory read requests that are sent to the read memory interface.

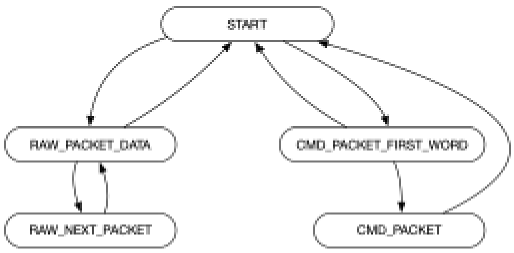


Figure 34. TX Memory Read State Machine

When non-raw packet requests are issued to the memory, information about the corresponding packet is written into a packet header FIFO that is four entries deep. The output of the packet header FIFO is used to generate the TT-Link header field of the non-raw packets that are sent to the MAC.

Sequence Number Updates

The TX queue controller also supports generating packets whose only purpose is to send an update of the local RX sequence number to the remote node. If this feature is enabled, then the TX queue controller tracks the time since the last non-raw packet was transmitted. If the time since the last sent sequence number is greater than some programmable threshold, then the TX queue controller issues a TX packet that contains the local RX sequence number, without any payload.

The TX sequence number is set to zero for all sequence number update packets.

Memory Read Tags

Each request to read payload data from memory is accompanied by a 37-bit tag that describes additional information about the request. This tag is used when the corresponding TX data is returned from the memory. Table 41 shows the format of the TX memory read tag.

Field	Width	Function
packet_len	14	packet length in bytes (only used in raw mode)
start_byte	4	start byte within first 16-byte sub-word (only used in raw mode)
beat_size	8	number of bytes requested from current 128-byte word (only used in raw mode)
eop	8	end of packet flag indicating which 16-byte sub-word contains the end of the packet
sop	1	start of packet flag
raw	1	flag indicating whether this is a raw packet or a non-raw packet
last_pkt	1	flag indicating this is the last packet of a sequence

Table 41. TX Memory Read Tag Format

Read Data Handling

The data that is read from memory is returned to the TX queue controller by the memory read interface. A TX state machine handles any additional massaging of the data stream before sending downstream toward the MAC.

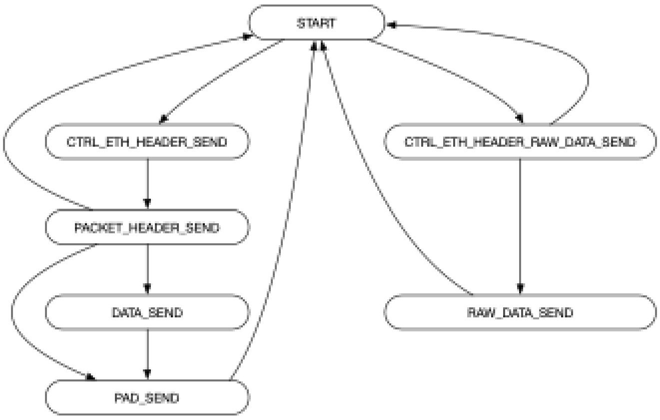


Figure 35. TX State Machine

PTP Support

The TX queue controller can be configured to inform the MAC to insert timestamps in order to support the IEEE 1588 Precision Time Protocol.

When two-step timestamp insertion is used, the MAC returns a timestamp value back to the Ethernet Controller. Each timestamp value is pushed into a Timestamp FIFO that must be read out by software. The timestamp FIFO has space for 16 timestamp values.

TX Queue Arbitration

The TX queue controllers arbitrate for access to the MAC TX data path, using a priority arbiter. Each queue can be assigned a separate priority level. Normally, access to the data path is granted on a per-packet basis: if multiple requests are made to the data path in the same clock cycle, the highest priority queue will have its packet transmitted toward the MAC.

TX Packet Editing

Any advanced networking headers are added to the TX packet following the arbitration stage; this includes the insertion of VLAN tags and/or headers for Layer 3 & 4 networking. The TX packet config table, shown in Table 40, is used to determine what changes are made to the packets passing through each TX queue controller.

Limitations

The TX packet editing capabilities have a number of significant limitations with respect to the advanced networking features.

IPv4 Header

All mandatory fields of the IPv4 header, with the exception of the Total Length field and the Header Checksum field, are taken from the statically programmed TX packet config table. The Total Length and Header Checksum fields are calculated and set by the hardware. Use of the IPv4 Options field is not supported.

IPv6 Header

All mandatory fields of the IPv6 header, with the exception of the Payload Length field, are taken from the statically programmed TX packet config table. The Payload Length field is calculated and set by the hardware. Extension headers are not supported.

UDP Header

All fields of the UDP header, with the exception of the Length and the Checksum fields, are taken from the statically programmed TX packet config table. The Length field is calculated and inserted by the hardware for all UDP packets. The Checksum field is optionally calculated and set by the hardware.

TCP Header

All fields of the TCP header, with the exception of the Checksum field, are taken from the statically programmed TX packet config table. The Checksum field is optionally calculated and set by the hardware. Use of the TCP Options field is not supported.

Though not directly related to the TX packet editing function, it is worth noting here that the RX Classification Engine does not support removal of the TCP header when operating in TT-Link mode.

TX Packet Buffer

Following the packet editing stage, completed packets ready for transfer to the MAC are stored in a dedicated buffer that allows storage for a complete packet. This buffer also manages the crossing of packets from the HSIO clock domain to the MAC clock domain.

Since the OmegaCore MAC/PCS is unable to handle back-pressure while a packet is in flight, it is necessary to wait for an entire packet to be written into the TX packet buffer before starting to transmit the packet to the MAC. Therefore, the TX packet buffer must be large enough to store the largest possible packet that can be transmitted. The TX packet buffer is implemented as a 66-entry FIFO, and with a 1024-bit data path, meaning that the TX packet buffer can store more than 8 KB of TX packet data.

MAC TX Interface

The final block in the Ethernet Controller's TX data path is the MAC TX Interface, which converts legacy signaling into the format required by the OmegaCore MAC/PCS. For example, the OmegaCore MAC implements three separate data path interfaces (800G, 200/400G, 100G), and the MAC TX Interface is used to demux the single Ethernet Controller TX data path into the three separate MAC data paths.

9.2.8. RX Data Path

MAC RX Interface

The first block in the Ethernet Controller's RX data path is the MAC RX Interface, which converts the OmegaCore MAC/PCS signals to the legacy signals used by the Ethernet Controller. For example, the OmegaCore MAC implements three separate data path interfaces (800G, 200/400G, 100G), and the MAC RX Interface is used to mux the three separate MAC data paths to the single Ethernet Controller RX data path.

RX Packet Buffer

Packets received from the MAC are stored in an RX packet buffer before being sent to the RX classification engine. Aside from managing the crossing of packets from the MAC clock domain to the HSIO clock domain, the RX packet buffer also manages handling of packets marked by the MAC as being in error. By default, the RX packet buffer will drop any errored packets, but it can be configured to pass through all packets.

The RX packet buffer is implemented as a 66-entry FIFO, and with a 1024-bit data path, this means that the RX packet buffer can store more than 8 KB of RX packet data. Since the MAC provides packets in cut-through operation (i.e., without buffering the entire packet), the Ethernet Controller's hardware does not know whether or not the packet being received has an error until the entire packet has been received. Therefore, the RX packet buffer receives and stores a complete packet before it can be determined whether or not to forward the packet to the rest of the RX data path.

RX Classification Engine

The RX classification engine performs various packet processing functions based on the content of each received packet, including filtering and routing of packets. Full documentation of the RX classification engine is provided in a separate document.

RX Packet Routing

Received packets are routed to one of the four RX queue controllers. There are two main methods for determining which queue controller receives packets: address-class and the RX classification engine.

Address-based routing uses the MAC destination address to decide how to route the packets. The MAC addresses are considered according to three groups: broadcast addresses, multicast addresses, and unicast addresses. Each address class contains a setting defining which RX queue controller is the destination for packets with that address class.

Within the address-based routing scheme it is also possible to distinguish between remote update traffic and normal data (forward) traffic, assuming the use of packet mode; remote update traffic can be routed to a different queue than normal data traffic.

RX Queue Controller

The RX queue controller is the main block in the Ethernet Controller that manages the reception of Ethernet frames from the MAC RX data path. There are four RX queue controllers. Access to the HSIO fabric is managed by the Ethernet memory write interface.

Each RX queue controller must be configured to operate in either "packet mode", in which case it is expected that the TT-Link header is present, or "raw mode".

RX State Machine

The RX state machine manages the reception of the various types of packets. The states are shown in Figure 36.

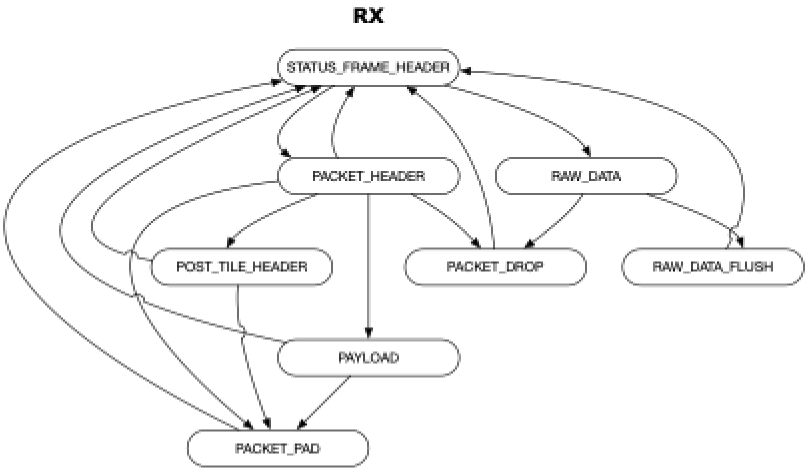


Figure 36. RX State Machine

For most packet types, the entire payload (not including any header) is written to memory. There are two exceptions to this: sequence number update packets and bad packets. Sequence number update packets have no payload, and so nothing is written to memory. Any packet that is detected as having an error — either indicated by the MAC or with a bad sequence number — is dropped without being written to memory.

A request to write to memory is performed for every 128-byte word that needs to be written to memory. Each request to write data to memory is accompanied by a 4-bit tag that describes additional information about the request. This tag is used when the corresponding write responses are returned from memory. Table 42 shows the format of the RX memory write tag.

Field	Width	Function
valid	1	flag indicating that this is a valid write request
packet_start	1	flag indicating the start of a packet
packet_end	1	flag indicating the end of a packet
last_packet	1	flag indicating this is the last packet of a sequence

Table 42. RX Memory Write Tag Format

The data path through the RX queue controller splits into a separate branch for remote update packets. Instead of being sent in the same path as normal data packets, remote update packets are sent to toward the HSIO fabric via a separate 64-bit path.

Ethernet Write Memory Interface

The Ethernet memory write interface converts the memory-style data write requests from the RX queue controllers into AXI write transactions. This block also includes an arbiter for managing access to the AXI interface.

Note that there is not necessarily a 1:1 correspondence between the number of packets received and the number of AXI transactions issued by the memory write interface. A single packet may result in one, two, or even three separate AXI write transactions.

This interface assigns the cmd_snoop bit in the TT-link header to the AWUSER[8] bit in outgoing AXI write request.

Address Remapping

The final stage of the RX data path, for both normal data traffic and remote update traffic, is an address remapping block that can remap the memory addresses before the AXI transactions reach the HSIO fabric.

Packet Drops

There are multiple points in the RX data path where packets can be dropped. This section summarizes the various packet drop scenarios.

MAC/PCS Internal Drops

The MAC/PCS will not drop individual packets. If, during the processing of a packet the MAC detects some kind of error with an individual packet, it will pass that packet on to the Ethernet Controller. However, if the Ethernet link goes down — for example, due to the PCS losing synchronization lock — then any packets that were in flight inside the MAC/PCS will be dropped.

Bad Packets from MAC

The MAC is capable of detecting certain errors with packets and indicating those errors to the Ethernet Controller. The following conditions are the most common reasons for the MAC indicating packet errors:

- packets that fail the Ethernet FCS (CRC)
- packets that are smaller than the minimum size

Packets marked as bad by the MAC are dropped by RX packet buffer.

Rejected Packets (RX Classification Engine)

The RX classification engine has the ability to drop packets according to various criteria.

Filtered Packets

The packet filtering function of the RX classification engine can be configured to drop packets with certain characteristics. For example, packets targeting certain address ranges or port numbers could be intentionally dropped, or packets arriving a rate higher than some threshold might be dropped.

Invalid Headers

The RX classification engine detects certain types of errors in packet headers. By default any packet with a header error will be dropped.

Bad Checksums

Some headers have checksums to verify the content of the packet. The RX classification engine will by default drop any packets with mismatching checksums.

Bad Sequence Numbers

When operating in packet mode, each RX queue controller verifies that the sequence number of the incoming packet matches what is expected. If the sequence number mismatches, then the RX queue controller will drop that packet and all packets until the sequence number matches the expectation.

Raw Mode Buffer Full

When operating in raw mode, the RX queue controller manages a buffer in memory for the storage of the incoming payload data. If that buffer is not configured to wrap around when full, then the RX queue controller will drop any packets that would cause overflow of the buffer.

9.2.9. Ethernet Flow Control

The Ethernet Controller supports transmitting and receiving 802.3 Pause and 802.1Q Priority Flow Control packets.

Generation of Flow Control Packets

The MAC/PCS can be configured to transmit 802.3 Pause packets and/or 802.1Q priority flow control packets. These packets are used to indicate to a remote transmitter that it should pause all or some of its data transfer.

The generation of these packets is managed by software: a configuration register can be written to request the sending of an individual flow control packet, and it is the MAC/PCS hardware that initiates transmission of the requested packets. There are nine control bits in total: one bit for generation of 802.3 Pause packets, and eight bits for the generation of the eight 802.1Q priority classes.

Reception of Flow Control Packets

The MAC/PCS can be configured to forward 802.3 Pause packets and/or 802.1Q priority flow control packets to the Ethernet Controller. Separately, the MAC/PCS can also be configured to send indications of the reception of Pause/PFC packets to the Ethernet Controller without sending the entire packet itself. The Ethernet Controller hardware has no ability to immediately or directly halt the generation of TX traffic when a Pause / PFC packet is received.

When the indication of the reception of a Pause/PFC packet is received by the Ethernet Controller, the associated pause quanta value is pushed into a Pause/PFC FIFO. The Pause/PFC FIFO has space for 8 entries and must be read out by software in order to drain the FIFO.

Automatic Response to Flow Control Packets

The OmegaCore MAC/PCS has the ability to automatically respond to received 802.3 Pause packets by pausing the transmission of packets for the amount of time indicated in the received Pause packet. The MAC/PCS cannot automatically pause TX traffic in response to received 802.1Q priority flow control packets.

9.3. Ethernet MAC/PCS

The Keraunos Ethernet Controller is designed to operate with the OmegaCore MAC/PCS IP from AlphaWave. For more details on the operation and design of the OmegaCore MAC/PCS, see the datasheet provided by AlphaWave.

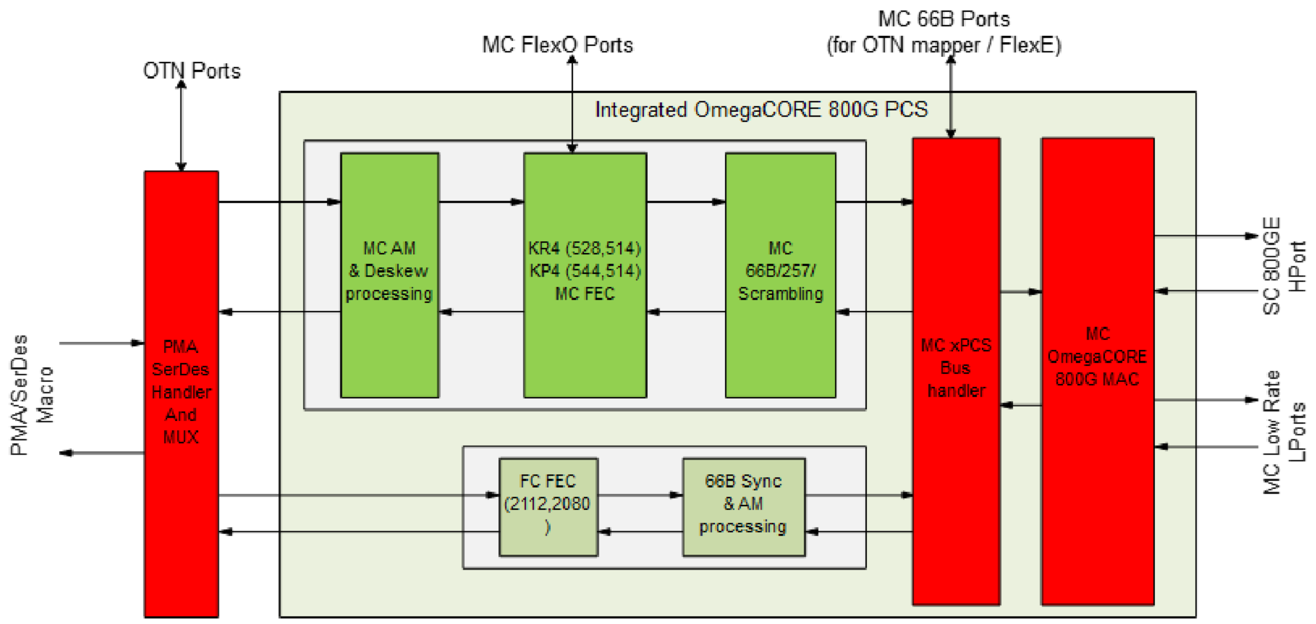


Figure 37. MAC/PCS Block Diagram

Summary of MAC/PCS features:

- 1024-bit application data path interface, running on 1.6 GHz clock
- no support for intra-packet flow control on application TX interface; application must provide full packet when MAC is ready
- no support for back-pressuring flow control on application RX interface; application must be ready to accept data at all times
- RX packet flow is cut-through: packet starts being output as soon as data is available, not after full packet has finished being received
- handles appending and dropping of CRC and small frame (<64B) paddingG
- two links of four lanes each of 128-bit data at interface to/from Serdes, running on max 830.078125 MHz clock

9.4. Register Infrastructure

Figure 38 shows the register infrastructure within the Ethernet Controller.

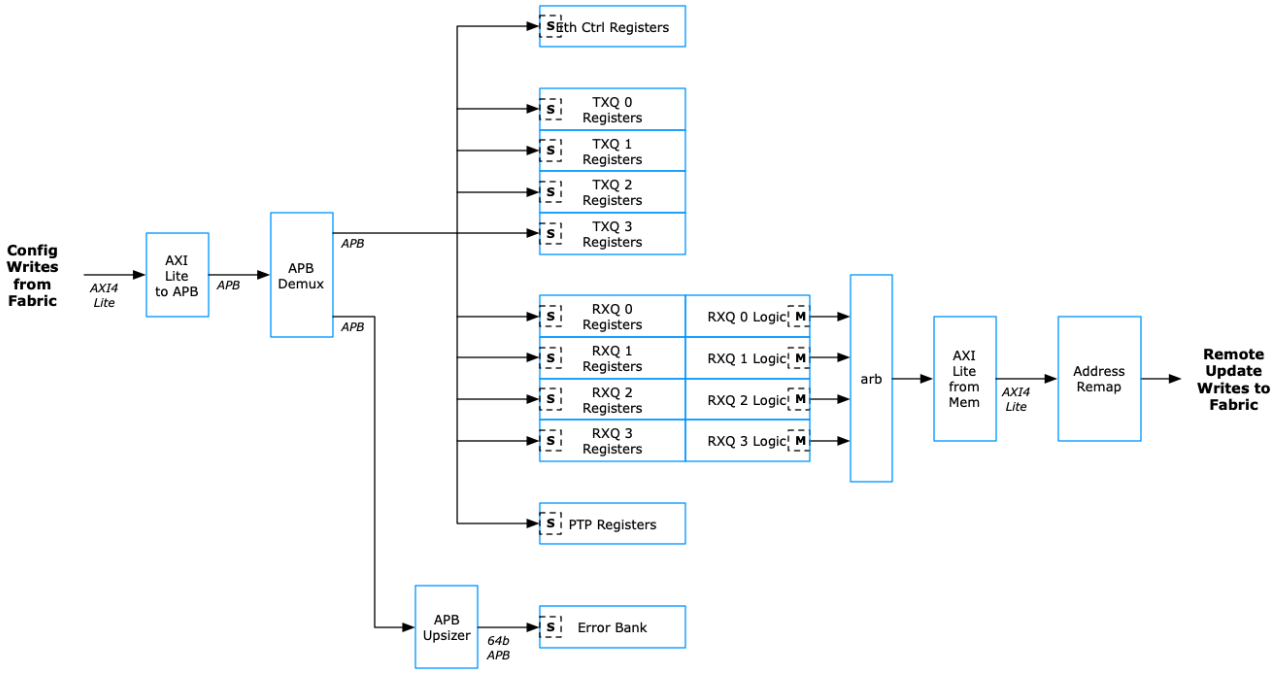


Figure 38. Register Infrastructure

9.5. Loopback Modes

Though not part of the Ethernet Controller, it is noted here that the downstream IPs support a number of loopback modes, as shown in Figure 39.

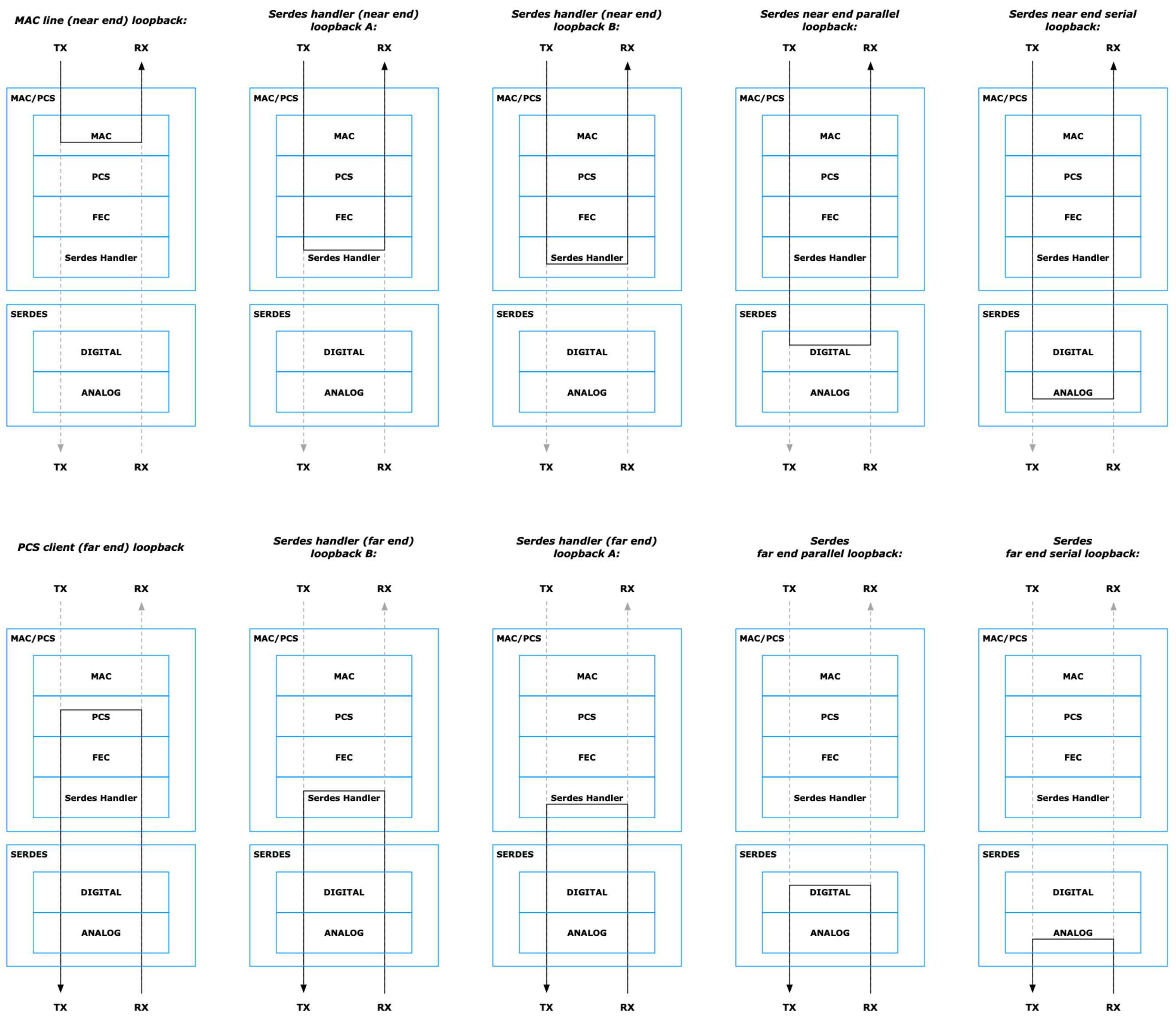


Figure 39. Loopback Modes

9.6. Profiler Logger

The Ethernet Controller includes an instance of the Profiler Logger IP, which can be used to log various events within the Ethernet Controller for debug and performance monitoring purposes. The Profiler Logger is described in greater detail elsewhere.

The following events are logged by the Profiler Logger instance in the Ethernet Controller:

- start of TX packet into TX packet buffer
- end of TX packet into TX packet buffer
- start of RX packet out of RX packet buffer
- end of RX packet out of RX packet buffer
- start of AXI TX read request
- end of AXI TX read response
- start of AXI RX ctrl write request
- end of AXI RX write response
- start of AXI RX write request
- end of AXI RX ctrl write response
- SW-initiated event

9.7. Error Handling

The Ethernet Controller has three separate domains for handling and reporting errors. RAS errors (Section 9.8) and Interrupts (Section 9.10.2) are discussed in separate subsections.

Unexpected errors in the operation of the Ethernet Controller hardware can be recorded and reported in status registers. The following error conditions can be individually enabled for reporting:

- TX packet buffer AFIFO overflow
- TX packet buffer Pkt Start FIFO overflow
- TX packet buffer Checksum FIFO overflow
- TX packet buffer Pkt End FIFO overflow
- RX packet buffer AFIFO overflow
- RX packet buffer Pkt Info FIFO overflow
- RX classification PE input buffer overflow
- RX classification PE output buffer overflow
- Late PTP PTI Update
- Late PTP Timestamp Update
- Pause/PFC FIFO overflow
- Timestamp FIFO overflow
- Read Mem IF Tag FIFO overflow
- Write Mem IF Count FIFO overflow

The occurrence of an error can be forwarded to the interrupt output of the Ethernet Controller.

9.8. RAS Error Bank

The Keraunos RAS functionalty is described in greater detail elsewhere, but it is noted here that the Ethernet Controller includes one instance of a RAS error bank. This error bank manages the signaling of memory-related parity errors that are reported by the memory wrappers. See Section 9.11.3 for more details on the memories in the Ethernet Controller.

9.9. Latency

Table 43 shows some typical one-way latencies through the Ethernet data paths. The values shown here include the following components of latency:

- latency to read payload memory from TL1 memory in local HSIO tile (TX)
- latency for Ethernet Controller to create and send packets to MAC
- latency through the MAC/PCS (TX and RX)
- latency through the serdes (TX and RX)
- latency for Ethernet Controller to receive and process packets from MAC
- latency to write payload memory to TL1 memory in local HSIO tile (RX)

The values shown here do NOT include:

- any latency associated with the physical connectivity (wire / backplane) between two Ethernet nodes

Start Point	End Point	Mode	Latency [ns]
Request to transmit received by TXQ	Remote TL1 memory write start	800G, small packets	387 ns
TX packet start into MAC	RX packet start out of remote MAC	800G, small packets	338 ns

Table 43. Sample Latencies

9.10. Software Interface

9.10.1. Address Map

Table 44 shows the full address map for the Ethernet Controller.

Region	Offset Range	Size
TX Queue 0	0000 - 0FFF	4 KB
TX Queue 1	1000 - 1FFF	4 KB
TX Queue 2	2000 - 2FFF	4 KB
TX Queue 3	3000 - 3FFF	4 KB
RX Queue 0	4000 - 4FFF	4 KB
RX Queue 1	5000 - 5FFF	4 KB
RX Queue 2	6000 - 6FFF	4 KB
RX Queue 3	7000 - 7FFF	4 KB
Control	8000 - 8FFF	4 KB
PTP Timer	9000 - 9FFF	4 KB
RX Classification	C000 - FFFF	16 KB

Table 44. Address Map

9.10.2. Interrupts

The Ethernet Controller provides a single wired interrupt output that is sent to processors elsewhere in the chip.

The following events can be used as sources for the Ethernet Controller interrupt generation:

- Ethernet error condition (see Section 9.7)
- timer value match
- dropping of bad packet by RX packet buffer
- RXQ packet reception complete (not asserted for pkts with bad sequence numbers, for sequence number update packets, or for register write / remote update packets); separate indication for each RX queue controller

9.11. Implementation

9.11.1. Clocks

Name	Max Func Freq [MHz]	Logic
HSIO clock	1650	HSIO clock, used for most Ethernet Controller logic
MAC clock	1600	MAC/PCS clock, used for logic close to MAC/PCS logic
REF clock	100	Reference clock, used for PTP timer and other low-frequency logic

Table 45. Clocks

HSIO Clock Speed Reduction

Keraunos may need to support lowering the frequency of the HSIO clock from the maximum 1.65 GHz, for example to reduce power consumption. However, the structure of the hardware is such that if the frequency of the HSIO clock is reduced below a certain threshold, then errors may start to occur in the RX data path due to the inability of the hardware to process the incoming packets quickly enough.

There are two buffers in the RX data path that are susceptible to overflow error if the HSIO clock is too slow (and if the rate of incoming packets corresponds to the maximum data rate): the RX packet buffer, and the output FIFO in the packet editing subblock of the RX classification engine. Each of these buffers have overflow error outputs that will indicate the occurrence of any overflow error. Many factors affect the lower limit to which the HSIO clock speed can be reduced without introducing errors, including the chosen data rate and the size of packets.

Note that if the transmitter(s) can be constrained to reduce the rate of packets being transmitted to match the reduction of the HSIO clock speed in the transmitter, then there should be no concern of overflow. Reduction in the TX packet rate could be achieved through various means, such as reducing the transmitter's clock speed or implementing one of the flow control mechanisms.

Table 46 shows some recommended lower limits for the frequency of the RX device's HSIO clock in various scenarios, assuming that there is **no corresponding reduction in the rate of remote TX traffic**.

Mode	Max Packet Size	800G
SW-initiated packet mode	256 B	1300 MHz
SW-initiated packet mode	1 KB	1000 MHz
SW-initiated packet mode	4 KB	1000 MHz

Table 46. Minimum HSIO Clock Frequency

9.11.2. Resets

Input Resets

Table 47 lists the hardware reset signals that are input to the Ethernet Controller.

Signal Name	Description	Clock Domain	Affected Logic
ref_prim_reset_ni	cold reset from SMU	REF clock	reset synchronizers
cold_hsio_reset_ni	Ethernet Controller domain reset	HSIO clock	most logic in Ethernet Controller
cold_hsio_reset_ptp_ni	Ethernet PTP domain reset	REF clock	PTP timer
prof_hsio_reset_ni	Profiler domain reset	HSIO clock	Profiler Logger
errbnk_hsio_reset_ni	Error Bank domain reset	HSIO clock	Error Bank

Table 47. Input Resets

Internal Reset Generation

Figure 40 shows how the main resets in the Ethernet Controller are generated.

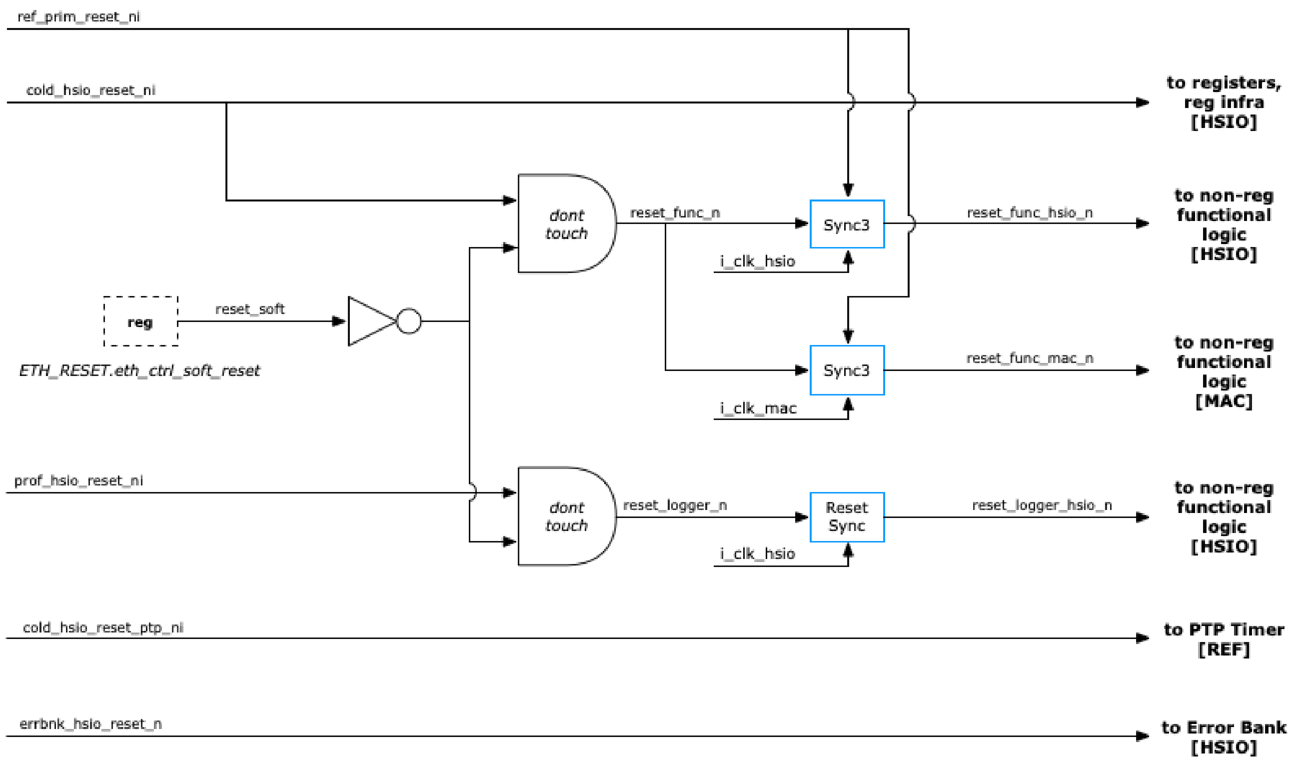


Figure 40. Reset Generation

9.11.3. Memories

There are memory macros located in the sub-blocks of the Ethernet Controller. Table 48 shows the full set of memories, including a number of different size values for each. The "logical" size of the memory is the base functional requirement, before accounting for any parity/ECC bits. The "nominal" size of each memory includes the parity/ECC bits, before accounting for the actual breakdown of the memory into smaller pieces for physical reasons.

Function	Type	# Instances	Logical Size	Nominal Size	Physical Impl Size	Total # of Bits	Data Protection
TXQ Request Buffer	two-port	4	1024 words x 128b	1024x136	1 x 1024x136	147456	parity
TX Packet Buffer	two-port	1	66 words x 1120b	66x1190	9 x 66x136	80784	parity
RX Packet Buffer	two-port	1	66 words x 1101b	66x1170	9 x 66x132	78408	parity

Table 48. Memories

Chapter 10. Profiling

The Keruanos Profiler block generates a stream of profiling records that are written to memory. These records provide passive visbility to the events taking place in the chiplet. Each record is timestamped with the Grendel System Time and can thus be correlated to all other events in the system.

10.1. Features and Usage

Keraunos profiling block provides the following features:

- Interfaces with CCE, TT Ethernet Controller, and MACPCS buses and logs data movement records
- Generates timestamped and custom data movement records
- Supports record storage in a specified ping-pong buffer located in any backing memory (Host DRAM via PCIE, HSIO SRAM, or Mimir GDDR).
- Enables software-triggered record logging
- Notifies host software when buffer full/almost full via MSI/specified notification address

The profiling traces can be used for the following purposes:

1. Primary usage: Identify stalls and scheduling issues useful for software tuning.
    1. Guide tuning of the following software knobs:
        - Number of TX queues to use to saturate datapaths on Tx and Rx
        - Number of cores to use to saturate to saturate datapaths on Tx and Rx
        - Use 1 Tx link vs more Tx links to go to a remote Grendel
    2. Guide software optimization
        - By getting execution time in SW after some hardware events like RX write in SRAM, etc.
    3. Identify ethernet link “busy”ness
2. Secondary usage: To debug CCE SW programming issue

10.2. Profiling Record Formats

Keraunos Profiler generates 16-byte profiling records. Two types of records are supported as described below:

Primary record format:

Bits	Field	Notes
[127]	Type	Primary Record = 1’b1
[126:124]	Source ID	BitWidth = log2(#Loggers), Loggers in HSIO tile = 5
[123:119]	Record ID	BitWidth = log2(#NumEvents), Max Events per Logger = 32
[118:64]	Record specific details	
[63:0]	Timestamp	Use timestamp if primary type, else uses secondary data

Table 49. Primary record format

Secondary record format:

Bits	Field	Notes
[127]	Type	Secondary record = 1’b0
[126:0]	Record specific details	

Table 50. Secondary record format

10.3. Keraunos Profiler

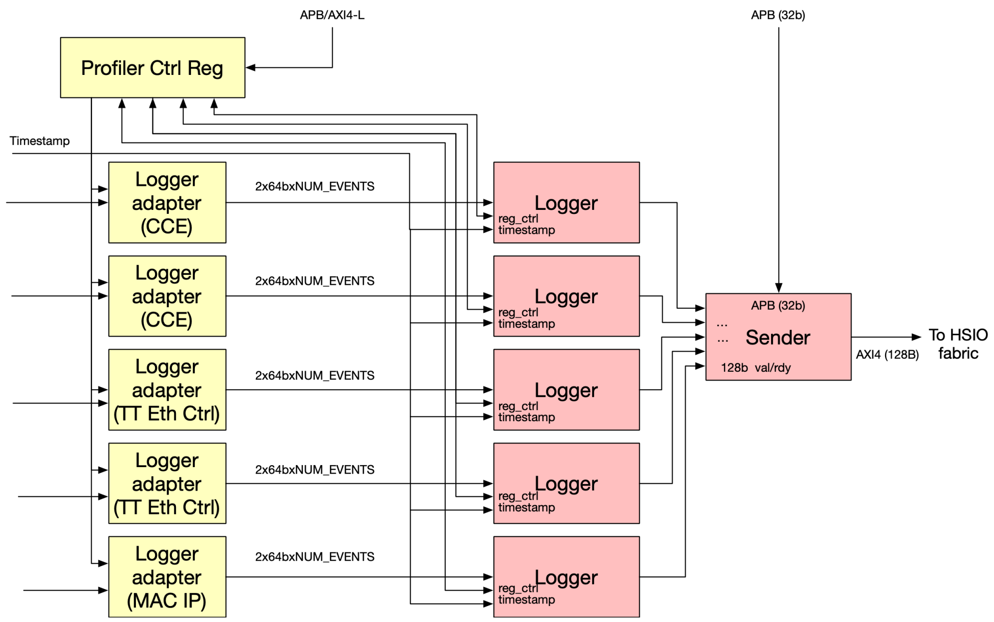


Figure 41. Keraunos Profiler

10.3.1. Components

Keraunos profiler consists of the Logger, Sender, Profiler Control Register, and Logger Adapter. Keraunos profiler is instantiated in each HSIO tile.

Logger

Logger accepts 16B primary and secondary records from multiple events and forwards them to the Sender. There can be up to 32 events per logger. Each logger has an input FIFO of configurable depth to store records per event and a round robin arbiter that streams data to output FIFO. More details on the logger can be found in Profiler Specification.

Sender

Sender accepts 16B records from multiple loggers and forwards record batches via AXI write transactions. As shown in Figure 41, Keraunos Profiler has 5 Loggers per Sender. Sender aggregates data up to 4KB, and sends out AXI burst transactions to a buffer located in any backing memory (Host DRAM via PCIE, HSIO SRAM, or Mimir GDDR). More details on the logger can be found in Profiler Specification Sender has the following register interface:

S.No.	Register	Size	Description
1	`base_addr`	64	Base address of the ping-pong buffer.
2	`size`	32	Size of the ping-pong buffer. The buffer is divided into a top and bottom half for ping-pong operation.
3	`top_full_notification_addr`	64	Address for notification when the top half of the buffer is full.
4	`top_full_notification_data`	64	Data for notification when the top half of the buffer is full.
5	`bottom_full_notification_addr`	64	Address for notification when the bottom half of the buffer is full.
6	`bottom_full_notification_data`	64	Data for notification when the bottom half of the buffer is full.
7	`status`	32	Status of the ping-pong buffer. Contains flags like `cb_top_full`, `cb_bottom_full`, `cb_top_empty`, `cb_bottom_empty`.
8	`control`	32	Control register to enable the sender, enable notifications, clear buffer, trigger flush etc.

Table 51. Sender Registers

Profiler Control Register

Profiler control register block holds control/status registers for loggers as well as registers for software-triggered records. At a high-level, it consists of the following registers. Note Profiler Control Registers could be implemented as a separate block or be part of any existing register block.

S.No.	Register	Size	Description
1	`sw_primary_data`	64	Primary data field for software-triggered profiler events. Bit 63 specifies record type (primary/secondary).
2	`sw_secondary_data`	64	Secondary data field for software-triggered profiler events. To be used only if record type is secondary.
3	`sw_record_issue`	1	Initiates a software-triggered record when set.
4	`enable_events`	32	Bitmask to enable or disable specific events.
5	`dropped_events`	32	Status register indicating which events have dropped records due to FIFO overflow.
6	`timestamp_select`	1	Selects the timestamp source, DFD or PTP.
7	`cmd_snoop filter`	1	Filter for log events when cmd_snoop bit (AXUSER[8]) is set. Only used in CCE.

Table 52. Profiler Control Registers

Logger Adapter

Logger adapter is specific to Logger instance in each IP and performs the following functions:

1. Taps onto various buses and extracts records to send to the Logger.
2. Initiates software-triggered records.
3. Adds source ID for records.

10.3.2. Implementation

The Keraunos Profiler is not a implemented as a single, unified block. Instead, its components are distributed across several hardware partitions as follows:

- **CCE (x2), TT Ethernet Controller (x2), and MACPCS partitions**: Each of these partitions contains a Logger wrapper, which includes the Profiler Control Registers, Logger Adapter, and a Logger.
- **HSIO fabric partition**: This partition contains the Sender.

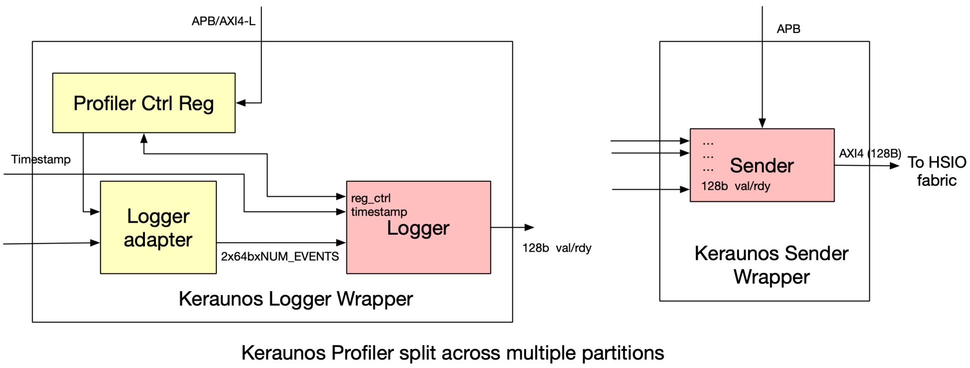


Figure 42. Keraunos Profiler Implementation

10.4. Profiling Events

10.4.1. CCE

There are two categories of buses on which events are logged in CCE:

1. AXI interfaces to SRAM (Total 16 events)

  - DMRISC SPM to SRAM: Useful for debug on tt-fabric kernels access to SRAM
  - DMA to SRAM: Useful to identify any stalls
  - ETH IF to SRAM and NOC IF to SRAM: Useful to identify any stalls and tt-fabric software optimization
  - Two modes as all traffic on this is non-burst:
  - Triggered on all requests (used for debug)
  - Triggered only on cmd_snoop (used for volume runs)
2. CCE to TT Ethernet controller interface (Total 8 events)
  - Trigger on Request accepted by TT Ethernet Controller and Remote Ethernet endpoint ack

The full list of events is described in Table 53.

Event ID	Interface	Event	Fields
0	N/A	Software-triggered event.	N/A
1	DMRISC SPM to SRAM	AXI read request start. Trigger on AR request.	`ARID[9:0]`, `ARADDR[22:0]`, `ARSIZE[2:0]`, `ARLEN[5:0]`
2	DMRISC SPM to SRAM	AXI read response end. Trigger on R response (Only `rlast`).	`RID[9:0]`, `RRESP[2:0]`
3	DMRISC SPM to SRAM	AXI write request start. Trigger on AW request.	`AWID[9:0]`, `AWADDR[22:0]`, `AWSIZE[2:0]`, `AWLEN[5:0]`
4	DMRISC SPM to SRAM	AXI write response end. Trigger on B response.	`BID[9:0]`, `BRESP[2:0]`
5	DMA to SRAM	AXI read request start. Trigger on AR request.	`ARID[9:0]`, `ARADDR[22:0]`, `ARSIZE[2:0]`, `ARLEN[5:0]`
6	DMA to SRAM	AXI read response end. Trigger on R response (Only `rlast`).	`RID[9:0]`, `RRESP[2:0]`
7	DMA to SRAM	AXI write request start. Trigger on AW request.	`AWID[9:0]`, `AWADDR[22:0]`, `AWSIZE[2:0]`, `AWLEN[5:0]`
8	DMA to SRAM	AXI write response end. Trigger on B response.	`BID[9:0]`, `BRESP[2:0]`
9	ETH IF to SRAM	AXI read request start. Trigger on AR request.	`ARID[9:0]`, `ARADDR[22:0]`, `ARSIZE[2:0]`, `ARLEN[5:0]`
10	ETH IF to SRAM	AXI read response end. Trigger on R response (Only `rlast`).	`RID[9:0]`, `RRESP[2:0]`
11	ETH IF to SRAM	AXI write request start. Trigger on a) AW request OR b) AW request with `cmd_snoop` bit set.	`AWID[9:0]`, `AWADDR[22:0]`, `AWSIZE[2:0]`, `AWLEN[5:0]`, `AWUSER[8]`
12	ETH IF to SRAM	AXI write response end. Trigger on B response.	`BID[9:0]`, `BRESP[2:0]`
13	NOC IF to SRAM	AXI read request start. Trigger on AR request.	`ARID[9:0]`, `ARADDR[22:0]`, `ARSIZE[2:0]`, `ARLEN[5:0]`
14	NOC IF to SRAM	AXI read response end. Trigger on R response (Only `rlast`).	`RID[9:0]`, `RRESP[2:0]`
15	NOC IF to SRAM	AXI write request start. Trigger on a) AW request OR b) AW request with `cmd_snoop` bit set.	`AWID[9:0]`, `AWADDR[22:0]`, `AWSIZE[2:0]`, `AWLEN[5:0]`, `AWUSER[8]`
16	NOC IF to SRAM	AXI write response end. Trigger on B response.	`BID[9:0]`, `BRESP[2:0]`
17	CCE to TT Eth controller (TX Queue 0)	TT Eth controller request accepted.	`src addr[19:0]`, `dest addr[19:0]`, `length[13:0]`, `cmd_snoop[0]`
18	CCE to TT Eth controller (TX Queue 0)	TT Eth controller (TX Queue 0..3) Remote Ethernet endpoint ack.	No data

Event ID	Interface	Event	Fields
19	CCE to TT Eth controller (TX Queue 1)	TT Eth controller request accepted.	`src addr[19:0]`, `dest addr[19:0]`, `length[13:0]`, `cmd_snoop[0]`
20	CCE to TT Eth controller (TX Queue 1)	TT Eth controller (TX Queue 0..3) Remote Ethernet endpoint ack.	No data
21	CCE to TT Eth controller (TX Queue 2)	TT Eth controller request accepted.	`src addr[19:0]`, `dest addr[19:0]`, `length[13:0]`, `cmd_snoop[0]`
22	CCE to TT Eth controller (TX Queue 2)	TT Eth controller (TX Queue 0..3) Remote Ethernet endpoint ack.	No data
23	CCE to TT Eth controller (TX Queue 3)	TT Eth controller request accepted.	`src addr[19:0]`, `dest addr[19:0]`, `length[13:0]`, `cmd_snoop[0]`
24	CCE to TT Eth controller (TX Queue 3)	TT Eth controller (TX Queue 0..3) Remote Ethernet endpoint ack.	No data

Table 53. CCE Profiling Events

10.4.2. TT Ethernet Controller

The full list of events supported by TT Ethernet Controller is described in Table 54.

Event ID	Interface	Event	Fields
0	AXI TX Read	Request start	`queue_num[1:0]` (from ID), `AXI len[7:0]`, `AXI addr[43:0]`
1	AXI TX Read	Response end (on `rlast`)	`queue_num[1:0]` (from ID)
2	AXI RX Write (Data)	Request start (on `AWVALID`)	`queue_num[1:0]` (from ID), `AXI len[7:0]`, `AXI addr[43:0]`, `last` (from `AWUSER`)
3	AXI RX Write (Data)	Response (on `BVALID`)	`queue_num[1:0]` (from ID)
4	AXI RX Write (Ctrl)	Request start (on `AWVALID`)	`queue_num[1:0]` (from ID), `AXI len[7:0]`, `AXI addr[43:0]`, `last` (from `AWUSER`)
5	AXI RX Write (Ctrl)	Response (on `BVALID`)	`queue_num[1:0]` (from ID)
6	TX packet (before TX packet buffer)	Start	`seq_num[15:0]`, `tx_pkt_config_select[3:0]`, `pkt_dest_addr[15:0]`, `MAC_DA[15:0]`
7	TX packet (before TX packet buffer)	End	`EOP_status[9:0]`
8	RX packet (after RX packet buffer)	Start	`seq_num[15:0]`, `pkt_dest_addr[15:0]`, `MAC_DA[15:0]`
9	RX packet (after RX packet buffer)	End	`EOP_status[19:0]`
10	Software	SW-initiated	Custom data

Table 54. TT Ethernet Controller Profiling Events

10.4.3. MACPCS

The full list of events supported by MAC/PCS is as described in Table 55.

Event ID	Interface	Event	Fields
0	TX (Channel 0)	Start of packet (SOP)	
1	TX (Channel 0)	End of packet (EOP)	
2	RX (Channel 0)	Start of packet (SOP)	
3	RX (Channel 0)	End of packet (EOP)	
4	Pause (Channel 0)	Send of pause_req	
5	Pause (Channel 0)	Receive of pause_ack	
6	PFC (Channel 0)	Send of pfc_req	

Event ID	Interface	Event	Fields
7	PFC (Channel 0)	Receive of pfc_ack	
8	TX (Channel 1)	Start of packet (SOP)	
9	TX (Channel 1)	End of packet (EOP)	
10	RX (Channel 1)	Start of packet (SOP)	
11	RX (Channel 1)	End of packet (EOP)	
12	Pause (Channel 1)	Send of pause_req	
13	Pause (Channel 1)	Receive of pause_ack	
14	PFC (Channel 1)	Send of pfc_req	
15	PFC (Channel 1)	Receive of pfc_ack	

Table 55. MAC/PCS Profiling Events

Chapter 11. SerDes

Keraunos-E100 uses the Alphawave ZeusCore 112G SerDes. This SerDes is chosen for its XLR (extra long reach) functionality, to provide the most flexibility in how we connect the topology between Grendel packages, and because it is Alphawave's offering with the lowest BER (bit error rate). The low BER is a requirement for some datacenter applications of Grendel.

11.1. Harvesting Muxes

Keraunos implements harvesting muxes as shown in Figure 1 to enable harvesting of HSIO tiles independently of the SERDES. Datapath muxing is implemented using pipeline-MUX1H(Mux-OneHot). TX MUX selection is controlled by MACPCS's control register and RX MUX selection is controlled by SerDes's control register. Clock muxing is implemented muxed in the direction from SerDes to MACPCS.

11.2. Lane Swapping

Keraunos supports lane swapping in the configurations listed in Figure 43.

Lane Count per Link	Max MAC Count	ANLT	Fixed	Lane Swap/Swizzle Set - must be inclusive in set	Notes
1	2	Yes	Yes	MAC0: {L0, L1, L2, L3} MAC1: {L4, L5, L6, L7}	Only one lane used/neogitated per 4x lane set
2	2	Yes	Yes	MAC0: {L0, L1, L2, L3} MAC1: {L4, L5, L6, L7}	Only two lanes used/neogitated per 4x lane set
4	2	Yes	Yes	MAC0: {L0, L1, L2, L3} MAC1: {L4, L5, L6, L7}	All four lanes within the range used per 4x lane set
8	1	Yes, limited swap	Yes	MAC0: {L0, L1, L2, L3}, {L4, L5, L6, L7}	Lanes swappable within each set

Figure 43. Keraunos Lane Swap Configurations

Chapter 12. System Address Map

Keraunos-E100 is designed to reuse the Quasar data movement kernels with no changes. The software kernel programming model for both Quasar and Keraunos-E100 uses the Grendel system address map for all addressing.

Address translation between the Grendel system address map and the Keraunos-100 local address map is done in two places in each HSIO tile:

- CCE boundary to HSIO fabric
- Quasar NOC boundary to HSIO fabric

SMC and the Ethernet hardware in each HSIO tile except for the Quasar NOC and CCE use Keraunos-E100 local addressing. [Local Address Map] lists the addresses for the on-chiplet hardware resources in the Keraunos-E100 local address map.

Chapter 13. I/O List

13.1. JTAG Signals (Two sets for IEEE 1838 support)

I/O	Notes
TCK	Primary TAP, Max Frequency 100MHz
TMS	
TDI	
TDO	
TRSTN	
TCK_S	Secondary TAP, Max Frequency 100MHz
TMS_S	
TDI_S	
TDO_S	
TRSTN_S	

13.2. HSIO Signals

13.2.1. D2D Signals

Keraunos-E100 contains 5 D2D links. Each link has the following signals, where n is the link number from 0 to 4:

I/O	Function	Notes
D2Dn_TX0_D[15:0]	D2D link n TX slice 0 data bumps	
D2Dn_CLK_TX0_P	D2D link n TX slice 0 clock bump	
D2Dn_CLK_TX0_N	D2D link n TX slice 0 clock bump	
D2Dn_TX0_AUX	D2D link n TX slice 0 AUX bump used for redundancy	
D2Dn_TX0_FEC	D2D link n TX slice 0 FEC bump used for redundancy	
D2Dn_TX1_D[15:0]	D2D link n TX slice 1 data bumps	
D2Dn_CLK_TX1_P	D2D link n TX slice 1 clock bump	
D2Dn_CLK_TX1_N	D2D link n TX slice 1 clock bump	
D2Dn_TX1_AUX	D2D link n TX slice 1 AUX bump used for redundancy	
D2Dn_TX1_FEC	D2D link n TX slice 1 FEC bump used for redundancy	
D2Dn_TX2_D[15:0]	D2D link n TX slice 2 data bumps	
D2Dn_CLK_TX2_P	D2D link n TX slice 2 clock bump	
D2Dn_CLK_TX2_N	D2D link n TX slice 2 clock bump	
D2Dn_TX2_AUX	D2D link n TX slice 2 AUX bump used for redundancy	
D2Dn_TX2_FEC	D2D link n TX slice 2 FEC bump used for redundancy	
D2Dn_RX0_D[15:0]	D2D link n RX slice 0 data bumps	
D2Dn_CLK_RX0_P	D2D link n RX slice 0 clock bump	
D2Dn_CLK_RX0_N	D2D link n RX slice 0 clock bump	
D2Dn_RX0_AUX	D2D link n RX slice 0 AUX bump used for redundancy	

I/O	Function	Notes
D2Dn_RX0_FEC	D2D link n RX slice 0 FEC bump used for redundancy	
D2Dn_RX1_D[15:0]	D2D link n RX slice 1 data bumps	
D2Dn_CLK_RX1_P	D2D link n RX slice 1 clock bump	
D2Dn_CLK_RX1_N	D2D link n RX slice 1 clock bump	
D2Dn_RX1_AUX	D2D link n RX slice 1 AUX bump used for redundancy	
D2Dn_RX1_FEC	D2D link n RX slice 1 FEC bump used for redundancy	
D2Dn_RX2_D[15:0]	D2D link n RX slice 2 data bumps	
D2Dn_CLK_RX2_P	D2D link n RX slice 2 clock bump	
D2Dn_CLK_RX2_N	D2D link n RX slice 2 clock bump	
D2Dn_RX2_AUX	D2D link n RX slice 2 AUX bump used for redundancy	
D2Dn_RX2_FEC	D2D link n RX slice 2 FEC bump used for redundancy	

13.2.2. Ethernet SERDES Signals

Keraunos-E100 contains 5 Ethernet links. Each link has the following signals, where n is the link number from 0 to 4:

I/O	Function	Notes
ETHn_TX_LN[7:0]_P	Ethernet link n TX lane positive data	
ETHn_TX_LN[7:0]_N	Ethernet link n TX lane negative data	
ETHn_RX_LN[7:0]_P	Ethernet link n RX lane positive data	
ETHn_RX_LN[7:0]_N	Ethernet link n RX lane negative data	
ETHn_REF_P	Ethernet link n external reference clock positive input	
ETHn_REF_N	Ethernet link n external reference clock negative input	
ETHn_ATEST	Ethernet link n analog test mux	

13.3. LSIO Signals

I/O	Testmode Function	Normal Function	Notes
GPIO[63:0]	Scan in/out	See SMC spec	
GPIO[64]	Scan clock	See SMC spec	
GPIO[65]	CLKMON_OUT (clock observation out)	See SMC spec	
GPIO[66]	CLKMON_IN (clock observation in)	See SMC spec	

Chapter 14. Electrical Specifications

14.1. Power Supply Specifications

Power Supply	Voltage	Bump	Category	Notes
VDD_1P80_IO	1.80V	VDD_1P80_IO	GPIO	ref_clock, 1.8V IOs (PMBus, AVSBus, SMBus, UART, I3C/I2C)
VDD_0P75_SYS	0.75V	VDD_0P75_SYS	SMU	SMU: Digital and SRAMs on the same supply
VDD_0P85_SOC_KER	0.85V	VDD_0P85_SOC	SOC	Logic supply for all blocks in HSIO tiles, SMN, QNP Mesh
VDD_0P75_DIG	0.75V	VDD_0P75_DIG_SYS	SMU	SMU: Fixed digital supply for sensors & PLLs
		VDD_0P75_DIG_SOC	SOC	SoC Digital (PLL/sensor): Digital logic inside PLL, PVT, and Vdroop sensors
		VDD_0P75_DIG_FUSE	FUSE (OTP)	OTP read & write (VDD for OTP comes at least 1 us after VDD for SYS)
VDD_1P20_ALG	1.20V	VDD_1P20_ALG_FUSE	FUSE (OTP)	OTP read & write (VDDH for OTP comes at least 1 us after VDD for OTP)
VDDQ_0P75_BOW	0.75V	VDDQ_0P75_BOW_<0-4>	BoW (D2D) PHY	BoW VDDQ (D2D 0.40V-0.75V output drivers)
VDD_1P20_ALG_BOW	1.20V	VDD_1P20_ALG_BOW_<0-4>	BoW (D2D) PHY	BoW VDDA (D2D 1.2V Analog)
VDD_0P75_BOW	0.75V	VDD_0P75_BOW_<0-4>	BoW (D2D) PHY	BoW VDDPHY (D2D single 0.75V VR for digital and analog)
VDD_0P75_PCIE_ETH_KER	0.75V	VDD_0P75_PCIE_ETH	PCIe/Ethernet SerDes	PCIE Tile Logic, Support Logic for Ethernet SerDes IP
		VPDIG_0P75_PCIE	PCIe	PCIE IP Digital Core
		VDD_P_0P75_ETH	Ethernet SerDes	Ethernet SerDes IP Digital Core (P_VDD for Ethernet SerDes)
		VP_0P75_PCIE	PCIE	PCIE IP Analog Core
		VDDL_PA_0P75_ETH_<0-4>	Ethernet SerDes	Ethernet SerDes IP Analog Core (PA_VDDL for Ethernet SerDes)
VDDH_1P20_PCIE_ETH_KER	1.20V	VPH_1P20_PCIE	PCIe	PCIE IP IO
		VP_OD_1P20_PCIE	PCIe	PCIE IP Analog Core
		VDDH_PA_1P20_ETH_<0-4>	Ethernet SerDes	Ethernet SerDes IP Analog Core (PA_VDDH for Ethernet SerDes)
VSS	GND	VSS	VSS	Ground for all supplies
	Analog GND	AVSS_OTP	VSS	Analog ground for VSS

14.2. Clock Specifications

Table 56 details all the clocks used in the Keraunos chiplet.

Name	Signal Name	Frequency (TT/SSPG)	DFS	Clock Source	Comment
Reference clock	ref_clk	100 MHz/100MHz	No	External pin	
JTAG clock	jtag_clk	100 MHz/100MHz	No	External pin	TCK in JTAG pins
System Management Clocks					
SMU (SMC/SEP) clock	smu_clk	800 MHz/800 MHz	No	CGM0 in SMU	
SMN clock	smn_clk	400 MHz/400 MHz	No	CGM1 in SMU	Clock for SMN, HSIO MAC/PCS/Serdes Cfg clock

PVT Sensor clock	pct_clk	200 MHz/200 MHz	No	CGM2 in SMU	
D2D Clocks					
Die-to-Die clock		2000 MHz/2000 MHz	No	PLL in BoW PHY	Die-to-Die Interface
QNP clock	qnp_clk	1500 MHz/1200 MHz	No*	CGM5/6 in SMU	Clock for QNP mesh (NOC2AXI in D2D). *Can use CGM5/6 + mux for DFS
HSIO Tile Clocks					
CCE clock	cce_clk	1650 MHz/1650MHz	Yes	CGMs in HSIO tile	Clock for CCE, 2 CGMs+mux for DFS
HSIO clock	hsio_clk	1650 MHz/1650 MHz	No*	CGM3/4 in SMU	Clock for HSIO fabric, TT Eth Ctrl, SRAM, DFD, Profiler, Error bank. *Can use CGM3/4 + mux for DFS
MAC/PCS clock	mac_clk	1600 MHz/1600 MHz	No	CGM in HSIO tile	Clock for Ethernet MAC/PCS
Ethernet SerDes lane clocks	serdes_lane_clk	645-831 MHz/645-831 MHz	No	Generated by SerDes	Clock for MAC/PCS <-> Ethernet SerDes interface
Ethernet SerDes reference clock	serdes_ref_clk	156.25 MHz/156.25 MHz	No	Bump on HSIO[2] Ethernet SerDes	Reference clock for all Ethernet SerDes, Differential clock on bumps of HSIO[2] SerDes and routed to all other SerDes. Driven by cystal oscillator on PCB. Frequency of 312.5 MHz is acceptible, but no advantage and will be divided internally to 156.25 MHz..
PCIE Clocks					
PCIE Clock	pcie_dclk	1000 MHz/1000 MHz	No	CGM in PCIe tile	PCIe tile main clock (PCIE controller, NOC-PCIE)
PCIE SerDes AHB Clock	pcie_hclk	500 MHz/500 MHz	No	CGM in PCIe tile	PCIE SERDES AHB clock, divide by 2 from PCIE clock
PCIE PIPE Clock	pclk	1000 MHz/1000 MHz	No	PLL in SerDes	Clock for PIPE interface

Table 56. Keraunos Clocks

14.2.1. PLLs

All PLLs in Keraunos including the ones in SMU are CGM PLLs.

The clock source column in Table 56 details the PLL placement for each of the clocks in Keraunos. The following clocks are present in each HSIO tile but there is no skew requirement between HSIO tiles for them. Thus the global portion of the clock tree is allowed to be unbalanced:

- MAC/PCS clock
- CCE core clock

Additionally, though it is functionally sufficent to have one PLL source for each of the above clocks in SMU, PD is allowed to choose to put some of the PLLs locally in each HSIO tile to simplify the global clocking.

14.2.2. DFS

Keraunos primarily supports relatively slow-speed DFS for the CCE core clock, which enables the CCE DMRISC cores to operate in a lower frequency when they are not doing useful work (e.g. spinning in a busy-wait loop). This is achieved by using two CGMs multiplexed with a glitch-free mux. CCE core clock PLLs are placed each of the HSIO tile, allowing DMRISC cores (of CCE0/1) in an HSIO tile to switch frequency independently of the DMRISC cores in other HSIO tiles.

Keraunos also supports slow-speed DFS for HSIO clock by using the existing CGM3, CGM4, and a glitch-free clock mux block in SMC. Similarly, Keraunos also supports slow-speed DFS for QNP clock by using the existing CGM5, CGM6, and a glitch-free clock mux block in SMC. There is not strong use-case for this DFS capability on both HSIO clock and QNP clock but can be taken advantage as it is already present in the SMC IP.

14.2.3. Clock Domains

Figure 44 illustrates the clock domains and the placement of CDCs in the Keraunos SOC. Note that only the various clock domains used in the PCIE and HSIO tiles are highlighted. The detailed clock domains and CDCs for HSIO are shown in Figure 45, and that for PCIE tile can be found in the Keraunos PCIE tile Specification (Chapter 2, Figure 1).

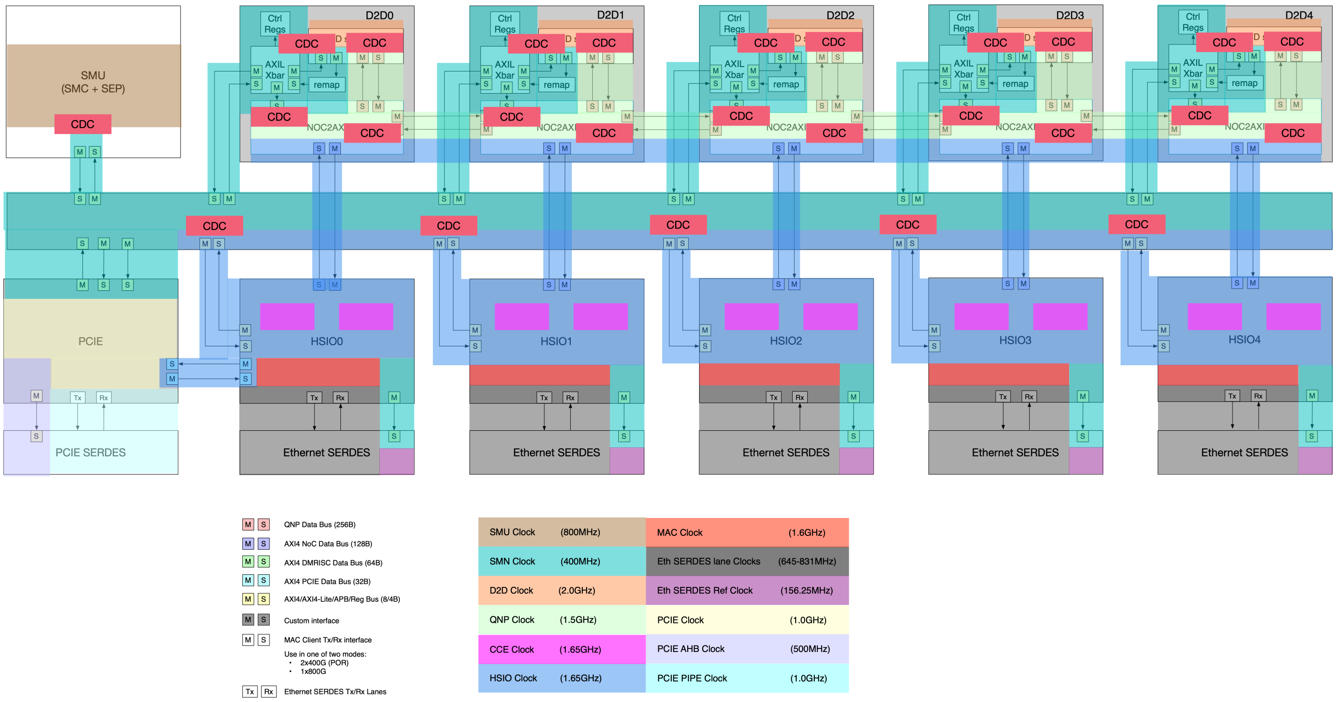


Figure 44. Keraunos Clock Domains

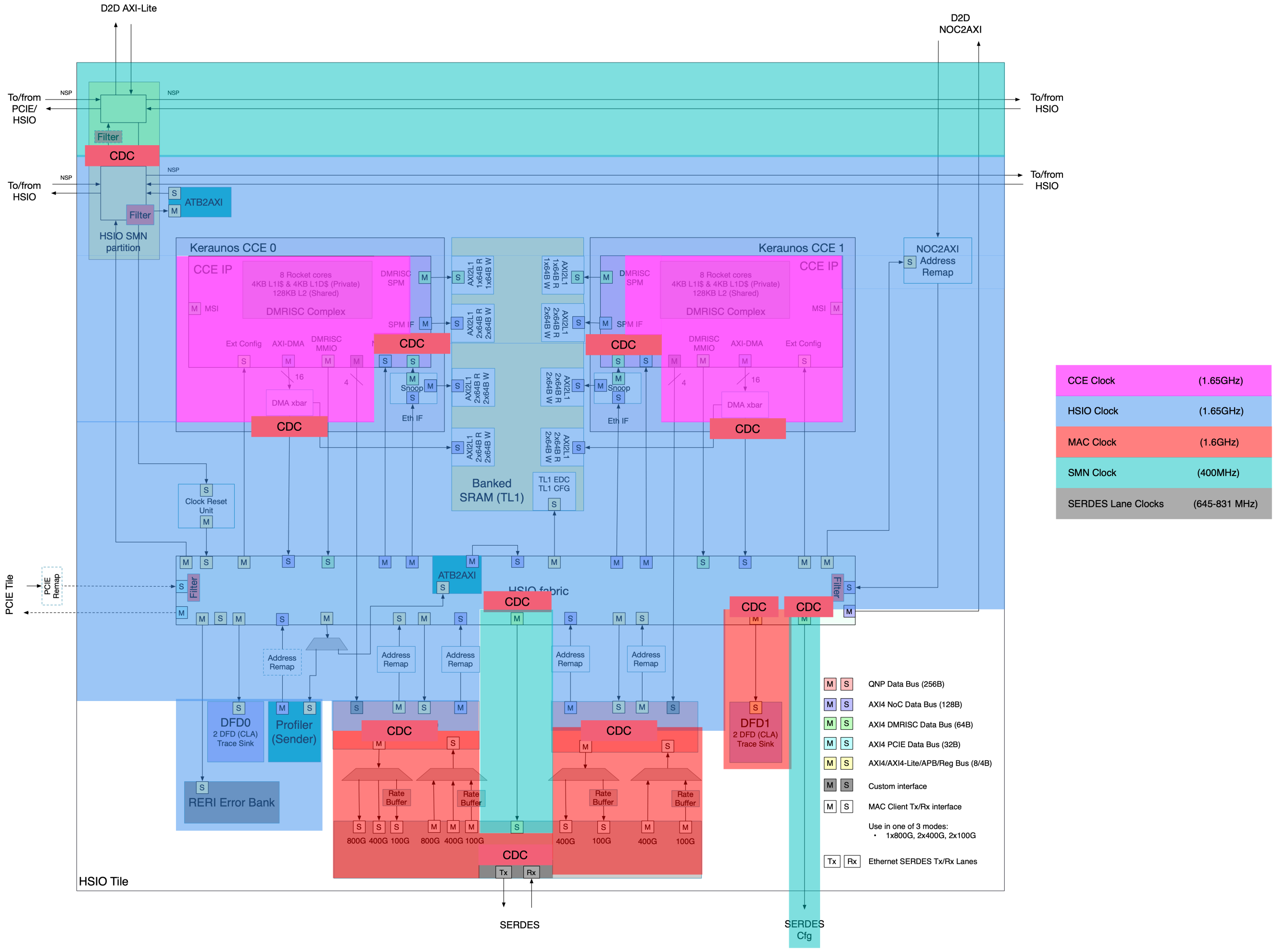


Figure 45. Keraunos HSIO Clock Domains

14.2.4. Clock gaters

Each clock supports a functional clock gater, placed next to the source CGM. Clock gater is driven by the debug clock stop logic in SMU. For clocks in SMU, hsio_clk and qnp_clk, clock gaters are placed in SMU while for clocks in HSIO tile, mac_clk and cce_clk, clock gaters are in HSIO clock reset unit.

14.2.5. Reference clock muxes

Each clock has a glitch-free clock mux to select between the reference clock and the generated clock from the CGM. These clock muxes enable the force reference clock functionality needed for reset assertion/deassertion. For clocks in SMU, hsio_clk and qnp_clk, ref clock muxes are placed in SMU while for clocks in HSIO tile, mac_clk and cce_clk, ref clock muxes are in HSIO clock reset unit.

Chapter 15. GPIO

15.1. PTP TimeSync to GPIO

GPIOs are expected to pass from sub-modules to top-level SMC module which will handle GPIO multiplexing. This section describes the connection of PTP debug GPIO attachment to the SMC.

15.1.1. PTP GPIO Attachment to SMC

Figure 46 shows the connectivity between two GPIOs and HSIO PTP timestamps through HSIO Mux.

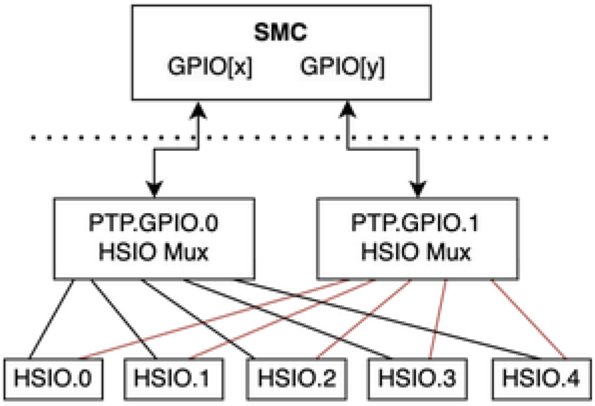


Figure 46. Connectivity between SMC and HSIO PTP timestamps

Keraunos and SMC must have: . 2x GPIOs to muxed out . Should allow for output driver, output enable and input - total of 6x wires (3x wires per functional mux in SMC * 2x GPIOs)

15.1.2. SMC Functional Mux Assumption

It is assumed that there will be a functional IO mux in the SMC and PTP IO will be muxes with an some other functionality. A drawing of this functional mux is shown below, for reference.

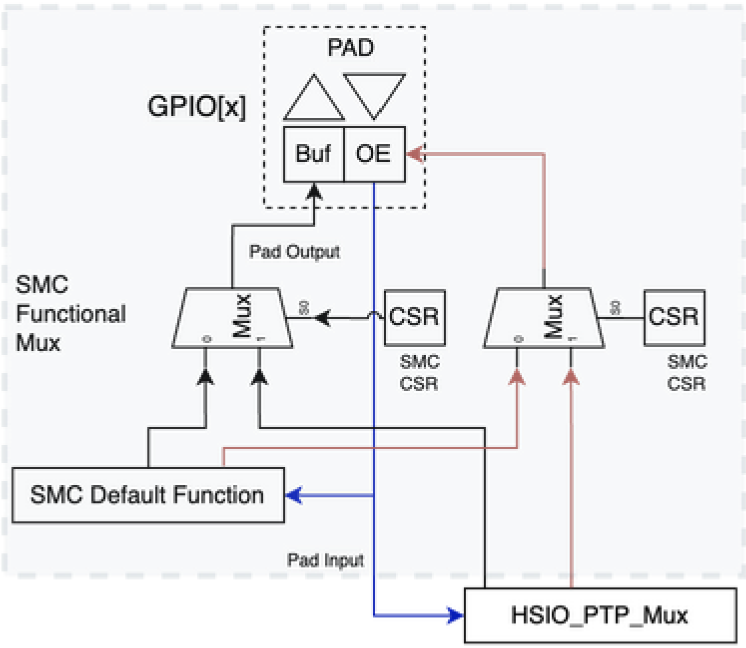


Figure 47. SMC Functional Mux Assumption

The SMC being the management unit should have CSR bits to control functional muxes. . CSR.GPIO[x].Output.Select = 1'b0 - SMC default function, 1'b1 - PTP Control . CSR.GPIO[x].Output.Enable = 1'b0 - SMC default function, 1'b1 - PTP Control . CSR.GPIO[y].Output.Select = 1'b0 - SMC default function, 1'b1 - PTP Control . CSR.GPIO[y].Output.Enable = 1'b0 - SMC default function, 1'b1 - PTP Control

As specified in Section 4.10 in Grendel System Management Architecture Specification, the two PTP IOs are muxed with UART_1_TX and UART_1_RX, and connected to BP_GPIO_15 and BP_GPIO_16.

15.1.3. HSIO PTP Mux

The goal of the HSIO PTP Mux is to push out a ToD adjusted counter value or current timestamp value. This feature is used for PTP bring up (both HW and SW), accuracy measurements, and debug. It will be used by all teams and needs to be multiplexed out to a header with GND with at least two simultaneous used GPIOs.

The value out of the PTP TS will be internally selected but we want to drive out a selectable bit from either ToD selectable bit, ToD+Adjusted selectable bit, or whatever other fields Alphwave will give us on the 1588/PTP timestamp. We want the bit to be selectable per port for both ports within a given HSIO tile.

Figure 48 shows the logical diagram of the HSIO PTP Mux.

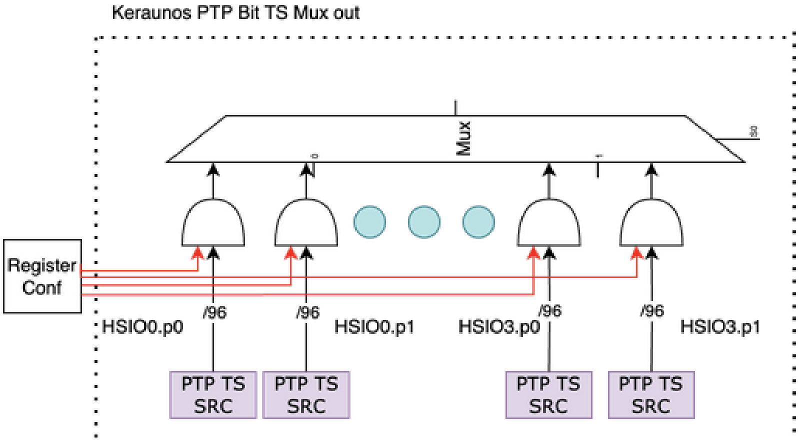


Figure 48. HSIO PTP Mux

HSIO PTP Mux Microarchitecture

As shown in Figure 49, HSIO PTP Mux is implemented in a distributed manner across HSIO tiles to minimize long wires running across the chiplet. PTP timestamp from HSIO4 is masked and sent to HSIO3, where it is ORed with the masked HSIO3 PTP timestamp, and further sent to the HSIO2. The same is repeated until HSIO0, which drives the final PTP GPIO outputs to SMC. The masked PTP timestamps pass through a 1-stage pipeline register at each HSIO tile boundary to meet timing. To ensure PTP timestamp transmission from each HSIO tile is same, additional pipeline registers are added within each HSIO tile as needed.

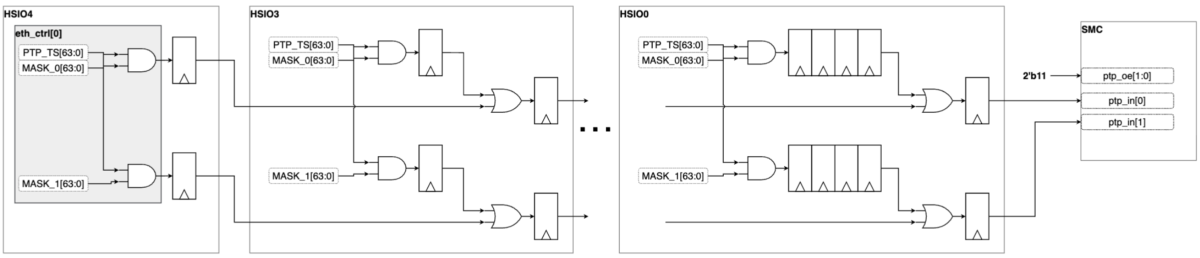


Figure 49. HSIO PTP Mux

Since MACPCS uses timestamp32s32ns for ToD, timestamp32s32ns is also used as the PTP timestamp to GPIO.

Chapter 16. PTP

Keraunos-E100 supports IEEE 1588 precision time protocol. Each Ethernet controller can be configured to be a PTP master or a PTP slave. Keraunos-E100 also supports loading PTP time to the PTP masters from the SMC as well as using the PTP time from the Ethernet controller as the source for the Grendel system time used by SMC and by the debug and profiling hardware.

Chapter 17. Security

Keraunos-E100 is compatible with the Grendel Security Specification.

Keraunos supports the following security features:

17.1. SEP and ISS

Keraunos includes SEcurity Processor (SEP), which acts as the Root of Trust (RoT) at the highest security level in SoPs where Keraunos is the primary chiplet. As detailed in Grendel Security Specification (Section 3.1), SEP is in charge of secure boot, secure lifecycle control, user authentication, secret key services, etc. SEP is isolated from the rest of SoC by its physical boundary.

As defined in Grendel Security Specification, ISS (Isolation SubSystem) is a domain that runs platform management tasks whose confidentiality and integrity only depends on itself and SEP. In Keraunos, the SMC is an ISS.

17.2. Runtime Security

This describes the additional security support provided by the chiplet in addition to that provided by the SEP and SMC instances.

17.2.1. Support for SEP and ISS Security

SEP and SMC (which is ISS) have private assets within the SEP and SMC instances, and also outside of these instances. All private assets inside of SEP/SMC are protected by SEP/SMC's inbound filters. Access to SEP/SMC private assets outside of SEP/SMC rely on NOC traffic filters provided by the SMN and HSIO fabrics in Keraunos (defined in xref:[]). SMN and HSIO fabrics propagate SEP and ISS source IDs with their traffic so that source ID check can be done at the NOC traffic filters. This includes sending over source ID across D2D to other chiplets. The source ID allocation for Keraunos is defined in Table 57.

Initiator	Source ID
SEP	4'b1111
SMC	4'b0011
Others	4'b0000

Table 57. Source ID allocation for Keraunos

17.2.2. HSIO Virtualization

Keraunos allows HSIO tiles to be allocated to different VMs at the same time. All transactions initiated from an HSIO tile, which include traffic initiated by CCEs, traffic from Ethernet Rx or logs from Profiler, are tagged with a group ID when exiting the HSIO tile. NOC traffic filters in HSIO fabric are used to only allow transactions within their own group ID. Note, group IDs are allocated at runtime depending on how HSIO tiles are allocated to different VMs.

Chapter 18. DFT

Keraunos-E100 implements the common Tenstorrent testability architecture. Please refer to the Grendel testability architecture spec [pending].

Chapter 19. Debug

19.1. CLA (Core Logic Analyzer)

19.1.1. CLA Summary

The CLA is a general purpose debug engine that provides control/observe oflogic curcuitry. It can be used for debug and trace for logic within an IP of interest. It is used for system-level debug and is instantiated into IPs and uses the same clock as the IP it is observing. It is useful for both functional and electrical issue debug.

The CLA takes a debug bus as an input. The debug bus is a cone of MUXing that takes a large number of interesting signals and MUXes it down to a 64b bus, that is supported by the CLA. The CLA is a programmable state machine that can be used for a number of triggering/tracing capabilities. The CLA is coded in a parameterized manner, so it can be scaled appropriately for various applications.

The CLA contains a number of general purpose counters that can be used to trigger on time thresholds, count events and more. The CLA is programmed through a register interface, and is accessible from JTAG or from a codestream. The debug bus is pipelined through the CLA to form a trace bus for storing a history of activity. The trace is compressed to save trace bandwidth and capacity. The trace can be made selectable by programming the CLA to only store during specific clock cycles or windows. The trace is timestamped, based on the GlobalTimestamp bus. The Globaltimestamp must increment on a fixed clock and not be subject to power management operations.

The CLA has cross-triggering inputs and outputs that allow it to be connected to all other CLAs in the system, allowing for coordinating debug activity across the entire system. This includes a nuber of activities, such as:

When coupled with cross-triggering, CLAs can track activity across the system - An example of this is where one CLA signals the occurence of an event through a cross-trigger, indicating to another CLA that it can begin looking for a related event.* Coordinating a package-wide clock stop - Doing a stop clock and scan is a proven technique for debug. CLAs can trigger a clock stop at a time of interest. To perform a package-wide clock stop, cross triggers are employed. Each CLA can cause a chiplet-wide clock stop, but the package-wide cross triggering can be used to stop all the chiplets within a package.

CLAs are connected to storage structures to be able to store a history of activity. There is flexibility in where the CLA trace is stored. Optinally, there is some dedicated SRAM. While dedicated SRAM is expensive, it is also completely nonintrussive to debug. Due to the cost of dedicated SRAM, is its desireable to dump to other storage strucures, including:

19.1.2. CLA Trace

The CLA supports dumping trace to storage structures. This trace is retreived for analysis with waveform viewing capabilities. * Main memory * Existing cache structures * Off-chip to a logic analyzer via PCIe or other

What trace storage is available for any particular CLA varies, depending on where instance is in the architecture.

A timestamp value is sent to each CLA to use for triggering and trace. This allows for being able to trigger on a time value. The timestamp values are also used within the trace. This is to support combining traces from anywhere in the system with everything properly ordered for debug across the context of the entire system.

CLAs are programmed wuth Event-Action Pairs (EAPs). EAPs define a mapping of decodes of the debug bus and the internal state of the CLA to actions. Actions can be internal, like incrementing a counter or changing the node (state) of the CLA. Actions can also be external, like firing a clock stop or starting trace stores.

19.1.3. CLA Instances in Keraunos

The table below shows information on all the CLAs in the Grendel system.

CLA Instance(s)	Clock Domain	Trace Destination	Interrupt Destination	Comments
SMC	soc_clock (400 MHz)	32 kB SRAM, Spill to GDDR7?	APLIC	
HSIO Slice 0 (1/HSIO slice x 5 HSIO tiles)	Ethernet	Shared w/Slice 1; 64kB SRAM		
HSIO Slice 1 (1/HSIO slice x 5 HSIO tiles)	Ethernet	Shared w/Slice 0; 64kB SRAM		

CLA Instance(s)	Clock Domain	Trace Destination	Interrupt Destination	Comments
HSIO Slice 0 (1/HSIO slice x 5 HSIO tiles)	MAC	Shared w/Slice 1; 64kB SRAM		
HSIO Slice 1 (1/HSIO slice x 5 HSIO tiles)	MAC	Shared w/Slice 0; 64kB SRAM		
PCIe Controller	PCIe	Dedicated 32kB SRAM, Spill to DDR?	APLIC	

19.1.4. CLA Register spaces

19.1.5. Timestamping

CLAs use timestamping for tracing, so that traces from all CLAs can be combined to form a coherent view of activity across the system. Timestamping is maintained in the CLA and is driven from RefClk

19.1.6. Cross-Triggering

The goals of debug cross triggering include:

- Ability to trigger from/to any CLA(s) in the package
- Minimal, fixed latency operation
- Reliable
- Useable from reset de-assert

There are two cross triggers in the chiplet that are connected to all CLAs in the chiplet. They are also expected to be connected between chiplets as well, so they can be utilized across the entire package.

Cross triggering within the Keraunos chiplet is accomplished over a logic network. Cross triggering between chiplets within a package is accomplished using a wire-OR scheme. The wire-OR is simple, requires no initialization or training and is somewhat low latency. It can also be employed at the package level to provide external control/onserve of cross triggers.1G1Gkkkk

19.1.7. Cross-Triggering Schemes

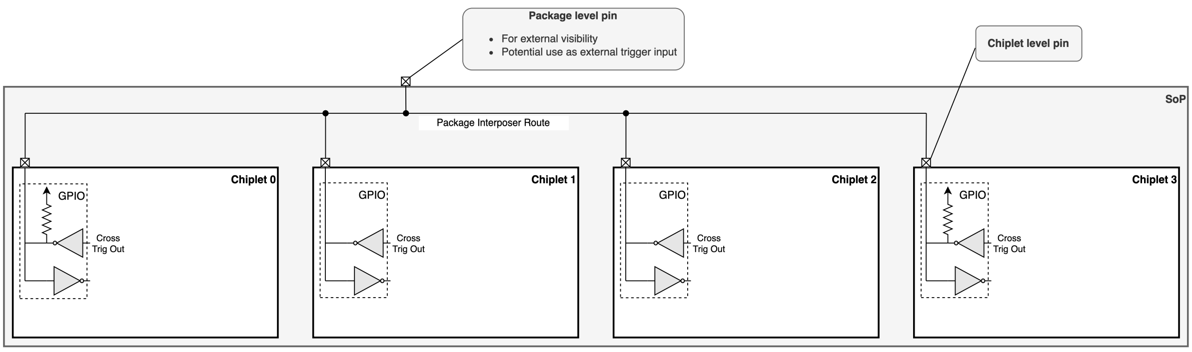
Chiplet Internal Cross Triggering

Cross triggering within Keraunos connects all CLAs together through a network that ORs together the cross triggers from all CLAs in the chiplet.

Functionally, any CLA asserting a cross trigger onto the logic network will see its own cross trigger reflected back to itself. This has implications in CLA usage and programming.

Package-Level Cross Triggering

It is intended that cross triggering between chiplets is supported through a wire-OR scheme.



The illustration above shows the logic sheme used to enable this.

The illustration below shows the cross trigger connectivity and functionality at the top-level within a chiplet. This block lives in the

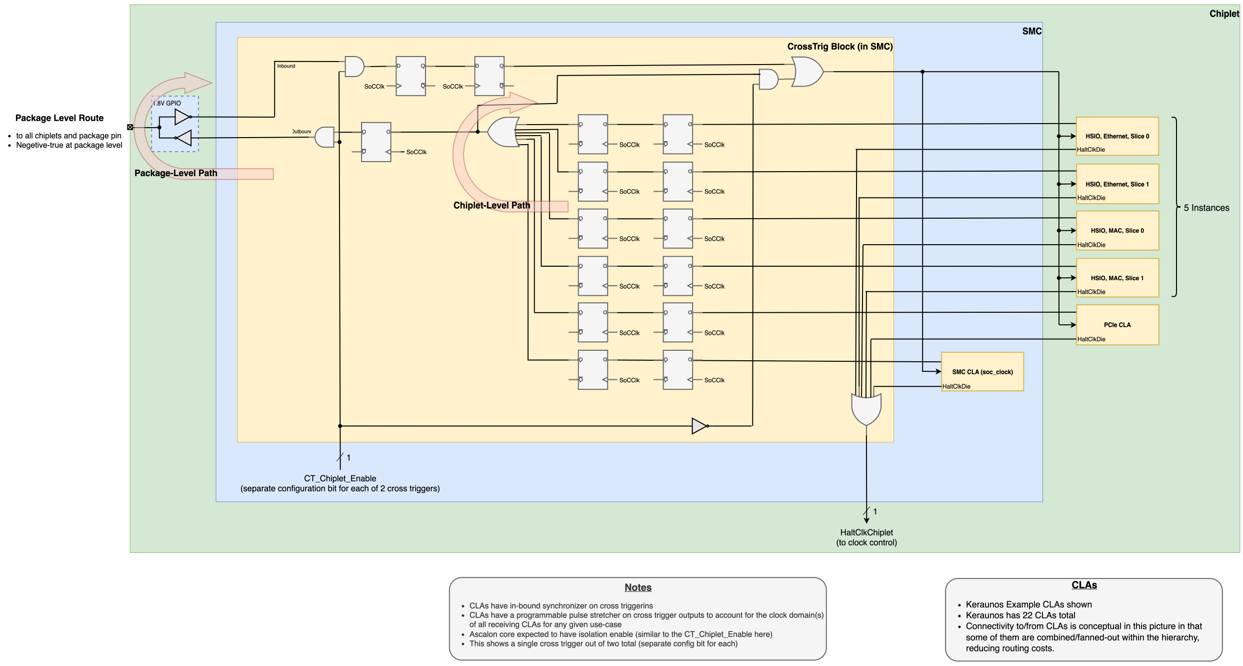
SMC of each chiplet. The Athena chiplet is shown as an example, including an arbitrary number of additional participants up to "Subsystem M-1". Each source clock domain is handled as a single inbound and outbound cross trigger, regardless of the number of CLAs it contains. For example, each Ascalon cluster has 16 CLAs (1 per core and 1 per SC block, each with 8 instances).

There is a separate programmable pulse stretcher in each clock domain in order to ensure that cross triggers froma faster clock domain are obseved in the triggering clock domain. The number of bits of stretch (n) is 5, which is enough to cover the full CPU core frequency (2.75 GHz) down to RefClk (100 MHz).

The RefClk flop is to limit the switching reequency, before all the outbount triggers are ORed together before going to the pad. Because we do not expect to meet timing to RefClk between all chiplets, there are 2 RefClk domain synchronizer stages on the in-bound path.

Functionally, any chiplet asserting a cross trigger onto the network will see its own cross trigger reflected back to itself. This has implications in CLA usage and programming.

The following picture is shown as an example, with Athena clusters, the SMC and extra, hypothetical blocks just as an example.



Considerations for cross triggers at the chiplet top-level:

- This block lives in the SMC, which is physically close to the GPIOs that are employed for package-level cross triggering.
- The RefClk synchronizer in the SMC takes the liberty of putting an OR gate in between the stages. While it is recognized that this additional propagation delay detracts from the metastability avoidance, that should be mitigated by the long RefClk period and low activity factors of cross triggers. The motive for this is to reduce latency.
- Each Ascalon cluster treated as a single source since all it's CLAs are in the same clock domain
- Inbound cross triggers synchronized to local clock internally.
- Outbound cross triggers stretched sufficiently to ensure they are received by all chiplets.
- Inbound and outbound cross triggers have a mask to isolate from package level cross triggers. This enables simultaneous use of package-level cross triggering and separate local cross triggering (Ex: within an Ascalon cluster)
- The number of ports is parameterized in RTL
- The number of ports for each chiplet is optimized for flexibility across all future SoPs. ==== Debug Bus

19.2. Scan and Aray Dump (Structural State Access)

Structural state access here refers to the ability to extract the state of the system after stopping functional clocks, which freezes the state of the system. This access is over the JTAG port. Once a clock stop is performed, the system cannot operate functionally again without doing a reset.

The state we refer to here falls into the categories of flop state and array state.

19.2.1. Flop State Access

- Accessed via structural single-chain scandump and covers almost ALL flops in design. Some flops are not scannable for various reasons. One example is with flops that control external power supplies, where making them scannable would cause the power supply voltage to erratically fluctuate as arbitrary data is shifted through during scan.
- Scan infrastructure is supported for manufacturing test. While manufacturing test has more efficient ways of doing scan (scan compression/decompression), it is basically there for free (although requires extra verification for system-level debug usage).

The functional clock stops within a chiplet have multiple types. One type is localized to a block, another is across the entire chiplet. In order to perform functional clock stops across multiple chiplets (or multiple packages), the cross trigger infrastructure must be used. In this usage, a CLA on each chiplet can fire a die-wide clock stop for it's die. Cross triggering is used to signal from the triggering CLA in one die (master), over to other CLAs (slaves) that are firing clock stops in their die. The latency of the clock stop in the slave die could be significant, depending on the type of cross triggering that is employed.

19.2.2. Array State Access

Array state is extracted after doing a scan dump. Arrays that have MBIST use their MBIST engines to extract the contents over JTAG. MBIST is inserted by the DFT flow in the back-end. For latch arrays, we have a scheme for extracting their contents using the "latch array dump controller". The latch array dump controller is an RTL wrapper that is added to each latch array. A single instance of latch arraay dump controller can handle multiple arrays.

- MBIST Arrays – Accessed while functional clocks running and covers most of array macros. MUST include mechanism for extracting array state after a scandump. This includes the ability to protect array state through a scandump from a TDR bit.
- Latch-Based Arrays – Accessed using new mechanism similar in nature to MBIST. Only applied to latch arrays deemed important for debug. MUST include mechanism for extracting array state after a scandump. This includes the ability to protect array state through a scandump from a TDR bit.

19.2.3. Scan and Array Dump Flow

The state access flow consists of the following sequence:

- Clock(s) Stop
  - Functional clock stop (from JTAG or CLA(s))
- Prepare for Scan Dump
  - If stop from a CLA: Set clock_stop_global TDR - This keeps the OCC disabled though scan, since the CLA clock stops will wiggle during scan.
  - Array Protect Before Scan
    - Write MBIST array write-protect TDR(s) to preserve MBIST array contents through scan
      - **Array protect TDR must be reset on powergood, since it needs to be stable through scan operations. The key point is that it does NOT reset on a warm reset (BH bug).**
    - Write latch array write-protect TDR(s) to preserve latch array contents
      - **Array protect TDR is NOT must be reset on powergood, since it needs to be stable through scan operations. The key point is that it does NOT reset on a warm reset (BH bug).**
- Scan Dump - Flop State Extract
  - Configure scan chains to include desired domains.
  - Perform single-chain scan to capture flop state
- MBIST Initialization Disable
  - Write to MBIST array initialization disable TDR since no need for MBIST init.
    - MBIST array initialization TDR is NOT scannable, since it needs to be stable through scan operations.
- Array State Extract
  - Perform warm reset to get functional clocks running again in preparation for MBIST array access
  - Read MBIST array contents for each array
  - Read latch array contents for each array

19.3. Register Access

Register Access from JTAG provides back-door access to critical system state that is important for debug. JTAG access to internal state can be entirely independent of mission-mode functionality, or it can leverage mission-mode functionality. Having entirely JTAG access to system state provides visibility to that state that is independent of any hang condition. However, it is very costly to have this type of access to everything. An alternative is to have a translation block that allows a single JTAG endpoint to gain control of a mission-mode system bus. With this scheme, the entire address space of the system bus is now visible through this single JTAG endpoint. This provides a large amount of visibility, without a high cost. The downside is that if the system bus becomes hung, this visibility can be lost.

Debug software consumes a Register Description Language (RDL) file. The RDL file content is an object modeling language used to specify and implement software accessible hardware registers and memories.

19.4. Boot Flow Debug

19.5. SMC Debug

Signal	Width	Description	Comments
Alias remap hit	64	When traffic passes through alias remapping, track which region was hit. 3 bits per remap, two remaps for read and write, 4 remaps total	
ATB sink	24	Telemetry data valid	
AVS states	217	States of the AVSBus state machine	

Note: The above is still under discussion and is not finalized. See: docs.google.com/spreadsheets/d/1-QnbEUE77pgLxh_j1Ie3yqOhnv0D2h4-dmFD4Y5VR50/edit?gid=1105269734#gid=1105269734

19.6. CCE Debug

19.7. Clock Observabiity

For clock observability the DFT team has implemented a block to facilitate this.

This block supports the following:

- Clock select MUXing.
- A ripple counter for measuring clock frequencies.
- Clock divider, to reduce the input clock frequency down to what the ripple counter can tollerate.
- Clock pulse counting based on TCK.

19.8. Analog Test Bus/Digital Test Bus

- The ATB would be routed to the dispatch engines in the NW and SE corners. The PD team believes noise should not be a concern since this is a differential pair. This way we are not missing any D2D instance's susceptibility to jitter. @Michael Smith - this was proposed after you had to drop.
- This would require 2 bumps for each of the arrows (4 bumps totals) because the VDDPHYs of the vertical and horizontal edge are physically separate.
- The select for the ATB will default to tri-state and be driven only if explicitly programmed. The registers for this are missing in the RTL today. @Tejpal Singh / @Deepinder Singh - to keep track.*

This is still being finalized and will be updated after that.

19.8.1. D2D

19.8.2. Ethernet

19.9. Chicken Bits

Chapter 20. JTAG

Keraunos-E100 implements JTAG-based testability that conforms with the Grendel testability spec [pending].

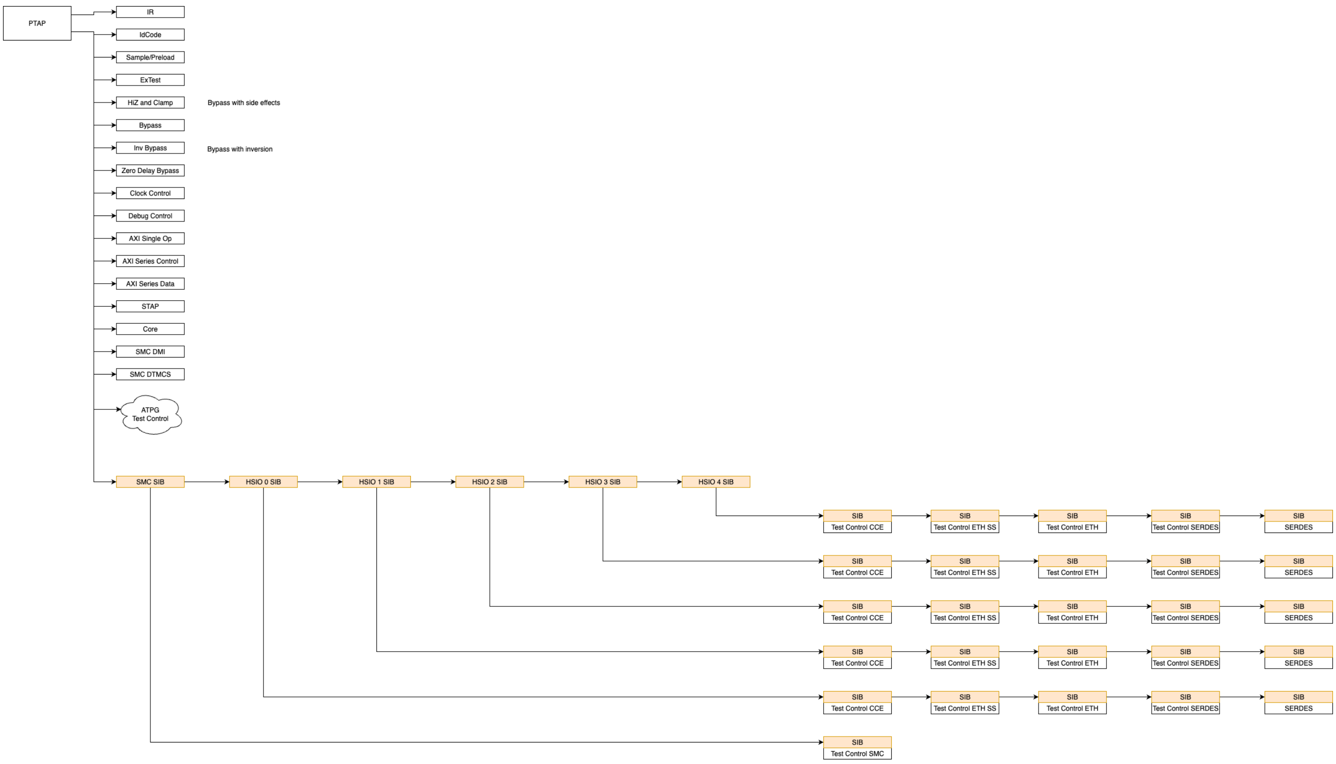
The following capabilities are supported:

- IEEE 1149.1 compliant boundary scan for all I/Os except JTAG (and power-on-reset if applicable)
- Primary and secondary TAP controllers for IEEE 1838 support
- Single internal scan chain for all onboard JTAG components with IEEE 1687-compliant iJTAG semantics (partitioned into hierarchy with SIB bit per segment)

A BSDL description of the boundary scan chain will be provided. An ICL description of the iJTAG network will be provided.

20.1. iJTAG Network

This will continue to be refined to align with the microarchitecture and physical design. The network will only be finalized at tapeout because the ordering is dictated by physical design constraints.



The detailed iJTAG network toplogy is presented in Appendix A as an ICL file. This file is designed to be an executable architectural specification in that the RTL should be implemented to match the topology and DV can directly use the ICL file in test generation to verify RTL compliance.

Chapter 21. Keraunos RAS

When a chiplet in the Grendel SOP detects soft/hard errors, the error is recorded and the event is signaled to the RAS Agent so that it can take appropriate action for the error.

The RVI RAS Error-record Register Interface (RERI) standard provides a framework to record and signal RAS errors. There are many options and possibilities in implementing the RERI specification.

The Grendel chiplets supports both IP specific error records and signaling in addition to the RERI specification with various layouts and error events are indicated by the version of the IP (TT and otherwise) included in the chiplet. Details about the RAS Architecture components, modules, and paradigms applied to the chiplets comprising the Grendel SOP are found in the Tenstorrent RAS Architecture. The information for each of Grendel SOP's IPs or chiplets may be duplicated here to provide a single reference point.

References . RERI Specification . RAS - RVI and TT

21.1. Keraunos E100 Chiplet RAS

The Keraunos E100 chiplet has 7 major RAS related components:

- Chiplet Harness
- AlphaWave OmegaCore MAC/PCS
- SERDES
- TT Ethernet controller
- CCE
- TL1
- D2D
- NOC

The chiplet harness RAS error events are duplicated here for completeness. Though all the components provide their own status registers for errors, they also provide a series of buses that are sampled to create TT RAS Atoms, Element, and Error Events. The resulting Error Events are recorded in the chiplet's error banks.

The error outputs are as shown in [:tab-keraunos-error-outputs].

IP	Signal
AlphaWave Omegacore MAC/PCS	`ocore_mac_cl_800ge_sram_ecc_err`
	`ocore_mac_cl_400ge_sram_ecc_err`
	`ocore_mac_cl_100ge_sram_ecc_err`
	`ocore_pcs_400ge_sram_ecc_err`
	`ocore_pcs_100ge_sram_ecc_err`
	`ocore_pcs_100gbe_sram_ecc_err`
	`ocore_pcs_generic_sram_ecc_err`
SERDES	`serdes_sram_ecc_sbit_err`
	`serdes_sram_ecc_dbit_err`
Ethernet Controller	`txq_ctrl_req_buf_parity_err`
	`tx_pkt_buf_parity_err`
	`rx_pkt_buf_parity_err`
CCE	i_wdt_timeout
	dcache_ecc_uncorrectable
	l2_banks_parity_uncorrectable[15:0]
	l2_dir_parity_uncorrectable[3:0]
	context_switch_parity_error

IP	Signal
TL1	i_tl1_fatal_err_irq
	i_tl1_critical_err_irq
	i_tl1_correctable_err_irq
	i_tl1_req_fp_overflow_Ored
	i_tl1_req_fp_underflow_Ored
	i_tl1_req_fp_underflow_Ored
	i_tl1_req_fp_underflow_Ored
D2D	`o_noc_sec_fence_irq_0`
	`o_intp_id_transaction_count_rtz_0`
	`o_n2a_ecc_error_0_0`
	`o_n2a_ecc_error_1_0`
	`o_n2a_ecc_error_2_0`
	`o_n2a_niu_timeout_intp_0`
NOC	`o_timeout_irq` (HSIO fabric)
	`o_smn_tile_timeout_irq` (HSIO SMN tile)

Table 58. Errors from Keraunos components

21.1.1. Keraunos Error Signaling

This section documents the various error banks in Keraunos and its organization. Keraunos has 5 error banks in each HSIO tile and 1 error bank in D2D tile.

HSIO Error Banks and Signaling

There are five error bank instances in HSIO tile:

1. Keraunos CCE0: CCE Error Bank records errors in CCE0 and SRAM.
2. Keraunos CCE1: CCE Error Bank records errors in CCE1 and SRAM.
3. TT Ethernet Controller 0: TT Ethernet Controller Error Bank records errors in TT Ethernet Controller 0
4. TT Ethernet Controller 1: TT Ethernet Controller Error Bank records errors in TT Ethernet Controller 1
5. MAC/PCS: MAC/PCS Error Bank records errors in MAC/PCS, SERDES, and HSIO fabric

There is only one TL1 shared between CCE0 and CCE1. To prevent the same error from being recorded in both CCEs sharing the TL1, the CCE provides the ability to mask TL1 events using a configuration bit programmable via MMR. This should be used to prevent the same error from being recorded in both CCEs.

Figure 50 shows the organization of the Error Banks in the HSIO tile, along with the configuration ports to the Error Banks and the RAS signaling via HSIO RAS signal reducer.

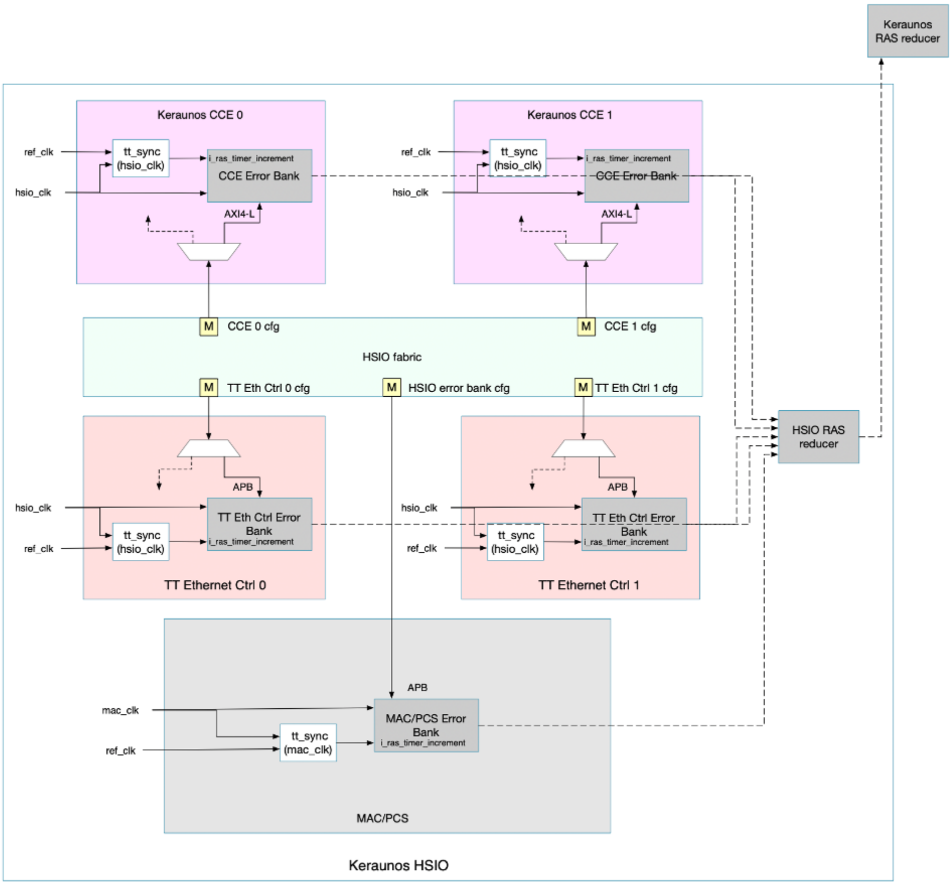


Figure 50. HSIO Error Banks

Keraunos SOC RAS signaling

Figure 51 shows the RAS signaling from D2D Error banks and HSIO RAS reducers to SMU.

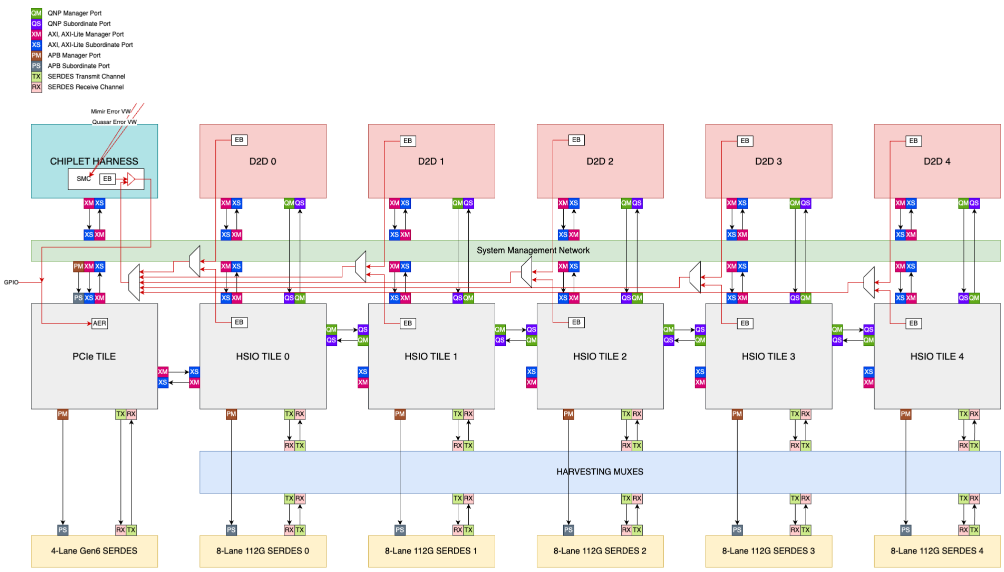


Figure 51. Keraunos SOC RAS Signaling

Every chiplet has a GPIO signal for uncorrectable and uncontained errors. A MMR controls which error signal or combination of signals results in asserting this GPIO output. The reset configuration is to assert the GPIO output for the assertion of the MSB of the error signal group. The output of this is connected to the PCIe AER error input.

21.1.2. Poison and SLVERR

The Keraunos NoC does not support poison in the Athena time frame. Therefore, the ethernet controller must turn AXI.SLVERR into poison that is written into the controller with an error event and turn memory poison into an AXI.SLVERR competition.

21.1.3. Keraunos RAS and Error Reporting Details

the actual signals for the TX/RX filtering logic related errors are BlackHole based placeholders for the moment until details are available to document.

the error bank registers must exist in the MMR address space and be accessible to both hart based and external micro-controller based RAS Agents.

Figure 52 depicts the various RAS components and organization of the RAS resources for the Keraunos chiplet.

The two uses of Keraunos have the same error events and banks, but the error reporting is different. For the Athena Grendel configuration, the Keraunos chiplet reports via D2D virtual wires to the Quasar chiplet which aggregates and forwards them to the Athena chiplet. However, in the PCIe Grendel configuration, the Keraunos chiplet receives error signals via D2D virtual wires from Quasar (which aggregated its own and Mimir errors) to be terminated the PCIe end-point AER error inputs.

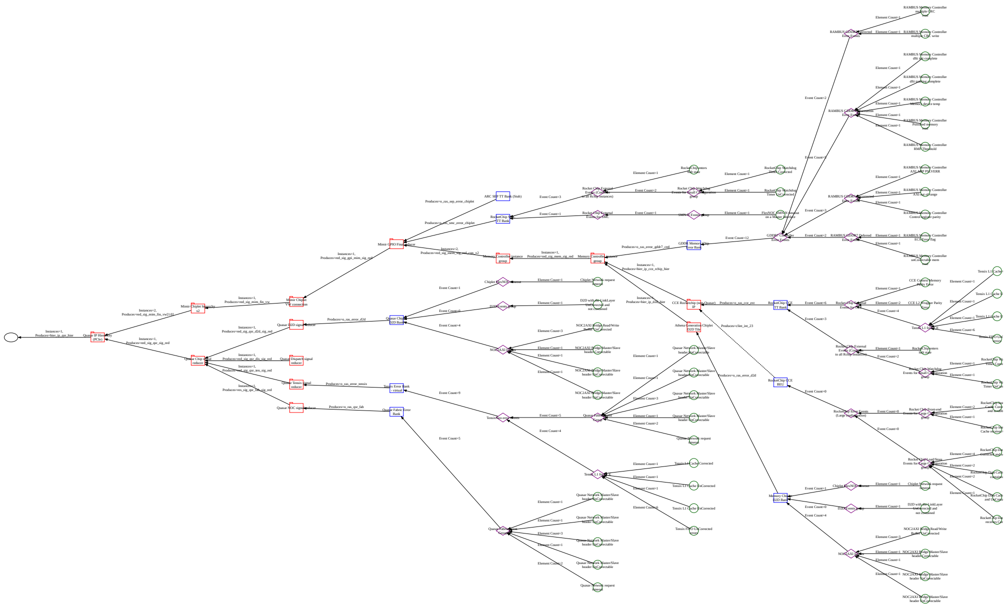


Figure 52. Keraunos RAS Banks and Events Graph

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.sep.stub

Name:
tt_sep_stub_error_bank

Description:
ARC SEP TT Bank (Stub)

Type:
stub</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Output:
"o_ras_sep_error_chiplet"</td>
<td>Count:
0</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.smc.chiplet.rchip.tt

Name:
tt_smc_error_bank

Description:
RocketChip SMC TT Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 3, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 4, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_smc_error_chiplet"</td>
<td>Count:
4</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.mchip.gddr.7.x4.ath

Name:
tt_gddr7_error_bank

Description:
GDDR Memory Chip Error Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_100

Description:
Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>100Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Mimir chiplet

Locator (ip):
Memory RAMBUS GDDR 7

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 5, 'N_REMOTE_RECORDS': 2, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 12, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_error_gddr7_cntl"</td>
<td>Count:
12</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.cce.chiplet.ath

Name:
tt_chiplet_cce_error_bank

Description:
RocketChip CCE TT Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_100

Description:
Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>100Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
BEU Interrupt Select

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
DMA Rchip

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 3, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 9, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_cce_ext"</td>
<td>Count:
9</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.cce.rchip.beu.ath

Name:
bnk.cce.rchip.beu.ath

Description:
RocketChip CCE BEU

Type:
rc_beu</td>
<td>Enum:
RAS_TIME_NONE

Description:
No valid time and TSV=0 for all error records in this bank.

Fields:
<table><tr><td>[63:0]</td></tr><tr><td>undefined</td></tr></table></td>
<td>Type:
BEU Interrupt Select

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
DMA Rchip RVI Core

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 5, 'N_REMOTE_RECORDS': 2, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 12, 'TYPE': 0}</td></tr></table>
Output:
"clint_int_23"</td>
<td>Count:
0</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.mchip.d2d.ath

Name:
tt_mimir_d2d_error_bank

Description:
Memory Chiplet D2D Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Mimir chiplet

Locator (ip):
Memory RAMBUS GDDR 7

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 5, 'N_REMOTE_RECORDS': 2, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 5, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_error_d2d"</td>
<td>Count:
5</td>
</tr>
</table>

Bank	Time	Initialization	Instantiation	Events
Identifier: bnk.qsr.d2d **Name:** tt_d2d_error_bank **Description:** Quasar Chiplet D2D Bank **Type:** tt_reri_rdl	**Enum:** RAS_TIME_HYBRID_100 **Description:** Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. `tsv` = 1 for all error records in this bank. **Fields:** [23:8]: 100Mhz Timer [7:0]: sequence	**Type:** TT RERI Error Bank **Values:** control_i: {'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}	**Locator (chip):** The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Parameters:** bank: {'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 5, 'N_REMOTE_RECORDS': 2, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1} event_reducer: {'N_EVENTS': 5, 'TYPE': 0} **Output:** "o_ras_error_d2d"	**Count:** 5
Identifier: bnk.tensix.neo **Name:** tt_qsr_tensix_error_bank **Description:** Tensix Error Bank - virtual **Type:** tt_virtual	**Enum:** RAS_TIME_NONE **Description:** No valid time and TSV=0 for all error records in this bank. **Fields:** [63:0]: undefined		**Locator (chip):** The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Output:** "o_ras_error_tensix"	**Count:** 9

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.fab.qsr

Name:
tt_qnp_error_bank

Description:
Quasar Fabric Error Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_100

Description:
Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>100Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Quasar chiplet

Locator (ip):
Accelerator TT QNP

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 2, 'N_REMOTE_RECORDS': 0, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 5, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_qsr_fab"</td>
<td>Count:
5</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.sep.chiplet.arc.tt

Name:
tt_sep_error_bank

Description:
ARC SEP TT Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 3, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 4, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_sep_error_chiplet"</td>
<td>Count:
4</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.smc.chiplet.rchip.tt

Name:
tt_smc_error_bank

Description:
RocketChip SMC TT Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 3, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 4, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_smc_error_chiplet"</td>
<td>Count:
4</td>
</tr>
<tr>
<td>Identifier:
bnk.smc.rchip.beu.lrg

Name:
bnk.smc.rchip.beu.lrg

Description:
SMC RocketChip BEU

Type:
rc_beu</td>
<td>Enum:
RAS_TIME_NONE

Description:
No valid time and TSV=0 for all error records in this bank.

Fields:
<table><tr><td>[63:0]</td></tr><tr><td>undefined</td></tr></table></td>
<td>Type:
BEU Interrupt Select</td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Output:
"clint_int_23"</td>
<td>Count:
0</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.enet.mac_pcs.ker

Name:
tt_ker_mac_pcs_error_bank

Description:
Keraunos Tile MAC/PCS Error Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Keraunos chiplet

Locator (ip):
IO TT Ethernet Tile

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 4, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 18, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_error_enet_hsio"</td>
<td>Count:
18</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.enet.ctl.ker

Name:
tt_ker_enet_ctl_error_bank

Description:
Keraunos Enet Controller Error Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_400

Description:
Upper portion increments at 400Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>400Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Keraunos chiplet

Locator (ip):
IO TT Ethernet Tile

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 4, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_400', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 1, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_error_enet_ctl</td>
<td>Count:
1</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.cce.chiplet.ath

Name:
tt_chiplet_cce_error_bank

Description:
RocketChip CCE TT Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_100

Description:
Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><td>[23:8]</td><td>[7:0]</td></tr><tr><td>100Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
BEU Interrupt Select

Values:
<table><tr><td>control_i</td></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
DMA Rchip

Parameters:
<table><tr><td>bank</td><td>event_reducer</td></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 3, 'N_REMOTE_RECORDS': 1, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 9, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_cce_ext"</td>
<td>Count:
9</td>
</tr>
</table>

<table>
<tr><th>Bank</th><th>Time</th><th>Initialization</th><th>Instantiation</th><th>Events</th></tr>
<tr>
<td>Identifier:
bnk.enet.d2d.ath

Name:
tt_d2d_error_bank

Description:
Ethernet Chiplet D2D Bank

Type:
tt_reri_rdl</td>
<td>Enum:
RAS_TIME_HYBRID_100

Description:
Upper portion increments at 100Mhz while the lower portion at event occurrence and reset with each timer tick. <code>tsv</code> = 1 for all error records in this bank.

Fields:
<table><tr><th>[23:8]</th><th>[7:0]</th></tr><tr><td>100Mhz Timer</td><td>sequence</td></tr></table></td>
<td>Type:
TT RERI Error Bank

Values:
<table><tr><th>control_i</th></tr><tr><td>{'cece': 1, 'ces': 1, 'eid': 0, 'else': 1, 'uecs': 3, 'ueds': 2}</td></tr></table></td>
<td>Locator (chip):
The Keraunos chiplet

Locator (ip):
IO TT Ethernet

Parameters:
<table><tr><th>bank</th><th>event_reducer</th></tr><tr><td>{'IMPL_ID': 0, 'LAYOUT': 1, 'LOCATOR_ENUM': 'RAS_LOCATOR_32_32_E1', 'N_ERROR_RECORDS': 5, 'N_REMOTE_RECORDS': 2, 'TIME_ENUM': 'RAS_TIME_HYBRID_100', 'VENDOR_ID': 'RAS_VEN_TT', 'VERSION': 1}</td><td>{'N_EVENTS': 5, 'TYPE': 0}</td></tr></table>
Output:
"o_ras_error_d2d"</td>
<td>Count:
5</td>
</tr>
</table>

Table 59. IP RAS Bank Description

There are 5 (4+1) Keraunous tile instances. Each tile instance generates unique error events with a an element descriptor indicating the instance number for the associated tile. If these are static, the tile disabled by harvesting will not record error events. If this is a dynamic assignment, then the harvesting may be hidden for error reporting.

<table>
<tr><th>Structure</th><th>Protections</th><th>Atoms</th></tr>
<tr>
<td>Locator (chip):
The chip identifier is deferred until instantiation.

Locator (ip):
SMC Rchip RVI Core

Description:
Controller Core

Description Enum:
RAS_DES_CON_COR</td>
<td>Description:
assertion

Detection:
Multiple

Correction:
0

Domain:
Single IP</td>
<td>Activation:
any

Activation Signal:
act_any

Observation:
rising edge detected

Observation Signal:
halt_and_catch_fire

Atom Responses:
stop everything</td>
</tr>
</table>

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_first_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_second_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** multiple **Detection:** Multiple **Correction:** Multiple **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** flexnoc_smn_timeout **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory Read Pipeline Data **Description Enum:** RAS_DES_CON_MEM_REA_PIP_DAT	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_2 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory Write Pipeline Data **Description Enum:** RAS_DES_CON_MEM_WRI_PIP_DAT	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_1 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory **Description Enum:** RAS_DES_CON_MEM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** No domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_20 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory **Description Enum:** RAS_DES_CON_MEM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** No domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_19 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory Unified Device **Description Enum:** RAS_DES_CON_MEM_UNI_DEV	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_16 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory Read Device Data **Description Enum:** RAS_DES_CON_MEM_REA_DEV_DAT	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_3 **Atom Responses:** deliver poison
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory **Description Enum:** RAS_DES_CON_MEM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** No domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_31 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory **Description Enum:** RAS_DES_CON_MEM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_21 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory **Description Enum:** RAS_DES_CON_MEM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_0 **Atom Responses:** set SLVERR

Structure	Protections	Atoms
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** No domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_18 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** No domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_23 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Controller Memory Read Device Data **Description Enum:** RAS_DES_CON_MEM_REA_DEV_DAT	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** interrupt_5 **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Unified L2 Array Data **Description Enum:** RAS_DES_MEM_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** cluster_ctrl_regs_i_ecc_parity_status_context_switch_parity_error_next **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Cache Unified L2 Array Data **Description Enum:** RAS_DES_CAC_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** l2_banks_err **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Cache Unified L2 Array Tag **Description Enum:** RAS_DES_CAC_UNI_L2_ARR_TAG	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** l2_dir_err **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** cor_err_irq **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** crit_err_irq **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** fatal_err_irq **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Data L1 FIFO Data **Description Enum:** RAS_DES_MEM_DAT_L1_FIF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 128 bit total with 64 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fifo_parity_err_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_overflow_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_underflow_Ored **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Integer FIFO **Description Enum:** RAS_DES_MEM_INT_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_int_overflow_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_error_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_nan_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Controller Core **Description Enum:** RAS_DES_CON_COR	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single IP	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** halt_and_catch_fire **Atom Responses:** stop everything

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_first_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_second_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Instruction L1 Array Data **Description Enum:** RAS_DES_CAC_INS_L1_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 16 bit total with 8 x 2 sub domain with I = 1	**Activation:** read entry **Activation Signal:** act_read_entry **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Instruction L1 Array Tag **Description Enum:** RAS_DES_CAC_INS_L1_ARR_TAG	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 26 bit total with 13 x 2 sub domain with I = 1	**Activation:** read set **Activation Signal:** act_read_set **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Instruction L1 Master Response **Description Enum:** RAS_DES_CAC_INS_L1_MAS_RES	**Description:** transaction protocol **Detection:** Transaction **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** trans data **Activation Signal:** act_trans_data **Observation:** assertion occurs **Observation Signal:** tl.resp.corrupt **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. demand 2. read entry **Activation Signal:** act_read_entry & act_op_demand **Observation:** decoded correctable **Atom Responses:** correct in-line
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is clean **Activation Signal:** act_read_entry & act_state_cln **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** read set **Activation Signal:** act_read_set **Observation:** decoded correctable **Atom Responses:** correct in-line
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is a hit 3. entry is clean **Activation Signal:** act_read_entry & act_state_cln & act_read_hit **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is modified **Activation Signal:** act_read_entry & act_state_mod **Observation:** decoded uncorrectable **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is a hit **Activation Signal:** act_read_entry & act_read_hit **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip RVI Core **Description:** Cache Data L1 Master Response **Description Enum:** RAS_DES_CAC_DAT_L1_MAS_RES	**Description:** transaction protocol **Detection:** Transaction **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** trans response **Activation Signal:** act_trans_response **Observation:** assertion occurs **Observation Signal:** tl.resp.corrupt **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Fabric TIU Master Response Timeout **Description Enum:** RAS_DES_FAB_TIU_MAS_RES_TIM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** SidebandManager_timeout_mainFault **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** D2D Master Logical **Description Enum:** RAS_DES_D2D_MAS_LOG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** Single Channel domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** intr_link_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_intp_id_transaction_count_rtz **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_noc_sec_fence_irq **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Read Buffer Data **Description Enum:** RAS_DES_BRI_FAB_REA_BUF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_0 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_1 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_2 **Atom Responses:** none
Locator (chip): The Mimir chiplet **Locator (ip):** Memory RAMBUS GDDR 7 **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_niu_timeout_intp **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Fabric TIU Master Response Timeout **Description Enum:** RAS_DES_FAB_TIU_MAS_RES_TIM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** SidebandManager_timeout_mainFault **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** D2D Master Logical **Description Enum:** RAS_DES_D2D_MAS_LOG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** Single Channel domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** intr_link_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_intp_id_transaction_count_rtz **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_noc_sec_fence_irq **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Read Buffer Data **Description Enum:** RAS_DES_BRI_FAB_REA_BUF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_1 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_2 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_niu_timeout_intp **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request Address **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_ADD	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_1 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_intp_id_transaction_count_rtz **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_noc_sec_fence_irq **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_2 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_niu_timeout_intp **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_niu_timeout_intp **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** cor_err_irq **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** crit_err_irq **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** fatal_err_irq **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Data L1 FIFO Data **Description Enum:** RAS_DES_MEM_DAT_L1_FIF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 128 bit total with 64 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fifo_parity_err_Ored **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_overflow_Ored **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_underflow_Ored **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Integer FIFO **Description Enum:** RAS_DES_MEM_INT_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_int_overflow_Ored **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_error_Ored **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet SERDES **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_nan_Ored **Atom Responses:** none
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request Address **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_ADD	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_0 **Atom Responses:** none
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_1 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_intp_id_transaction_count_rtz **Atom Responses:** none
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_noc_sec_fence_irq **Atom Responses:** none
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_ecc_error_2 **Atom Responses:** none
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_qnp_niu_timeout_intp **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Quasar chiplet **Locator (ip):** Accelerator TT QNP **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_niu_timeout_intp **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** dss_arc_clk_dccm_mpm_sec **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** dss_arc_clk_dccm_mpm_ded **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Instruction L1 Array Data **Description Enum:** RAS_DES_CAC_INS_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** dss_arc_clk_iccm0_mpm_sec **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Instruction L1 Array Data **Description Enum:** RAS_DES_CAC_INS_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** dss_arc_clk_iccm0_mpm_ded **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Controller Core **Description Enum:** RAS_DES_CON_COR	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single IP	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** halt_and_catch_fire **Atom Responses:** stop everything
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_first_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_second_timeout **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** multiple **Detection:** Multiple **Correction:** Multiple **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** flexnoc_smn_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Instruction L1 Array Data **Description Enum:** RAS_DES_CAC_INS_L1_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 16 bit total with 8 x 2 sub domain with I = 1	**Activation:** read entry **Activation Signal:** act_read_entry **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Instruction L1 Array Tag **Description Enum:** RAS_DES_CAC_INS_L1_ARR_TAG	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 26 bit total with 13 x 2 sub domain with I = 1	**Activation:** read set **Activation Signal:** act_read_set **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Instruction L1 Master Response **Description Enum:** RAS_DES_CAC_INS_L1_MAS_RES	**Description:** transaction protocol **Detection:** Transaction **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** trans data **Activation Signal:** act_trans_data **Observation:** assertion occurs **Observation Signal:** tl.resp.corrupt **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. demand 2. read entry **Activation Signal:** act_read_entry & act_op_demand **Observation:** decoded correctable **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is clean **Activation Signal:** act_read_entry & act_state_cln **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** read set **Activation Signal:** act_read_set **Observation:** decoded correctable **Atom Responses:** correct in-line
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is a hit 3. entry is clean **Activation Signal:** act_read_entry & act_state_cln & act_read_hit **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Data **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is modified **Activation Signal:** act_read_entry & act_state_mod **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Array Tag **Description Enum:** RAS_DES_CAC_DAT_L1_ARR_TAG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 42 bit total with 21 x 2 sub domain with I = 1	**Activation:** 1. read entry 2. entry is a hit **Activation Signal:** act_read_entry & act_read_hit **Observation:** decoded uncorrectable **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Cache Data L1 Master Response **Description Enum:** RAS_DES_CAC_DAT_L1_MAS_RES	**Description:** transaction protocol **Detection:** Transaction **Correction:** 0 **Domain:** Single Cache Line domain	**Activation:** trans response **Activation Signal:** act_trans_response **Observation:** assertion occurs **Observation Signal:** tl.resp.corrupt **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Memory Unified L2 Array Data **Description Enum:** RAS_DES_MEM_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** read entry **Activation Signal:** act_read_entry **Observation:** decoded correctable **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** SMC Rchip RVI Core **Description:** Memory Unified L2 Array Data **Description Enum:** RAS_DES_MEM_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** read entry **Activation Signal:** act_read_entry **Observation:** decoded uncorrectable **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet Slave Buffer Data **Description Enum:** RAS_DES_CON_ETH_SLA_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800_axi*sram_ecc_sbit_err **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet Slave Buffer Data **Description Enum:** RAS_DES_CON_ETH_SLA_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800_axi*sram_ecc_dbit_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*800ge*sram_ecc_sbit_err **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*400ge*sram_ecc_sbit_err_0 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*400ge*sram_ecc_sbit_err_1 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_sbit_err_0 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_sbit_err_1 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_sbit_err_2 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_sbit_err_3 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*800ge*sram_ecc_dbit_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*400ge*sram_ecc_dbit_err_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*400ge*sram_ecc_dbit_err_1 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_dbit_err_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_dbit_err_1 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_dbit_err_2 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet MAC Buffer Data **Description Enum:** RAS_DES_CON_ETH_MAC_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*mac_cl*100ge*sram_ecc_dbit_err_3 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*400ge*sram_ecc_sbit_err_0 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*400ge*sram_ecc_sbit_err_1 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_sbit_err_0 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_sbit_err_1 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_sbit_err_2 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_sbit_err_3 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_sbit_err_15_0 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_sbit_err_31_16 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_sbit_err_47_32 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_sbit_err_57_48 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*generic*sram_ecc_sbit_err_15_0 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*generic*sram_ecc_sbit_err_31_16 **Atom Responses:** correct in-line
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*400ge*sram_ecc_dbit_err_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*400ge*sram_ecc_dbit_err_1 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_dbit_err_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_dbit_err_1 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_dbit_err_2 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100ge*sram_ecc_dbit_err_3 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_dbit_err_15_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_dbit_err_31_16 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_dbit_err_47_32 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800pcs100gba*sram_ecc_dbit_err_57_48 **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*generic*sram_ecc_dbit_err_15_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller Ethernet PCS Buffer Data **Description Enum:** RAS_DES_CON_ETH_PCS_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 514 bit total with 257 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** *oepak800*pcs*generic*sram_ecc_dbit_err_31_16 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Bridge IO Buffer Data **Description Enum:** RAS_DES_BRI_IO_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** serdes_sram_ecc_sbit_err **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Bridge IO Buffer Data **Description Enum:** RAS_DES_BRI_IO_BUF_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** serdes_sram_ecc_dbit_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller IO Receive Buffer Data **Description Enum:** RAS_DES_CON_IO_REC_BUF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 16 bit total with 8 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** rx_pkt_buf_parity_err **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller IO Transmit Buffer Data **Description Enum:** RAS_DES_CON_IO_TRA_BUF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 16 bit total with 8 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** tx_pkt_buf_parity_err **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet Tile **Description:** Controller IO Transmit Request Data **Description Enum:** RAS_DES_CON_IO_TRA_REQ_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 16 bit total with 8 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** assertion occurs **Observation Signal:** txq_ctrl_req_buf_parity_err **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Unified L2 Array Data **Description Enum:** RAS_DES_MEM_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** cluster_ctrl_regs_i_ecc_parity_status_context_switch_parity_error_next **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Cache Unified L2 Array Data **Description Enum:** RAS_DES_CAC_UNI_L2_ARR_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** l2_banks_err **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Cache Unified L2 Array Tag **Description Enum:** RAS_DES_CAC_UNI_L2_ARR_TAG	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** l2_dir_err **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** cor_err_irq **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** crit_err_irq **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Array Data **Description Enum:** RAS_DES_MEM_ARR_DAT	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 64 bit total with 32 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** fatal_err_irq **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Data L1 FIFO Data **Description Enum:** RAS_DES_MEM_DAT_L1_FIF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 128 bit total with 64 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fifo_parity_err_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_overflow_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_fp_underflow_Ored **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Integer FIFO **Description Enum:** RAS_DES_MEM_INT_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_req_int_overflow_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_error_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Memory Floating Point FIFO **Description Enum:** RAS_DES_MEM_FLO_POI_FIF	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Multiple domains	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** i_tl1_fp_nan_Ored **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Controller Core **Description Enum:** RAS_DES_CON_COR	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single IP	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** halt_and_catch_fire **Atom Responses:** stop everything

Structure	Protections	Atoms
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_first_timeout **Atom Responses:** none
Locator (chip): The chip identifier is deferred until instantiation. **Locator (ip):** DMA Rchip **Description:** Execution Timer Instruction **Description Enum:** RAS_DES_EXE_TIM_INS	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_wdt_second_timeout **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Fabric TIU Master Response Timeout **Description Enum:** RAS_DES_FAB_TIU_MAS_RES_TIM	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** SidebandManager_timeout_mainFault **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** D2D Master Logical **Description Enum:** RAS_DES_D2D_MAS_LOG	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** Single Channel domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** intr_link_err **Atom Responses:** none

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_intp_id_transaction_count_rtz **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_noc_sec_fence_irq **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Read Buffer Data **Description Enum:** RAS_DES_BRI_FAB_REA_BUF_DAT	**Description:** even parity **Detection:** 1 **Correction:** 0 **Domain:** 32 bit total with 16 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_0 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_1 **Atom Responses:** correct in-line

Structure	Protections	Atoms
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request Control **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ_CON	**Description:** SECDED (even.even) **Detection:** 2 **Correction:** 1 **Domain:** 116 bit total with 58 x 2 sub domain with I = 1	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_ecc_error_2 **Atom Responses:** none
Locator (chip): The Keraunos chiplet **Locator (ip):** IO TT Ethernet **Description:** Bridge Fabric Master Request **Description Enum:** RAS_DES_BRI_FAB_MAS_REQ	**Description:** assertion **Detection:** Multiple **Correction:** 0 **Domain:** Single Transaction domain	**Activation:** any **Activation Signal:** act_any **Observation:** rising edge detected **Observation Signal:** o_n2a_niu_timeout_intp **Atom Responses:** none

Table 60. Micro-Architectural Protection Table

Error event details are provided in Appendix RAS

21.2. Keraunos Error Event Details

The error events for Keraunos are documented below.

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.arc.dat.l1:ce

Description:
ARC Data Cache Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
Update the structure with in-line corrected data</td>
<td>Structure Name:
Cache Data L1 Array Data

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
dss_arc_clk_dccm_mpm_sec

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.arc.dat.l1:uec

Description:
ARC Data Cache UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td>
<td>Structure Name:
Cache Data L1 Array Data

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
dss_arc_clk_dccm_mpm_ded

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.arc.ins.l1:ce

Description:
ARC UnCorrected Execution Error

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
Update the structure with in-line corrected data</td><td>Structure Name:
Cache Instruction L1 Array Data

Structure Value:
RAS_DES_CAC_INS_L1_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
dss_arc_clk_iccm0_mpm_sec

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.arc.ins.l1:uec

Description:
ARC Instruction Cache Corrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td><td>Structure Name:
Cache Instruction L1 Array Data

Structure Value:
RAS_DES_CAC_INS_L1_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
dss_arc_clk_iccm0_mpm_ded

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.halt:uec

Description:
RocketChip enters halt state

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the IP</td>
<td>Structure Name:
Controller Core

Structure Value:
RAS_DES_CON_COR

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
halt_and_catch_fire

Protection Description:
Condition (assertion)

Protection Domain:
Single IP

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Stop generating new requests and issuing current and future interface completions

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.wd:ce

Description:
RocketChip Watchdog Timer Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
1. Restart the application
2. Clear the resource generating the event</td>
<td>Structure Name:
Execution Timer Instruction

Structure Value:
RAS_DES_EXE_TIM_INS

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_wdt_first_timeout

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.wd:uec

Description:
RocketChip Watchdog Timer UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the application</td>
<td>Structure Name:
Execution Timer Instruction

Structure Value:
RAS_DES_EXE_TIM_INS

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_wdt_second_timeout

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.art.smn.timeout:uec

Description:
FlexNOC SMN-N timeout on a Master Interface

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td><td>Structure Name:
Bridge Fabric Master Request

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
flexnoc_smn_timeout

Protection Description:
Multiple

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rch.ins.l1:ce

Description:
RocketChip Instruction Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event</td>
<td>Structure Name:
Cache Instruction L1 Array Data

Structure Value:
RAS_DES_CAC_INS_L1_ARR_DAT

Activation Description:
Read a specific entry

Activation Signal:
act_read_entry

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
Parity (SED) with even

Protection Domain:
16 bit total with 8 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_VIRTUAL

Status:
<table><tr><td>AIT</td><td>IT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>3</td><td>1</td><td>1</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_SWB

Information Format:
<table><tr><td>[15:0]</td><td>[31:16]</td><td>[47:32]</td><td>[63:48]</td></tr><tr><td>way</td><td>set</td><td>bank</td><td>part</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
1. Invalidate entire array/structure
2. Replay/re-issue request</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rch.ins.l1:ce

Description:
RocketChip Instruction Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event</td>
<td>Structure Name:
Cache Instruction L1 Array Tag

Structure Value:
RAS_DES_CAC_INS_L1_ARR_TAG

Activation Description:
Read the entry and all ways associated with the set

Activation Signal:
act_read_set

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
Parity (SED) with even

Protection Domain:
26 bit total with 13 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_VIRTUAL

Status:
<table><tr><td>AIT</td><td>IT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>3</td><td>1</td><td>1</td><td>0</td><td>4</td></tr></table>

Information ENUM:
RAS_INFO_SWB

Information Format:
<table><tr><td>[15:0]</td><td>[31:16]</td><td>[47:32]</td><td>[63:48]</td></tr><tr><td>way</td><td>set</td><td>bank</td><td>part</td></tr></table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
1. Invalidate entire array/structure
2. Replay/re-issue request</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rch.ins.l1:uec

Description:
RocketChip Instruction Cache receives Corrupt

Type:
rc_beu

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. Restart the application
2. FW translates native error record to RERI and wite into an error bank
3. Clear the resource generating the event</td>
<td>Structure Name:
Cache Instruction L1 Master Response

Structure Value:
RAS_DES_CAC_INS_L1_MAS_RES

Activation Description:
A transactions data received

Activation Signal:
act_trans_data

Observation:
An signal is asserted

Observation Signal:
tl.resp.corrupt

Protection Description:
Protocol (transaction)

Protection Domain:
Single Cache Line domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_VIRTUAL

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>3</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
Do not fill with provided data</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.rch.lsu.l1:ce

Description:
RocketChip Data Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event
</td>
<td>
Structure Name:
Cache Data L1 Array Data

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_DAT

Activation Description:
1. Demand operation
2. Read a specific entry

Activation Signal:
act_read_entry & act_op_demand

Observation:
Data and Check decoding indicates that data is CORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
1. Invalidate affected entry
2. Replay/re-issue request
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>

Identifier:
evt.rch.lsu.l1:ce

Description:
RocketChip Data Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event

</td>
<td>

Structure Name:
Cache Data L1 Array Data

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_DAT

Activation Description:
1. Read a specific entry
2. The accessed entry is clean/duplicate

Activation Signal:
act_read_entry & act_state_cln

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:

<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>6</td></tr>
</table>

Information ENUM:
RAS_INFO_NONE

Information Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

</td>
<td>

Response (Structure):
No response

Response (Logic):
Invalidate affected entry

</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rch.lsu.l1:ce

Description:
RocketChip Data Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event</td>
<td>Structure Name:
Cache Data L1 Array Tag

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_TAG

Activation Description:
Read the entry and all ways associated with the set

Activation Signal:
act_read_set

Observation:
Data and Check decoding indicates that data is CORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
42 bit total with 21 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>1</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
1. Invalidate affected entry
2. Replay/re-issue request</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.rch.lsu.l1:ce

Description:
RocketChip Data Cache Corrected and scrubbed

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event
</td>
<td>
Structure Name:
Cache Data L1 Array Tag

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_TAG

Activation Description:
1. Read a specific entry
2. Read the entry that is a match for the set
3. The accessed entry is clean/duplicate

Activation Signal:
act_read_entry & act_state_cln & act_read_hit

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
42 bit total with 21 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_SWB

Information Format:
<table>
<tr><td>[15:0]</td><td>[31:16]</td><td>[47:32]</td><td>[63:48]</td></tr>
<tr><td>way</td><td>set</td><td>bank</td><td>part</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
1. Invalidate affected entry
2. Replay/re-issue request
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.rch.lsu.l1:ude

Description:
RocketChip Data Cache UnCorrected contained

Type:
rc_beu

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event</td>
<td>Structure Name:
Cache Data L1 Array Data

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_DAT

Activation Description:
1. Read a specific entry
2. The accessed entry is clean/duplicate

Activation Signal:
act_read_entry & act_state_cln

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>1</td><td>0</td><td>0</td><td>6</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
Invalidate affected entry</td></tr>
</table>

Event	Element	Response
Identifier: evt.rch.lsu.l1:ude **Description:** RocketChip Data Cache UnCorrected contained **Type:** rc_beu **Error Category:** UnCorrected Error Critical (UEC) **Response (Agent):** 1. FW translates native error record to RERI and wite into an error bank 2. Clear the resource generating the event	**Structure Name:** Cache Data L1 Array Data **Structure Value:** RAS_DES_CAC_DAT_L1_ARR_DAT **Activation Description:** 1. Read a specific entry 2. The accessed entry is modified/unique **Activation Signal:** act_read_entry & act_state_mod **Observation:** Data and Check decoding indicates that data is UnCORRECTABLE **Protection Description:** ECC (SECDED) with even.even **Protection Domain:** 32 bit total with 16 x 2 sub domain with I = 1 **Transaction:** RAS_XACT_EXPLICIT_READ **Address:** RAS_ADDR_SUPERVISOR_PHYSICAL **Status:** AIT: 1, IV: 0, SV: 0, TT: 4 **Information ENUM:** RAS_INFO_NONE **Information Format:** [63:0]: value **Supplemental ENUM:** RAS_SUPPL_NONE **Supplemental Format:** [63:0]: value	**Response (Structure):** No response **Response (Logic):** Invalidate affected entry

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.rch.lsu.l1:uec

Description:
RocketChip Data Cache UnCorrected and UnContained

Type:
rc_beu

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. Restart the application
2. FW translates native error record to RERI and wite into an error bank
3. Clear the resource generating the event
</td>
<td>
Structure Name:
Cache Data L1 Array Tag

Structure Value:
RAS_DES_CAC_DAT_L1_ARR_TAG

Activation Description:
1. Read a specific entry
2. Read the entry that is a match for the set

Activation Signal:
act_read_entry & act_read_hit

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
42 bit total with 21 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
1. Invalidate affected entry
2. Replay/re-issue request
</td>
</tr>
</table>

Event	Element	Response
Identifier: evt.rch.lsu.l1:uec **Description:** RocketChip Data Cache UnCorrected and UnContained **Type:** rc_beu **Error Category:** UnCorrected Error Critical (UEC) **Response (Agent):** 1. Restart the application 2. FW translates native error record to RERI and wite into an error bank 3. Clear the resource generating the event	**Structure Name:** Cache Data L1 Master Response **Structure Value:** RAS_DES_CAC_DAT_L1_MAS_RES **Activation Description:** A transcation response received **Activation Signal:** act_trans_response **Observation:** An signal is asserted **Observation Signal:** tl.resp.corrupt **Protection Description:** Protocol (transaction) **Protection Domain:** Single Cache Line domain **Transaction:** RAS_XACT_EXPLICIT_READ **Address:** RAS_ADDR_SUPERVISOR_PHYSICAL **Status:** AIT: 1, IV: 0, SV: 0, TT: 4 **Information ENUM:** RAS_INFO_NONE **Information Format:** [63:0]: value **Supplemental ENUM:** RAS_SUPPL_NONE **Supplemental Format:** [63:0]: value	**Response (Structure):** No response **Response (Logic):** Do not fill with provided data

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rch.lsu.l1:uec_cor

Description:
RocketChip Data Cache receives Corrupt

Type:
rc_beu

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. Restart the application
2. FW translates native error record to RERI and wite into an error bank
3. Clear the resource generating the event</td>
<td>Structure Name:
Cache Data L1 Master Response

Structure Value:
RAS_DES_CAC_DAT_L1_MAS_RES

Activation Description:
A transcation response received

Activation Signal:
act_trans_response

Observation:
An signal is asserted

Observation Signal:
tl.resp.corrupt

Protection Description:
Protocol (transaction)

Protection Domain:
Single Cache Line domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>1</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
Do not fill with provided data</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.rch.uni.mem:ce

Description:
RocketChip Scratchpad Memory Corrected

Type:
rc_beu

Error Category:
Corrected Error (CE)

Response (Agent):
Update the structure with in-line corrected data
</td>
<td>
Structure Name:
Memory Unified L2 Array Data

Structure Value:
RAS_DES_MEM_UNI_L2_ARR_DAT

Activation Description:
Read a specific entry

Activation Signal:
act_read_entry

Observation:
Data and Check decoding indicates that data is CORRECTABLE

Protection Description:
Parity (SED) with even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
Correct the data in-line
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.rch.uni.mem:uec

Description:
RocketChip Scratchpad Memory UnCorrected

Type:
rc_beu

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. Restart the application
2. FW translates native error record to RERI and wite into an error bank
3. Clear the resource generating the event
</td>
<td>
Structure Name:
Memory Unified L2 Array Data

Structure Value:
RAS_DES_MEM_UNI_L2_ARR_DAT

Activation Description:
Read a specific entry

Activation Signal:
act_read_entry

Observation:
Data and Check decoding indicates that data is UnCORRECTABLE

Protection Description:
Parity (SED) with even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_SUPERVISOR_PHYSICAL

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
Indicate Corrupt in the TileLink
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.axi.arr:ce

Description:
Ethernet Controller AXI Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet Slave Buffer Data

Structure Value:
RAS_DES_CON_ETH_SLA_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800_axisram_ecc_sbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.axi.arr:uec

Description:
Ethernet Controller AXI Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet Slave Buffer Data

Structure Value:
RAS_DES_CON_ETH_SLA_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800_ axisram_ecc_dbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td valign="top">Identifier:
evt.enet.ctl.mac.800ge:ce

Description:
Ethernet Controller MAC 800ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td valign="top">Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl800gesram_ecc_sbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td valign="top">Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.400ge:ce

Description:
Ethernet Controller MAC 400ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl400gesram_ecc_sbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.400ge:ce

Description:
Ethernet Controller MAC 400ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl400gesram_ecc_sbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.100ge:ce

Description:
Ethernet Controller MAC 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_sbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.100ge:ce

Description:
Ethernet Controller MAC 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_sbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.100ge:ce

Description:
Ethernet Controller MAC 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_sbit_err_2

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.100ge:ce

Description:
Ethernet Controller MAC 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_sbit_err_3

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.800ge:uec

Description:
Ethernet Controller MAC 800ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl800gesram_ecc_dbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.400ge:uec

Description:
Ethernet Controller MAC 400ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl400gesram_ecc_dbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.400ge:uec

Description:
Ethernet Controller MAC 400ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl400gesram_ecc_dbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.100ge:uec

Description:
Ethernet Controller MAC 100ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_dbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.mac.100ge:uec

Description:
Ethernet Controller MAC 100ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_dbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.mac.100ge:uec

Description:
Ethernet Controller MAC 100ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800mac_cl100gesram_ecc_dbit_err_2

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>

Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>

Identifier:
evt.enet.ctl.mac.100ge:uec

Description:
Ethernet Controller MAC 100ge Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only

</td>
<td>

Structure Name:
Controller Ethernet MAC Buffer Data

Structure Value:
RAS_DES_CON_ETH_MAC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
*oepak800*mac_cl*100ge*sram_ecc_dbit_err_3

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:

<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>

Information ENUM:
RAS_INFO_NONE

Information Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

</td>
<td>

Response (Structure):
No response

Response (Logic):
No response

</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.400ge:ce

Description:
Ethernet Controller PCS 400ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs400gesram_ecc_sbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.400ge:ce

Description:
Ethernet Controller PCS 400ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs400gesram_ecc_sbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100ge:ce

Description:
Ethernet Controller PCS 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_sbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.enet.ctl.pcs.100ge:ce

Description:
Ethernet Controller PCS 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td><td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_sbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100ge:ce

Description:
Ethernet Controller PCS 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_sbit_err_2

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100ge:ce

Description:
Ethernet Controller PCS 100ge Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_sbit_err_3

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100gba:ce

Description:
Ethernet Controller PCS 100gba Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_sbit_err_15_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100gba:ce

Description:
Ethernet Controller PCS 100gba Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_sbit_err_31_16

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100gba:ce

Description:
Ethernet Controller PCS 100gba Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_sbit_err_47_32

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.enet.ctl.pcs.100gba:ce

Description:
Ethernet Controller PCS 100gba Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td><td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_sbit_err_57_48

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.generic:ce

Description:
Ethernet Controller PCS Generic Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcsgenericsram_ecc_sbit_err_15_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.generic:ce

Description:
Ethernet Controller PCS Generic Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcsgenericsram_ecc_sbit_err_31_16

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
Correct the data in-line

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.400ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs400gesram_ecc_dbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.400ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs400gesram_ecc_dbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_dbit_err_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_dbit_err_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_dbit_err_2

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100ge:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gesram_ecc_dbit_err_3

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>**Identifier:**
evt.enet.ctl.pcs.100gba:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>**Structure Name:**
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
*oepak800*pcs*100gba*sram_ecc_dbit_err_15_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>**Response (Structure):**
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100gba:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_dbit_err_31_16

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.100gba:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_dbit_err_47_32

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.100gba:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcs100gbasram_ecc_dbit_err_57_48

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.ctl.pcs.generic:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcsgenericsram_ecc_dbit_err_15_0

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.ctl.pcs.generic:uec

Description:
Ethernet Controller MAC Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only
</td>
<td>
Structure Name:
Controller Ethernet PCS Buffer Data

Structure Value:
RAS_DES_CON_ETH_PCS_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
oepak800pcsgenericsram_ecc_dbit_err_31_16

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
514 bit total with 257 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.serdes:ce

Description:
Ethernet SERDES Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Bridge IO Buffer Data

Structure Value:
RAS_DES_BRI_IO_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
serdes_sram_ecc_sbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>
Identifier:
evt.enet.serdes:uec

Description:
Ethernet SERDES Array Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response
</td>
<td>
Structure Name:
Bridge IO Buffer Data

Structure Value:
RAS_DES_BRI_IO_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
serdes_sram_ecc_dbit_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>
Response (Structure):
No response

Response (Logic):
No response
</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.tx.rx:uec

Description:
Ethernet TX/RX Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller IO Receive Buffer Data

Structure Value:
RAS_DES_CON_IO_REC_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
rx_pkt_buf_parity_err

Protection Description:
Parity (SED) with even

Protection Domain:
16 bit total with 8 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.tx.rx:uec

Description:
Ethernet TX/RX Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller IO Transmit Buffer Data

Structure Value:
RAS_DES_CON_IO_TRA_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
tx_pkt_buf_parity_err

Protection Description:
Parity (SED) with even

Protection Domain:
16 bit total with 8 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.enet.tx.rx:uec

Description:
Ethernet TX/RX Array UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Controller IO Transmit Request Data

Structure Value:
RAS_DES_CON_IO_TRA_REQ_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
An signal is asserted

Observation Signal:
txq_ctrl_req_buf_parity_err

Protection Description:
Parity (SED) with even

Protection Domain:
16 bit total with 8 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>

Identifier:
evt.rchip.cm.wra:uec

Description:
CCE Context Memory Parity Error

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only

</td><td>

Structure Name:
Memory Unified L2 Array Data

Structure Value:
RAS_DES_MEM_UNI_L2_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
cluster_ctrl_regs_i_ecc_parity_status_context_switch_parity_error_next

Protection Description:
Parity (SED) with even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:

<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>

Information ENUM:
RAS_INFO_NONE

Information Format:

<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:

<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>

</td><td>

Response (Structure):
No response

Response (Logic):
No response

</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.l2.wra:uec

Description:
CCE L2 Wrapper Parity Error

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Cache Unified L2 Array Data

Structure Value:
RAS_DES_CAC_UNI_L2_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
l2_banks_err

Protection Description:
Parity (SED) with even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.l2.wra:uec

Description:
CCE L2 Wrapper Parity Error

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Cache Unified L2 Array Tag

Structure Value:
RAS_DES_CAC_UNI_L2_ARR_TAG

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
l2_dir_err

Protection Description:
Parity (SED) with even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.tensix.l1:ce

Description:
Tensix L1 Cache Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
Update the structure with in-line corrected data</td>
<td>Structure Name:
Memory Array Data

Structure Value:
RAS_DES_MEM_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
cor_err_irq

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.tensix.l1:uec

Description:
Tensix L1 Cache UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the IP</td><td>Structure Name:
Memory Array Data

Structure Value:
RAS_DES_MEM_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
crit_err_irq

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.tensix.l1:uec.p

Description:
Tensix L1 Cache UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the IP</td>
<td>Structure Name:
Memory Array Data

Structure Value:
RAS_DES_MEM_ARR_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
fatal_err_irq

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
64 bit total with 32 x 2 sub domain with I = 1

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>

Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only

</td>
<td>

Structure Name:
Memory Data L1 FIFO Data

Structure Value:
RAS_DES_MEM_DAT_L1_FIF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_req_fifo_parity_err_Ored

Protection Description:
Parity (SED) with even

Protection Domain:
128 bit total with 64 x 2 sub domain with I = 1

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:

<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>6</td></tr>
</table>

Information ENUM:
RAS_INFO_NONE

Information Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:

<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>

</td>
<td>

Response (Structure):
No response

Response (Logic):
No response

</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Memory Floating Point FIFO

Structure Value:
RAS_DES_MEM_FLO_POI_FIF

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_req_fp_overflow_Ored

Protection Description:
Condition (assertion)

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>6</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td><td>Structure Name:
Memory Floating Point FIFO

Structure Value:
RAS_DES_MEM_FLO_POI_FIF

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_req_fp_underflow_Ored

Protection Description:
Condition (assertion)

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>6</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Memory Integer FIFO

Structure Value:
RAS_DES_MEM_INT_FIF

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_req_int_overflow_Ored

Protection Description:
Condition (assertion)

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>6</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td><td>Structure Name:
Memory Floating Point FIFO

Structure Value:
RAS_DES_MEM_FLO_POI_FIF

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_fp_error_Ored

Protection Description:
Condition (assertion)

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>6</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.tensix.l1.fifo:uec

Description:
Tensix FIFO UnCorrected errors

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Memory Floating Point FIFO

Structure Value:
RAS_DES_MEM_FLO_POI_FIF

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
i_tl1_fp_nan_Ored

Protection Description:
Condition (assertion)

Protection Domain:
Multiple domains

Transaction:
RAS_XACT_IMPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>6</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

Event	Element	Response
Identifier: evt.rchip.halt:uec **Description:** RocketChip enters halt state **Type:** tt_reri **Error Category:** UnCorrected Error Critical (UEC) **Response (Agent):** Restart the IP	**Structure Name:** Controller Core **Structure Value:** RAS_DES_CON_COR **Activation Description:** Always active **Activation Signal:** act_any **Observation:** A signal or group of signals raising edge is captured **Observation Signal:** halt_and_catch_fire **Protection Description:** Condition (assertion) **Protection Domain:** Single IP **Transaction:** RAS_XACT_NONE **Address:** RAS_ADDR_NONE **Status:** AIT: 0, IV: 0, SV: 0, TT: 0 **Information ENUM:** RAS_INFO_NONE **Information Format:** [63:0]: value **Supplemental ENUM:** RAS_SUPPL_NONE **Supplemental Format:** [63:0]: value	**Response (Structure):** Stop generating new requests and issuing current and future interface completions **Response (Logic):** No response

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.wd:ce

Description:
RocketChip Watchdog Timer Corrected

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
1. Restart the application
2. Clear the resource generating the event</td>
<td>Structure Name:
Execution Timer Instruction

Structure Value:
RAS_DES_EXE_TIM_INS

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_wdt_first_timeout

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.rchip.wd:uec

Description:
RocketChip Watchdog Timer UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the application</td>
<td>Structure Name:
Execution Timer Instruction

Structure Value:
RAS_DES_EXE_TIM_INS

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_wdt_second_timeout

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.chiplet.noc.tim:uec

Description:
Chiplet Network request timeout

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the affected sub-system only</td>
<td>Structure Name:
Fabric TIU Master Response Timeout

Structure Value:
RAS_DES_FAB_TIU_MAS_RES_TIM

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
SidebandManager_timeout_mainFault

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>Identifier:
evt.d2d.ath:uce

Description:
D2D with the LinkLayer UnCorrected and not contained

Type:
uc_spoof

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
1. FW translates native error record to RERI and wite into an error bank
2. Clear the resource generating the event</td><td>Structure Name:
D2D Master Logical

Structure Value:
RAS_DES_D2D_MAS_LOG

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
intr_link_err

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
Single Channel domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td><td>Response (Structure):
No response

Response (Logic):
No response</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>
Identifier:
evt.n2a.data:uec

Description:
NOC2AXI Bridge Read/Write Buffer UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain
</td><td>
Structure Name:
Bridge Fabric Master Request Control

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ_CON

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_intp_id_transaction_count_rtz

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>4</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td><td>
Response (Structure):
No response

Response (Logic):
No response
</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr><td>
Identifier:
evt.n2a.data:uec

Description:
NOC2AXI Bridge Read/Write Buffer UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain
</td><td>
Structure Name:
Bridge Fabric Master Request Control

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ_CON

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_noc_sec_fence_irq

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_EXPLICIT_READ

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>4</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td><td>
Response (Structure):
No response

Response (Logic):
No response
</td></tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.n2a.data:uec

Description:
NOC2AXI Bridge Read/Write Buffer UnCorrected

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td>
<td>Structure Name:
Bridge Fabric Read Buffer Data

Structure Value:
RAS_DES_BRI_FAB_REA_BUF_DAT

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_n2a_ecc_error_0

Protection Description:
Parity (SED) with even

Protection Domain:
32 bit total with 16 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.n2a.head:ce

Description:
NOC2AXI Bridge Master/Slave header Correctable

Type:
tt_reri

Error Category:
Corrected Error (CE)

Response (Agent):
No response</td>
<td>Structure Name:
Bridge Fabric Master Request Control

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ_CON

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_n2a_ecc_error_1

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
116 bit total with 58 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
Correct the data in-line

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.n2a.head:uec

Description:
NOC2AXI Bridge Master/Slave header UnCorrectable

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td>
<td>Structure Name:
Bridge Fabric Master Request Control

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ_CON

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_n2a_ecc_error_2

Protection Description:
ECC (SECDED) with even.even

Protection Domain:
116 bit total with 58 x 2 sub domain with I = 1

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table><tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table><tr><td>[63:0]</td></tr><tr><td>value</td></tr></table></td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

<table>
<tr><th>Event</th><th>Element</th><th>Response</th></tr>
<tr>
<td>Identifier:
evt.n2a.mas:uec

Description:
NOC2AXI Bridge Master/Slave header UnCorrectable

Type:
tt_reri

Error Category:
UnCorrected Error Critical (UEC)

Response (Agent):
Restart the coherent domain</td>
<td>Structure Name:
Bridge Fabric Master Request

Structure Value:
RAS_DES_BRI_FAB_MAS_REQ

Activation Description:
Always active

Activation Signal:
act_any

Observation:
A signal or group of signals raising edge is captured

Observation Signal:
o_n2a_niu_timeout_intp

Protection Description:
Condition (assertion)

Protection Domain:
Single Transaction domain

Transaction:
RAS_XACT_NONE

Address:
RAS_ADDR_NONE

Status:
<table>
<tr><td>AIT</td><td>IV</td><td>SV</td><td>TT</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Information ENUM:
RAS_INFO_NONE

Information Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
Supplemental ENUM:
RAS_SUPPL_NONE

Supplemental Format:
<table>
<tr><td>[63:0]</td></tr>
<tr><td>value</td></tr>
</table>
</td>
<td>Response (Structure):
No response

Response (Logic):
No response</td>
</tr>
</table>

Table 61. Keraunos Definition Table

Chapter 22. Performance

The following KPIs will be verified in simulation and emulation.

22.1. Throughput KPIs

KPI	Value
D2D Read Throughput Per Link	110GB/s
D2D Write Throughput Per Link	110GB/s
D2D Read + Write Throughput Per Link	220GB/s
800G Ethernet Tx Throughput Per Link	100GB/s
800G Ethernet Rx Throughput Per Link	100GB/s
800G Ethernet Tx + Rx Throughput Per Link	200GB/s
CCE Read Throughput through BoW D2D Per Instance	110GB/s
CCE Write Throughput through BoW D2D Per Instance	110GB/s
CCE Read + Write Throughput through BoW D2D	220GB/s
CCE Read Throughput To A Single 800G Ethernet Link	100GB/s
CCE Write Throughput To A Single 800G Ethernet Link	100GB/s
CCE Read + Write Throughput To A Single 800G Ethernet Link	200GB/s
CCE Composite Read Throughput through BoW D2D	550GB/s
CCE Composite Write Throughput through BoW D2D	550GB/s
CCE Composite Read + Write Throughput through BoW D2D	1100GB/s
CCE Composite Read Throughput through Ethernet	500GB/s
CCE Composite Write Throughput through Ethernet	500GB/s
CCE Composite Read + Write Throughput through Ethernet	1000GB/s
800G Ethernet Link To Link Tx Throughput	100GB/s
800G Ethernet Link To Link Rx Throughput	100GB/s
800G Ethernet Link To Link Tx + Rx Throughput	200GB/s
Total 800G Ethernet Link To Link Tx Throughput	500GB/s
Total 800G Ethernet Link To Link Rx Throughput	500GB/s
Total 800G Ethernet Link To Link Tx + Rx Throughput	1000GB/s
800G Ethernet Link Tx Packet Throughput	24.4M packets/s (4KB frame); 97.6M packets/s (1KB frame)
800G Ethernet Link Rx Packet Throughput	24.4M packets/s (4KB frame); 97.6M packets/s (1KB frame)
800G Ethernet Link Tx + Rx Packet Throughput	48.8M packets/s (4KB frame); 195.2M packets/s (1KB frame)
Total 800G Ethernet Tx Packet Throughput	122M packets/s (4KB frame); 488M packets/s (1KB frame)
Total 800G Ethernet Rx Packet Throughput	122M packets/s (4KB frame); 488M packets/s (1KB frame)
Total 800G Ethernet Tx + Rx Packet Throughput	244M packets/s (4KB frame); 976M packets/s (1KB frame)
PCIe Read Throughput	32GB/s
PCIe Write Throughput	32GB/s
Total PCIe Read + Write Throughput	64GB/s
Ethernet Read + PCIe Read Offchip Through Shared D2D	110GB/s
Ethernet Write + PCIe Write Offchip Through Shared D2D	110GB/s

Chapter 23. Composition

Keraunos-E100 is attached to Quasar's short edge for the PCIe card products that support scaleout. The D2D connectivity and grid coordinates in those products is shown below.

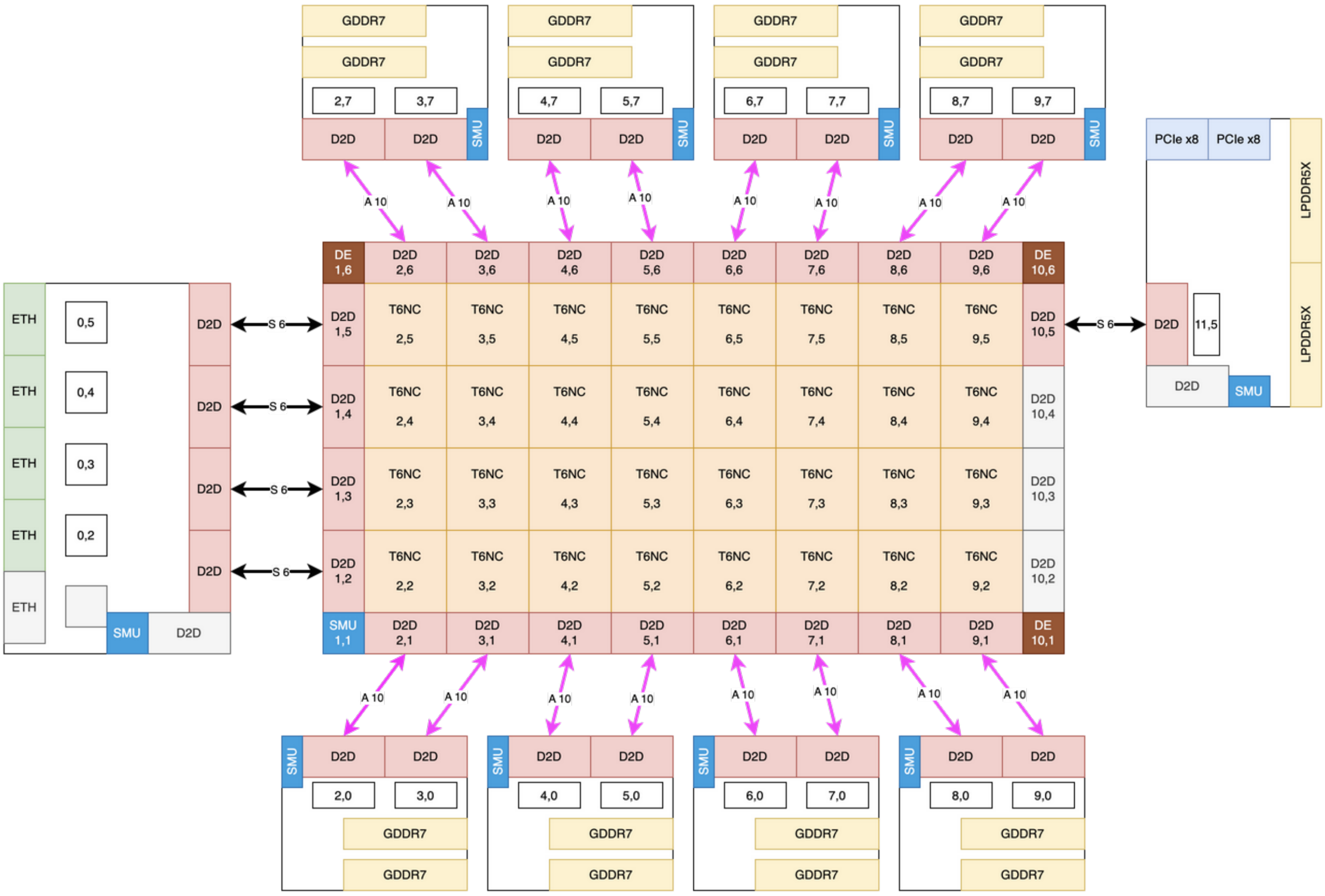


Figure 53. Keraunos-E100 in PCIe card SoP

Keraunos-E100 is attached to Quasar's long edge for the Halo products. In this case, there is more than one Keraunos-E100 in the package and we connect these using a straight-through path through Quasar's NOC between the two Keraunos-E100 chiplets.

Figure 54. Keraunos-E100 in Halo SoP

Chapter 24. Unsupported Features

24.1. RDMA

Keraunos-E100 does not include dedicated hardware support for RDMA beyond what is supported in Blackhole. There are potential use cases where allowing Keraunos-E100 to directly read/write Quasar memory can help performance. Similarly, there are use cases where accelerating protocols like TTPoE or RoCEv2 via dedicated hardware can reduce latency. We defer adding hardware for such RDMA to a future version of Keraunos-E100. We expect we can support any such RDMA protocol in software using CCE but that there will be software overhead in the Keraunos-E100 implementation over an optimized hardware implementation.

24.2. Cross-chiplet switching

Keraunos-E100 limits Ethernet switching to the links contained within the Keraunos-E100 chiplet. There are use cases, like scaleup connectivity and top-of-rack switching, where allowing Ethernet switching to be expanded to a much larger set of links can be useful. We defer such switching functionality to a future version of Keraunos.

24.3. Near-IO compute

Keraunos-E100 does not provide AI compute resources in each HSIO tile. There is value in having such near-IO compute, e.g., for offloading operations like AllReduce and for doing format conversions during data movement. In this generation of Grendel, we implement such capability only in the switch product, which includes both Quasar and Keraunos. In this switch, Quasar provides the AI compute and format conversion capabilities.

We will evaluate adding such near-IO compute in the next version of the Keraunos chiplet. In Keraunos-E100, near-IO compute is limited to the operations that can be carried out by the RocketCore processors in the CCE. Simple integer operations are possible, but the throughput is not expected to be sufficient to do data manipulation on the fly without adding latency. Data manipulation for AI data formats like BF16 is not possible directly within Keraunos-E100.

24.4. Harvesting

Keraunos-E100 requires all connected D2D links to work. It also requires all connected Ethernet SERDES lanes and all connected PCIe PHY lanes to work. For use cases where only a subset of the SERDES and PHYs and D2D links are connected, the unconnected SERDES and PHY and D2D links can be defective.

Keraunos-E100 does not have any hardware features to enable harvesting of partially functional silicon dice. The only portions that can be defective are the unused portions.

Chapter 25. Keraunos test plan

25.1. Performance tests

25.1.1. Latency

The tests use 128B transaction to measure latency.

ID	Src	Dst	Measurement	Notes
0	D2D0	HSIO0 SRAM		
1	HSIO0 SRAM	D2D0		
2	D2D1	HSIO0 SRAM		
3	HSIO2 SRAM	D2D3		
4	D2D0	HSIO4 SRAM		
5	HSIO0 SRAM	MAC0		
6	MAC0	HSIO0 SRAM		
7	HSIO0 SRAM	HSIO1 SRAM		
8	HSIO2 SRAM	HSIO0 SRAM		
9	HSIO0 SRAM	HSIO3 SRAM		
9	HSIO4 SRAM	HSIO0 SRAM		
10	HSIO0 CCE0 DMRISC	HSIO0 SRAM		

Table 62. Latency tests

25.1.2. Bandwidth

Three traffic patterns can be tests:

- Uniform transaction sizes ranging from 16B to 16KB
- Multiple beat large size transaction (16KB) followed by one small size transaction (64B)
- random generated various size transactions with normal distribution

ID	Traffic 1	Traffic 2	Traffic 3	Traffic 4	Traffic 5	Traffic 6
0	D2D0 → HSIO0 SRAM					
1	D2D0 → HSIO0 SRAM	D2D1→HSIO1 SRAM				
2	D2D0 → HSIO0 SRAM	D2D1→HSIO1 SRAM	D2D2 → HSIO2 SRAM	D2D3→HSIO3 SRAM		
3	D2D0 → HSIO0 SRAM	D2D1→HSIO1 SRAM	D2D2 → HSIO2 SRAM	D2D3→HSIO3 SRAM		
4	D2D0 → HSIO0 SRAM	D2D1→HSIO1 SRAM	D2D2 → HSIO2 SRAM	D2D3→HSIO3 SRAM	D2D4→HSIO4 SRAM	D2D0 →PCIE
5	HSIO0 SRAM → MAC0					
6	HSIO0 SRAM → MAC0	HSIO1 SRAM → MAC0				
7	MAC0 → HSIO0 SRAM					
8	HSIO0 CCE0 DMRISC→HSIO SRAM					

ID	Traffic 1	Traffic 2	Traffic 3	Traffic 4	Traffic 5	Traffic 6
9	D2D0→HSIO0 SRAM					
	HSIO0 SRAM → MAC0					
	MAC0 → HSIO0 SRAM					
	HSIO0 SRAM → D2D0					
10	D2D0→HSIO0 SRAM	D2D1→HSIO1 SRAM				
	HSIO0 SRAM → MAC0	HSIO1 SRAM → MAC1				
	MAC0 → HSIO0 SRAM	MAC1 → HSIO1 SRAM				
	HSIO0 SRAM → D2D0	HSIO1 SRAM → D2D1				
11	D2D0→HSIO0 SRAM	D2D1→HSIO1 SRAM				
	HSIO0 SRAM → MAC0	HSIO1 SRAM → MAC1				
	MAC0 → HSIO0 SRAM	MAC1 → HSIO1 SRAM				
	HSIO0 SRAM → D2D0	HSIO1 SRAM → D2D1				
12	D2D0→HSIO0 SRAM	D2D1→HSIO1 SRAM	D2D2→HSIO2 SRAM	D2D3→HSIO3 SRAM	D2D4→HSIO4 SRAM	
	HSIO0 SRAM → MAC0	HSIO1 SRAM → MAC1	HSIO2 SRAM → MAC2	HSIO3 SRAM → MAC3	HSIO4 SRAM → MAC4	
	MAC0 → HSIO0 SRAM	MAC1 → HSIO1 SRAM	MAC2 → HSIO2 SRAM	MAC3 → HSIO3 SRAM	MAC4 → HSIO4 SRAM	
	HSIO0 SRAM → D2D0	HSIO1 SRAM → D2D1	HSIO2 SRAM → D2D2	HSIO3 SRAM → D2D3	HSIO4 SRAM → D2D4	
13	D2D0→HSIO1 SRAM	HSIO1 SRAM → D2D0				
14	D2D0→HSIO1 SRAM	HSIO1 SRAM → D2D0				
	D2D1→HSIO0 SRAM	HSIO0 SRAM → D2D1				
15	D2D0→HSIO0 SRAM	D2D1→HSIO1 SRAM		D2D3→HSIO3 SRAM		
	HSIO0 SRAM → MAC0	HSIO1 SRAM → MAC1		HSIO3 SRAM → MAC3		
	MAC0 → HSIO0 SRAM	MAC1 → HSIO1 SRAM		MAC3 → HSIO3 SRAM		
	HSIO0 SRAM → D2D0	HSIO1 SRAM → D2D1		HSIO3 SRAM → D2D3		

Table 63. Bandwidth tests

25.2. Ethernet tests

Ethernet tests are also used to test the functionality of different packet format

Test ID	Priority	Flow #	Packet Type	Ingress	Egress	Test Bench	Outcome	TT Config	Flow Purpose	Traffic Patterns	Overall Purpose
0	0	0	SVLAN	HSIO.0.RX	D2D.0	HSIO0←→D2D/CCE	CCE.0: Route to D2D		Accept SVLAN and send D2D	1. Flows back to back random starting order for each burst	Accept both a VLAN tagged and untagged packet at the same time while noisy other switch traffic doesn't cause problems
		1	IPV4_UCAST_UDP	HSIO.0.RX	HSIO.0.TX		CCE.1 Route back out port		Accept untagged UDP and send back to port	2. Burst size should be 1-8 sets random	
		2	802.3_STP	HSIO.0.RX	Drop at TCAM		TT Ethernet Port should drop STP		Drop Spanning Tree before going to CCE	3. Minimum IPG	
1	1	0	IPV4_MCAST_UDP	HSIO.0.RX	D2D.0	HSIO0←→D2D/CCE	CCE.0: Route to D2D		Accept Multicast event and send to D2D	1. Flows back to back random starting order for each burst	Have multicast, unicast and broadcast be received at the same time
		1	L2 PTP Event	HSIO.0.RX	HSIO.0.TX		CCE.1 Route back out port		Accept PTP event send PTP back out with timestamp update to same port through CCE	2. Burst size should be 1-8 sets random	
		2	IPV4_UCAST_UDP	HSIO.0.RX	HSIO.0.TX		CCE.1 Route back out port		Loop unicast packet back through CCE	3. Minimum IPG	
		3	IPV4_BCAST_UDP	HSIO.0.RX	Drop at TCAM		TT Ethernet Port should drop STP		Drop broadcast packet at TT Ethernet TCAM		
2	2	0	RoCEv2	D2D.0	HSIO.0	HSIO0←→D2D/CCE	RoCEv2 Quasar-→ port		Sourced RoCEv2 traffic outbound		
		1	RoCEv2	HSIO.0	CCE.0		RoCEv2 source terminate to CCE		RoCEv2 traffic inbound		
		2	SVLAN	HSIO.0	CCE.0		SVLAN traffic we need to drop		Simple VLAN Tag Drop at TT Ethernet TCAM		
		3	ARP-B	HSIO.0	D2D.0		ARP Broadcast to host				
		4	ARP-R	D2D.0	HSIO.0		ARP Reply from host				
3	1	0	TTLink_L2	D2D.0	HSIO.0	HSIO0←→D2D/CCE	Quasar-→ port TTLink TX				Prove that TTLink traffic can coexist with ttraffic that isn't TTLink. Preserve the headers for packets even with RAW and non-RAW mode if possible
		1	TTLink_L2	HSIO.0	CCE.0		TTLink source terminate to CCE				
		2	SVLAN	HSIO.0	CCE.0		SVLAN traffic we need to drop		Simple VLAN Tag Drop at TT Ethernet TCAM		

Test ID	Priority	Flow #	Packet Type	Ingress	Egress	Test Bench	Outcome	TT Config	Flow Purpose	Traffic Patterns	Overall Purpose
		3	CV/SVLAN	HSIO.0	D2D.0		ARP Broadcast to host				
		4	IPV4_BCAST_UDP	D2D.0	HSIO.0		ARP Reply from host				
4	0	0	TT_Link_L2_L4	HSIO.0	CCE.0	HSIO0←→D2D/CCE	Packet loops back to same port with header				
		1	TT_Link_L2_L4	CCE.0	HSIO.0		Packet loops back to same port with header				
		2	L4 PTP Event	HSIO.0	CCE.0		Captured by CCE				
		3	L4_PTP_General	HSIO.0	CCE.0		Captured by CCE				
5	0	0	TT_Link_L2	HSIO.0	CCE.0	HSIO0←→D2D/CCE					
		1	TT_Link_L2	HSIO.0	CCE.0						
6	0	0	TT_Link_L2	HSIO.0	CCE.0	HSIO0←→D2D/CCE	Packet loops back to same port with header				Basic TT Link L2 forwarding from HSIO.0 to CCE and back out HSIO.0
		1	TT_Link_L2	HSIO.0	CCE.0		Packet loops back to same port with header				
		2	L2 PTP Event	HSIO.0	CCE.0		L2 PTP frame is given to CCE and timestamp is updated				

Chapter 26. Export Restrictions

There is no encryption or AI capability in this chiplet. This chiplet does have aggregate data transfer capability that exceeds 150GB/s.

Chapter 27. Registers

Appendix A: ICL Description of iJTAG Network

This is a work in progress and is incomplete. This will also evolve as we receive the IP from the vendors and close the microarchitecture.